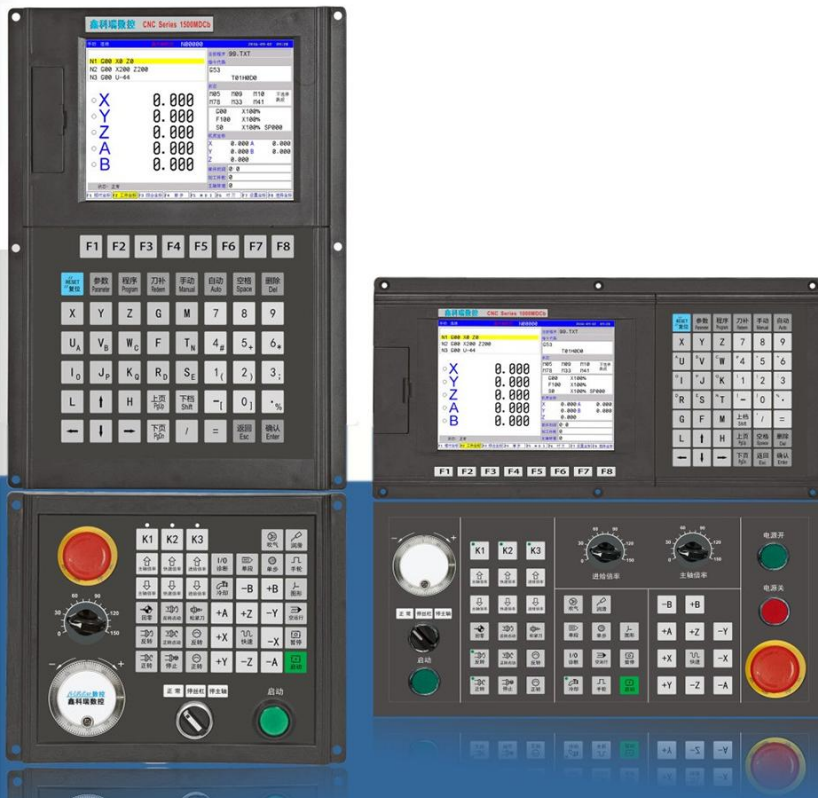


NEWKer



Making the most ideal and practical
product to service CNC world

NEW15/18iM Controller

Operation Manual

CHENGDU NEWKer CNC-TECHNOLOGY CO.,LTD

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Chapter 1 Preface

This system is our company's newly developed universal CNC system for milling machines, drilling machines, and boring machines. The control circuit adopts the latest industrial-grade high-speed ARM processor and large-scale FPGA technology, with multi-layer PCB boards. The entire machine uses highly integrated chips and surface-mounted components, making the structure more compact and reasonable, better ensuring system reliability and stability. It features high-speed real-time control (maximum rapid traverse speed up to 240 m/min, maximum feed interpolation speed up to 30 m/min) and high precision. The system employs 800x600 dot matrix TFT LCD with adaptive brightness technology, featuring uniform LED backlight with long service life, overcoming the drawback of display brightness being affected by ambient temperature. Full-screen Chinese menu display makes operation extremely simple and convenient.

This system is a universal fully digital CNC system supporting 3-axis to 8-axis coordinated motion (with milling machines as typical application), featuring bus-based closed-loop control. It has powerful functions and rich instruction set, with programming codes conforming to ISO international standards. It directly controls AC servo systems, and when paired with our company's dual-channel AC servo drives, offers even better cost performance. This manual provides detailed instructions on programming and operating this CNC system.

Important Notes:

- 1. All functions for A, B, C, Xs, and Ys axes are only effective when the 4th to 8th axis options are configured.**
- 2. Improper operation of this CNC system may cause accidents. Please read this manual carefully before use.**
- 3. All absolute position functions require the system to be configured as an absolute position system. All bus functions require the system to be configured as a bus system.**
- 4. The "Start" button on the system panel is for debugging only (set as "enabled" or "disabled" via Other Parameter P9-D14 bit, must be set to "0" during normal use). An external "Start" button must be installed during system assembly. Otherwise, safety accidents may occur due to limited lifespan of the panel button!!! Therefore, prolonged use of the panel "Start" button is strictly prohibited, and our company shall not be responsible for any safety consequences resulting from such use!!!**

Chapter2 System installation and connection

2.1 System electrical specification

- 32bits high performance industrial grade ARM+DSP+FPGA
- 128M User storage space
- 800x600 TFT LCD adaptive brightness, LED backlight LCD
- Touch type key board to have excellent operational sensitivity.
- RS232 communication port
- USB port
- Highly anti-interference of switching power supply.
- Two-way spindle to variable frequency and speed governing
- Manual pulse generator
- The band switch trim the feeding speed and spindle speed

2.2 System technical index

- Controllable axis: X Y Z A B XS YS 8 axis
- interpolation: straight line X Y Z A B XS YS 8 axis, arc X Y Z any two axis
- Pulse equivalency: 0.001mm
- Maximum speed: 240m/min
- Processing speed: 0.01-30000mm/min
- Minimal input unit: 0.001mm
- Programme size range: ± 99999.999 mm
- Programme coordinate system definition: IOS-840 international standard
- Programme code: IOS-841 international standard
- Mean Time Between Failure (MTBF): Greater than 6000 hours

2.3 Environment of operation

- Power supply: AC 220V (+10%,-15%), frequency 50Hz $\pm 1\%$
- Power source ≤ 100 W
- Running temperature: 5~45°C, relative humidity: 40~80%
- Storage and transportation temperature: 0~55°C, relative humidity less than 90%(40°C)

2.4 System installation and connection

Before installation, first check whether the required hardware components including the CNC system, drive power supply, servo motors, optical encoders, and motorized tool turret are complete, undamaged and properly matched.

The CNC system must be firmly installed with sufficient space around it to ensure air circulation. The control panel should be positioned for convenient operation while avoiding

contact with machining chips that may cause burns.

Separate high-voltage and low-voltage wiring. The CNC system and drive power supply should be separated from the machine's high-voltage circuits as much as possible. Keep all signal cables away from AC contactors to reduce interference. It is best to connect optical encoders, limit switches and reference point signals directly to the CNC system without passing through the high-voltage cabinet. Power cables must be properly grounded.

Securely connect all plugs and tighten fastening screws. Never plug or unplug drive, motor or signal cable connectors while powered.

When installing the CNC system, protect the panel from damage by hard or sharp objects. If painting is required, remove the CNC system first to avoid staining the panel.

The CNC system should be kept away from strong magnetic fields, high-voltage interference sources, flammable materials, explosives and other hazardous substances.

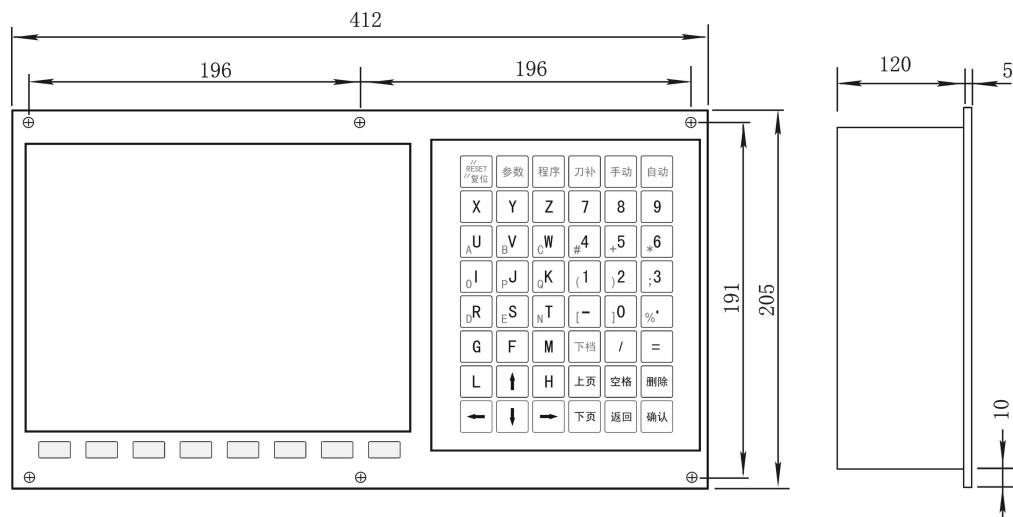
Precautions:

Must be installed in an electrical cabinet with proper lightning protection

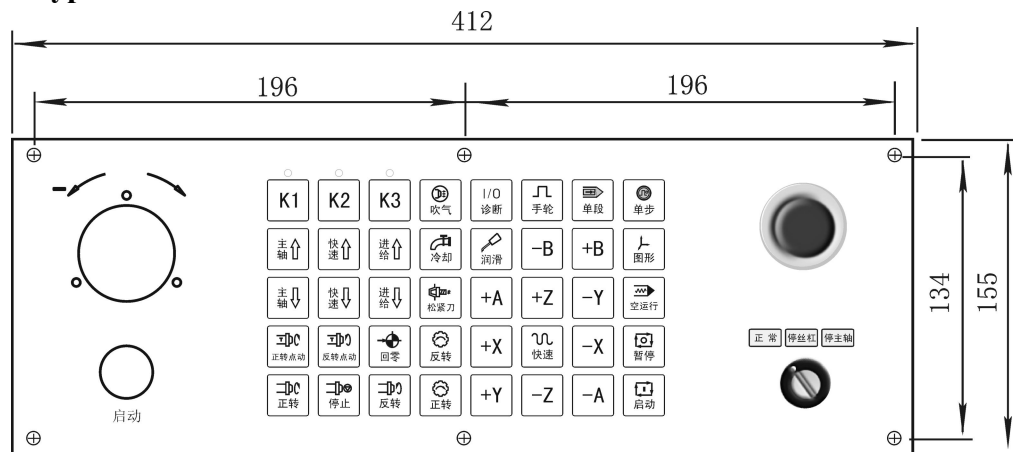
Must be securely mounted to prevent loosening from vibration

Do not install on or near flammable materials to prevent fire hazards

2.5 System installation dimension(412x205x120)

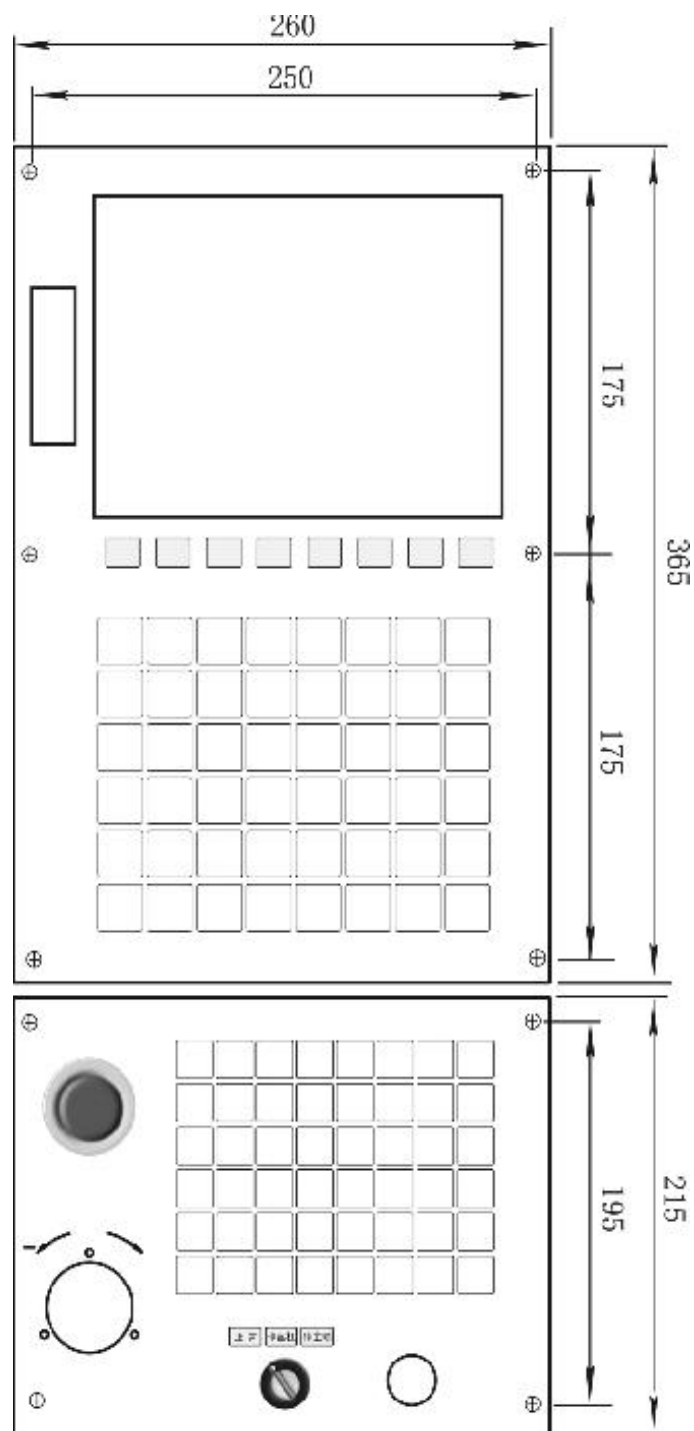


A type:



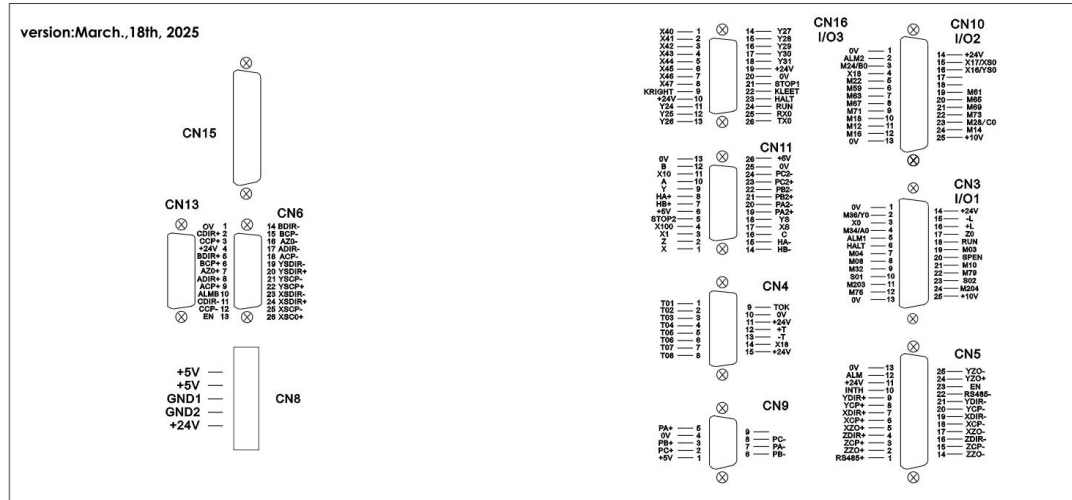
C type:

3、NEW15iM controller

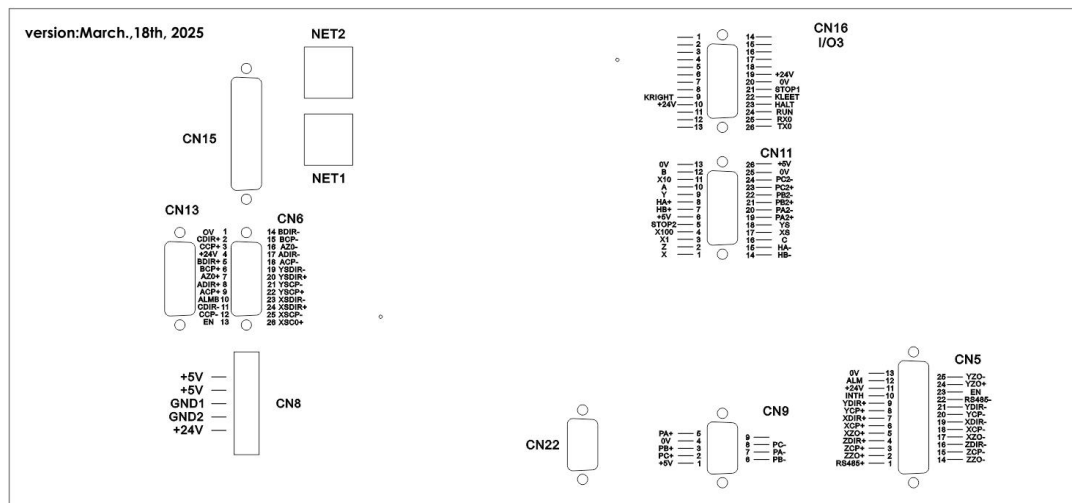


2.6 System rear view

1、NEW1000MDc/Mic controller



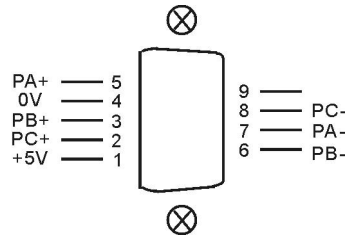
2、NEW18iM controller



Pay attention: Switching power supply L N must through isolation transformer and insert to AC 220V, current 0.5A.

2.7 Interface connection graph

2.7.1 Spindle encoder CN9 male socket(DB9)



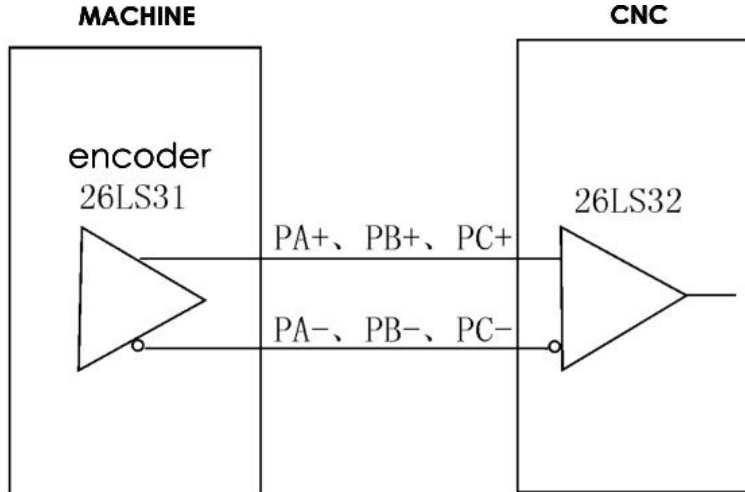
CN9 DB9(pin) spindle encoder				
signal	pin	I/O	function	availability
0V	4	OUT	0V	0V
+5V	1	OUT	+5V	+5V
PA+	5	IN	+A signal	5V
PA-	7	IN	-A signal	
PB+	3	IN	+B signal	5V
PB-	6	IN	-B signal	
PC+	2	IN	+Z signal	5V
PC-	8	IN	-Z signal	

Note:

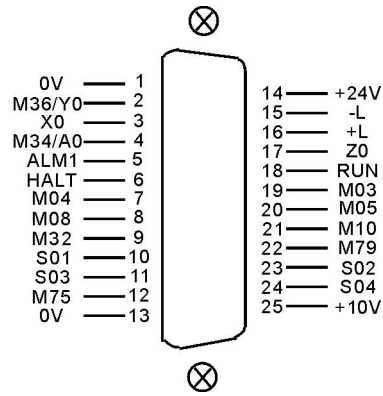
The encoder output signal adopts differential line driver output mode with +5V power supply.

Signal cables must use twisted-pair shielded cables, with length not exceeding 20 meters.

The input signal of encoder PA PB PC:



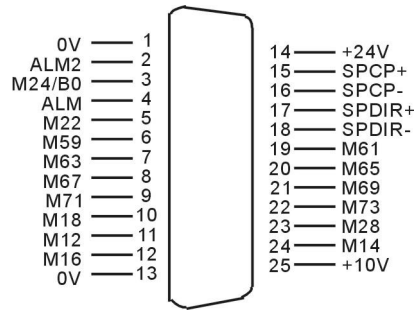
2.7.2 I/O 1 signal connection CN3 female socket(DB25)



CN3 I/O 1 Machine signal DB25 female socket				
Signal	Pin	I/O	Function	Effective level
0V	1	OUT	Power GND	0V
+24V	14	OUT	24V power	+24V
M36/Y0	2	IN	User define input 9/Y(C) axis homing signal	0V
X0	3	IN	X axis homing signal	0V
Z0	17	IN	Z axis homing signal	0V
-L	15	IN	Negative limit switch	0V
+L	16	IN	Positive limit switch	0V
M34/A0	4	IN	User define input 8/A axis homing signal	0V
ALM1	5	IN	Spindle servo alarm1	0V
HALT	6	IN	External <i>Pause</i> button	0V
RUN	18	IN	External <i>Run</i> button	0V
M03	19	OUT	Spindle CW	0V
M04	7	OUT	Spindle CCW	0V
M05/SPEN	20	OUT	Spindle stop/spindle EN	0V
M08	8	OUT	Coolant	0V
M10	21	OUT	Chuck tighten/loose	0V
M32	9	OUT	Lubrication	0V
M79	22	OUT	Tailstock	0V
S01	10	OUT	1 st Spindle 1st gear	0V
S02	23	OUT	1 st Spindle 2nd gear	0V
M203/S03	11	OUT	2 nd spindle CW/1 st Spindle 3rd gear	0V
M204/S04	24	OUT	2 nd spindle CCW/1 st Spindle 4th gear	0V
M75	12	OUT	Caxis control signal	0V
+10V	25	OUT	1 st spindle analog output	0~10V
0V	13	OUT	Analog output GND	0V

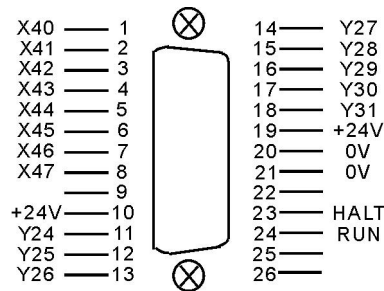
2.7.3 I/O 2 signal connection CN10 female socket(DB25)

Description: On the 2028G mainboard, CN10 pins 15 and 16 serve as the homing signals for the 7th and 8th axes respectively, while pin 4 is reserved for X18 input. Pins 17 and 18 are unconnected.



CN10 I/O 2 Machine signal DB25 femal socket				
Signal	Pin	I/O	Function	Effective level
0V	1	OUT	Power ground	0V
+24V	14	OUT	24V power	+24V
SPCP+(X17/XS0)	15	OUT	Spindle pulse signal+	0V
SPCP-(X16/YS0)	16	OUT	Spindle pulse signal-	
SPDIR+/null	17	OUT	Spindle direction signal+	0V
SPDIR-/null	18	OUT	Spindle direction signal-	
ALM(X18)	4	IN	Servo drive alarm	0V
ALM2	2	IN	Lubrication error	0V
M24/B0	3	IN	User defined Input 7/B axis homing signal	0V
M22	5	IN	M01 Spindle orientation ready	0V
M59	6	OUT	Huff	0V
M61	19	OUT	User defined Output 1	0V
M63	7	OUT	User defined Output 2	0V
M65	20	OUT	User defined Output 3	0V
M67	8	OUT	User defined Output 4	0V
M69	21	OUT	User defined Output 5	0V
M71	9	OUT	User defined Output 6	0V
M73	22	OUT	User defined Output 7	0V
M18	10	IN	User defined Input 1	0V
M28/C0	23	IN	User defined Input 2/C axis homing signal	0V
M12	11	IN	User defined Input 3	0V
M14	24	IN	User defined Input 4	0V
M16	12	IN	User defined Input 5	0V
+10V	25	OUT	The 2 nd analog output	0~10V
0V	13	OUT	Analog output GND	0V

2.7.4 I/O 3 signal connection CN16 female socket(DB25)



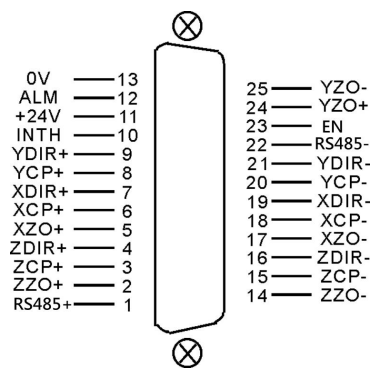
CN16 I/O3 Machine signal DB26 femal socket				
Signal	Pin	I/O	Function	Effective level
0V	20、21	OUT	Power ground	0V
+24V	10、19	OUT	24V power	+24V
X40	1	IN	Spare input 0	0V
X41	2	IN	Spare input 1	0V
X42	3	IN	Spare input 2	0V
X43	4	IN	Spare input 3	0V
X44	5	IN	Spare input 4	0V
X45	6	IN	Spare input 5	0V
X46	7	IN	Spare input 6	0V
X47	8	IN	Spare input 7	0V
RIGHT	9	IN	Right override switch	0V
Y24	11	OUT	Spare output 0	0V
Y25	12	OUT	Spare output 1	0V
Y26	13	OUT	Spare output 2	0V
Y27	14	OUT	Spare output 3	0V
Y28	15	OUT	Spare output 4	0V
Y29	16	OUT	Spare output 5	0V
Y30	17	OUT	Spare output 6	0V
Y31	18	OUT	Spare output 7	0V
STOP	21	IN	External E-stop	0V
LEFT	22	IN	Left override switch	0V
RUN	24	IN	External Run signal	0V
HALT	23	IN	External Stop signal	0V
TXD	25	OUT	RS232 send	
RXD	26	IN	RS232 receive	

2.7.5 CN4 wiring(DB15)

CN4 I/O 4 signal DB15				
Signal	Pin	I/O	Function	Effective level
0V	10	OUT	Power GND	0V
+24V	11/15	OUT	24V power supply	+24V
Y18	12	OUT	+T spare output	0V
Y19	13	OUT	-T spare output	0V

X18	14	IN	X18 spare input	0V
X00	1	IN	T1 spare input	0V
X01	2	IN	T2 spare input	0V
X02	3	IN	T3 spare input	0V
X03	4	IN	T4 spare input	0V
X04	5	IN	T5 spare input	0V
X05	6	IN	T6 spare input	0V
X06	7	IN	T7 spare input	0V
X07	8	IN	T8 spare input	0V
X21	9	IN	TOK spare input	0V

2.7.6 X/Y/Z servo driver control signal connection CN5 male socket(DB25)



CN5 Servo driver control signal DB25 male socket				
Signal	Pin	I/O	Function	Effective level
XCP+/-	6/18	OUT	X axis pulse	5V
XDIR+/-	7/19	OUT	X axis direction	5V
YCP+/-	8/20	OUT	Y axis pulse	5V
YDIR+/-	9/21	OUT	Y axis direction	5V
XZO+/-	5/17	IN	X axis motor zero	5V
ZCP+/-	3/15	OUT	Z axis pulse	5V
ZDIR+/-	4/16	OUT	Z axis direction	5V
ZZO+/-	2/14	IN	Z axis motor zero	5V
YZO+/-	24/25	IN	Y axis motor zero	5V
ALM/0V	12/13	OUT	Servo drive alarm	0V
+24V/EN	11/23	OUT	+24V/EN	24V
INTH	10	OUT	Servo drive alarm reset	0V
RS485+/-	1/22		Absolute RS485	

2.7.7 A/B servo drive CN6 female socket(DB15)

CN6 DB15 A/B servo drive				
signal	pin	I/O	function	availability
ACP+/-	1/9	OUT	A axis pulse	5V
ADIR+/-	2/10	OUT	A axis direction	5V

BCP+/-	4/12	OUT	B axis pulse	5V
BDIR+/-	5/13	OUT	B axis direction	5V
AZ0+/-	3/11	IN	A axis motor zero	5V
BZ0+/-	6/14	IN	B axis motor zero	5V
ALM/0V	7/8	IN	Servo drive alarm/0V	0V
+24V	15	OUT	+24V power supply	24V

2.7.8 A/B/C servo drive(2028G motherboard) CN6 female socket(DB26)

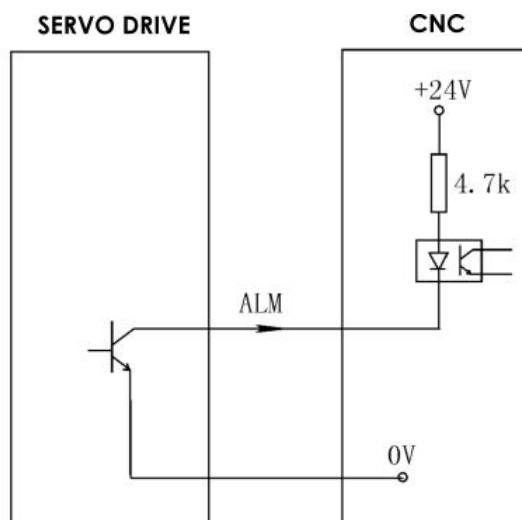
CN6 DB26 A/B/C servo drive				
signal	pin	I/O	function	availability
ACP+/-	9/18	OUT	A axis pulse	5V
ADIR+/-	8/17	OUT	A axis direction	5V
AZ0+/-	7/16	IN	A axis motor zero	5V
BCP+/-	6/15	OUT	B axis pulse	5V
BDIR+/-	5/14	OUT	B axis direction	5V
+24V/EN	4/13	OUT	+24V/EN	5V
CCP+/-	3/12	OUT	C axis pulse	5V
CDIR+/-	2/11	OUT	C axis direction	5V
XSCP+/-	26/25	OUT	XS axis pulse	5V
XSDIR+/-	24/23	OUT	XS axis direction	5V
YSCP+/-	22/21	OUT	YS axis pulse	5V
YSDIR+/-	20/19	OUT	YS axis direction	5V
ALMB/0V	10/1	IN	Servo drive alarm/0V	0V

Note:

1. Drive signal cables must use twisted-pair shielded cables, with a maximum length of 20 meters.

2. The ALM (alarm) signal input mode (NO/NC) is determined by Other Parameter P17.

***Servo drive alarm ***



2.7.9 External switch, MPG CN11 female socket (DB15/bus type DB26)

CN11 DB15 MPG DB26(female socket)				
Signal	Pin	I/O	Function	Effective level
0V	13	OUT	Power GND	0V
+5V	6	OUT	5V power	+5V
PA+	8	IN	+A signal	5V
PA-	15	IN	-A signal	
PB+	7	IN	+B signal	5V
PB-	14	IN	-B signal	
STOP	5	IN	External Emergency stop	0V
H_B	12	IN	B axis selection	0V
H_X100	4	IN	100 ratio signal	0V
H_X10	11	IN	10 ratio signal	0V
H_X1	3	IN	1 ratio signal	0V
H_A/HALT	10	IN	A axis select/stop signal	0V
H_Z	2	IN	Z axis select signal	0V
H_Y/RUN	9	IN	Y axis select/run signal	0V
H_X	1	IN	X axis select signal	0V
H_C	16	IN	C axis select signal	0V
H_XS	17	IN	XS axis select signal	0V
H_YS	18	IN	YS axis select signal	0V
Z2_A+/-	19/20	OUT	2 nd encoder A signal	
Z2_B+/-	21/22	OUT	2 nd encoder B signal	
Z2_C+/-	23/24	OUT	2 nd encoder C signal	
0V/+5V	25/26	OUT	power	

Pay attention: The inner power supply are all +5V of all signal, do not access voltage higher than 5V.

2.7.9.1 Usage for MPG(Manual pulse generator)

Note: When "Other Parameter" P1 is set to 1, a standard MPG can be connected.

A, X, Y, Z, X1, X10, X100 are used to select the feed axis and gear position (speed multiplier).

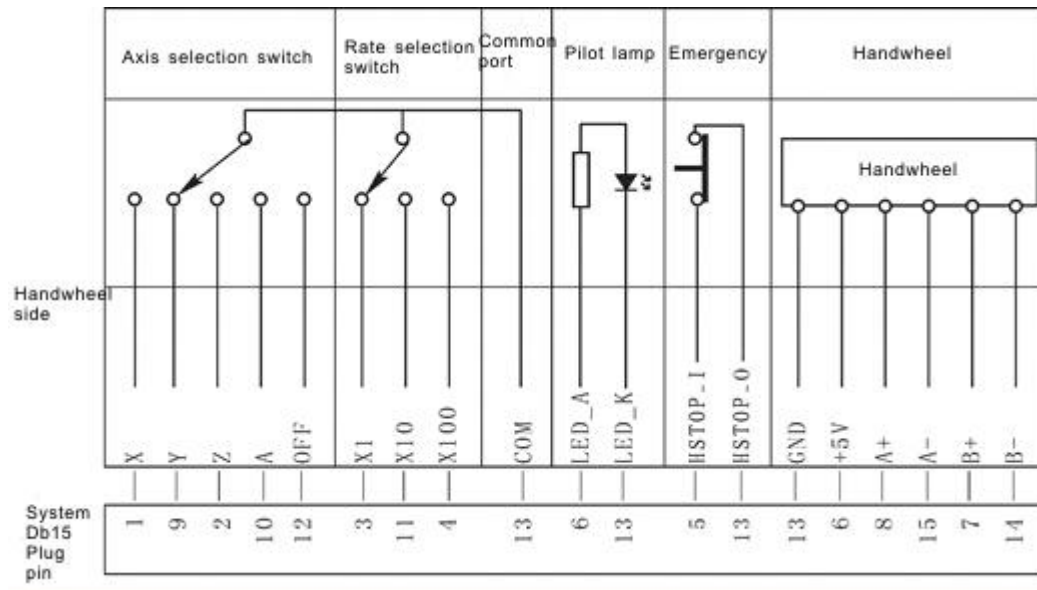
P33 and P34 under "Other Parameters" must also be set to 0 in this mode.

Signal Definitions:

PA+ / PA- → Quadrature pulse A input (Phase A)

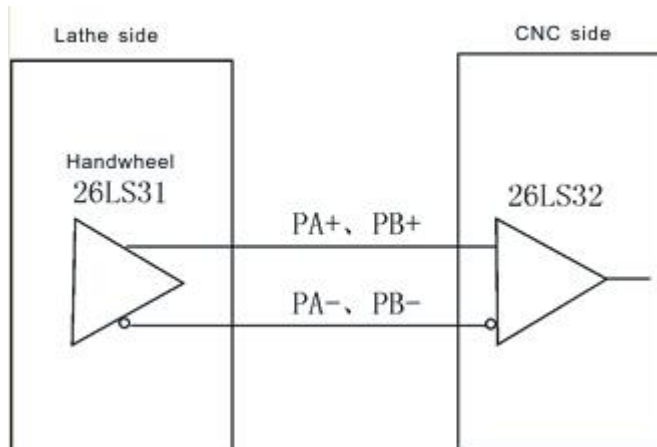
PB+ / PB- → Quadrature pulse B input (Phase B)

MPG connect diagram:

**Pay attention:**

1. The output signal of handwheel adopts line output, the power supply is +5V.
2. Just connect PA+ PB+ if adopt voltage output.
3. Manual pulse generator needn't "Enter" button, if there is a "Enter" button, use the line to short the ends of switch.
4. The inner power supply are all +5V of all signal, do not access voltage higher than 5V.

The input signal of handwheel:

**2.7.9.2 Usage of terminal band switch behind sub panel**

Signal	CN15 Pin	I/O	Function	Effective level
0V	29、39	OUT	Power GND	0V
+5V	11、26	OUT	5V power supply	+5V
PA+	41	IN	A Signal +	5V
PA-	12	IN	A Signal -	
PB+	27	IN	B Signal +	5V
PB-	42	IN	B Signal -	

STOP	44	IN	External E-Stop	0V
KRUN	14	IN	Run	0V
KLEFT	15	IN	Left override switch	0V
KRIGHT	30	IN	Right override switch	0V
DS0	8	IN	Spindle baud byte 1	0V
DS1	38	IN	Spindle baud byte 2	0V
DS2	23	IN	Spindle baud byte 3	0V
DS3	37	IN	Spindle baud byte 4	0V
DK0	7	IN	Servo axis band byte 1	0V
DK1	22	IN	Servo axis band byte 2	0V
DK2	9	IN	Servo axis band byte 3	0V
DK3	24	IN	Servo axis band byte 4	0V
LRUN	43	OUT	Program run output	0V

When both Axis Parameters P1 and P2 are set to "1", the band switch can be used. DS0, DS1, DS2, DS3 are the input signals for spindle speed override gears, with 4-bit control for a total of 16 gears.

DK0, DK1, DK2, DK3 are the input signals for feed rate override gears, with 4-bit control for a total of 16 gears.

Warning:

All signals are internally powered by +5V — DO NOT connect any voltage higher than 5v

2.7.9.3 external E-stop button usage

Pin CN3-18 can be configured as an external start button for program auto-run.

Pin CN3-6 can be configured as an external stop button for program pause.

2.7.10 Axis position feedback signal CN13 male socket(DB26)

CN13 Position feedback or linear scale signal DB26male socket				
Signal	Pin	I/O	Function	Effective level
+5V/0V	9/10	OUT	+5V/0V	
+24V/0V	26/23	OUT	+24V/0V	
DX+/DX-	25/24		RS485+/RS485-	
XA+/-	18/8	IN	X axis A signal	5V
XB+/-	17/7	IN	X axis B signal	5V
YA+/-	14/4	IN	Y axis A signal	5V
YB+/-	13/3	IN	Y axis B signal	5V
ZA+/-	16/6	IN	Z axis A signal	5V
ZB+/-	15/5	IN	Z axis B signal	5V
AA+/-	12/2	IN	A axis A signal	5V
AB+/-	11/1	IN	A axis B signal	5V
BA+/-	22/21	IN	B axis A signal	5V
BB+/-	20/19	IN	B axis B signal	5V

Other Parameters P200 to P220 configure this function. After an alarm occurs, press "G" in the diagnostic interface to:

Zero both command position and feedback position

Clear the following error alarm

Notes:

Encoder/scale signals use differential line driver output (+5V power supply).

Signal cables must use twisted-pair shielded cables (max length: 20m).

This is an optional function.

2.7.11 bus type controller Ethernet CN22 socket(DB15)

335G motherboard CN22 bus type Ethernet signal DB15(6 pair)				
Signal	Pin	I/O	Function	Effective level
TXA-/TXA+	1/2(1 pair)		bus A signal	
TXB-/TXB+	3/4(2 pair)		Bus B signal	
+24V/0V	5/6(3 pair)	OUT	+24V/0V	Power supply
+24V/0V	7/8(4 pair)	OUT	+24V/0V	
+5V/0V	11/12(5pair)	OUT	+5V/0V	
485+/485-	10/15(6pair)		485+/485-	

2026G/5353G motherboard CN22 bus type Ethernet DB15(6pair)				
Signal	Pin	I/O	Function	Effective level
T422+/T422-	1/2(1pair)		Bus A signal	
R422+/R422-	3/4(2pair)		Bus B signal	
X22/X20	11/6(3pair)	IN	X22/X20	
+5V/EN	12/7(4pair)	OUT	+5V/EN	Power supply
+24V/0V	13/8(5pair)	OUT	+24V/0V	
+24V/0V	14/9(6pair)	OUT	+24V/0V	

2.7.12 Bus type controller servo drive pulse signal socket CN21(DB26)

CN21 servo drive pulse signal DB26				
Signal	Pin	I/O	Function	Effective level
DX+/DX-	1/10		RS485+/RS485-	
ACP+/ACP-	2/3	OUT	A axis pulse	
ADIR+/ADIR-	4/5	OUT	A axis direction	
XSCP+/XSCP-	6/7	OUT	XS axis pulse	
XSDIR+/XSDIR-	8/9	OUT	XS axis direction	
XCP+/XCP-	11/12	OUT	X axis pulse	
XDIR+/XDIR-	13/14	OUT	X axis direction	
ZCP+/ZCP-	15/16	OUT	Z axis pulse	
ZDIR+/ZDIR-	17/18	OUT	Z axis direction	
C/YCP+/-	19/20	OUT	Y axis pulse	
C/YDIR+/-	21/22	OUT	Y axis direction	
SP/YSCP+/-	23/24	OUT	YS axis pulse	
SP/YSDIR+/-	25/26	OUT	YS axis direction	

Important Notice for RS485 Operation:

When using RS485 functionality (connected to Pin 1 & Pin 10), the system 0V reference must be connected to the driver signal 0V ground.

2.7.13 Bus type Ethernet socket CN24,CN25

*CN24 is designated for EtherCAT drive network interface

*CN25 is reserved for OPC functionality

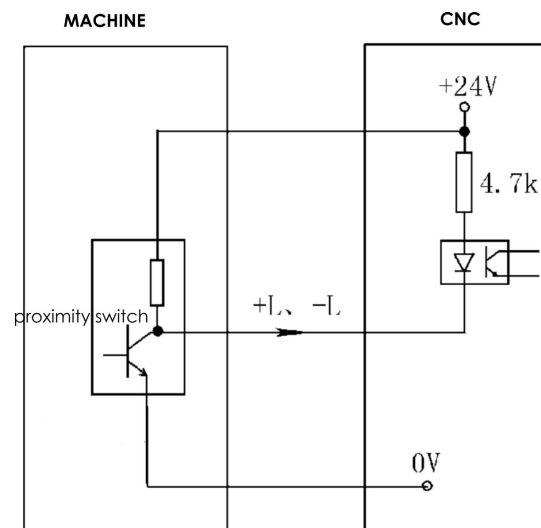
*When connecting pulse-type drives, CN24 may be alternatively configured for OPC functionality

*Configuration is selected via Miscellaneous Parameter P350a

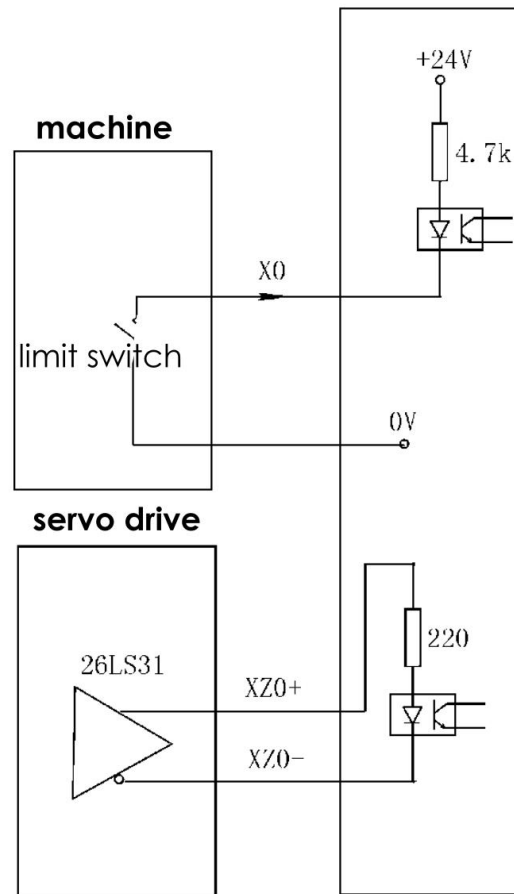
2.8 machine electronic parts installation**2.8.1 hard limit**

[e.g. X, Z axis positive hard limit(+L),same as -L]

Use NPN type hall switch

**2.8.2 mechanic zero**

[e.g. X axis(Z axis same)]



When the floating zero function is disabled in system settings, homing requires detection of both the proximity switch signal and the motor Z-pulse signal. This is achieved by setting Axis Parameter P46 to "00000000".

The homing function requirements are configured through Axis Parameter P45:

The system provides six startup homing modes:

- 1) Set to 1: No prompt or restriction after power-on.
- 0) Set to 0: Prompt mode - displays dialog reminding operator to perform homing after power-on (no enforcement).
- 8) Set to 8: Enforced mode - blocks program execution until homing is completed, displays "Feed axis not referenced" if skipped.
- 9) Set to 9: Strict enforcement mode - prevents any axis movement until homing is completed, displays "Feed axis not referenced" if skipped.

Homing detection method (Axis Parameter P46):

- 0) After triggering limit switch → Reverse → Detect switch release → Seek Z-pulse
 - 1) After triggering limit switch → Reverse → Detect switch release only
 - 2) After triggering limit switch → Continue forward → Detect switch release → Seek Z-pulse
- Other values: After triggering limit switch → Continue forward → Detect switch release only
- Note: For precision, modes 0 or 2 are recommended as they verify the encoder Z-pulse. Switch-only homing (modes 1/others) relies solely on switch accuracy.

Homing direction/sequence (Axis Parameter P47 - bitwise):

- D2: X-axis direction (1=negative, 0=positive)
- D4: Z-axis direction (1=negative, 0=positive)

D8: Sequence (1=Z-first, 0=X-first)

Limit switch type (Axis Parameter P48 - bitwise):

D0: X-axis (1=NC, 0=NO)

D2: Z-axis (1=NC, 0=NO)

Additional settings:

- **Z-pulse search distance: Axis Parameters P49-P53**

- **Post-homing offset: Axis Parameters P54-58 (executed at G00 speed)**

(Note: Offset values depend on machine reference point installation and coordinate system configuration)

Important:

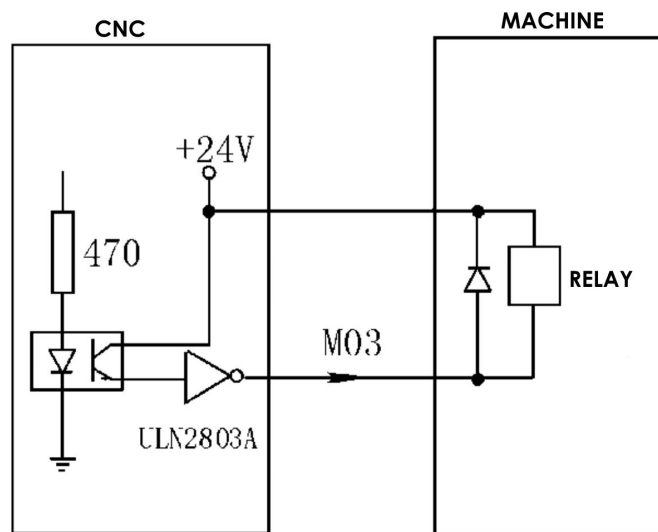
1) Deceleration switches may use NPN Hall sensors

2) Installation must ensure deceleration travel length $\leq 25\text{mm}$

2.8.3 machine electronic control signals

Taking M03 as an example (same applies to M04, M05, M08, M10, M32, M79, M75, M59, M61, M63, M65, M67, M69, M71, M73, S01-S04)

M03 output signal diagram:



As shown in the diagram, when the system issues M03, it completes a circuit with 24V DC, activating the intermediate relay. One set of the relay's NO (Normally Open) contacts energizes the coil circuit of the spindle forward rotation AC contactor, initiating spindle forward rotation.

All output signals are active-low (0V effective).

Critical Notes:

1. For inductive loads (relays, contactors):

Parallel diodes must be installed to absorb back-EMF and prevent system damage.

RC snubber circuits are additionally required for electromagnetic contactors.

2. ULN2803A chip output mapping:

U28: M59, M61, M63, M65, M67, M69, M71, M73

U29: M03, M04, M05, M08, M10, M79, M32, M75

U30: +T, -T, S01, S02, S03, S04, LRUN, INTH

3. Custom signal restrictions:

M70/M71 and M72/M73: When configured as chuck/center pin control signals, they cannot be repurposed for other custom functions.

M65/M67/M69: When used as runtime alarm outputs:

M65: Program stop output

M69: Program running output

M67: Alarm output

These signals become unavailable for other custom assignments.

2.8.3.1 spindle control

System Outputs Two Channels of Spindle (First and Second Spindle) Control Signals, Relevant Parameters as Follows:

Axis Parameters

P13-2: Sets spindle braking time (i.e., M05 output hold time).

Shorter setting → Faster braking.

Longer setting → Slower braking.

Unit: 10ms.

P13-1: Sets spindle brake signal type (1 = Long signal, 0 = Short signal).

P65: Enables/disables spindle position feedback detection (encoder signal).

1 = Detect, 0 = Do not detect.

P200: Spindle encoder pulses per revolution (4× line count).

P200-1: Spindle encoder line count alarm threshold (diagnostic value, valid when >10).

P59: Spindle start during gear change (1 = Yes, 0 = No).

P60: Motor speed during gear change (1/100 rpm).

P61: Rotation direction during gear change (0 = Forward, 1 = Reverse).

P62: Stop time during gear change (10ms).

P63: Low-speed rotation time during gear change (10ms).

P64: Spindle stop delay time (10ms).

P220: First spindle full-pulse control channel (Forward: 82, Reverse: 92).

P221: First spindle pulses per revolution (full-pulse control).

P222: First spindle acceleration/deceleration rate (rpm/2ms).

P223: Second spindle full-pulse control channel (Forward: 84A/85B/88Ys, Reverse: 94A/95B/98Ys).

P224: Second spindle pulses per revolution (full-pulse control).

P225: Second spindle acceleration/deceleration rate (rpm/2ms).

P226: Second spindle speed ratio numerator (following first spindle).

1 = Invalid, 1000 + Numerator = Follow first spindle.

P227: Second spindle speed ratio denominator (following first spindle).

1 = Invalid, 1000 + Denominator = Follow first spindle.

P228: First spindle start/stop controls second spindle (0 = Yes, 1 = No).

Speed Parameters

P48: Maximum spindle speed (1st gear/high gear, at 10V command). Unit: rpm.

P49: Maximum low gear (2nd gear) speed or second spindle max speed (at 10V command).

Unit: rpm.

P50: Maximum 3rd gear speed (at 10V command). Unit: rpm.

P51: Maximum 4th gear speed (at 10V command). Unit: rpm.

P52: Second spindle maximum speed (at 10V command). Unit: rpm.

Other parameters

P13: Spindle-chuck interlock.

0 = No relation between chuck state and spindle rotation.

1 = Spindle can only rotate when chuck is clamped (safety interlock).

Note: Tailstock cannot be operated while spindle is running.

Safety recommendation: Enable interlock (1).

Control signal through M32

2.8.3.2 lubrication control(M32/M33)

The system can enable the automatic lubrication function by setting "Other Parameter" P4.

After enabling this function:

"Other Parameter" P6 sets the lubrication interval time (unit: seconds)

"Other Parameter" P5 sets the lubrication duration (unit: milliseconds)

The control signal is output through M32.

2.8.3.3 tool clamping/unclamping control(M10/M11)

The tool clamping/unclamping control parameters of this system are as follows:

Other Parameters

P13: Sets whether spindle rotation interlocks with tool clamping state.

0: No relation between tool clamping state and spindle rotation.

1: Interlock enabled (spindle can only rotate when tool is clamped).

P20: Sets tool clamping control method (depends on machine configuration).

0: Single signal control (M10 controls both clamping/unclamping).

M10 active: Tool clamps.

M10 inactive: Tool unclamps.

1: Dual signal control (M10 and M71).

M10 active + M71 inactive: Tool clamps.

M71 active + M10 inactive: Tool unclamps.

(M10 outputs M10, M78 outputs M71).

P22: External tool clamping control button/footswitch.

0: Disabled.

1: Enabled (toggle signal M16 changes state per activation: clamp↔unclamp).

P24: Tool clamp/unclamp signal hold time (short signal mode).

0: Long signal.

Unit: Seconds (s)

Note:M16 is a multi-function signal - only one function can be used at a time.

2.8.3.4 controller alarm signal: ALM, ALM1, ALM2, Door witch erro M12, E-stop

Other Parameters

P7: Door switch detection

- 0: No door switch
- 1: Door switch installed

(Safety recommendation: Enable detection)

P8: Door switch type

- 0: Normally open (NO)
- 1: Normally closed (NC)

P17: Servo alarm signal type (CN5 pin 12 ALM)

- 0: NO
- 1: NC

P18: Spindle alarm signal type (CN3 pin 5 ALM1)

- 0: NO
- 1: NC

P19: Machine alarm signal type (CN19 pin 2 ALM2)

- 0: NO
- 1: NC

P27: Emergency stop (STOP) switch type (CN11)

- 0: NO
- 1: NC

(Safety recommendation: Set to NC)

Emergency Stop (STOP):

When pressed during emergencies:

1. All machine motions stop
2. System resets with "EMERGENCY STOP" display
3. Requires button release to continue
4. When Other Parameter P29 is active:
 - M67 output signal activates (alarm output)
 - Can be used for safety protection (e.g. power cutoff)

Machine Alarm:

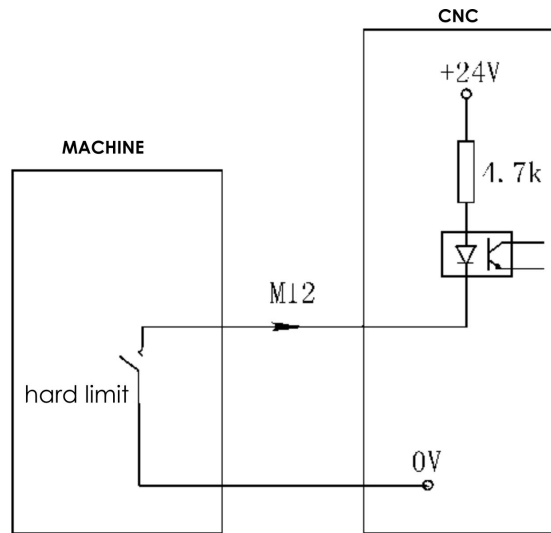
When Other Parameter P29 is active:

- M67 output signal activates during alarms

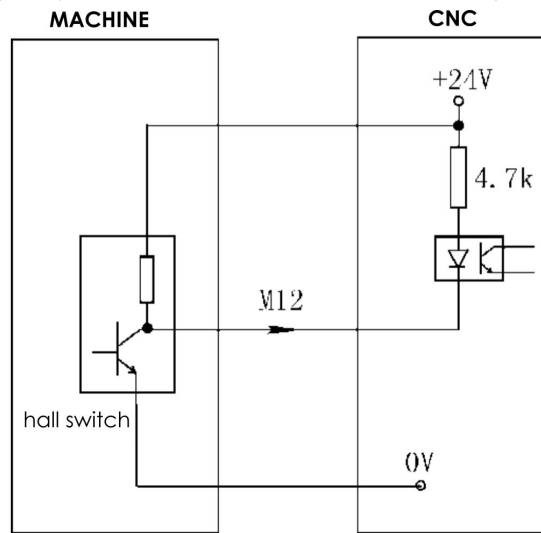
2.8.3.5 user defined input signal

M12(M14, M16,M18,M28,M22,M24,external RUN, external halt, external E-stop are same)

M12 switch wiring diagram(use hard limit switch)



M12 switch wiring diagram(use NPN hall switch as detect switch)



Note:

1. M12, M14, M16, M18, M28 are multifunctional signal, only one could be used.
2. All input signal are low level 0V effective

2.8.4 Absolute type controller setting steps

1. Turn on the power supply.
2. Set XYZA axis motor encoder type in Other parameter P300.
3. Set electrical gear in Axis parameter.

Numerator = 10×reduction ratio

Denominator = ball screw pitch

Axis parameter P11=00000001; soft limit effectively;

Axis parameter P23=11111011; home point with floating mode;

4. Set the other parameter in system

P300=01111100; the absolute value function;

P301=92; P302=91; P303=90;

P304/P305/P306/P307=131072;

$P309/P310/P311/P312 = (1/\text{electronic gear}) * 10000000$

5. check P56 parameter in driver(NEWKer driver)

P1=1, enter the password;

P56=1/2/3/4 corresponds to the X/Y/Z/A, such as the Z axis drive is set to 3;

Press "Enter" for a while to save parameter in EP-status;

6. Run M500 in MDI, the small circle in front of machine coordinate should become green, that means read the absolute encoder data successfully;

7. Check coordinate feedback direction:

1) move coordinate in manual mode and remember the machine coordinate value;

2) Run M500(read all axis), or M501/M502/M503/M504(corresponds to the X/Y/Z/A axis) in MDI, check if the absolute value coordinate is still before and after running M501-M504, then run axis again and run M500 again, if coordinate are always change. Go to Other parameter P309-P312, change the sign of value in P309-P312.

Sample:

If X Servo motor is connect with ball screw directly. And ball screw pitch is 5mm.

1. Axis parameter: electronic gear denominator P17=10, electronic gear numerator P18=5;

2. Other parameter $P309 = 5/10 * 10^7 = 5000000$;

A>Move X coordinate to 10.3456, then run M501 in MDI, check if coordinate is 10.34**, then move X coordinate to 25.1212, and run M501 in MDI again, to check if X coordinate is 25.12**. then move X coordinate to 30.4510, then run M501 in MDI again, and check if coordinate is 30.45**. then go to Step 8.

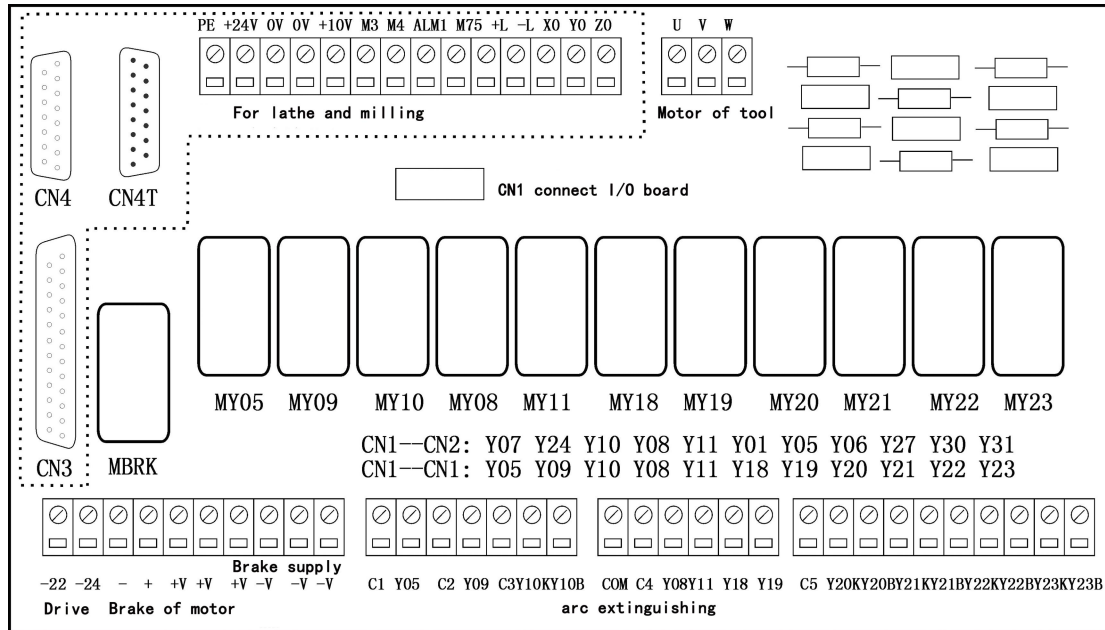
B>Move X axis to 10.3456, then run M501 in MDI, check if coordinate is 10.34**. if it is not. then move X coordinate to 25.1212, and run M501 in MDI again, to check if X coordinate is 25.12**. it changes less than 25 or bigger than 25. then change Other parameter=-5000000. then repeat Step A>;

8. Setting the zero point of machine: After finish step 4, move all axis to zero point, then set other parameter P314-P318, and input E into P314-P318, and press enter to clear the coordinates, finally, the current point is set as the zero point.

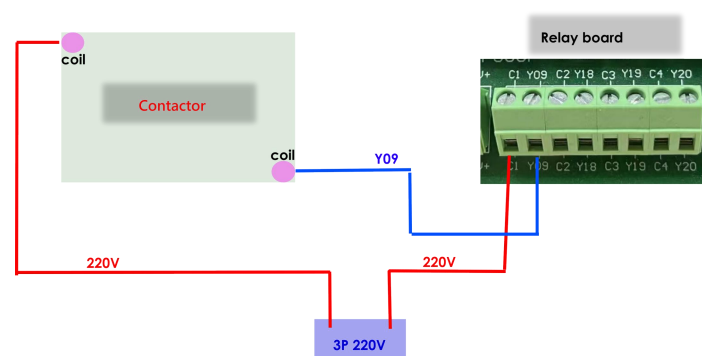
9. Manual move every axis to the limit position of machine to set the soft limit value in axis parameter;

10. Set the No.41 parameter in other parameter and back up the current system parameter

2.8.5 Relay board of controller(optional)

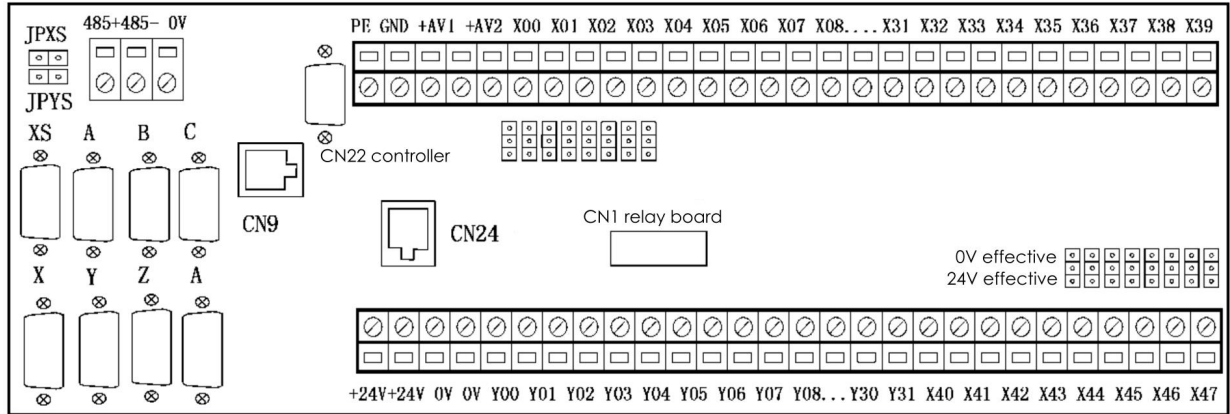


1. CN1 connects to relay board CN1. Pins CN1-1 to CN1-11 control relays: Y05, Y09, Y10, Y08, Y11, Y18, Y19, Y20, Y21, Y22, Y23 correspond to: M71, M79, M10, M32, M08, +T, -T, S04, S03, S02, S01.
2. _22 connects to driver CN3-22 pin, _24 connects to driver CN3-24 pin.
3. - and + connect to the brake motor.
4. -V and +V connect to external brake power supply.
5. The PE terminal must be securely connected to the machine's grounding bar to prevent interference.
6. Dotted line section indicates connections for standard systems:
 - CN3 and CN4 connect to system CN3 and CN4
 - CN4T connects to the motorized tool turret signal socket
7. Jumpers JX1/JX2 select whether relays MY20/MY21 control M03/M04 or S04/S03.
8. Common terminals:
 - C1: Common for M71/Y05
 - C2: Common for M79/Y09
 - C3: Common for M10/Y10
 - C4: Common for M32/Y08, M08/Y11, +T/Y18, -T/Y19
 - C5: Common for S04(M03), S03(M04)
 - C5: Common for S02/Y22, S01/Y23
9. relay board common terminal and output example

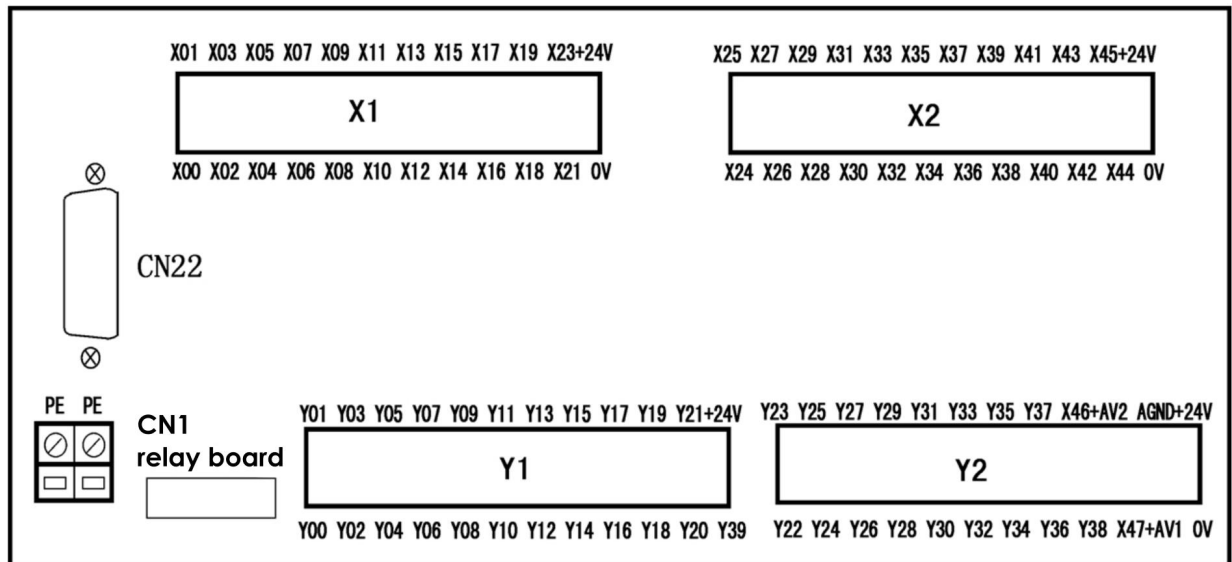


Bus type controller I/O board

(1) 335G I/O board



(2) 2026G I/O board



- 1) CN22 connects to controller;
- 2) CN24 connects to POWERlink bus driver;
- 3) CN9 connects to RS485 communication port;
- 4) CJ1, CJ2 connect to input/output signals;
- 5) CJ3, CJ4 connect to standard RS485 communication;
- 6) XYZABC/XS/YS connect to pulse-type servo drivers;
- 7) JPXS, JPYS jumpers: When connecting pulse-type drivers, if alarm signal is normally closed, install JPXS/JPYS when XS/YS axis is absent (remove corresponding jumper when XS/YS axis exists);
- 8) +AV1, +AV2 are dual-channel analog 0-10V output signals, 0V is signal ground;
- 9) CN1 connects to relay board CN1, pins CN1-1 to CN1-11 control relays:
Y05, Y09, Y10, Y08, Y11, Y18, Y19, Y20, Y21, Y22, Y23 correspond to:
M71, M79, M10, M32, M08, +T, -T, S04, S03, S02, S01;

Special note:

PE terminal must be securely connected to grounding bar to prevent interference.

1. Inputs

X0-X07: Motorized turret T01-T08;

X20: XYZA driver alarm ALM;

X21: Motorized turret position signal TOK;

X22: B/C/XS/YS driver alarm ALMB;

X23: Spindle inverter alarm;

X24: Low lubricant alarm ALM2;

X09: Negative limit switch -L;

X10: Positive limit switch +L;

X08/X11/X12/X13/X26/X25/X17/X16: A/Y/X/Z/B/C/XS/YS axis home signals
A0/Y0/X0/Z0/B0/C0/XS0/YS0;

X27/M22; X28/M18; X29/M12; X30/M14; X31/M16;

X14/X15/X18/X19/X40-X47: User-defined input points;

2. Outputs

Y16: Drive enable EN (also brake output for bus-integrated drive motors);

Y17: Drive alarm clear INTH;

Y30(SPEN): Spindle servo enable;

Y31(SPEN2): Second spindle servo enable;

Y00/M61; Y01/M63; Y02/M65; Y03/M67; Y04/M69; Y05/M71;

Y06/M73; Y07/M59; Y08/M32; Y09/M79; Y10/M10; Y11/M08;

Y12/M05; Y13/M04; Y14/M03; Y15/M75; Y16/EN; Y17/INTH;

Y18/+T; Y19/-T; Y20/S04; Y21/S03; Y22/S02; Y23/S01;

Y28/M203; Y29/M204; Y30/SPEN; Y31/SPEN2;

Y24-Y27: User-defined output points;

2.8.6 pulse type servo drive socket X/Y/Z/A/B/C/XS/YS(DB15)**2.8.6.1 Bus type controller with pulse type servo drive wiring chart**

XYZABC/XS/YS pulse type servo drive DB15 (7pair)				301 servo drive	302 servo drive
Singal	Pin	I/O	Function	DB25	DB25
CP+/CP-	1/9(1pair)	OUT	Pulse	6/18	6/18
DIR+/-	2/10(2pair)	OUT	Direction	7/19	7/19
EN/INTH	3/4(3pair)	OUT	EN/ servo drive alarm reset	23/10	302G:23/10 302H:14/==
+24V/0V	11/13(4pair)	OUT	24V power supply/0V	11/21	302G:11/21 302G:2/25
BP+/-	15/8(5pair)	IN	Encoder feedback B	4/16	302G:null 302H:8/1
AP+/-	14/7(6pair)	IN	Encoder	3/15	320G:null

			feedback A		302H:21/20
ALMB-/+	6/5(7pair)	IN	Servo drive alarm NC	13/12	13/12
ALMK+/-	12/13(7pair)	IN	Servo drive alarm NO		
485+/-				CN2-5/6	5/17
BK+/-				22/24	22/24
2 nd channel pulse type servo drive DB15 pin(4pair)					
CP+/CP-	1/9(1pair)	OUT	Pulse		3/15
DIR+/DIR-	2/10(2pair)	OUT	Direction		4/16
BP+/BP-	15/8(3pair)	IN	Encoder feedback B		302G:null 302H:23/9
AP+/AP-	14/7(4pair)	IN	Encoder feedback A		302G:null 302H:11/10
ALMB-/ALMB+	6/5(shortcircuit)	IN	Servo drive alarm NC		

2.8.6.2 servo drive (powerlink) bus type controller ethernet port(CN24)

Connect to power link servo drive ethernet port

CN24 servo drive Ethernet port signal(2pair)				
Signal	Pin	I/O	Function	Corresponding port
TXA-/TXA+	1/2	IN	A signal	2/1
TXB-/TXB+	3/6	IN	B signal	3/6

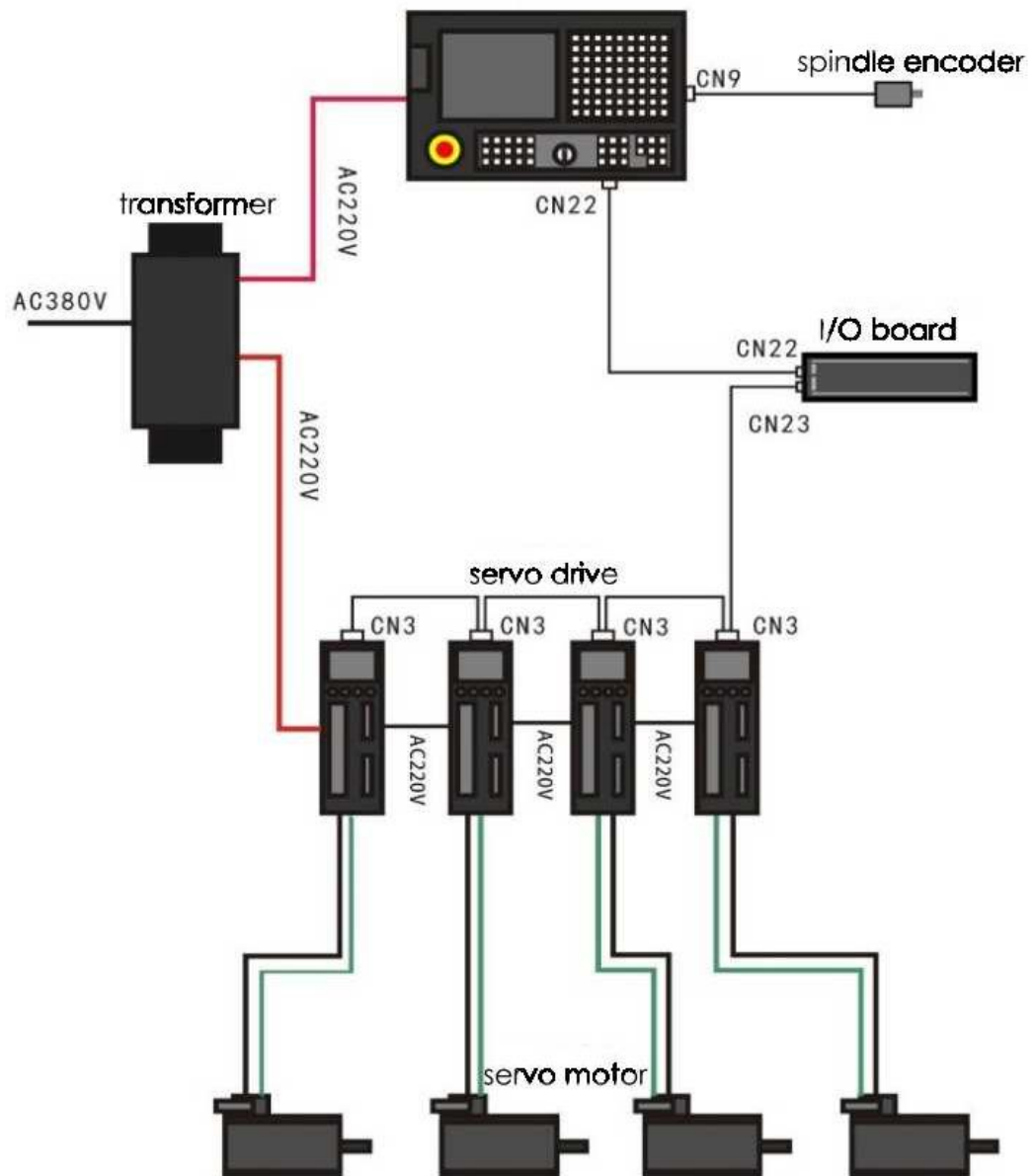
2.8.6.3 RS485 communication socket CN9

CN9 RS485 communication socket(2pair)				
signal	Pin	I/O	function	
485+/485-	4/5		Signal	
0V	1		Signal GND	

Note: Signal cables must use twisted-pair shielded cables.

2.8.7 bus type controller diagram

bus type controller servo drive wiring diagram



Chapter 3 Operation

3.1 Summary

When using this CNC system, as long as the master of the system parameters, the program edit, manual operation, automatically running, it can be very convenient to operate.

3.2 Operation panel

This system panel is composed of the main panel and side sub-panel. The main panel is used for parameter setting and program editing and the sub-panel is for tool setting and processing operations.

3.3 Keyboard description

3.3.1 Rate increase or decrease

(1) Rapid override(G)

There are six gears in rapid override form 5% to 100%, by adjusting the key of rapid override is for the following instruction: G00,G26,G28,G611,G613, rapid feed fixed cycle, rapid manual feed.

(2) Feed override(F)

There are sixteen gears in feed override from 0% to 150%, by adjusting the key of feed override is for the following instruction:G01,G02,G03, the feed override of the fixed cycle and manual run effectively.

(3) Spindle override(S)

There are sixteen gears in spindle override from 5% to 150%, by adjusting the key of spindle override is for the speed of the first spindle.

3.3.2 Usage for intervention switch

It is used to start the execution of processing program.

3.3.3 Others keys

Keyboards	Functions
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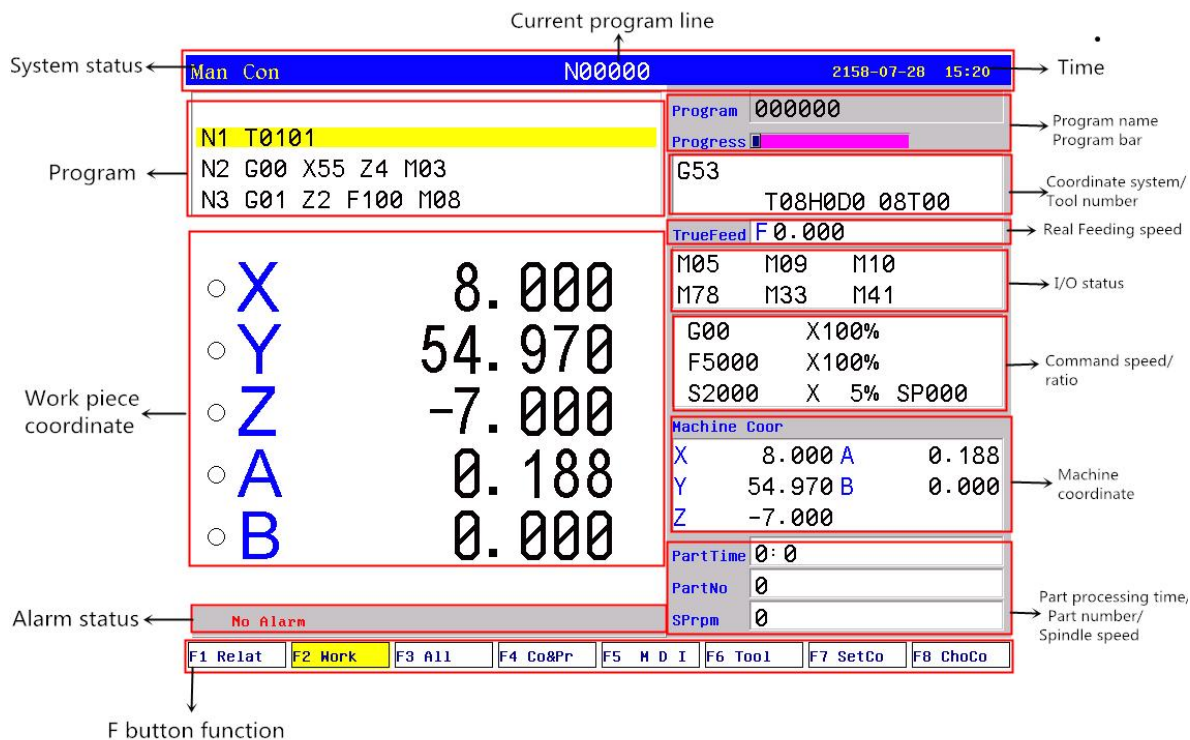
Letter key Number key	ABCSEFGHIJKLMNOPQRSTUVWXYZ123456789 .-: for program instructions, parameters' edition; number keys are used for inputting data and selecting sub-menu.
Edit key	“↑、↓、→、←、Del、PgUp、PgDn”for programming, direction keys can be used for selecting menu.
+X,-X +Y,-Y +Z,-Z +A,-A	It is used to feed manually.
Function key	“Esc” returning to upper level or stop a operation “Enter” selecting sub-menu and changing a newline “Del” delete program “program”entering program edition “parameter” entering parameter setting “diagnosis” entering diagnosis I/o function “manual” entering manual status “handwheel” for starting or stopping handwheel function “Tool” for confirming current tool 's position in machine too coordinates system. “Redeem” for amending tool change errors “Auto” entering automatic status “MDI” entering MDI function “”selecting auto-coordinates/diagram machining “”for single segment or constant work “” for coordinates mode or diagram mode speedy simulating “”for manual increment or manual continuous

Control key	“   ” spindle cw, ccw rotation “” coolant on/off “” for the shift between electric tool carrier and gang tool carrier “” for the shift between hand-driven continuous high speed and low speed. “” all axes return to home point “” for spindle chuck on/off “” for thumbstall on/off “ ” handwheel gear selection “ ” adjusting spindle speed “ ” adjusting feed speed “ ” adjusting G00 speed
Feed key	    +Y +A -Y -A For X、Y、Z 、A axes direction feed

3.4 Manual operation

The system adjusts one-level menu operation, intuitive, convenient, shortcut, prompt comprehensive information.

Powering the system is to enter the main interface



The left part of the first line on the screen is the running mode of system(manual continuous, manual incremental, auto continuous, auto single block, auto idle running, hand wheel), middle part is the current line of processing program, right part is date and time.

The left top area is program part, the second line is current program line;

The left button area is work piece coordinate display;

The second area from bottom is the information display, like alarm;

The bottom area is the F1-F8 function keys.

The right area the display of program name, instructions, machine status and machine coordinate etc;

Menu operation

Press “program” to enter into program management, then edit, modify, check, delete, copy program;

Press “parameter” to enter into parameter management, then check and modify parameter of controller;

Manual mode is used to modify machine coordinate and management of machine auxiliary performance, tool setting, specify start point of work piece and manual process. Manual mode cannot used to run program.

3.4.1 The key of manual operation

(1) “F”: Taking mm/min as the unit to set the manual feed speed, the input range is from 1 to 30000mm/min. And the input method according to data input method in parameter.

- (2) “”: Switching cycle from “manual continuous” to “manual increment”
- (3) “S”: Set the speed of the first spindle. The range is from 0 to 99999, the max depends on the No.36 parameter in speed parameter.
- (4) “I”: Modify the increment in manual increment
- (5) “ ”: Press once to increase or decrease 10% feed speed when the No.1 axis parameter is 0, the range is from 0% to 150%,16 gears totally.
- (6) “ ”: Press once to increase or decrease G00 or manual rapid override 20%.The range is from 5% to 100%,16 gears totally.
- (7) “ ”: Press once to increase or decrease the spindle override 10% when the No.2 parameter in axis parameter is 0. The range is form 5% to 150%,16 gears totally.
- (8) “ ”: To switch cycle “0.001” “0.01” “0.1” or “0.1” “0.01” “0.001” in the handwheel function.
- (9) “Diagnosis”: Enter the diagnosis of input or output.
- (10) “F7 Setup”: To set a value(G54-G59) in workpiece coordinate(G54-G59);Use “MDI” to set G54-G59 in machine coordinate(G53).
- (11) “Auto”: Select automatic mode.
- (12) “Manual”: Select manual mode.
- (13) Spindle controlled: “ ||  ||  || ”Controlling spindle on clockwise, counterclockwise, stop, correspond to instructions M03,M04,M05. When No.56 parameter in the axis parameter is “8”then press “spindle on counterclockwise” means counterclockwise inching turning.
- (14) “Cooling”: Coolant on or off correspond to instructions M08,M09.
- (15) “Chuck”: Chuck tightens or loose correspond to instructions M10,M11.
- (16) “Manual speed controlled”: Press “1” “2” “3” “4” “5” “6” “7” “8” “9” to set feed override “F30” “F60” “F120” “F250” “F500” “F1000” “F1500” “F2000” “F2500” “F3000”.
- (17) “Huff”: control huff on or off, correspond to instructions M59, M58.
- (18) “Switch manual continuous or increment”: Press  to manual continuous or increment, it displays I=XXXX.XXX when it is manual increment.
- (19) “Back to home point”: Press  and X or Z, the X or Z axis goes back to the home point automatically; Press “0”X axis firstly and then Z axis; Press “Esc” to cancel the construction. The speed controlled by No.31 No.33 parameter in speed parameter, the direction is determined by No.28 parameter in axis parameter.

(20) “Tool carrier controlled”: Press  to change next tool automatically if it is gang tool carrier; After changing next tool it will be stop if it is electric tool carrier; Which tool has changed is going to be redeem. Press “T” and number to change tool directly

(21) “Coordinates feed”: Press “↑ ↓ ← →” correspond to feed A axis and Z axis’s positive or negative direction.

(22) “Switch speed”: Press  to switch the speed to system speed which is changed by No.1 No.2 parameter in speed parameter when it is in coordinate feed, loosen it that will be the previous speed. If set the speed higher than the speed in parameter, it will be the set speed to feed.

(23) “Switch coordinates’ display”: Press “PgUp” or “PgDn” to switch the display which correspond to “relative” “absolute” “machine”.

(25) “Partno clear”: Press Del and Enter to clear quantity of processed work piece.

(26) User-defined “K1”: Turn on/off Y24

(27) User-defined “K2”: Turn on/off Y25

(28) User-defined “K3”: Turn on/off Y26

(29) “Incremental coordinate”: Press “Setup” to fix or set 0 after select “relative” coordinate.

(30) search center point: convert into G54-G59, press”K”. according to prompt information, controller will calculate value and save the offset of G43-G59.

PS: Machine coordinate clear: Press “E” in parameter and then press “Enter”.

3.4.2 Manual continuous

Continuous operation is to press the time as the basis, Press to feed, up to stop feeding. Making sure the axis and using “+X、-X、+Y、-Y、+Z、-Z、+A、-A” to feed, the speed of feed is determined by display on the interface(F) times the rate.

When continuous starting, press “” to switch the speed to No.1 No.2 No.3 parameter value in speed parameter. If set the speed higher than the speed in parameter, the feed speed will be No.1 No.2 No.3 parameter in speed parameter times rapid override.

In order to facilitate the user single axis cutting in the manual function, setting the manual speed in manual status. Press “F” and input the speed.

When the hard limit point beyond positive and negative feed running axis two direction at, stop the feed and prompt to feed reverse direction.(the same as hereinafter)

The manual maximum speed is limited by No.4 parameter in speed parameter, when setting the speed is higher than the value of parameter, then will be the No.4 parameter.

When No.38 other parameter is 8, “” is change into a switch, press once to turn on (no more to always press), press again to turn off.

3.4.3 Manual increment

This operation is to set the value of increment as the basis, press “↑ ↓ ← →” once to run a value of increment. It will prompts “I=0010.000” in manual increment represent for the value of increment is 10mm, press “I” to revise and Enter. But also

press “handwheel” and “  ”to switch the value into 0.001mm 0.01mm 0.1mm.

The speed is the speed on display(F) times the rate.

3.4.4 Back to machine's home point (reference point)

There are two ways to back to home point in this system, not only the switch for home point, but also can set floating point, the methods as follows:

Switch for home point:

Back to home point operation is to feed every axis to machine's home point position in turn. When the parameter of feeding axis which back to home point is 0, the axis of coordinate detects the home point and return to the pulsing signal of “Zero”, the data of machine's coordinate will be 0 automatically.

Switch on the power supply of the system, release alarm and the button of emergency after the CNC is power off, the need to back to home point to set machine's coordinate correctly.

Instruction:

1. The system requires for backing to the home point every time when it is power on, the requirement can be set by No.38 parameter in axis parameter, it can be prompt or force;
2. The way and type of detecting signal can be set by No.39 parameter in axis parameter, so detect the switch of home point is effective, also detect the Z pulsing signal of electrical motor after detecting the switch of home point (precision higher), detect forward or reverse for Z pulsing signal of electrical motor.
3. The direction for backing to home point can be set by No.40 parameter in axis parameter, D2 D3 D4 correspond to X Y Z axis, 0 is forward, 1 is reverse.
4. The type of the switch for home point can be set by No.41 parameter in axis parameter, D0 D1 D2 correspond to X Y Z axis, 0 is always on, 1 is always off.
5. The maximum length of detecting Z pulse of electrical motor can be set by No.37 No.38 No.39 parameter in axis parameter, the value must less than the pulse of electrical motor run a cycle.
6. The shifting distance after backing to home point can be set by No.46 No.47 No.48 parameter in axis parameter, rapid move coordinate to the value of parameter after backing to home point.

No switch for home point:

To set floating point to make sure, turn on corresponding function of floating point by No.33 parameter in axis parameter, setting No.34 No.35 No.36 parameter to make sure the floating point of X Y Z axis, the home point of machine

The steps to set floating point as follows:

1. Setting the No.33 parameter in axis parameter to enable corresponding floating point. For example: Turn X axis on is "00001000". (turn all of them on is 00111000.)
2. Moving X axis to designate position so that set floating point.
3. Press "Parameter", "Axis parameter" and select No.34 parameter, "Enter", popup a dialog box of X axis' floating point coordinate. Import the value of setting machine coordinate.

If it is 0, the machine coordinate of X axis now is the home point of X axis. The machine backs to this position every time when backing to the home point.

If it is 15, the current machine coordinate of X axis is 15.000, the distance to machine's home point is 15mm.

The method to set floating point of Y Z axis is the same as the above to set X axis.

Operation for backing to the home point:

At the manual condition, press " "and select X Y Z A axis to back to the home point in dialog box. Or import 0 to make the axis back in order, the cycle will turn to green in front when backing to the home point successfully, defeat otherwise.

If stop in the process, press "Stop" or "Reset" to stop backing to the home point.

Pay attention:

Every time when power on the system, it must back to the home point to make sure the accuracy of machine process. The system power off unusually or in an accident, it must back to the home point, otherwise could cause error.

3.4.5 Hand wheel(MPG)

Two types: hand held and panel, No.1 parameter in other parameter to set.

1.Panel:

Press "hand wheel" and "X" "Y" "Z" "A" to select an axis, "  " to adjust the gear.

2.Hand held:

Press "hand wheel" and operate the switch of axis selection to select an axis, operate the axis and switch of hand wheel override to adjust the gear.

Instruction

The hand wheel is mainly used for "Tool", the speed and the hand wheel feed of one measure is related to rotate the hand wheel fast or low. The speed is not too fast best when the system cooperate with stepping motor.

Hand wheel's pulse generator speed to be lower than 200r/min(The hand wheel to 100 pulse a cycle), the hand wheel's acceleration is controlled by No.24 parameter in speed parameter(the bigger the faster). The maximum speed is controlled by No.26(X axis) No.27(Y axis) No.28(Z axis) in speed parameter.

Hand wheel is invalid in auto-coordinates diagram machining, it only works in working coordinates.

3.4.6 Work piece coordinate system

1. Press "MDI" button to select the corresponding work piece coordinate system (G54-G59);
2. Moving the axis which is going to set tool to the specified location;
3. Press "Setup" to import the corresponding coordinate of work piece coordinate system..

3.5 Auto operation

Auto refers to processing the editing program of workpiece. This system can start at arbitrary point, and also can start at arbitrary line or with arbitrary tool. Starting arbitrary line or with arbitrary tool must use absolute coordinate to edit the program. Auto operation can't move the manual coordinate.

Running program selection: In the program interface, press “↑ ↓” to move the cursor to a program which is going to be carry out, press “C” to select the program to run automatically.

Press “Auto” to enter automatic mode from manual mode.

Switch display of coordinates: Press “Pgup” “Pgdn” to switch the interface into “absolute coordinate” “relative coordinate” All coordinate”.

3.5.1 Automatic process

“Single or Continuous”: Press “” to switch cycle.

“Continuous”: The program continue to execute every program segment(program line) to end or the instruction of stop to stop.

“Single ”: The program just execute one program line and end, wait another operation or press “Run” again to execute one next program line.

“Coordinate or Figure”: Press “” to switch cycle.

“Automatic coordinate”: The axis of coordinate will display with value.

“Automatically figure”: The axis of coordinate will display with a figure. There are two kinds of figure, horizontal milling machine and slant-bed machine, No.3 parameter in tool parameter to control.

“Automatic idle run ”: The program is speedy simulate, the axis of coordinate can't move.

3.5.2 Processing at arbitrary program line or with arbitrary tool

A. Run at actual line

At the automatic process condition, press “—” to pop up a dialog box, import a number of line, press “Enter” to confirm, the line will start from the defined actual line.

Pay attention:

1. The line is the actual line in the program, not the “N” stand for the line. The system process to the line you import with a speed which is set by No.6 parameter in speed parameter(G01/G02/G03), then process the program normally.

2. The line of default is the line of suspend the program last time, to facilitate user’s operations.

3. At the interface of coordinate to use “N” to search line and press “Reset” to back to the beginning of program.

B. Run at the marked line

The system has a function to run at the marking line. At automatic process condition, press “N” to prompt a dialog box to import the marking line, press “Enter” to confirm. Press “Run” to process program at the line you import(mark).

Pay attention:

The line is not the actual line, is the “N” stand for the line. The system process to the line you import with a speed which is set by No.6 parameter in speed parameter(G01/G02/G03), then process the program normally.

3.5.3 Start program

Press “Auto” to switch to automatic mode to process program, two methods as follows.

- (1) Press “Run”
- (2) Switch on the Run of external signal.

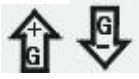
3.5.4 Stop processing program

Five methods as follows to stop:

- (1) The instruction of program M00 M01 M02 M30 M20.
- (2) Press “” to run a current line and stop.
- (3) Pause switch on the panel.
- (4) Switch on the Halt of external signal.
- (5) Press “Reset” to stop all the actions of program.(Like spindle, tools and others)

3.5.5 Real-time control in automatic process

(1) “”: Press once to increase or decrease 10% feed speed when the No.1 axis parameter is 0, the range is from 0% to 150%,16 gears totally; When the No.1 axis parameter is 1, external band switch takes in control, Adjust the speed of process arbitrarily in the process according to the different situation.

(2) “”: Press once to increase or decrease G00 or manual rapid override

20%.The range is from 5% to 100%,16 gears totally. Adjust the rapid override arbitrarily according to the different situation.

(3) “ ”: Press once to increase or decrease the spindle override 10% when the No.2 parameter in axis parameter is 0. The range is form 5% to 150%,16 gears totally. When the No.2 axis parameter is 1, external band switch takes in control, Adjust the speed of spindle arbitrarily in the process according to the different situation.

(4) Stop in the process: At the continuous mode in process condition, press “” to stop running after executing a current program line, wait for operating.

(5) Suspend in the process: Turn the intervention switch right or middle and switch on external stop signal of Halt, the processing program will stop; Press “Reset” to exit automatic process mode and the program line is going to back to the first of the processing program.

(6) Keep feeding: When the process is suspending, press “Manual” to keep feeding automatically, also can adjust the coordinate, press “Auto” and “Run” to run to the point of suspend automatically to end.

(7) Exit process: Press “Reset” when processing, suspending or keep feeding.

(8) Enable macro program by input point during processing. For example: if X7 is valid, then run PrgramUser9, and add following PLC:

```

X07  M122
----|-----|------(S)M300      ;if X7 is valid, run M300;
X07  M300 M122
----|-----|-----|------(S)M1014    ;if X7 is valid, reset current program
M300  M122  X07
----|-----|/|-----|/|------(S)M1069    ;Enter in manual mode;
M300  M122  M01 X07
--|-----|/|-----|-----|/|------(S)M1079    ;Execute ProgramUser9;
                                   |--(R)M300      ;reset M300

```

M300: intermediate register, the same M300 is prohibit in PLC;

M1014: reset current program;

M1069: manual mode

M1070-M1079: corresponds to ProgramUser0-9.

3.5.6 Operation of MDI

At the manual or automatic coordinate conditions, press “M” to get into the processing mode of MDI. Processing a program line that you import in “MDI”, press “Esc” to give up and exit when importing, press “Run” to carry out the program line that you import.

3.5.7 Operation of Handwheel

Press “Handwheel” at automatic mode, the program of turn handwheel is

processing automatically, the speed is related to the speed of “F”, feed override and turn handwheel fast or slow. This mode is for trying to process in running program usually.

Pay attention: The acceleration, deceleration and maximum speed of running handwheel are controlled by No.23-No.29 parameter, use the acquiescent acceleration, deceleration and the speed of G00 when the parameter is set to be invalid.

3.5.8 Function of DNC

The storage space of user is 32Mbit in this system, use DNC to process when the processing program is greater than 32M or the remainder storage space. Switch on RS232 or USB to realize the function of DNC in this system.

A. Instruction for RS232-DNC

1. Use the dedicated communication wire to connect the computer and the system to set the corresponding communication interface and speed by the system.
2. Use the dedicated communication software of this system by computer to set the corresponding communication interface and speed. Press “Send CNC program file”, select the program file to process linked, enter the status of sending program file.
3. To enter the interface of program file in NC system, press "L" to enter the status of linked process, now the upper right corner of the display interface is "RS232--DNC", press “Run” to running carry out linked process in the automatic status.
4. Turn “Intervention switch” to middle or right to stop the running system in the process of linked process, press “Stop” or “Reset” to exit the status of linked process.

Pay attention: 1. The baud rate is related to operational environment when using serial port to send files.

2. The communication cable can't more than 10 meters length.

3. Only the dedicated communication software of this system can send program in user's computer. To set the sending speed of PC as the NC, defeat otherwise.

B. Instruction of USB-DNC

USB-DNC is realized by U-disk, switch on U-disk and system, select program to execute in U-disk.

Press “B” to open U-disk in program interface, select corresponding program to press “C” to execute program, press “Auto” to get into automatical mode and press “Run” to process the program.

Pay attention: 1. Don't unplug U-disk in the process of USB-DNC, defeat processing otherwise.

2. Back to the system program interface from U-disk interface after finish USB-DNC.

3. After selecting the program, it is best to press “P” to compile once to make sure the program is right before executing program of USB-DNC.

3.6 Operate safety, Alarm

3.6.1 Emergency stop



Press “” when emergency accidents happening, the system will stop all the actions of machine and shows “Emergency stop” on the interface. Wait for the button up. M67 imports effective signal when No.29 parameter in other parameter to be set effectively.



Press “” in the process or running machine, system coordinate and machine’s position may change, make sure the system coordinate again before processing, it is best to carry out operation of backing to the home point to make coordinate same as the machine’s position.

The button can be external which is controlled by No.27 parameter in other parameter to set it normal open or close.

3.6.2 Reset system

Press “Reset” to stop current operation in anytime when the system is running, especially stop all the actions of machine(spindle, tools and so on) in automatical or manual mode, but the coordinate won’t lose, so needn’t to back to the home point.

3.6.3 Alarm

The screen shows error information and twinkles when the machine has alarm, the program is stop running, the coordinate stop moving, check the reason for alarm and clear troubles to run again. The signal M67 is effective when No.29 parameter in other parameter is “1”.

(1) X、Y、Z、A axis are positive hard limited: X、Y、Z、A axis are in the positive position which is limited .

(2) X、Y、Z、A axis are negative hard limited: X、Y、Z、A axis is in the negative position which is limited .

(3) Spindle and inverter (frequency changer) alarm: The alarm signal of machine’s spindle is effective.(ALM1)

(4) No.0 alarm: The alarm signal of machine’s spindle is effective.(ALM2)

(5) X, Y, Z, A axis driver alarm: The alarm signal of servo drivers is effective. (ALM). Press “B” to import INTH signal to reset the servo drivers in diagnosis mode.

(6) No.5 alarm for door switch: The alarm signal of M12(door switch) is effective.

(7) +5V is under voltage: Supply voltage is low, +5V of the system is low.

(8) Emergency stop: Press the button of emergency stop.

3.7 Parameter operation

At any status conditions, press “parameter” to enter the status to set the parameter. Parameter in this system includes “processing parameter” “speed parameter” “axis parameter” “tool parameter” “other parameter” “coordinate” “password”, 7 kinds totally.

Man Con		鑫科数控 N00000		2019-03-14 16:55			
1,Cycle d of G73 (mm):		5.000		Program LATH			
2,Cycle d of G83 (mm):		5.000		Progress			
3,Cycle Q-direction of G76G87:		1		G53			
4,Cycle Q-direction of G87:		1		T01H0D0			
5,Cycle spindle angle of G76G8		0		TrueFeed F 0.000			
6,Mode of G84G74(0 is High spe		0.000		M05 M09 M10			
7,Cycle d of G84G74 (mm):		0.000		M78 M33 M41			
17,Running program need Sp run		0		G00 X100%			
18,Set M20 the time of auto-ru		-1		F500 X120%			
19,Set part count:		0		S800 X110% SP000			
21,G01/G02/G03 line delay(ms)I		0		Machine Coor			
22,G00 line delay(ms)I>100I:		0		X -493.784 A 40.267			
23,Smooth acceleration/deceler		70		Y 6.059 B 0.000			
				Z 812.906			
				PartTime 0: 0			
				PartNo 0			
				SPrpm 0			
No Alarm							
F1 User	F2 Speed	F3 Axis	F4 Tool	F5 Other	F6 Coor	F7 PASSWD	F8 CANCEL

Press “F1-F7” correspondingly to enter corresponding interface after enter the parameter interface, press “↑ ↓” to select the number of parameter and press “Enter” to prompt a dialog box to import data and press “Enter” again to fix parameter successfully.

Instructions for parameter as follows:

3.7.1 User parameter (processing)

★ 1,Cycle d of G73 (mm)

Used for setting tool retriev value d of fixed loop G73;

★ 2,Cycle d of G83 (mm)

Used for setting tool retriev value d of fixed loop G83;

★ 3,G76 Offset Direction (Q Value)(G17:1 is+X,2 is-X,3 is+Y,4 is-Y)

Offset Q direction of fixed loop G76;

★4,G87 Offset Direction (Q Value) (G17:1 is+X,2 is-X,3 is+Y,4 is-Y)

Offset Q direction of fixed loop G87;

★5,Cycle spindle angle of G76G87(0.1degree)

Spindle orientation angle of G76 and G87;

★6,Mode of G84G74(0 is High speed to d,8 is to R)

Chips removal mode if G84/G74.

★7,Cycle d of G84G74 (mm)

Tool retrieve value d of G84/G74;

★8,Cycle G73G83 Pause time before each tool lift (ms) [It's effective when the absolute value is greater than 5, and when it's set to a negative number, it means that the tool lift is in G01 mode, otherwise it's in G00 mode]

★17,Running program need Sp run [1 mean Yes,0 mean No]

For interlock between program and spindle, to set if detect rotation of spindle or not while program running.

If set as 1, the spindle needs rotate when program running(Need detect the encoder rotation when it is M03); Set as 0, then no need to detect.

★18,Set M20 the time of auto-running

The loop times of M20 in program, infinite loop when it is minus.

★19, Program trial run model [2: stop at the current line, 4: automatically jump to a line, +32: press the run button to run and release to stop, 512: start from the current line]

★21,Inter-segment Delay for G01/G02/G03(ms)[>100]

The delay time between G01/G02/G03, this parameter solves the overcut problem in the corner.

★22,Delay after G00(ms)[>100]

★23,MPG smooth acceleration/deceleration constant[50-100]

The smooth acceleration constant when using MPG, range: 50-100, smaller value corresponds the faster acceleration, but more vibration.

★33,M03/M04/S detect the spindle speed(0:M69 relay, 8:encoder feedback)

M03/M04/S code detect if spindle speed reach already. 0 means to detect M69 auxiliary relay, 8 means to detect encoder feedback

- ★ 34, The error (RPM) of the spindle speed is detected by the encoder feedback.

The error of spindle speed feedback from encoder.

- ★ 200, System protect times [≥ 2 minutes]

★ When under main interface and no dialog, and if $P18 > 2$ minutes, controller will enter in screen protection, any key to quit.

202, System internal parameters

Coordinate axis display (1: normal, 4: no X, 8: no Y, 16: no Z, 32 : only X, 64 : only Y, 128 : only Z, 256 : XY)

- ★ 203, Whether strict inspection G41/G42

This is a special parameter, 23103490: enables the pause button, 2131970: strictly checks, 34818: does not strictly check, 6326274 : fast pre-reading processing.

- ★ 205, System internal parameters (+8: the interpolation tapping must wait for the pulse to be sent; 32: the axis displays four decimal places; 96: the axis displays four decimal places; 160: HX-G instruction; +256: the milling machine H adopts absolute mode; +4096: the milling machine cycle will not be executed if the coordinate point is not programmed)

- ★ 210, Enable graphics display area (8: manual, 0: Automatic)

Enable graph displaying. 8: a fixed display area is enabled, and the others indicate that the display area is automatically calculated.

- ★ 211, Graphic display area X-axis negative end

X axis negative end of graph displaying;

- ★ 212, Graphic display area X-axis positive end

X axis positive end of graph displaying;

- ★ 213, Graphic display area Y-axis negative end

Y axis negative end of graph displaying;

- ★ 214, Graphic display area Y-axis positive end

Y axis positive end of graph displaying;

- ★ 215, Graphic display area Z-axis negative end

Z axis negative end of graph displaying;

- ★ 216, Graphic display area Z-axis positive end

Z axis positive end of graph displaying;

- ★ 230, Select executive program through input

(+4...+128): X26-X31, (+256...+32768): X16-X23

Executing program by input points, for example, if $P230 = +4 + 8 = 12$, when X26 or

X27 is effective to choose the program HIDEFILEX26 or X26 or HIDEFILEX27 or X27.

D2=1, means +4, if X26 is effective, then execute Program X26 or HIDEFILEX26;

D3=1, means +8, if X27 is effective, then execute Program X27 or HIDEFILEX27;

D4=1, means +16, if X28 is effective, then execute Program X28 or HIDEFILEX28;

D5=1, means +32, if X29 is effective, then execute Program X29 or HIDEFILEX29;

D6=1, means +64, if X30 is effective, then execute Program X30 or HIDEFILEX30;

D7=1, means +128, if X31 is effective, then execute Program X31 or HIDEFILEX31;

D8=1, means +256, if X16 is effective, then execute Program X16 or HIDEFILEX16;

D9=1, means +512, if X17 is effective, then execute Program X17 or HIDEFILEX17;

D10=1, means +1024, if X18 is effective, then execute Program X18 or HIDEFILEX18;

D11=1, means +2048, if X19 is effective, then execute Program X19 or HIDEFILEX19;

D12=1, means +4096, if X20 is effective, then execute Program X20 or HIDEFILEX20;

D13=1, means +8192, if X21 is effective, then execute Program X21 or HIDEFILEX21;

D14=1, means +16384, if X22 is effective, then execute Program X22 or HIDEFILEX22;

D15=1, means +32768, if X23 is effective, then execute Program X23 or HIDEFILEX23;

★ 231, "Delete" mode[0:backward deletion,1:Forward delete]

★ 232, Whether to detect spindle zero position before tapping [18:Yes,0:No]

Check spindle home position before tapping;

★ 233, G06 circle teaching function[0:No,1:Yes]

★ 234, Program backtrack function[+8: MPG,+512:MPG for cursor movement]

Set +8 means use MPG to run program forward or backward;

★ 235, Automatically generate comments when teaching [0:Yes,1:No]

★ 236, Type of circular arc center when teaching [0:G5/G6 IJK format, 1:G2/G3 R format]

Set it to 0 means that the G5/G6 IJK format. 1 means the G2/G3 R format.

★ 237, Teaching command generation method [0: normal, 1: that each straight line processing line generates G01 code when teaching]

★ 238, Automatically insert a row when teaching the middle row [0:insertion, 1:modification]

★ 239, Press "XYZ" When teaching in the middle row[0:insertion, 1:modification]

★ 240, Same coordinate recording mode when teaching [0: do not record,1: recording]

★ 307, M18xx/M28xx/WAT waits for the longest alarm time(ms)[>=10]

The max waiting time for M18xx/M28xx/WAT code, if over the time, controller will alarm.

For example: when this parameter is set to 2000, the program WAT+X16 will alarm if no X16 signal is detected after 2 seconds.

★ 400, Translate DXF file to G code[1:Sort, 4:Tool start point sorting, 8:No sort]

1) if P400=1, controller will sequence end to end;

2) if P400=4, controller will sequence end to end, and judging distance between ends to tool start point by parameter, and choose to start tool from nearby end;

3) if P400=8, controller will not sequence. When generate DXF file from AutoCAD, please pay attention to drawing sequence, because DXF file save date according to drawing sequence, the g code generated by controller will also execute according to drawing sequence.

★ 401, Translate DXF file to G code X-axis coordinate.(mm)

X coordinate of the tool starting point in the DXF file.

★ 402, Translate DXF file to G code Y-axis coordinate.(mm)

Y coordinate of the tool starting point in the DXF file.

★ 500, Customize G74 equal to ProgramG No.[101-170(101-150 Non-modal, 151-170 modal)]

Set the macro program number corresponding to the custom instruction G74.

For example, if it's set to 101, it means that the G74 instruction executes the macro program ProgramG101. Parameters P501-P504 have similar functions.

★ 501, Customize G81 equal to ProgramG No.[101-170(101-150 Non-modal, 151-170 modal)]

★ 502, Customize G82 equal to ProgramG No.[101-170(101-150 Non-modal, 151-170 modal)]

★ 503, Customize G83 equal to ProgramG No.[101-170(101-150 Non-modal, 151-170 modal)]

★ 504, Customize G84 equal to ProgramG No.[101-170(101-150 Non-modal, 151-170 modal)]

★ 551, 5-Axis CNC Machine Configurations (10--99)

[10: B tool-A platform, 11: AC tool, 12: AC platform, 13: A tool-C platform, 14: A tool-B platform; 15: BC platform]

★ 552, RTCP compensation reverse and coordinate system coincidence (reverse

10/14: B+4, A+8; 11/12/13: A+4, C+8; 15: B+4, C+8; coordinate system coincidence +16:[12A axis, 13C axis, 14/15B axis] corresponding programming and workpiece coordinate system coincidence; +32: 12/15C axis corresponding programming and workpiece coordinate system coincidence)

★ 553, Coordinate-1 of Machine tool in the Center of 5-Axis CNC Rotary platform (um)

Machine tool type is 10/12: Y coordinate of the A-axis rotary center; 13: X coordinate of the C-axis rotary center; 14: X coordinate of the B-axis rotary center; 15: X coordinate of the B-axis rotary center.

★ 554, Coordinate-Z of Machine tool in the Center of 5-Axis CNC Rotary platform (um)

Machine tool type is 10/12: Z coordinate of the A-axis rotary center; 13: Z coordinate of the C-axis rotary center; 14: Z coordinate of the B-axis rotary center; 15: Z coordinate of the B-axis rotary center.

★ 555, Measuring tool length at the center of 5-axis CNC rotary platform.(mm)

Machine type is 10: measuring the tool length of the A-axis rotary center; 12/15: measuring the tool compensation value H1 of the 5-axis CNC rotary platform center; 13: measuring the tool length of the C-axis rotary center; 14: measuring the tool length of the B-axis rotary center.

★556, Machine coordinates (X or Y-axis) of the C-axis rotation center in AC-axis or BC-axis rotary platform 5-axis CNC machines.(um)(machine type is 12 or 15)

★557, Y-axis offset of the AC-axis centerline in AC-axis.(um) (machine type is 11)

★558, X-axis offset between the C-axis and spindle centerline in AC-axis.(um) (machine type is 11)

★ 560, Is there an over-speed alarm for RTCP

Set 1: Automatically reduce the running speed without alarm, 0: no alarm

★561,Maximum X-axis speed in manual mode with 5-axis RTCP enabled. (mm/min)[>1]

★562,Maximum Y-axis speed in manual mode with 5-axis RTCP enabled. (mm/min)[>1]

★563,Maximum Z-axis speed in manual mode with 5-axis RTCP enabled. (mm/min)[>1]

★564,Maximum X-axis speed in auto mode with 5-axis RTCP enabled.(mm/min)[>1]

★565,Maximum Y-axis speed in auto mode with 5-axis RTCP enabled.(mm/min)[>1]

- ★566, Maximum Z-axis speed in auto mode with 5-axis RTCP enabled.(mm/min)[>1]
- ★600, Internal feedrate override for non-G00 commands[>=20 valid]
- ★600-0, Maximum waiting time for M7[>=10](ms)
- ★601, X-axis M7 (>10: following error is less than this value; -6: feedback pulses stop changing; -8: system pulse output completed)
- 602, Y-axis M7 (>10: following error is less than this value; -6: feedback pulses stop changing; -8: system pulse output completed)
- 603, Z-axis M7 (>10: following error is less than this value; -6: feedback pulses stop changing; -8: system pulse output completed)
- ★604, 4th axis M7 (>10: following error is less than this value; -6: feedback pulses stop changing; -8: system pulse output completed)
- ★605, 5th axis M7 (>10: following error is less than this value; -6: feedback pulses stop changing; -8: system pulse output completed)
- ★606, 6th axis M7 (>10: following error is less than this value; -6: feedback pulses stop changing; -8: system pulse output completed)
- ★607, 7th axis M7 (>10: following error is less than this value; -6: feedback pulses stop changing; -8: system pulse output completed)
- ★608, 8th axis M7 (>10: following error is less than this value; -6: feedback pulses stop changing; -8: system pulse output completed)
- ★700, Alarm when G31 fails to detect a signal [0: Feed hold, 1: Alarm, +1024: No alarm]
- ★701, Intermittent execution of M880-M929
Execute during coordinate refresh. (10000/10049)
Execute ProgramUser0-49 at 100ms intervals.(-10000 to -10049)
- ★702, Default program coordinate system (3-9:G53-G59, 10-57:G54.1-48)

3.7.2 Speed parameter

- ★1, X-axis G00 speed(mm/min)
Rapid traverse speed (G00) of the X-axis during automatic operation. Maximum:
30,000(mm/min)
- ★2, Y-axis G00 speed(mm/min)
- ★3. Z-axis G00 speed(mm/min)
- ★4. A-axis G00 speed(mm/min)

★ 5 , 5th- axis G00 speed (mm/min)

5a, 6th- axis G00 speed (mm/min)

5b, 7th- axis G00 speed (mm/min)

5c, 8th- axis G00 speed (mm/min)

5d, Zs-axis G00 speed (mm/min)

5e, As-axis G00 speed (mm/min)

5f, Bs-axis G00 speed (mm/min)

5g, Cs-axis G00 speed (mm/min)

★ 6. Maximum manual feed speed(mm/min)

Set the maximum running speed limit of the feed axis in manual mode.

Note: It's recommended to set this value to 50% of G00 speed to ensure the safety of manual operation.

6-1, Maximum manual feed speed for X-Axis (mm/min) [≥ 50]

6-2, Maximum manual feed speed for Y-Axis(mm/min) [≥ 50]

6-3, Maximum manual feed speed for Z-Axis (mm/min) [≥ 50]

6-4, Maximum manual feed speed for 4th-Axis (mm/min) [≥ 50]

6-5, Maximum manual feed speed for 5th-Axis (mm/min) [≥ 50]

★ 7, Maximum auto feed speed (mm/min)

Maximum speed for automatic operation. Max: 30000 (mm/min)

To ensure the maximum speed efficiency of each axis, this parameter value can be set higher than the G00 speed to ensure that each axis reaches the maximum speed during linkage (limited to the composite speed).

★ 8, Default speed for G01/G02/G03(mm/min)

When the first interpolation instruction (G01/G02/G03) in the program does not have a given speed, the speed called by the instruction during automatic operation. Max: 5000(mm/min)

★ 9, Dry run speed(mm/min)

The speed at which the program is executed during dry run. Max:30000(mm/min)

★ 10, Manual feed axis speed(mm/min)

The feed axis speed in manual mode, range: $\leq$ Maximum manual feed speed(mm/min)

★ 11, Manual spindle speed(rpm)

The spindle speed in manual mode.(rpm)

Note: In manual mode, press the “S” key to set and this parameter value will be automatically refreshed.

★ 12, Feed axis starting speed(mm/min)

★ 13, Feed axis maximum mutation speed(mm/min)

★ 14, Limit the speed of G1G2G3 single axis[1: Yes,0: No]



It's used to set whether to limit the maximum running speed of each axis during interpolation. When this parameter is set to 1, it means limit, that is, the maximum running speed of each axis during interpolation will be limited by the parameter, and 0 means the opposite.

- ★ 15, X-Axis G1G2G3 max speed(mm/min)
- ★ 16, Y-Axis G1G2G3 max speed(mm/min)
- ★ 17, Z-Axis G1G2G3 max speed(mm/min)
- ★ 18, A-Axis G1G2G3 max speed(mm/min)
- ★ 19, 5th-Axis G1G2G3 max speed(mm/min)
- 19a, 6th-Axis G1G2G3 max speed(mm/min)
- 19b, 7th-Axis G1G2G3 max speed(mm/min)
- 19c, 8th-Axis G1G2G3 max speed(mm/min)
- ★ 20, X-axis acceleration/deceleration constant((mm/min)/s)
- ★ 21, Y-axis acceleration/deceleration constant((mm/min)/s)
- ★ 22, Z-axis acceleration/deceleration constant((mm/min)/s)
- ★ 23, A-axis acceleration/deceleration constant((mm/min)/s)

X-axis acceleration and deceleration constant. The larger the value, the faster the acceleration and deceleration. Range:1-99999.

Note: The value depends on machine configuration, incorrect setting may lead to machine malfunction. Usually, the bigger load, the smaller value. If installed with stepper motor, it can not be over 15000.

- ★ 24, 5th- axis acceleration and deceleration constant((mm/min)/s)
- 24a, 6th-axis acceleration and deceleration constant ((mm/min)/s)
- 24b, XS-axis acceleration and deceleration constant ((mm/min)/s)
- 24c, Ys-axis acceleration and deceleration constant ((mm/min)/s)
- 24d, Zs-axis acceleration/deceleration constant((mm/min)/s)
- 24e, As-axis acceleration/deceleration constant((mm/min)/s)
- 24f, Bs-axis acceleration/deceleration constant((mm/min)/s)
- 24g, Cs-axis acceleration/deceleration constant((mm/min)/s)
- ★ 25, Auto run acceleration/deceleration constant[1-500]

Used to set the acceleration and deceleration constant of automatic operation, with a setting range of 1-500. When the parameter value is set within the range of 1-500, the automatic operation acceleration and deceleration processing speed is based on this parameter, otherwise it's based on the acceleration and deceleration parameter value of each axis in the user parameter.

This parameter is mainly set to distinguish between manual and automatic acceleration and deceleration. This parameter is only set when the acceleration and deceleration required by the two states are greatly different. It's usually set to invalid.

★ 26, MPG acceleration/deceleration constant

The acceleration/deceleration time constant speed when use MPG, range is 500-32000, bigger value means faster acceleration/deceleration.

★ 27, MPG testing acceleration/deceleration constant[>500]

★ 28, MPG testing G00 speed(mm/min)[>10]

The idle running speed when use MPG in auto mode. Ineffective when lower than 10.

★ 28-1, MPG start-up speed during auto operation(mm/min)[>5]

★ 29, MPG X-axis limit speed(mm/min)

★ 30, MPG Y-axis limit speed(mm/min)

★ 31. MPG Z-axis limit speed(mm/min)

★ 32. MPG A-axis limit speed(mm/min)

Used to set max axis speed when use MPG, range: 100-max manual speed. The value of this parameter setting is related to the machine configuration. It's recommended that this value does not exceed 4000 to ensure safe operation.

★ 33, MPG 5th-axis speed limit (mm/min)

33a, MPG 6th-axis speed limit (mm/min)

33b, MPG 7th-axis speed limit (mm/min)

33c, MPG 8th-axis speed limit (mm/min)

33-1, Thread machining Z axis acceleration/deceleration constant [Servo drive set to 0]

33-2, Thread machining X axis acceleration/deceleration constant [Servo drive set to 0]

Used to set the speed increase and decrease time constant of the X axis during thread processing. The setting range is 300-90000. The larger the value, the faster the speed decreases.

Note: This parameter setting is related to the machine tool configuration. When stepping, the setting can be smaller to ensure safe operation.

If this parameter is set to less than 300, it means no speed increase or decrease. Usually, the servo can be set to no speed increase or decrease to ensure efficiency, set to 0.

33-3, Servo motor thread cycle X-axis retract speed override(0.1 times)

33-4, Thread machining synchronization Solution (Ethercat:20000)

★ 34, Acceleration/deceleration constant type[0:line, 8: curve]

Used to set acceleration/deceleration constant type of axis, 8 means curve type(S shape), 0 means linear type.

Depending on driver types, usually stepper motor is curve type, servo motor is linear type.

★ 35, S-curve initial acceleration/deceleration constant[>=10]

The bigger vale means faster acceleration

★ 36, S-curve Jerk Constant[>=10]

The bigger vale means faster acceleration.

★ 37, S-curve maximum Acc/Dec Constant[>=500]

The bigger vale means faster acceleration.

★ 38, X-axis homing positive direction speed(mm/min)

The running speed of the X axis when it returns to the reference point and touches the reference point switch in the positive direction(mm/min),range: less than the X axis G00 speed.

★ 39, X-axis homing negative direction speed(mm/min)

When the X axis returns to the reference point, the speed of the motor Z pulse signal is detected after the reference point switch is disconnected. Range: 20-500.

Note: This parameter value will affect the accuracy of returning to the reference point. The smaller the value, the higher the accuracy.When the value is set, do not change it again, otherwise it will also affect the reference point position.

★ 40, Y-axis homing positive direction speed(mm/min)

41-0, the distance that the C axis runs before returning to the encoder zero reference point (1/1000 degrees)

When the system spindle is set to C axis, M800 instructs the C axis to run the distance (1/1000 degree) before returning to the encoder zero position, and P41 is the speed (mm/min) of the C axis returning to the encoder zero position as instructed by M800.

★ 41, Y-axis homing negative direction speed(mm/min)

★ 42, Z-axis homing positive direction speed(mm/min)

★ 43, Z-axis homing negative direction speed(mm/min)

★ 44, A-axis homing positive direction speed(mm/min)

★ 45, A-axis homing negative direction speed(mm/min)

★ 46, 5th-axis homing positive direction speed (mm/min)

★ 47, 5h-axis homing negative direction speed (mm/min)

47-1, 6th-axis homing positive direction speed (mm/min)

47-2, 6th-axis homing negative direction speed (mm/min)

47-3, 7th-axis homing positive direction speed (mm/min)

47-4, 7th-axis homing negative direction speed (mm/min)

47-5, 8th-axis homing positive direction speed (mm/min)

47-6, 8th-axis homing negative direction speed (mm/min)

★ 48-1, G96 spindle speed lower limit under constant linear speed control (rpm)

★ 48, Spindle high gear range max speed(rpm)(r/min)

Set the maximum spindle speed, corresponding to the speed when the command voltage is 10V.

Note: When there are high and low gear and multi-gear functions, this parameter is the maximum spindle speed of high gear and first gear.

★ 49, Spindle low gear range (stage 2) max speed(rpm)

★ 50, Spindle third gear range max speed(rpm)

★ 51, Spindle fourth gear range max speed(rpm)

★ 52, Second Spindle max speed(rpm)

Set the maximum speed of second spindle, corresponding to the speed when the command voltage is 10V.

52-0, Spindle Jog fixed speed(0.1rpm) [>2]

52-1, Third Spindle max speed (rpm)

52-2, Fourth Spindle max speed (rpm)

52-3, Third Spindle manual Speed (rpm)

52-4, Fourth Spindle manual Speed (rpm)

52-5, First Spindle min speed(rpm) or min voltage (mV, with “-” in front to indicate voltage) [S0 test value]

Method 1: The min speed of the first spindle, for example: when it's set to 100, it means the speed at S0 is 100 rpm.

Method 2: The min voltage of the first spindle, for example: when it's set to -100, it means the output at S0 is 100mV.

52-6, First Spindle max speed(rpm) or max voltage (mV, with “-” in front to indicate voltage) [S max speed test value]

Method 1: The max speed of the first spindle, for example: when it's set to 3000, it means the speed is 3000 rpm when outputting 10V.

Method 2: The max voltage of the first spindle, for example: when it's set to -10000, it means the output is 10V at the maximum speed.

52-7, Second Spindle min speed(rpm) or min voltage (mV, with “-” in front to indicate voltage) [SS0 test value]

Method 1: The min speed of the second spindle, for example: when it's set to 100, it means the speed at SS0 is 100 rpm.

Method 2: The min voltage of the second spindle, for example: when it's set to -100, it means the output at SS0 is 100mV.

52-8, Second Spindle max speed(rpm) or max voltage (mV, with “-” in front to indicate voltage) [SS max speed test value]

Method 1: The max speed of the second spindle, for example: when it's set to 3000, it means the speed is 3000 rpm when outputting 10V.

Method 2: The max voltage of the second spindle, for example: when it's set to -10000, it means the output is 10V at the maximum speed.

★ 53, G02/G03 Circular backlash compensation mode

- 0 means A: the larger the backlash value, the faster the compensation speed;
- 8 means B: the compensation speed is determined by the parameter;
- +4 means: arc programming IJK is the coordinate from the end point to the center of the circle;
- +256 the helical interpolation speed does not exceed F.

When D3=0, A type: The backlash value is compensated at one time. When performing arc reverse backlash compensation, the compensation speed of the compensation axis is affected by the reverse backlash value. The larger the reverse backlash value, the faster the compensation speed is to ensure that the tool does not pause. However, the compensation speed of the compensation axis will not exceed 10000mm/min.

+8(D3 bit), B mode: The backlash value is compensated in sections. When performing arc reverse backlash compensation, the compensation speed of the compensation axis is determined by the parameter value;

+4(D2 bit): IJK is coordinate from the end point to centre of circle in arc programme, plus 4 means the IJK is to the end point in G02 G03, otherwise is to start point.

★54, Circular Interpolation type-B backlash compensation speed (mm/min)

54-1, Circular Interpolation type-B backlash compensation starting speed (mm/min)[>10]

54-2, Circular Interpolation type-B backlash compensation Acc/Dec Constant (mm/min)/s[>10]

★55, Activate the speed function [1: Yes, 0: No]

It's used to set whether the controller needs to pre-process the speed of short lines. If it's set to "1", it will be processed, and if it's set to "0", it will not be processed.

55-1, Deceleration termination speed at direction reversal (mm/min)

The end speed when the program is running and decelerating in the reverse direction.

★56, Hard limit emergency stop critical speed(mm/min)

★57, MPG stop speed (mm/min) [>100]

Set the speed when MPG stop, bigger value means faster stop.

★58, Spindle RPM during tap retraction(rpm)[>1]

The encoder follows the spindle speed when the tap retraction. For example, when it's set to 100, it means that when the spindle speed drops to 100r/min, the Z axis starts to retract in the reverse direction.

58-1, Spindle backlash compensation in tapping cycles (pulse number)

For example, when it's set to 200, it means that the spindle will reverse 200 pulses before the Z axis retract, and then the Z axis starts to retract in the opposite

direction.

58-2,Z-axis Pre-retraction value before Spindle reversal in tapping(um) [10-5000]

Z-axis backlash compensation pulses before the spindle reverses in following and rigid tapping

For example, when it's set to 500, it means that the Z-axis retract 500um and the spindle starts to reverse, which is similar to the Z-axis backlash compensation.

58-3, Feed speed during tap retraction (mm/min) [≥ 60]

The Z-axis retract speed of rigid tapping is used when the requirements are relatively high. Set the Z-axis retract speed to a parameter fixed speed.

★ 60,Activation of smoothing function(+4:Manual;+8:MPG;+16:Program)

<1>P60=28 means that the smoothing function is enabled for manual/MPG/program processing;

<2>Smoothing function - The curve smoothing interpolation method is implemented through the Savitzky-Golay filter to smooth the broken line. The sliding average filter data smoothing is implemented based on Numpy.convolve, and then the log() and exp() function algorithms are used to implement smoothing filtering of data frames with a large number of data frames at the peaks and troughs of the curve. The main purpose is to solve the curve noise generated between straight lines or between straight lines and arcs;

<3>The smoothing time constant is set by parameters P61-P68. The larger the smoothing time constant value, the smoother it is, but the corner accuracy between straight lines or straight lines and arcs is worse;

<4>Enhanced smoothing is to smooth the pulses obtained through smoothing again. Setting the level means repeating the smoothing process again. It is mainly to solve the problem that the first-level smoothing coefficient is too large and may cause poor contour accuracy.

★ 61,Manual smoothing time constant[2-500],Recommended setting: 4

★ 62,MPG smoothing time constant[2-500],Recommended setting: 4

★ 63,Program smoothing time constant[2-500],Recommended setting: 6-12

★ 65,Manual super-smoothing time constant[2-50],Recommended setting: 4

★ 66,MPG super-smoothing time constant [2-50],Recommended setting: 4

★ 67,Program super-smoothing time constant [2-50],Recommended setting: 12

★ 68,Enhanced smoothing level[1-9], Recommended setting: 2-8

★ 101,X-Axis jump speed(mm/min)[>1]

★ 102,Y-Axis jump speed(mm/min)[>1]

★ 103,Z-Axis jump speed(mm/min)[>1]

★ 104,4th-Axis jump speed(mm/min)[>1]

★ 105,5th-Axis jump speed(mm/min)[>1]

It is used to set the initial speed of the feed axis when it is running, and the end speed when it is decelerating. When the feed axis is running at a speed lower than this value, it will reach the speed directly without the speed increase or decrease process. When the running speed is higher than this value, the speed increase starts from this value. P101-P108 indicates the starting speed of the X/Y/Z/A/B/C/Xs/Ys axis (mm/min) [>1].

Note: The size of this value is related to the machine tool configuration. Improper setting may cause equipment failure or accident.

Normally, it is not greater than 100 for stepper drive and 500 for servo.

★ 106,6th-Axis jump speed(mm/min)[>1]

★ 107,7th-Axis jump speed(mm/min)[>1]

★ 108,8th-Axis jump speed(mm/min)[>1]

★ 111,X-Axis max velocity step change(mm/min)[>1]

★ 112,Y-Axis max velocity step change(mm/min)[>1]

★ 113,Z-Axis max velocity step change(mm/min)[>1]

★ 114,4th-Axis max velocity step change(mm/min)[>1]

★ 115,5th-Axis max velocity step change(mm/min)[>1]

Used to set the maximum increment of each axis speed when the speed changes during multi-axis continuous trajectory interpolation. To ensure the maximum increment of each axis speed when interpolating the composite speed, when the speed increment value is greater than this value, there will be a speed increase (decrease) process. If it is less than this value, it will directly reach the end. It is mainly used to increase the continuity of multi-axis continuous trajectory interpolation. P111-P118 represents the maximum speed mutation of X/Y/Z/A/B/C/Xs/Ys (mm/min) [>1].

Example: When this value is 300, when the X-axis speed changes from F800 to F1600 in multi-axis continuous trajectory interpolation, the change process should be that the X-axis speed first changes from F800 to 1100, and then accelerates to F1600.

★ 116,6th-Axis max velocity step change(mm/min)[>1]

★ 117,7th-Axis max velocity step change(mm/min)[>1]

★ 118,8th-Axis max velocity step change(mm/min)[>1]

★ 200,G00 continuous run is valid[1:No,16: Yes]

Set whether G00 performs pre-reading processing. When set to 16, it means

pre-reading continuous motion processing.

★210,Thread machining Spindle speed fluctuation tolerance[0 :do not wait, >=2 valid]

★211,Rigid tapping Spindle entry speed(rpm)

★212,Rigid tapping Spindle retraction starting speed(rpm)

★213,Spindle Acc/Dec during rigid tapping entry (rpm/S)[>1]

★214,Spindle Acc/Dec during rigid tapping retraction (rpm/S)[>1]

★215,Spindle preload during rigid tapping entry (1/1000 laps)[>2]

★216,Spindle trailing compensation during rigid tapping entry(1/1000 laps)[>2]

★217,Spindle preload during rigid tapping retraction (1/1000 laps)[>2]

★218,Spindle trailing compensation during rigid tapping retraction (1/1000 laps)[>2]

★230,SP command in G00 speed(0.1rpm)

Set the positioning speed of executing SP command G00 when the spindle is positioned.

★231,SP mode[+4: F ,+8: G90/G91,+16: Send pulses from the current value]

+4: When executing the SP command, the positioning is carried out according to the current F speed [degrees/minute] (in G00 state, it's still carried out according to the spindle positioning speed parameter), otherwise it's carried out according to the spindle positioning speed parameter.

+8: When executing the SP command, it's controlled by the current G90/G91. When it is G90, it means that it will definitely be calculated, and when it is G91, it means that the incremental will not be calculated. Otherwise, it will be calculated and run in the nearest direction.

+16: The system's SP value display is determined by the current sent pulse, otherwise the display value is determined by the encoder feedback.

+8192: The spindle analog quantity is sent repeatedly every one second.

+16384: After executing M05, the analog quantity also stops outputting after the spindle stops.

★232,Spindle orient direction in interpolation[0: Positive,1: Negative]

0: SP command positive direction positioning, 1: SP command negative direction positioning,others: nearest direction positioning.

★233,SP home mode(1:pulse,2: control by the drive)

1: Output pulse system returns to zero, 2:Control by the drive (output M61 detects M22), 16 or 32: The spindle is positioned to the parameter P234 angle after

accurate stop, 16: Output M75 detects M22.

★234,Spindle orientation stop angle(0.1degree)

★235,Spindle homing speed(0.1 rpm)

3.7.3 Axis parameter

★1,Feed axis band switch[1 :Yes, 0: No]

This parameter is used to set the operation mode of X and Z axis magnification adjustment. 1 means to use external band switch for adjustment, and 0 means to use the "Feed↑" and "Feed↓" keys on the panel for adjustment.

Note: When this parameter is set to 1, the interface of the external band switch is the VDK0 (OFF), VDK1 (X100), VDK2 (X10), and VDK3 (X1) signals of the CN11.

★2,Spindle and G00 band switch[1:Spindle, 2:G00, 0: No]

Used to choose the spindle override switch, 1 means choose external ban switch, 0 means use "SP ↑" and "SP ↓" button.

Note: When this parameter is set to 1, the interface of the external band switch is the VDS0 (HCOM), VDS1 (Z), VDS2 (Y), and VDS3 (X) signals of the CN11.

★3,X-Axis negative travel limit(mm)

Set the negative/positive direction coordinate value of the X-axis soft limit. The value is based on the machine coordinate system. P3-P12 Set the negative/positive direction soft limit of the X/Y/Z/A/B/C/Xs/Ys axes.

★4,X-Axis positive travel limit(mm)

★5,Y-Axis negative travel limit(mm)

★6,Y-Axis positive travel limit(mm)

★7,Z-Axis negative travel limit(mm)

★8,Z-Axis positive travel limit(mm)

★9,A-Axis negative travel limit(mm)

★10,A-Axis positive travel limit(mm)

★11,5th-Axis negative travel limit(mm)

★12,5th-Axis positive travel limit(mm)

12a,6th-Axis negative travel limit(mm)

12b,6th-Axis positive travel limit(mm)

12c,7th-Axis negative travel limit(mm)

12d,7th-Axis positive travel limit(mm)

12e,8th-Axis negative travel limit(mm)

12f,8th-Axis positive travel limit(mm)

12g,Zs-Axis negative travel limit(mm)

★12h,Zs-Axis positive travel limit(mm)

12i,As-Axis negative travel limit(mm)

12j,As-Axis positive travel limit(mm)

12k,Bs-Axis negative travel limit(mm)

12l,Bs-Axis positive travel limit(mm)

12m,Cs-Axis negative travel limit(mm)

12n,Cs-Axis positive travel limit(mm)

13, Spindle brake Y12 signal mode [0: spindle brake, 1: spindle enable]

13-1,Spindle brake hold long signal[0: No, 1: Yes]

0 means that M05 outputs Y12 spindle brake, 1 means that M03/M04 outputs Y12 spindle enable signal.

When P13-1 is 0, the spindle brake is set to a short signal, and 1 is set to a long signal;

P13-2 sets the time when the spindle brake is a short signal.

13-2,Spindle brake time(10ms)

★15,Soft limited

D0=1: Spindle cooling starts automatically;D2: X axis;D3: Y axis; D4: Z axis;

D5: A axis; D6: B axis; D9: C axis; D10: Xs axis; D11: Ys axis;1 :not limit;

Set each axis in each bitIf the bit is set to 1, it means that the soft limit function of the axis is invalid, and if it is set to 0, it is valid.

D15=1: The relative coordinates are saved when the power is turned off.

Example: If the X-axis soft limit function is invalid, set this parameter to:
00000100

★16,X-Axis backlash compensation(um)

Used to set the reverse clearance value (radius value) of the transmission mechanism when the X-axis moves in the reverse direction. When the axis moves in the reverse direction, the system automatically calls this value for compensation. Unit: μm (micrometer).

P16-P20 Set the reverse clearance compensation of X/Y/Z/A/B/C/Xs/Ys.

★17,Y-Axis backlash compensation(um)

★18,Z-Axis backlash compensation(um)

★19,A-Axis backlash compensation(um)

★20, 5th-Axis backlash compensation(um)

20a,6th-Axis backlash compensation(um)

- 20b, 7th-Axis backlash compensation(um)
- 20c, 8th-Axis backlash compensation(um)
- 20d, Zs-Axis backlash compensation(um)
- 20e, As-Axis backlash compensation(um)
- 20f, Bs-Axis backlash compensation(um)
- 20g, Cs-Axis backlash compensation(um)

★ 21, X-Axis direction signal[1:normal, 0: reverse]

Used to change the direction of the X-axis motor. When this value is set to 0, the moving direction of the X-axis is equal to the opposite of the command direction. When this value is set to 1, the moving direction of the X-axis is equal to the command direction.

★ 22, Y-Axis direction signal[1:normal, 0: reverse]

★ 23, Z-Axis direction signal[1:normal, 0: reverse]

★ 24, A-Axis direction signal[1:normal, 0: reverse]

★ 25, 5th-Axis direction signal [1:normal, 0: reverse]

25a, Motor direction signal reverse (D1:J6; D2:XS; D3:YS; D4:ZS; D5:AS; D6:BS; D7:CS)

★ 23, X-Axis electronic gear numerator(1-999999)

★ 27, X-Axis electronic gear denominator(1-999999)

★ 28, X-Axis electronic gear denominator(1-999999)

★ 29, Y-Axis electronic gear numerator(1-999999)

★ 30, Y-Axis electronic gear denominator(1-999999)

★ 31, Z-Axis electronic gear numerator(1-999999)

★ 32, Z-Axis electronic gear denominator(1-999999)

★ 33, A-Axis electronic gear numerator(1-999999)

★ 34, A-Axis electronic gear denominator(1-999999)

★ 35, 5th-Axis electronic gear numerator (1-999999)

★ 36, 5th-Axis electronic gear denominator (1-999999)

36a, 6th-Axis electronic gear numerator (1-999999)

36b, 6th-Axis electronic gear denominator(1-999999)

36c, 7th-Axis electronic gear numerator (1-999999)

36d, 7th-Axis electronic gear denominator (1-999999)

36e, 8th-Axis electronic gear numerator (1-999999)

36f, 8th-Axis electronic gear denominator (1-999999)

36g,Zs-Axis electron gear numerator(1-999999)

36h,Zs-Axis electron gear denominator(1-999999)

36i,As-Axis electron gear numerator(1-999999)

36j,As-Axis electron gear denominator(1-999999)

36k,Bs-Axis electron gear numerator(1-999999)

36l,Bs-Axis electron gear denominator(1-999999)

36m,Cs-Axis electron gear numerator(1-999999)

36n,Cs-Axis electron gear denominator(1-999999)

Used to set XYZABCXsYs axis electronic gear ratio.

The numerator=10*reduction ratio;

The denominator = ball screw pitch;

If above parameter setting incorrect, axis will move incorrect, the command distance may not match with real movement.

★37,XYZ axis positive travel limit[0 open,1 close]

★38,XYZ axis negative travel limit[0 open,1 close]

Used to set XYZA axis limit type, 0 means normally open, 1 means normally close.

★39,Floating zero point and Y-Axis compensation

[D3:X;D4:Y;D5:Z;D6:A;D7:B;D8:C;D9:Xs;D10:Ys;0: mechanical zero point;1 float zero point]

Used to set the type of each axis home type, and it is a bit parameter.

D10|D9|D8|D7|D6|D5|D4|D3|D2|D1|D0

D3-D10 corresponds to X/Y/Z/A/B/C/Xs/Ys home type, 1 means float zero point, 0 means mechanical zero point switch;

D1|D0 set if Y axis coordinate reduction;

For example, if just set X axis as float zero point, set:00001000

39-1,X of reference point 1 (mm)

39-2,Y of reference point 1 (mm)

39-3,Z of reference point 1 (mm)

39-4,A of reference point 1 (mm)

★40,Machine coordinate setting with X-Axis floating zero point

This parameter sets the X-axis coordinate value of the floating zero point where the machine tool currently stops. When the machine is turned on and returns to zero, it returns to the floating zero point.

When the mechanical zero point is not used, it is set to a floating zero point.

When the zero return operation is performed, the system will quickly locate to the zero point position.

- ★41, Machine coordinate setting with Y-Axis floating zero point
- ★42, Machine coordinate setting with Z-Axis floating zero point
- ★43, Machine coordinate setting with A-Axis floating zero point
- ★44, Machine coordinate setting with 5th-Axis floating zero point
 - 44a, Machine coordinate setting with 6th-Axis floating zero point
 - 44b, Machine coordinate setting with 7th-Axis floating zero point
 - 44c, Machine coordinate setting with 8th-Axis floating zero point
- ★45, Feed axis homing requirements[1: no need, 0: prompt, 8: mandatory , 9 super mandatory, 15:all axes mandatory, 31:all axes super mandatory]

Used to set the system's handling of reference point return at startup. There are six reference point return handling methods:

Set to 1 (no need): The system does not prompt or impose any restrictions after each startup.

Set to 0 (Prompt): The system will pop up a dialog box prompting the operator to perform a reference point return operation after each startup, and no restrictions will be imposed thereafter.

Set to 8 (Mandatory): The system will pop up a dialog box prompting the operator to perform a reference point return operation after each startup. If a reference point return operation is not performed before automatic operation, the system will display a "Feed axis has not returned to the reference point" prompt and will not execute the program.

Set to 9 (Super mandatory): The system will pop up a dialog box prompting the operator to perform a reference point return operation after each startup. If a reference point return operation is not performed, the system will display a "Feed axis has not returned to the reference point" prompt and will not execute the movement.

Set to 15 (Mandatory): All axes are in mandatory mode; set to 31 (Super mandatory): All axes are in Super mandatory mode.

- ★46, Feed axis homing mode[0:Reverse detection, 1:Reverse non-detection , 2:Detect but not reverse, 3:No detection and no reverse]

Few methods for setting detection switch and zero pulse signal of motor encoder when each axis is backing to zero point:

Set as 0: When backing to zero point, crush to the reference switch, running reversal after the switch is off, then detect the zero pulse signal of motor encoder.

Set as 1: When backing to zero point, crush to the reference switch, running reversal after the switch is off.

Set as 2: When backing to zero point, crush to the reference switch, continue to

run after the switch is off, then detect the zero pulse signal of motor encoder.

Set as others: When backing to zero point, crush to the reference switch, continue to run after the switch is off.

According to the circuit to set, normally suggest to set as 0 or 2, if not to detect the zero pulse signal of motor encoder, the accuracy will be worse, it only base on switch of reference point.

46a, Feed axis homing shield detection encoder zero position

[+2:Yes,+4:X,+8:Y,+16:Z,+32:A,+64:B]

★47, Homing reverse direction parameter

[D2:X;D3:Y;D4:Z;D5:A; 0:Positive;1:Negative]

Exclusive setting of direction and sequence for every axis.

D10|D9|D8|D7|D6|D5|D4|D3|D2|D1|D0

D2 control X home direction; D3 control Y home direction; D4 control Z home direction; D5 control A home direction; D6 control B home direction, 1 means negative, 0 means negative.

For example, if set X axis home direction is negative, then P47=000000100.

D1 is a status control bit of chuck, 1 means detecting, 0 means without detecting.

If D1=1, after chuck clamping ready M10, system will detect M12, only if M12 is valid, will system execute next step. After chuck release ready M11, system will detect M14, only if M14 is valid, will system execute next step.

★48, Home switch normally closed parameter

[D0:X;D1:Y;D2:Z;D3:A;1:Close;0:Open; D6=1 MPG/Auto; D7=1:

Manual/Auto;D18=1:G28 to Zero]

Used to set home switch type[NC/NO].

D15|D14|D13|D12|D11|D10|D9|D8|D7|D6|D5|D4|D3|D2|D1|D0

1 means normal close, 0 means normal open. D0 control X home direction; D1 control Y home direction; D2 control Z home direction; D3 control A home direction; D4 control B home switch type; D5 control C home switch type; D14 control Xs home switch type; D15 control Ys home switch type.

For example, if X,Z home switch is NC, then P48=000000101.

If D6=1, controller will switch between MPG and Auto mode automatically.

If D7=1, controller switch between Manual mode and Auto mode automatically, controller will switch into manual mode after program finish, pressing Start button in manual mode will switch into auto mode automatically.

★49, X-Axis detection zero position length(100um)

Sets the range within which the motor encoder zero pulse signal is detected after

the switch is released when the X-axis returns to zero.

★ 50,Y-Axis detection zero position length(100um)

★ 51,Z-Axis detection zero position length(100um)

★ 52,A-Axis detection zero position length(100um)

★53,5th-Axis detection zero position length(100um)

★ 54,X-Axis offset after homing(10um)

This setting specifies the offset distance the X-axis must travel after detecting the servo motor zero pulse signal to complete the reference point return.Range: -99999 to +99999.

Note: The offset after reference point return is at G00 speed.

★ 55,Y-Axis offset after homing(10um)

★ 56,Z-Axis offset after homing(10um)

★ 57,A-Axis offset after homing(10um)

★58, 5th-Axis offset after homing(10um)

58-1, 6th-Axis offset after homing(10um)

58-2, 7th-Axis offset after homing(10um)

58-3, 8th-Axis offset after homing(10um)

★ 59,Spindle start during gear shifting[1: Yes;0 : No;64:Hold the spindle state]

Set whether to start the spindle when the spindle shifts between high and low gears. 1 means to start, 0 means not to start.

★ 60,Spindle motor speed during gear shifting(1/100rpm)

Set the speed of starting the spindle when the spindle shifts between high and low gears.

★ 61,Spindle rotation direction during gear shifting[0: M03,1: M04]

Set the direction of starting the spindle when the spindle high and low gears are switched, 0 means rotate CW, 1 means rotates CCW.

★ 62,Spindle stop time during gear shifting(10ms)

Set the time to start the spindle when the spindle shifts between high and low gears.

★ 63,Spindle low-speed rotation time during gear shifting(10ms)

Set the time to stop the spindle (M05) when the spindle shifts between high and low gears.

★ 64,Spindle stop delay time(10ms)

The delay time between canceling M03/M004 and output M05.

★64-1,Spindle reverse jog enabled[8:Valid]

★ 67, Spindle direction reversal delay time (100ms)

★ 68, Feed axis reversal delay time(ms)

Delay time after feed axis direction change, set to 0 if the requirement is high.

★ 69, Manual button homing method (1: all return to zero, 0: select return to zero)

★ 80,XZ-Axis coordinate plan [D2: Z-Axis workpiece, D3: X-Axis workpiece, D4: Z-Axis machine tool,D5: X-Axis machine tool,D6: Z-Axis rotary axis,D7: X-Axis rotary axis]

Used to set if X and Z is rotary axis and do coordinate plan. 1 means yes, 0 means no. The parameter is bit parameter.

D7|D6|D5|D4|D3|D2|D1|D0.

D2 control Z axis work piece coordinate plan; D3 control X axis work piece coordinate plan; D4 control Z axis machine coordinate plan; D5 control X axis machine coordinate plan; D6 control Z axis is rotary axis, D7 control X axis is rotary axis.

★ 81, Y-Axis function[0: rotary axis, 1: linear axis]

★ 82,Y-Axis rotary axis workpiece coordinate[0:No;1:normalization]

★ 83,Y-Axis rotary axis machine coordinate[0:No;1:normalization]

★ 96a, 6th-Axis function[0: linear axis, 1: rotary axis]

★ 96b, 6th-Axis rotary axis machine coordinate[0:No;1:normalization]

★ 96c, 6th-Axis rotary axis workpiece coordinate[0:No;1:normalization]

★ 97a,7th-Axis function[0: linear axis, 1: rotary axis]

★ 97b, 7th-Axis rotary axis machine coordinate[0:No;1:normalization]

★ 97c, 7th-Axis rotary axis workpiece coordinate[0:No;1:normalization]

★ 98a, 8th-Axis function[0: linear axis, 1: rotary axis]

★ 98b, 8th-Axis rotary axis machine coordinate[0:No;1:normalization]

★ 98c, 8th-Axis rotary axis workpiece coordinate[0:No;1:normalization]

★ 100, Internal parameter

whether to enable the 4th axis interpolation function[0: Yes, 1:No]

★ 101,A-axis function[0: rotary axis, 1: linear axis]

★ 102,4th-axis rotary axis normalization

Coordinate normalization options:

1 - Applies to workpiece coordinate system only

2 - Applies to machine coordinate system only

3 - Applies to both coordinate systems

★ 103, Auto release the output point or auxiliary relay [10000+Y output point number,

20000+M relay number]

When the coordinate axis is running, the corresponding output point is automatically released or the auxiliary relay sets the output point when the coordinate axis is running. A value greater than 10000 indicates setting the Y point, and a value greater than 20000 indicates setting the M relay number.

For example, when it is set to 10021, it indicates output Y21.

★104, Auto release the input point or auxiliary relay [10000+X input point number, 20000+M relay number, 30000+waiting time ms]

When setting a coordinate operation output point, set whether to detect the input point. A value greater than 10000 indicates detection of point X, greater than 20000 indicates detection of the M relay, and greater than 30000 indicates a delay setting time.

For example, setting 20005 means outputting point Y before detecting point X5 in position.

★105, Automatic clamping axis release (1:A axis; 2: X axis; 4: Y/C axis; 8: Z axis; 16:B axis; 32: C axis; 64:Xs axis; 128: Ys axis)

★110, Internal parameter

Whether the system internal parameters start the multi-axis interpolation function (5-12: 5th axis to 12th axis, 1: no)

★111, 5th-Axis function setting [0 for rotary axis, 1 for linear axis]

★112, 5th-axis rotary axis normalization [Workpiece coordinate + 1; Machine coordinate +2]

★113, 5th-Axis is displayed as C-Axis [0 :no, 1: yes]

★200,SP encode pulse(PPR)[4 times encode lines]

The pulse unit per turn of Spindle encoder, the parameter value=encoder resolution*4. For example: 4096 means a 1024-line encoder.

200-1,SP encoder line alarm threshold (diagnostics)[>10:Valid]

The tolerant pulse unit of spindle encoder feedback. Effective when more than 10.

200-2,Quadrature PPR (x4) of spindle(set as 0 if Encode:SP is 1:1, otherwise it must be integer >99)

Quadrature pulses per revolution (x4) of spindle, if the encoder is connected with spindle by 1:1, set parameter as 0, otherwise, the value should be integrator bigger than 99.

★201, Spindle position feedback detection (1 means yes, 0 means no)

★202, Angle detection during spindle orientation

★203, Angle detection error during spindle orientation

★210, Interpolated tapping axis[91:X,92:Y,93:Z,94:A,95:B,96:C,97:Xs,98:Ys]

The Z-Axis as the main axis for interpolation tapping. For example, 95: The Z axis and the 5th axis perform interpolation tapping.

210a, Y-Axis for tapping[91:X,92:Y,93:Z,94:A,95:B,96:C,97:Xs,98:Ys]

The Y-Axis as the main axis for interpolation tapping. For example, 95: The Y-Axis and the 5th-Axis perform interpolation tapping.

210b, X-Axis for tapping[91:X,92:Y,93:Z,94:A,95:B,96:C,97:Xs,98:Ys]

The X-Axis as the main axis for interpolation tapping. For example, 95: The X-Axis and the 5th-Axis perform interpolation tapping.

★220, Full-pulse control channel for 1st spindle

1 :analog, positive 81-88:XYZABCXsYs, reverse 91-98:XYZABCXsYs

1:Analog 0-10V output, 82 is full pulse forward rotation, 92 is full pulse reverse rotation.

★221, Pulses per revolution (PPR) for 1st spindle full-pulse control

The number of pulses per full revolution of the first spindle is the number of pulses per revolution of the motor.

★222, Acc/Dec control for 1st spindle full-pulse operation(rpm/2ms)

223, Full-pulse control channel for 2nd spindle

1:analog, positive 81-88:XYZABCXsYs, reverse 91-98:XYZABCXsYs

1:Analog 0-10V output, 81-88 is full pulse forward rotation, 91-98 is full pulse reverse rotation.

★224, Pulses per revolution (PPR) for 2nd spindle full-pulse control

★225, Acc/Dec control for 1st spindle full-pulse operation(rpm/2ms)

★226, Speed numerator for dual-spindle following

1:invalid;

1000+numerator :following the 1st spindle;

-(1000+numerator) : following the 2nd spindle

For example, 1002 means the 2nd spindle follows the 1st spindle with a numerator of 2, and -1002 means the 1st spindle follows the 2nd spindle with a numerator of 2.

★227, Speed denominator for dual-spindle following

1:invalid;

1000+denominator :following the 1st spindle;

-(1000+denominator) : following the 2nd spindle

For example, 1002 means the 2nd spindle follows the 1st spindle with a denominator of 2, and -1002 means the 1st spindle follows the 2nd spindle with a denominator of 2.

★228, The start/stop of the 1st spindle simultaneously controls the 2nd spindle
[0 :Yes, 1: No]

★229, The start/stop of the 2nd spindle simultaneously controls the 1st spindle
[0 :Yes, 1: No]

★250,Encoder bus bit width for Ethernet-based servo motors

★260,Full-pulse control channel for 3rd spindle

★261,Full-pulse control channel for 4th spindle

★262,Full-pulse control channel for 5th spindle

★263,Full-pulse control channel for 6th spindle

★264,Max speed of 3rd spindle(rpm)

★265,Max speed of 4th spindle(rpm)

★266,Max speed of 5th spindle(rpm)

★267,Max speed of 6th spindle(rpm)

★268,Pulses per revolution (PPR) for 3rd spindle full-pulse control

★269,Pulses per revolution (PPR) for 4th spindle full-pulse control

★270,Pulses per revolution (PPR) for 5th spindle full-pulse control

★271,Pulses per revolution (PPR) for 6th spindle full-pulse control

★272,Acc/Dec control for 3rd spindle full-pulse operation

★273, Acc/Dec control for 4th spindle full-pulse operation

★274, Acc/Dec control for 5th spindle full-pulse operation

★275, Acc/Dec control for 6th spindle full-pulse operation

★323,Whether the rotary axis disables shortest-path movement

(+4:X;+8:Y/C;+16:Z ;+32:A;+64:B)

When the coordinates are set to rotational axes, whether to avoid running at the nearest distance. Set D2-D6 to 1 indicates that the X/Y/Z/A/B axes do not run at the nearest distance.

★404,Spindle pulse control direction signal(0: reverse,1: normal)

The motor mode when the 1st spindle is set to full pulse, 0 means reverse, 1 means forward.

★405,Whether the spindle uses electronic gearing(0: Yes,1: No)

★406,Numerator of low-gear electronic gearing for spindle(1-32767)

- ★407, Denominator of low-gear electronic gearing for spindle(1-32767)
- ★408, Numerator of high-gear electronic gearing for spindle(1-32767)
- ★409, Denominator of low-gear electronic gearing for spindle(1-32767)
- ★410, Interpolation tap SP axis[91:X,92: Y/C,93: Z,94:A,95: B]
- ★410a, Y axis tap interpolation[91:X,92: Y/C,93: Z,94:A,95: B]
- ★410a, X axis tap interpolation[91:X,92: Y/C,93: Z,94:A,95: B]
- ★411, Tapping and shared axis control mode[2: follow encode;3:interpolation to SP]
- ★412, Spindle teeth number(<P413)

The first spindle and the encoder are not 1:1 with the number of spindle teeth of the reducer.

- ★413, Encode teeth number(>P412)
- ★414, Following axis (7/8/9,17/18/19,27/28/29/D5..D13=1/A follows X/Y/Z,C follows X/Y/Z,B follows X/Y/Z)

A follower axis is to set one or more coordinate axes to move simultaneously with the followed coordinate axis.

- 7 (or D5 = 1) indicates that the A axis follows the X axis;
- 8 (or D6 = 1) indicates that the A axis follows the Y axis;
- 9 (or D7 = 1) indicates that the A axis follows the Z axis;
- 17 (or D8 = 1) indicates that the B axis follows the X axis;
- 18 (or D9 = 1) indicates that the B axis follows the Y axis;
- 19 (or D10 = 1) indicates that the B axis follows the Z axis;
- 27 (or D11 = 1) indicates that the C axis follows the X axis;
- 28 (or D12 = 1) indicates that the C axis follows the Y axis;
- 29 (or D13 = 1) indicates that the C axis follows the Z axis.

- ★501, M123M125 follow-up command advance Z axis distance [>9 or <-9 effective] (um)
- ★502, M123M125 follow-up command advance roughing allowance [radius value] (um)
- ★503, M123M125 follow-up roughing maximum feed speed [>9 effective] (mm/min)
- ★600, Position control gain for tapping spindle in bus system [10001-30000] (gain value is the last four digits, recommended value: 20200)
- ★601, Spindle position control gain during tap retraction in bus system [10001-20000] (gain value is the last four digits, recommended value: 10800)

★602, Smoothing time constant for rigid tapping in bus system [classic method: 10002-10120, new method: 20002-20120; recommended value: 20100]

★602a, New solution for thread tracking machining (>10000 is valid, recommended value: 10020, actual smoothness value -10000)

★603, Positioning tolerance value for tapping feed axis in bus system (Feed axis command pulse number, recommended value is 5)

If it's set to 5, it means that when the error value is <5, the tapping line is in place.

★604, Positioning tolerance value for tap retraction feed axis in bus system (Feed axis command pulse number, recommended value is 20)

★605, Max positioning error of feed axis during interpolated tapping position loop control in bus system (Feed axis command pulse number, recommended to set 3 pitch pulses)

★606, Max positioning error of Spindle during interpolated tapping position loop control in bus system (Feed axis command pulse number, recommended to set 3 pitch pulses)

★607, Display tapping error pulse count

1: The error between the number of tapping axis feedback pulses and the number of spindle encoder pulses during tapping. A positive value indicates that the tapping axis is ahead, and a negative value indicates that the spindle is ahead. The display of this error value can be used to indicate the rigidity difference between the feed axis and the spindle.

★608, Precise display of tapping error pulse count

Used for rigid tapping debugging. 1: means waiting for the pulse to be sent out during tapping. It can accurately display the tapping error but the efficiency will be reduced. So it should be set to 0 after debugging.

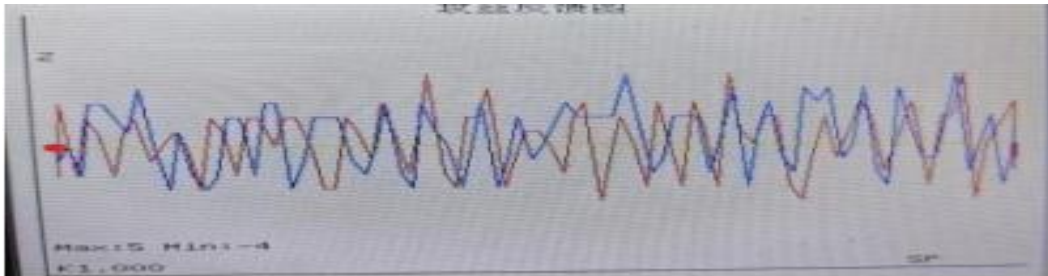
★610, Enable position feedback recording [+1 for straight line; +2 for arc; +4 for tapping; +8 for linear synchronization; +16 for C-axis and A-axis synchronization of G809 for machining; +64 for displaying graphics]

Enable the position feedback record function. This function can display the position graph of 1 straight line/+2 arc/+4 tapping/+8 linear synchronization/+16 turning G809 C-axis and A-axis synchronization. After the command is run, press "Feedback Graph" in the "Diagnosis Interface" to display the position feedback record graph.

<1> Rigid tapping display debugging method <P610=68>

1. Adjust the spindle and feed axis rigidity, requiring an absolute error of <10 (a positive value indicates the feed axis is ahead).

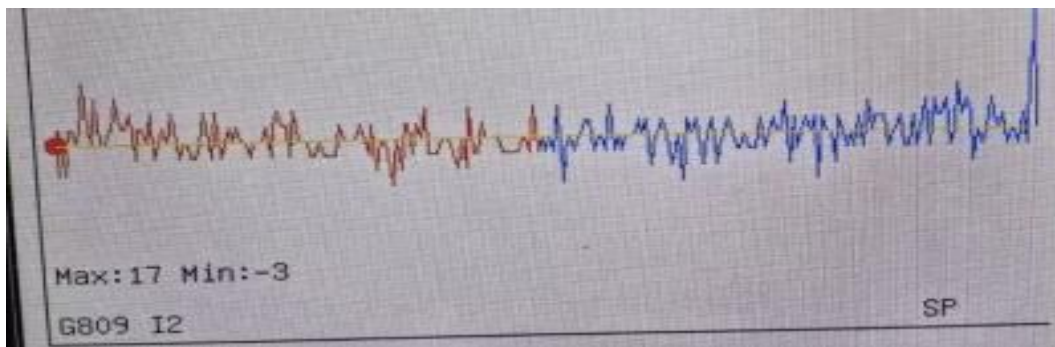
2. The yellow line is the reference line; the horizontal axis is the accumulated spindle pulse value; the vertical axis is the synchronization error; the red line is the tool feed line, and the blue line is the tool lift line. Try to ensure that the three lines coincide (above the yellow line indicates the feed axis is ahead).



<2> Phase-Flying debugging method <P610=80>

1. Adjust the C-axis and A-axis rigidity, requiring an absolute error of <10 (positive values indicate A-axis lead).

2. The yellow line is the reference line. The horizontal axis represents the accumulated spindle pulse value, and the vertical axis represents the synchronization error. A position above the yellow line indicates A-axis lead.



Note: This function requires the system to be connected to encoder feedback/tapping Z-axis feedback/flying square A-axis feedback!!!

3.7.4 Tool parameter

★ 1,C Tool radius compensation's establish(0 mean A,1 mean B)

The method to establish radius C compensation, 0 means A, 1 means B.

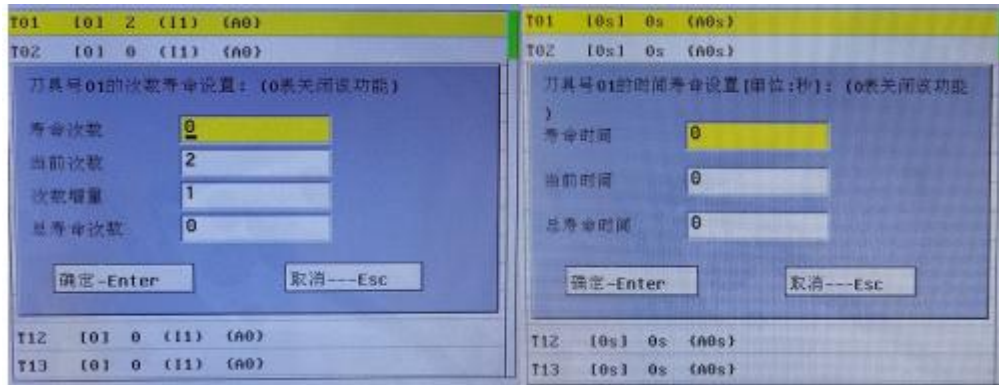
★ 2,C Tool radius compensation's cancel(0 mean A,1 mean B)

The method to cancel radius C compensation, 0 means A, 1 means B.

★ 8,Tool mode (0:M06,1:T change,+32768:Tool life management)

In tool compensation interface, press “Redeem” again to enter into tool life

management, press "Redeem" once again to back tool compensation.



The tool life management interface offers two management options:

<1> By tool usage count:

Each time a tool is changed, the "Current Time" automatically increments by one. If the set life count is reached, the system automatically compensates for the set "Length Increment" and "Radius Increment" (0 indicates no compensation). If the set total life count is reached (0 indicates no compensation), the system automatically issues an alarm. After the alarm, you must manually re-enter the total life count.

<2> By tool usage time:

The "Current Time" automatically calculates the machining time. If the set life count is reached, the system automatically compensates for the set "Length Increment" and "Radius Increment" (0 indicates no compensation). If the set total life count is reached (0 indicates no compensation), the system automatically issues an alarm. After the alarm, you must manually re-enter the total life count.

Set the radius C compensation cancellation method: 0 for type A, 1 for type B.

★9, Actual length of reference tool [Absolute input] (Press 'H' on the tool compensation interface to auto measure)

★32, Tool position signal/WAT signal filter [+256+512+1024: 2/4/8ms of Tool position signal, +2048+4096+8192: 2/4/8ms of WAT signal]

D+number means the bit from right to left:

D8=1 means filter detection signal of tool position by 2ms;

D9=1 means filter detection signal of tool position by 4ms;

D10=1 means filter detection signal of tool position by 8ms, D8-D10 can be 1 at the same time;

D11=1 means filter WAT waiting input signal by 2ms;

D12=1 means filter WAT waiting input signal by 4ms;

D13=1 means filter WAT waiting input signal by 8ms, D11-D13 can be 1 at the same time;

3.7.5 Other parameter

★1,MPG type (1:handheld, 2:sub-panel)

Set the MPG type for the system configuration; 1: handheld mode, 0:Sub- panel mode.

Note: When this parameter is set to 1, the MPG interface CN11 cannot use the swing switch to adjust the spindle or feedrate. That is, "Axis Parameters" P1 and P2 can only be set to "0." A, X, Y, Z, X1, X10, X100, and OFF select the feed axis and gear position, respectively.

★1a,MPG count reverse (0 : normal, 1:A/B signal exchange)

★2,Internal/external chuck (0:Internal chuck, 1:External chuck)

★3,use control switch

87/88: valid, 98/99/100: automatically valid, 87/87.3/98: press the run key to start, 87.3/89: MPG invalid, 0: closed; set a negative value to disable the second spindle

★4,Auto lubrication(0: Yes,1: No)

Set if machine installed with automatic lubricate system.

★5,Auto lubrication time(0.01s)

Set auto lubricate time, namely M32 hold time, unit:10ms.

★6,Auto lubrication interval(0.01s)

Set interval time between each lubrication, namely interval time between M32 valid. Unit: s.

★7,Door switch detection(M12)(0:No,1:Yes)

Set if detect door switch signal, if set as 0, means without door switch; if set as 1, means with door switch. It is suggest to detect door switch for safety sake.

Note: 1. door switch is detected through M12;

2. when M12 is valid, in manual mode, feed axis coordinate can move, but in auto mode, system will pause program automatically.

★8,Door switch(0:Open,1:Close)

Type of door switch signal, 0 means normally open, 1 means normally closed.

★9,Bit parameter

D1=1 clear to 0; D2=1 space ;D5=0 close spindle; D6 speed; D8=1 save M10 when power on;D10=1 arrange; D12=1 skip is invalid; D13=1 back to zero is invalid

The format: D15|D14|D13|D12|D11|D10|D9|D8|D7|D6|D5|D4|D3|D2|D1|D0.

D0 bit: Default as 1, can't change.

D1 bit: Set as 1 means the number of workpiece clear automatically when power on, set as 0 means keep.

D2 bit: Set as 1 means insert space into letters automatically when editing the program, set as 0 means not insert.

D3 bit: Default as 0, can't change.

D4 bit: Default as 0, can't change.

D5 bit: Set as 1 means not stop spindle rotation and cooling when pressing

RESET, 0 means stop, default is 0.

D6 bit: Set as 1 means each axis work according to own speed, work with nonlinear trajectory when executing G00, 0 means simultaneous, reach at the same time. Default is 1.

D7 bit: Default as 0, can't change.

D8 bit: Set as 1 means save the status M10/M11 of spindle loose or tight when power off, recover this status when power on again. Set as 0 means spindle will be tight automatically when power on. Default is 1.

D9 bit: Default as 1, can't change.

D10 bit: Set as 1 means the number of line is arranged automatically when programming.

D11 bit: Set as 1, the output analog of the first spindle is the same time from the first, second spindle channel, the function of the second spindle is valid.

D12 bit: Set as 1, shield the skip function, "/" in front of the segment means invalid.

D13 bit: Default as 0, can't change.

D14 bit: Default as 0, can't change.

D15 bit: Set as 1, the tool set show the relative value, otherwise it shows the absolute value.

Pay attention: This parameter includes the bit which can't be changed. If change this bit, it will cause some problems with controller.

★10, Auto parts counter[1: Yes, 0: No]

Used to count processed work piece quantity.

★11, Program edit sequence number increment

if user want to count program lines automatically, just need to set P11=1.

★12. Internal parameters

[6 for automatic entry, 7 for key entry, 16 for key quick entry, 8 for allow import, 9 for clear]

The LOGO interface setting is displayed at startup. 6 means automatically entering the main interface after the LOGO is displayed, 7 means pressing a key, 16 means automatically reading the motor position at startup and quickly entering the main interface by pressing a key, 8 means you can import a LOGO interface file, and 9 means clearing the LOGO file.

★13, Whether spindle rotation and tool clamping are interlocked(0 mean no)

Set if interlock between spindle and chuck.

0 means no relationship between spindle and chuck status;

1 means only if spindle is in M05 status, can chuck clamp or unclamp;

8 means only if spindle is in M05 status and spindle encoder keep still, can chuck clamp or unclamp;

16 means detect speed and interlock;

32 means when program running in auto mode and status is not pause, button of clamp can not work;

56 means super interlock. when program running in auto mode and status is not pause, button of clamp can not work;

The parameter setting depends on machine configuration and user preference, it is suggested to set as interlock in safety sake.

13-4, The 2nd spindle or 1st spindle third and fourth gear output [1: M203M204 output Y21Y20, 0: M43M44 output Y21Y20]

★14, Whether the auto coolant button is active[1:Yes,0:No]

Set if coolant button works in auto mode. The parameter setting depends on user preference, when set it as 1, operator can control coolant system at any time, to meet processing needs.

★17, XYZA-Axis Servo ALM/X20(0:Open,1:Close)

Set type of XYZA alarm, 0 means normally open, 1 means normal close.

17-1, BCXsYs-Axis Servo ALM/X22(0:Open,1:Close)

17-2, whether to configure an integrated driver (output Y16 brake signal when the drive is normal) [1:valid, 0: invalid]

17-3, Auxiliary relay M824--M831 function [0: 1--8 axes zero return state, 1: 1--8 axes running state]

Set to 0, auxiliary relays M824--M831 indicate the zero return status of 1-8 axes; when set to 1, they indicate the running status of 1-8 axes.

★18, Servo alarm signal for 1st spindle: ALM1/X23 (0:Open,1:Close)

Set the type of the system detection machine tool alarm signal (ALM1, pin 5 of CN3). 0 means normally open, and 1 means normally closed.

18a, Enable the 2nd spindle servo alarm: ALM3/M22(X27) [1:Yes,0:No]

18b, Servo alarm signal for 2nd spindle: ALM3/M22(X27) (0:Open,1:Close)

★19, Lubrication alarm: ALM2/X24(0:Open,1:Close)

Set the type of the system detection machine tool alarm signal (ALM1, pin 2 of CN10). 0 means normally open, and 1 means normally closed.

★20, Tool clamp/unclamp control signal(0:single,1:double M10/M71)

This parameter determines whether the system requires only one signal (one-way valve) or two signals (two-way valve) to control the release/tension of the release/tension tool. This parameter depends on the machine configuration.

Set to 0, only the M10 output signal controls the release/tension of the release/tension tool. When M10 is active, the system tightens the release/tension tool; when M10 is inactive, the system releases the release/tension tool.

Set to 1, M71 and M10 control the release/tension of the release/tension tool, respectively. When relay M10 is active and M71 is inactive, the system tightens the release/tension tool; when M71 is active and M10 is inactive, the system releases the release/tension tool. That is, M10 outputs M10, and M11 outputs M71.

Whether to check whether the release/tension is in place. Axis parameter P40-D1 is 1 for checking; 0 for not checking.

If it's set to detect, after clamping M10 into place, it will automatically detect whether M14 is valid, and will only execute the next action if it is valid; after releasing M11 into place, it will automatically detect whether M12 is valid, and will only execute the next action if it is valid.

★21, Whether the external button is filtered during external tool clamp/unclamp control (0:No,1:Yes M16)

★22, External tool clamping control(0: No, 1: Yes M16)

Used to set if there is an external switch controlling chuck clamping or unclamping, it is a loop signal, namely chuck status changes everytime when Signal becomes effective once(effective once, then unclamp; effective another 1 time, then clamp). when set as 0, means without external chuck control switch; set as 1, means with external chuck control, the signal is M16.

★23, Whether coolant flow detection alarm M14/X30 is enabled [0: Yes, 1: No]

23a, Coolant flow alarm detection signal [1: Open, 0: Closed]

23b, Coolant flow alarm persistence detection time(s)

★24, Air pressure detection alarm M16 / X31[0: No, 1: Yes]

24a, Air pressure alarm detection signal [0: Open, 1: Closed]

★25, Tool clamp/unclamp (M10/M11) short signal time

Set the holding time when the tool clamp/unclamp output control signals (M10/M71) are short signals. A value of 0 indicates a long signal.

★26, Panel emergency stop (0: Open, 1: Close)

Set the emergency stop switch on the system panel to either normally open (0) or normally closed (1). For safety reasons, it is recommended to set it to normally closed.

★27, MPG emergency stop(0: Open, 1: Close)

Set the emergency stop switch (MPG emergency stop or external emergency stop) on the system CN11-5 pin to normally open (0) or normally closed (1). This parameter can be used to set the emergency stop switch. For safety reasons, it is recommended to set it to normally closed.

★28, Run status output(0: Invalid, 1: Valid M69/Y04 run, M65/Y02 stop)

This parameter specifies whether the system outputs a run signal. When the machine is in operation, a setting of 0 indicates no run output signal. A setting of 1 indicates a run output signal, M69. M65 is output when the program stops.

Note: This signal output can be used as a machine status indicator, depending on the machine configuration.

★28a, Activate stop lamp when axes are stopped(Y02)[0: No, 1: Yes]

★29, Alarm output M67/Y03 (0: Invalid, 1: Valid)

Used to set if output alarm signal, namely when machine is in alarm status, if P29=0, controller will not output alarm signal, if P29=1, controller will output alarm signal Y03.

★29a, Whether the alarm output blinks intermittently[0: No, 1: Yes]

★30, Set language(1: Chinese, 0 : English)

★31, Machine tool I/O logic controller PLC[0: Stop, 1: Start]

Set the running status of the system's built-in PMC. 1 means start, 0 means stop, 32 means the PLC controls the output points simultaneously during diagnosis (for debugging), and 64 means the PLC does not control the diagnostic control output (for debugging).

Note: This parameter is usually used as a debugging parameter. It must be started in actual use, otherwise the machine tool will behave abnormally.

★32, High-speed I/O logic controller PLC for machine tools

Set the operating sequence for the system's built-in PLC. 1 means high-speed sequencing (scan cycle 8ms), 0 means high-speed sequencing (scan cycle 100ms), and 28 means ultra-high-speed sequencing (scan cycle 2ms).

Note: This parameter is typically used for debugging and must be enabled in actual use; otherwise, the machine will malfunction.

★33,Disable external stop button [0:No, 64:Yes]

★35,soft-limit without home as manual[1 Yes,0 No]

Set whether the soft limit function is enabled when the system is not in the zero return state. 1 means enabled, 0 means disabled.

Note: Setting this parameter is related to safe operating practices.

★36,Set system time[year-month-day-hour-minute]

Used to set system time, format: year-month-day-hour-minute.

★37,Serial communication baud rate(RS232) [0=7200; 1=9600; 2=14400; 3=19200; 4=38400; 5=57600; 6=115200]

Note: 1. A larger value results in faster communication, but also less stable communication.

2. The baud rate on both ends of the communication device must be set to the same.

★37-1,ModBus station number for OPC-enabled serial port[Odd parity: 10000+,even parity:20000+,no parity 30000+]

The station number set in RS232, if controller station number is 2, and check mode is odd check, then set as 10002.

★38,Manual quick keys can be latched[8:Yes]

Used for manual fast key usage. When this parameter is set to 8, manually press the "Fast" key to switch between fast and slow speeds. There will be a prompt on the screen when fast.

★39,Special parameter

★40,Special parameter

★41,Current parameter backup

Backup current parameter setting as default parameter. After setting machine ready, press P41 twice then set current parameter as default setting for future maintenance.

★42,Restore backup parameters

Restore the default parameter setting which is defined by pressing P41 last time.

★49,Enable external run/pause/emergency stop buttons[1: X43 is running,X44 is pause; 0: Invalid]

★50,Whether to position to the previous segment endpoint when starting from mid-program[8:Yes,0:No]

★51, Whether to return to safe point before moving to the previous segment endpoint when starting from mid-program (+4: Z, +8: X, +16: feed hold)

Set whether to return to the safety point coordinates first when starting from the mid-program and returning to the previous line coordinates. P52 sets the machine tool coordinates of the safety point.

+64: Coordinate offsets set during feed hold take effect immediately for the

current line.

+128: Enter MPG mode during feed hold, maintaining the MPG coordinates.

+256: After manual operation during feed hold, the system does not automatically return to the origin. This is used for manual intervention.

For example:

G00 Z0; Approaches the workpiece.

M00; After a pause, manually moves to the workpiece surface. Next, use G50 to set the starting coordinates.

G50 Z0; Switches to automatic mode and presses the start key to execute this line without returning to the intervention point.

G0 Z50

★52, Machine coordinates of the safe point to return to first

★120, Manual axis button direction reversal(4:X;8:Y;16:Z;32:A)

★140, Comparison between current user coordinates and reference point 1
(1001/2001/3001/4001 table X/Y/Z/A, less than reference point M77 is valid)

★141, Comparison between current user coordinates and reference point 2
(1002/2002/3002/4002 table X/Y/Z/A, less than reference point M78 is valid)

★150, X (mm) of reference point 1

★151, Y (mm) of reference point 1

★152, Z (mm) of reference point 1

★153, A (mm) of reference point 1

★160, X (mm) of reference point 2

★161, Y (mm) of reference point 2

★162, Z (mm) of reference point 2

★163, A (mm) of reference point 2

★200, X-Axis feedback operation alarm error (pulse)[>1]

When controller activated position feedback function, if tacking error during axis running is bigger than P200-P204a, system will alarm.

Bigger than 1 is valid. Press “G” to clear the feedback position ,clear position and the alarm.

★201, Y-Axis feedback operation alarm error (pulse)[>1]

★202, Z-Axis feedback operation alarm error (pulse)[>1]

★203, 4th-Axis feedback operation alarm error (pulse)[>1]

204, 5th-Axis feedback operation alarm error (pulse)[>1]

204a, 6th-Axis feedback operation alarm error (pulse)[>1]

204b, 7th-Axis feedback operation alarm error (pulse)[>1]

204c, 8th-Axis feedback operation alarm error (pulse)[>1]

204d, Zs-Axis feedback operation alarm error (pulse)[>1]

204e, As-Axis feedback operation alarm error (pulse)[>1]

204f, Bs-Axis feedback operation alarm error (pulse)[>1]

204g, Cs-Axis feedback operation alarm error (pulse)[>1]

★205,X-Axis feedback stop alarm error (pulse)[>1]

When controller activated position feedback function, if tacking error after axis stopping is bigger than P205-P209g, system will alarm.

Bigger than 1 is valid. Press “G” to clear the feedback position ,clear position, and the alarm.

★206,Y-Axis feedback stop alarm error (pulse)[>1]

★207,Z-Axis feedback stop alarm error (pulse)[>1]

★208,4th-Axis feedback stop alarm error (pulse)[>1]

★209, 5th-Axis feedback stop alarm error (pulse)[>1]

209a, 6th-Axis feedback stop alarm error (pulse)[>1]

209b, 7th-Axis feedback stop alarm error (pulse)[>1]

209c, 8th-Axis feedback stop alarm error (pulse)[>1]

209d, Zs-Axis feedback stop alarm error (pulse)[>1]

209e, As-Axis feedback stop alarm error (pulse)[>1]

209f, Bs-Axis feedback stop alarm error (pulse)[>1]

209g, Cs-Axis feedback stop alarm error (pulse)[>1]

★210,X-Axis feedback electronic gear numerator[Auto-calculation: L screw lead (μm), M encoder line count]

When the system is equipped with axis position feedback, this parameter represents the electronic gear numerator. You can also enter the lead screw lead and encoder line count. For example, if the lead screw is 6mm and the encoder has 2500 lines, enter: L6000M2500. P210-P214 represent the X/Y/Z/A/B/C/Xs/Ys axes.

Note:1) When using the LxxxxMxxxx format, you must first set the electronic gear numerator and denominator for each axis in the system.

2) This format must be used when reading absolute encoder data via the serial port.

★211,Y-Axis feedback electronic gear numerator[Auto-calculation: L screw lead (μm), M encoder line count]

★212,Z-Axis feedback electronic gear numerator[Auto-calculation: L screw lead (μm), M encoder line count]

★213,4th-Axis feedback electronic gear numerator[Auto-calculation: L screw lead (μm), M encoder line count].

★214, 5th-Axis feedback electronic gear numerator[Auto-calculation: L screw lead (μm), M encoder line count].

214a, 6th-Axis feedback electronic gear numerator[Auto-calculation: L screw lead (μm), M encoder line count].

214b, 7th-Axis feedback electronic gear numerator[Auto-calculation: L screw lead (μm), M encoder line count].

214c, 8th-Axis feedback electronic gear numerator[Auto-calculation: L screw lead (μm), M encoder line count].

214d, Zs-Axis feedback electronic gear numerator[Auto-calculation: L screw lead (μm), M encoder line count].

214e, As-Axis feedback electronic gear numerator[Auto-calculation: L screw lead (μm), M encoder line count].

214f, Bs-Axis feedback electronic gear numerator[Auto-calculation: L screw lead (μm), M encoder line count].

214g, Cs-Axis feedback electronic gear numerator[Auto-calculation: L screw lead (μm), M encoder line count].

★215, X-Axis feedback electronic gear denominator[Auto-calculation: L screw lead (μm), M encoder line count]

When the system is equipped with axis position feedback, this parameter represents the electronic gear denominator. You can also enter the lead screw lead and encoder line count. For example, if the lead screw is 6mm and the encoder has 2500 lines, enter: L6000M2500. P215-P219g represent the X/Y/Z/A/B/C/Xs/Ys axes.

★216, Y-Axis feedback electronic gear denominator[Auto-calculation: L screw lead (μm), M encoder line count]

★217, Z-Axis feedback electronic gear denominator[Auto-calculation: L screw lead (μm), M encoder line count]

★218, 4th-Axis feedback electronic gear denominator[Auto-calculation: L screw lead (μm), M encoder line count]

★219, 5th-Axis feedback electronic gear denominator[Auto-calculation: L screw lead (μm), M encoder line count]

219a, 6th-Axis feedback electronic gear denominator[Auto-calculation: L screw lead (μm), M encoder line count]

219b, 7th-Axis feedback electronic gear denominator[Auto-calculation: L screw lead (μm), M encoder line count]

219c, 8th-Axis feedback electronic gear denominator[Auto-calculation: L screw lead (μm), M encoder line count]

219d, Zs-Axis feedback electronic gear denominator[Auto-calculation: L screw lead (μm), M encoder line count].

219e, As-Axis feedback electronic gear denominator[Auto-calculation: L screw lead (μm), M encoder line count].

219f, Bs-Axis feedback electronic gear denominator[Auto-calculation: L screw lead (μm), M encoder line count].

219g, Cs-Axis feedback electronic gear denominator[Auto-calculation: L screw lead (μm), M encoder line count]

★299, Auto set the sign of coordinate value per revolution for absolute motors when configuring electronic gearing[0:No,1:Yes]

★300. Whether the feed axis is equipped with an absolute encoder motor

[X:D2,Y:D3,Z:D4,A:D5,B:D6,C:D7,Xs:D8,Ys:D9,Zs:D10,As:D11,Bs:D16,Cs:D17,0:No,1:Yes]

If the servo motor is with absolute encoder, it is necessary to set P300 accordingly.

Format: D15|D14|D13|D12|D11|D10|D9|D8|D7|D6|D5|D4|D3|D2|D1|D0

0 means without absolute encoder, 1 means with absolute encoder motor. D2 control X axis; D3 control Y axis; D4 control Z axis; D5 control A axis; D6 control B axis; D7 control C axis; D8 control Xs axis; D9 control Ys axis.

When the system is equipped with an absolute motor, this parameter selects whether the coordinate axis is equipped with an absolute encoder motor. P301 is the lower 16-bit memory address of the motor's multi-turn absolute encoder (e.g., set to 92), P302 is the upper 16-bit memory address of the absolute encoder's single-turn (e.g., set to 91), and P303 is the lower 16-bit memory address of the absolute encoder's single-turn (e.g., set to 90).

Note:

The system connects to the drive via RS485 communication via CN13 port. The communication protocol is as follows:

1) The drive station addresses for RS485 communication are: 1-8, corresponding to the X/Y/Z/A/B/C/Xs/Ys axes.

2) The baud rate for RS485 communication is 19200; the data bits are 8; the stop bit is 1; the format is RTU; D14=0D15=0 indicates odd parity, D14=1D15=0 indicates even parity, and D14=0D15=1 indicates no parity.

When the system is equipped with an absolute motor, this parameter selects whether the axis is equipped with an absolute encoder motor. P301 is the lower 16-bit memory address of the motor's absolute encoder multi-turn (e.g., set to 92), P302 is the upper 16-bit memory address of the absolute encoder single-turn (e.g., set to 91), and P303 is the lower 16-bit memory address of the absolute encoder single-turn (e.g., set to 90). Setting P303-1 to 1 means a special single-turn calculation method:

Single-turn value = upper 16 bits * 10000 + lower 16 bits.

The method for setting the special drive address is as follows:

- (1) P303 = -999999 (single-turn 32-bit/multi-turn 16-bit), P301 multi-turn, P302 single-turn;
- (2) P303 = -999977 (single-turn 32-bit/multi-turn 32-bit), P301 multi-turn, P302 single-turn;
- (3) P302 = -999966 (single-turn 16-bit/multi-turn 16-bit), P301 multi-turn, P303 single-turn;
- (4) P301 negative value (multi-turn 32-bit), such as: -5632 means multi-turn reads 32-bit data from 5632;
- (5) P303 = -999888, P302 = -999888 (signed 32-bit), P301 Address;
- (6) P303 = -999222 (signed 64 bits), P302 lower 32 bits, P301 upper 32 bits.

★301, Lower 16-bit memory address of multi-turn absolute encoder

★302, Upper 16-bit memory address of single-turn absolute encoder

★303, Lower 16-bit memory address of single-turn absolute encoder

303-1, Single-turn value calculation method[0:Normal, 1:Upper 16-bit*10000+lower 16-bit]

★304, Pulses per revolution (PPR) of X-Axis absolute encoder

★305, Pulses per revolution (PPR) of Y-Axis absolute encoder

★306, Pulses per revolution (PPR) of Z-Axis absolute encoder

★307, Pulses per revolution (PPR) of 4th-Axis absolute encoder

★308, Pulses per revolution (PPR) of 5th-Axis absolute encoder

308a, Pulses per revolution (PPR) of 6th-Axis absolute encoder

308b, Pulses per revolution (PPR) of 7th-Axis absolute encoder

308c, Pulses per revolution (PPR) of 8th-Axis absolute encoder

308d, Pulses per revolution (PPR) of Zs-Axis absolute encoder

308e, Pulses per revolution (PPR) of As-Axis absolute encoder

308f, Pulses per revolution (PPR) of Bs-Axis absolute encoder

308g, Pulses per revolution (PPR) of Cs-Axis absolute encoder

P304-P308 are used to set resolution of motor absolute encoder, for example, if motor encoder resolution is 17bit, then P304-P308g=131072; if motor encoder resolution is 23bit, then P304-P307=8388608.

★309, Coordinate distance value per revolution of X-Axis absolute encoder(nm)

★310, Coordinate distance value per revolution of Y-Axis absolute encoder(nm)

★311, Coordinate distance value per revolution of Z-Axis absolute encoder(nm)

★312, Coordinate distance value per revolution of 4th-Axis absolute encoder(nm)

★313, Coordinate distance value per revolution of 5th-Axis absolute encoder(nm)

313a, Coordinate distance value per revolution of 6th-Axis absolute encoder(nm)

313b, Coordinate distance value per revolution of 7th-Axis absolute encoder(nm)

313c, Coordinate distance value per revolution of 8th-Axis absolute encoder(nm)

313d,Coordinate distance value per revolution of Zs-Axis absolute encoder(nm)

313e,Coordinate distance value per revolution of As-Axis absolute encoder(nm)

313f,Coordinate distance value per revolution of Bs-Axis absolute encoder(nm)

313g,Coordinate distance value per revolution of Cs-Axis absolute encoder(nm)

When the machine installed with absolute encoder motor, P309-P313g are used to set coordinate distance value per revolution of axis absolute encoder, unit: nm. Negative value means reverse direction of coordinate counting.

For example, when the X-axis is directly connected with a 4mm lead screw, this value is set to 4000000 or -4000000.

★314,Multi-turn offset of X-Axis absolute encoder[Enter E to clear,EV to clear&Set]

When the machine installed with absolute encoder motor, P314-P18g are used to set motor absolute muti-turn offset value, enter E will clear absolute multi-turn data, normally it is used to set home position or solve data flow when multi-turn value is too big.

★315,Multi-turn offset of Y-Axis absolute encoder[Enter E to clear,EV to clear&Set]

★316,Multi-turn offset of Z-Axis absolute encoder[Enter E to clear,EV to clear&Set]

★317,Multi-turn offset of 4th-Axis absolute encoder[Enter E to clear,EV to clear&Set]

★318,Multi-turn offset of 5th-Axis absolute encoder[Enter E to clear,EV to clear&Set]

318a,Multi-turn offset of 6th-Axis absolute encoder[Enter E to clear,EV to clear&Set]

318b,Multi-turn offset of 7th-Axis absolute encoder[Enter E to clear,EV to clear&Set]

318c,Multi-turn offset of 8th-Axis absolute encoder[Enter E to clear,EV to clear&Set]

318d,Multi-turn offset of Zs-Axis absolute encoder[Enter E to clear,EV to clear&Set]

318e,Multi-turn offset of As-Axis absolute encoder[Enter E to clear,EV to clear&Set]

318f,Multi-turn offset of Bs-Axis absolute encoder[Enter E to clear,EV to clear&Set]

318g,Multi-turn offset of Cs-Axis absolute encoder[Enter E to clear,EV to clear&Set]

★319,16-bit address display for RS485 absolute/EtherCAT bus drive feedback

[231/232:Load 0.1%;282/283:Speed;284:Current;435:Torque 0.1%(Nm)]

Set the system to display the drive feedback status:

231/232: Displays motor load in 0.1% units;

282: Displays motor speed in P/s (EtherCAT);

283: Displays motor speed in r/min (Modbus 485);

284: Displays motor current in A (Modbus 485);

435: Displays motor torque in 0.1% (EtherCAT)/NM (Modbus 485);

★320, Denominator of coordinate distance value per revolution for X-Axis absolute encoder

When the system is equipped with an absolute motor, this parameter represents the denominator of the coordinate distance per revolution of the X-axis absolute encoder. That is, the coordinate distance per revolution of the X-axis absolute encoder = P309/P320.

For example, if the X-axis 4mm leadscrew reduction ratio is 3/1, P309 = 4000000 and P320 = 3.

P320-P324: X/Y/Z/A/B/C/Xs/Ys axes.

★321, Denominator of coordinate distance value per revolution for Y-Axis absolute encoder

★322, Denominator of coordinate distance value per revolution for Z-Axis absolute encoder

★323, Denominator of coordinate distance value per revolution for 4th-Axis absolute encoder

★324, Denominator of coordinate distance value per revolution for 5th-Axis absolute encoder

324a, Denominator of coordinate distance value per revolution for 6th-Axis absolute encoder

324b, Denominator of coordinate distance value per revolution for 7th-Axis absolute encoder

324c, Denominator of coordinate distance value per revolution for 8th-Axis absolute encoder

324d, Denominator of coordinate distance value per revolution for Zs-Axis absolute encoder

324e, Denominator of coordinate distance value per revolution for As-Axis absolute encoder

324f, Denominator of coordinate distance value per revolution for Bs-Axis absolute encoder

324g, Denominator of coordinate distance value per revolution for Cs-Axis absolute encoder

★350, Internet

[0:Close,1:Open, 8:Auto,12:OPC<TCP:Server>,40:OPC<UDP>,552:The stamping process adopts network communication]

350a, Ethercat Drive network port selection [0:NET1,1:NET2]

If set to 0, it means that the Ethercat drive communication selects network port 1, and if set to 1, it means that the Ethercat drive communication selects network port 2.

350b, LAN communication OPC function network port selection[0:NET1,1:NET2]

★351, Gateway IP address

Set as you want, it is better to keep first three numbers the same to address in LAN, for example: 192.168.1.1

★352, Local IP address

It is better to keep first three numbers the same to address in LAN, for example: 192.168.1.103

★353, Local MAC address

Make sure every piece should be different address, for example: 255:255:255:1

★354, Network server IP address

The IP address of computer, It is better to keep first three numbers the same to address in LAN, for example: 192.168.1.13

★361-366 set the Wi-Fi function, Wi-Fi function of USB internet: The network card of USB use EP-N8508GS and BL-LW05-AR5, others cannot supported currently. The install steps as follows:

1. Install ES browser on Android phone;
2. Open internet of ES browser -->Remote manager-->Set;
3. Set the number of port to be 3721 or 2121 or 2221
4. Set the root directory and select the internal storage of the cellphone
5. Setting up a management account with anonymity
6. Set the code to be GBK mode
7. Open the remote manager (put the IP address shown into the parameter No.561)

If there isn't Wi-Fi signal, could use Hotspot of cellphone to form LAN:

1. Open internet of ES browser -->Internet manager-->Create internet hotspot
2. Use defined hotspot, set the user name and code
3. Input the name and code to No.362 and No.363 parameter
4. Quit remote manager and enter remote manager (put the IP address shown into the parameter No.354)

The operation method of entering the FTP server folder on the controller:

- 1、 Press "N" in the program interface(or shift+N).
- 2、 After entering the FTP server folder, press the "N" key (or F6 key) to exit the FTP server folder according to the same operation above.

The advantage of using FTP to transfer files is that you can see the contents of PC folder on the system, and you can freely choose the files you want to transfer, which is more convenient to use. You can use anonymous login FTP server (no need

to set parameter P365, P366), you can also use FTP username and password, FTP username and password using the original wired network FTP username and password.

When the system is connected to the wireless WIFI and the wired Ethernet at the same time, when the FTP server is opened, the wireless WIFI is preferred, and the parameters in the system are as follows:

★361,FTP server IP address of WiFi hotspot[Input based on the value of the FTP server in the phone,As:192.168.2.206]

★362,WiFi hotspot user name

★363,WiFi hotspot password

★364,WiFi hotspot [8:Open,Others:Close]

★365,FTP user name

★366,FTP password

★380,Automatic tool X setting initial coordinate(mm)

★381,Automatic tool Y setting initial coordinate(mm)

★382,Automatic tool Z setting initial coordinate and return point(mm)

★383,Rapid negative movement speed for automatic tool setting(mm/min)

★384,Slow positive movement speed for automatic tool setting(mm/min)

★385,Z-axis coordinate value of workpiece top surface in current workpiece coordinate system after automatic tool setting(mm)

★386,Rapid positioning speed for automatic tool setting(mm/min)

★387,Automatic tool setting mode(1:Fixed tool setter,0:Floating tool setter)

★388,Minimum Z-axis machine coordinate value(mm)

★389,Z-Axis droop value(mm)

Define macro variables of the automatic tool setting gauge function are as follows (corresponding to the other parameters P380 - P389):

#380: The X axis machine coordinate of initial position when automatic tool setting;(Unit:mm)

#381: The Y axis machine coordinate of initial position when automatic tool setting;(unit:mm)

#382: The Z axis machine coordinate of initial position and returning point with automatic tool setting;(unit:mm)

#383: The negative speed of automatic tool setting;(mm/min)

#384: The positive speed of automatic tool setting;(mm/min)

#385: The Z axis coordinate of workpiece surface in current workpiece coordinate system after automatic tool setting;(mm)

#386: The speed which is rapid move to locating position with automatic tool setting;(mm/min)

#387: Automatic tool setting mode (1 means fixed point, 0 means floating point).

#388: The minimal machine coordinate value of Z axis (mm);

#389: The gap value of Z axis [The height which is the gauge surface relative to the workpiece surface(mm)];

Fixed point gauge means putting the gauge in a fixed position, every time the X Y Z axis are automatic running to the fixed point first in tool setting; But the floating point gauge search the tool setting gauge signal along negative of the Z axis.

The input point X25 is default to be the checking point of automatic tool setting gauge to input.

2.The instruction:

M880 (corresponding to ProgramUser0) automatic tool setting instruction;

M882 (corresponding to ProgramUser2),

M883 (corresponding to ProgramUser3) set the gap of Z axis.

3. Automatic tool setting steps:

1)Set the No.380--No.388 parameter in other parameter;

2)Set the No.389 parameter in other parameter to set the drop value of Z axis: this operation needs to be set only once.

A.Run M882 instruction in MDI to set the gap of Z axis;

B.Manual run Z axis to move the tool nose to the workpiece surface;

C.Run M883 instruction in MDI to automatic set the gap of Z axis No.389 parameter in other parameter;

3) MDI choose the workpiece coordinate system G54/G59;

4) Automatic tool setting: MDI running the M880 instruction, automatic set the Z axis offset of the current workpiece coordinate system.

★390, Total detection count of automatic tool setting(1--5)

★391, Distance from RTCP tool rotation center to tool holder (mm)

★400, Axis name display configuration [xxxxx2]

123456 represents X/Y/Z/A/B/C respectively.

For example, 123462 means the X/Y/Z/A/C axes, and 124502 means the X/Y/A/B axes.

★401, Feed axis pulse port configuration [xxxxxxxx0]

12345678 represents X/Y/Z/A/B/C/Xs/Ys respectively, such as 123478560, the 7th/8th/5th/6th axis pulses are output from B/C/XS/YS ports

★500,Machine tool No.

★501,Interface color display [1: White,8:Black]

★591,One-step definition for stepper configuration

Set the current parameters to the factory default stepper motor parameters, which are used for stepper devices installed on machine tools. This parameter mainly adjusts some speed parameters based on the motor characteristics.

★ 592, One-step definition for servo configuration

★ 600, System bus control method (4000-4099: EtherCAT; 6000: Modbus;
6001: 1-6axis Modbus; 6002: 3 axis Modbus)

★ 601, EtherCAT position loop gain coefficient (Recommended setting: 40)

★ 602, EtherCAT position loop gain coefficient for thread machining (Recommended setting: 40)

602a, New solution for thread tracking machining (EtherCAT set: 20000)

★ 603, etherCAT Position in position allowable difference (number of motor encoder pulses)

Recommended setting: 10005

For periodic position control ($0x6060 = 8$), the setting value must be greater than 10000. (In this case, the system's position loop gain factor has no effect).

For PID position loop control ($0x6060 = 9$), the setting value does not need to be increased; the setting value can be less than or equal to 800 (in this case, the system's position loop gain factor will take effect).

This parameter can be modified online in real time. For Maxim servos, the value of $0x6060$ will be displayed on the digital display, such as OP8rn or OP9rn.

For EtherCAT bus, when setting the position loop PID control on the system, the following two parameters must be modified:

P601, EtherCAT position loop gain coefficient;

P602, EtherCAT position loop gain coefficient for thread machining.

Note: It is recommended to use periodic position control ($0x6060 = 8$, P603 setting value should be greater than 10000).

P603 = 1-9999 indicates periodic speed control, without PDO mapping to $0x6081/6083/6084$.

P603 = 10001-12000 indicates periodic position control, without PDO mapping to $0x6081/6083/6084$.

P603 = 20001-29999 indicates periodic speed control, with PDO mapping to $0x6081/6083/6084$.

P603 = 30001-32000 indicates periodic position control, with PDO mapping to $0x6081/6083/6084$.

Note: $0x6081$ is the feedforward speed, $0x6083$ is the acceleration, and $0x6084$ is the deceleration. It is best to set the system to no mapping all of these. When using periodic speed control, the drive must accelerate and decelerate as quickly as

possible, otherwise oscillation will occur.

1. EtherCAT linear scale full closed-loop control method:

Other parameter settings:

P601 = 40; EtherCAT position loop gain coefficient

P602 = 40; EtherCAT position loop gain coefficient for thread machining

P603 = 5; EtherCAT positioning tolerance value

P631 = 1310720; X-axis closed-loop linear scale function enabled, alarm error value 1310720 (10 motor revolutions)

P641 = 131072; X-axis closed-loop linear scale feedback pulse electronic gear numerator [Scale feedback pulse number x numerator / denominator = motor encoder feedback pulse number; negative value indicates reverse feedback direction; P641-P658 are the same]

P651 = 10000; X-axis closed-loop linear scale feedback pulse electronic gear denominator

Example:

Motor encoder 1 revolution 131072, lead screw 10mm, scale accuracy 1u, 1:1 reducer

Numerator/denominator = Number of motor encoder feedback pulses * linear scale accuracy / [reduction ratio * lead screw distance]

$$= 131072 * 0.001 / 10 = 131072 / 10000$$

Linear scale AB command = Number of scale feedback pulses * numerator of linear scale feedback pulse electronic gear / denominator of linear scale feedback pulse electronic gear

2. EtherCAT linear scale AB Signal Compensation Function

Other Parameter Settings:

P601 = 40; EtherCAT position loop gain coefficient

P602 = 40; EtherCAT position loop gain coefficient for thread machining

P603 = 5; EtherCAT positioning tolerance value

P631 = -888888; Enable linear scale AB signal compensation on the X-Axis

P641 = 131072; Electronic gear numerator for X-Axis closed-loop linear scale feedback pulses [linear scale feedback pulse count x numerator / denominator = motor encoder feedback pulse count. A negative value indicates the feedback direction is reversed. Same for P641-P658]

P651 = 10000; Electronic gear denominator for X-Axis closed-loop linear

scale feedback pulses

Example: The motor encoder has a rotation of 131072. If the AB feedback is 10,000 pulses, the motor encoder feedback pulses will be 131072. Pulse number [compensation amount] = grating scale feedback pulse number x numerator / denominator = 10000 * 131072 / 10000 = 131072

★604,EtherCAT feedforward gain[10001-10100 or 20001-20100]

★605,etherCAT feedforward filter time constant [1--25]

★610,EtherCAT electronic gear common denominator (Recommended setting: 1000)

★611,X-Axis electronic gear numerator for EtherCAT (Recommended setting: 131072)

★612,Y-Axis electronic gear numerator for EtherCAT (Recommended setting: 131072)

★613,Z-Axis electronic gear numerator for EtherCAT (Recommended setting: 131072)

★614,4th-Axis electronic gear numerator for EtherCAT (Recommended setting: 131072)

★615,5th-Axis electronic gear numerator for EtherCAT (Recommended setting: 131072)

★616,6th-Axis electronic gear numerator for EtherCAT (Recommended setting: 131072)

★617,7th-Axis electronic gear numerator for EtherCAT (Recommended setting: 131072)

★618,8th-Axis electronic gear numerator for EtherCAT (Recommended setting: 131072)

618a,Zs-Axis electronic gear numerator for EtherCAT (Recommended setting: 131072)

618b,As/Bs/Cs-Axis electronic gear numerator for EtherCAT (Recommended setting: 131072)

★620,EtherCAT station number configuration [≥ 10 :effective](Recommended setting: 12345678)

★621,etherCAT Servo drive type

1 means Maxsine, Aecon/Invnt/Estun/STEP/Xinje, 2 means Sanyo Denki/Panasonic A5, 3 means Enpu ,4 means Yakotec/Thinkvo/Panasonic A6/Yuhai, 5 means bichannel Thinkvo, 6 means Tsino-dynatron, 7 means Zhenzheng/+3Eura/+4Dorna/Kinco+6DVS, 8 means Jotong/Weide)

Example: 777777773 means Eura driver; 777777774 means Dorna/Kinco

driver; 77777776 means DVS driver.

If the 9th byte is 5, it means delay 1 minutes before communication after power on, for example, P621=66666665 means when connect Tsino-dynatron driver, delay 1 minute before communication in order to make sure driver initializes normally.

621a, EtherCAT Spindle drive type

First two digits means 1/2nd spindle control mode

9: Closed-loop speed control mode, 1: Open-loop position control mode, next two digits: 1/2nd spindle drive type)

★ 622, X-Axis collision threshold via EtherCAT (0.1%)

★ 623, Y-Axis collision threshold via EtherCAT (0.1%)

★ 624, Z-Axis collision threshold via EtherCAT (0.1%)

★ 625, 4th-Axis collision threshold via EtherCAT (0.1%)

★ 626, 5th-Axis collision threshold via EtherCAT (0.1%)

★ 627, 6th-Axis collision threshold via EtherCAT (0.1%)

★ 631-638, Permissible alarm error value when X/Y/Z/4th/5th/6th/7th/8th-Axis closed-loop linear scale function is enabled [>2: Closed-loop function]

★ 641-648, X/Y/Z/4th/5th/6th/7th/8th-Axis closed-loop linear scale feedback pulse electronic gear numerator

★ 651-658, X/Y/Z/4th/5th/6th/7th/8th-Axis closed-loop linear scale feedback pulse electronic gear denominator

★ 700, EtherCAT power-on delay time [>2 valid] (*50ms)

★ 800, Internal parameters

Internal parameters (0: lathe, 1: milling machine, 69: machining center)

★ 900, User-defined dialog box [1: is invalid, 4: is valid, 12: is all]

Used to set if activate user-defined dialog when press Enter in main interface.

★ 901, Axis home sequence (>9) [5 bits, 1/2/3/4 is X/Y/Z/A, last bit must is 0]

Used to set sequence of each axis home. For example, if P901=4521360, it means axis go home sequence is 452136 axis.

★ 902, Internal parameters [2: keyboard 1, 4: keyboard 2, 8: keyboard 3]

902a, Shield the manual shift key +1-9 to switch the auxiliary relay status function [1: Yes, 0: No]

★ 903, Internal parameters [2: enabling SD card function, +16: the 5th axis welding follow-up processing function (priority), +32: the 4th axis welding follow-up processing function]

★ 904, Internal parameters [4: file decryption, 8: file encryption]

★ 905, Data retention without ferroelectricity [+64 for saving data only when the

system is idle, and not when the system is powered off]

Used to set sequence of each axis home. For example, if P901=4521360, it means axis go home sequence is 452136 axis.

★910,M18/M22/M24/M28 High speed input[1:Yes]

Used to set if define M18/M22/M24/M28 as high speed input of G31/G311.

★911,Whether enable M18 as the teaching, M28 as the recording

function[1:Yes,0:No]

Used to set if activate teaching function by input point M18/M24/M28.

★912,"Reset" to reset the output interface[1:Yes,0:No]

Used to set if reset output point when press Reset button.

★913, Whether auxiliary relays M768-M783 are used as extended custom alarms [16: Start, 0: Not start]

★915, Whether to display the current channel number (set to 85 to start)

85 means turning on the dual-channel system to display the channel number mark.

★916, Set to display the current channel number (set to 1--9 to be effective)

★994,CNC controller serial number

★995,Machine tool serial number

★996,User-defined serial number

3.7.6 Coordinate system

The parameter has the function of multiple coordinate system, includes 6 workpiece coordinate system and a lathe coordinate system G53. A processing program can set a workpiece coordinate system, workpiece coordinate system could move its original to change,the value of coordinate system in parameter is coordinate value of the original point (zero point) in the lathe coordinate system.

Using G54 to G59 to set 6 workpiece coordinate system, in coordinate system interface could modify the coordinate value of original of 6 workpiece coordinate system in lathe coordinate system.

Parameter:

1-0,Current group of workpiece coordinate[G54-G59]

1-1,X of work coordinates G54-G59(mm)

1-2,Y of work coordinates G54-G59(mm)

1-3,Z of work coordinates G54-G59(mm)

1-4,A of work coordinates G54-G59(mm)

1-5,B of work coordinates G54-G59(mm)

1-6,C of work coordinates G54-G59(mm)

1-7,Xs of work coordinates G54-G59(mm)

1-8,Ys of work coordinates G54-G59(mm)

- 2-0,Current group of workpiece coordinate[G54.1-G54.48]
- 2-1,X of work coordinatesG54.1-G54.48(mm)
- 2-2,Y of work coordinatesG54.1-G54.48(mm)
- 2-3,Z of work coordinatesG54.1-G54.48(mm)
- 2-4,A of work coordinatesG54.1-G54.48(mm)
- 2-5,B of work coordinatesG54.1-G54.48(mm)
- 2-6,C of work coordinatesG54.1-G54.48(mm)
- 2-7,Xs of work coordinatesG54.1-G54.48(mm)
- 2-8,Ys of work coordinatesG54.1-G54.48(mm)
- 1,X of work coordinates G54(mm)[Incremental input, E start entry means absolute]
- 2,Y of work coordinates G54(mm)
- 3,Z of work coordinates G54(mm)
- 4,A of work coordinates G54(mm)
- 6,X of work coordinates G55(mm)
- 7,Y of work coordinates G55(mm)
- 8,Z of work coordinates G55(mm)
- 9,A of work coordinates G55(mm)
- 11,X of work coordinates G56(mm)
- 12,Y of work coordinates G56(mm)
- 13,Z of work coordinates G56(mm)
- 14,A of work coordinates G56(mm)
- 16,X of work coordinates G57(mm)
- 17,Y of work coordinates G57(mm)
- 18,Z of work coordinates G57(mm)
- 19,A of work coordinates G57(mm)
- 21,X of work coordinates G58(mm)
- 22,Y of work coordinates G58(mm)
- 23,Z of work coordinates G58(mm)
- 24,A of work coordinates G58(mm)
- 26,X of work coordinates G59(mm)
- 27,Y of work coordinates G59(mm)
- 28,Z of work coordinates G59(mm)
- 29,A of work coordinates G59(mm)
- 30,B of work coordinates G59(mm)
- 41,X of work coordinates G54.1(mm)
- 42,Y of work coordinates G54.1(mm)
- 43,Z of work coordinates G54.1(mm)
- 44,A of work coordinates G54.1(mm)
- 45,B of work coordinates G54.1(mm)
- 46,X of work coordinates G54.2(mm)
- 47,Y of work coordinates G54.2(mm)
- 48,Z of work coordinates G54.2(mm)
- 49,A of work coordinates G54.2(mm)
- 50,B of work coordinates G54.2(mm)

51,X of work coordinates G54.3(mm)
52,Y of work coordinates G54.3(mm)
53,Z of work coordinates G54.3(mm)
54,A of work coordinates G54.3(mm)
55,B of work coordinates G54.3(mm)
56,X of work coordinates G54.4(mm)
57,Y of work coordinates G54.4(mm)
58,Z of work coordinates G54.4(mm)
59,A of work coordinates G54.4(mm)
60,B of work coordinates G54.4(mm)
61,X of work coordinates G54.5(mm)
62,Y of work coordinates G54.5(mm)
63,Z of work coordinates G54.5(mm)
64,A of work coordinates G54.5(mm)
65,B of work coordinates G54.5(mm)
66,X of work coordinates G54.6(mm)
67,Y of work coordinates G54.6(mm)
68,Z of work coordinates G54.6(mm)
69,A of work coordinates G54.6(mm)
70,B of work coordinates G54.6(mm)
71,X of work coordinates G54.7(mm)
72,Y of work coordinates G54.7(mm)
73,Z of work coordinates G54.7(mm)
74,A of work coordinates G54.7(mm)
75,B of work coordinates G54.7(mm)
76,X of work coordinates G54.8(mm)
77,Y of work coordinates G54.8(mm)
78,Z of work coordinates G54.8(mm)
79,A of work coordinates G54.8(mm)
80,B of work coordinates G54.8(mm)
81,X of work coordinates G54.9(mm)
82,Y of work coordinates G54.9(mm)
83,Z of work coordinates G54.9(mm)
84,A of work coordinates G54.9(mm)
85,B of work coordinates G54.9(mm)
86,X of work coordinates G54.10(mm)
87,Y of work coordinates G54.10(mm)
88,Z of work coordinates G54.10(mm)
89,A of work coordinates G54.10(mm)
90,B of work coordinates G54.10(mm)

3.7.7 Password

In this system in order to prevent from the parameter modification in accident, make sure the lathe working, the system adopts the parameter setting of classify the

authority. Divided into "CNC factory" and "Lathe factory", "User factory" three level authority. The "CNC factory" set for the function of the system, belong to internal parameter; "Lathe factory" set equipment configuration of lathe and mechanical index and some parameter about safety; "User factory" set processing technology, performance and the processing program.

The initial situation of three-level classification in this system: "CNC factory" is enabled, "Lathe factory" and "User factory" both are not enabled. If you want to enable the authority function, you must use the initial password to enable access function, then set the corresponding new access password to enable. Pay attention to the initial code is only be used once, the code will invalid after setting a new code, please remember the new code.

Pay attention: the code must be 6 digits, the code can be number and letter.

Parameter:

1. Is enable CNC Co.' password
2. Is enable Machine Co.' password
3. Is enable User' password
4. Modify CNC Corporation.'s password
5. Modify Machine Corporation.'s password
6. Modify User's password
7. Curry work time
8. Software version

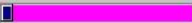
3.8 Set tool compensation

Press "Redeem" to enter interface of redeem in any interface, including "Radius compensation" "Length" "Clear all value" "Clear current value" "Set tool" "Posit tool" "Set", total 7 functions, correspond to press "F1-F7" to enter corresponding interface, press "Esc" to back the primary menu interface. Press "Redeem" to enter interface of redeem in any interface, including "Radius compensation" "Length" "Clear all value" "Clear current value" "Set tool" "Posit tool" "Set", total 7 functions, correspond to press "F1-F7" to enter corresponding interface, press "Esc" to back the primary menu interface.

3.8.1 Radius compensation

Press "F1" to enter radius compensation interface in redeem interface. The parameter is used to set adopt corner radius of the tool.

Setting method: Press "↑ ↓" to make cursor move to the corresponding tool and press "Enter" to pop up a dialog box, import corresponding tool radius(Absolute value), press "Enter" at last.

AutoCon Stop		高科数控		N00000		2019-04-08 10:43	
Press T Key Length make tool base on mainfac							
T01	H:	0.000	└	0.000┐			
T02	H:	0.000	└	0.000┐			
T03	H:	0.000	└	0.000┐			
T04	H:	0.000	└	0.000┐			
T05	H:	0.000	└	0.000┐			
T06	H:	0.000	└	0.000┐			
T07	H:	0.000	└	0.000┐			
T08	H:	0.000	└	0.000┐			
T09	H:	0.000	└	0.000┐			
T10	H:	0.000	└	0.000┐			
H11	H:	0.000	└	0.000┐			
H12	H:	0.000	└	0.000┐			
H13	H:	0.000	└	0.000┐			
					Program 99.TXT		
					Progress 		
					G53		
					T00H0D0		
					TrueFeed F 0.000		
					M05 M09 M10		
					M78 M33 M41		
					G00 X100%		
					F500 X100%		
					S0 X100% SP000		
					Machine Coord		
					X 206.579		
					Y -4.179		
					Z 200.000		
					PartTime 12:29		
					PartNo 62		
					SPrpm 0		
No Alarm							
F1 Radius F2 Length F3 ACLEA F4 CLEAR F5 SetTool F6 ToolSeat F7 Set F8 CANCEL							

3.8.2 Length compensation

Press “F2” to enter length of redeem interface. The parameter is used to modify the length which is adopt or reset the length.

Method of modifying the length:

Press “↑ ↓” to make cursor move to the corresponding tool number and press “Enter” to pop up a dialog box, import the modifying axis into the dialog box and import the modifying value(import 0.05 to plus 0.05, import -0.05 to reduce 0.05), press “Enter” to confirm. The system calculates current value of redeem after finishing setting.

Method of set tool automatically:

Make machine move to a position so that measure corresponding tool coordinate, press “↑ ↓” to make cursor move to corresponding tool number and press “A” to pop up a dialog box, import the reset axis into dialog box and import the value of measuring the workpiece of corresponding axis, press “Enter” to confirm. The length compensation of corresponding axis has been reset. The system automatic refresh current value of redeem after finishing setting. It also can be set in manual status, press “H” in manual status to set tool automatically.

Method of initializing the length compensation value and radius of tool:

Press “F3” or “F4” to initialize all the length compensation or current length compensation.

Pay attention: the length compensation can be positive or negative number, but the radius compensation only can be positive number.

3.8.3 Posit tool

Press “F6” to enter posit tool interface in redeem interface. The parameter is used to set the kind of tool when adopting radius compensation of tool.

Method of setting: Press “↑ ↓” to make cursor move to corresponding tool number and press “Enter” to pop up a dialog box, import the code of corresponding tool kinds and press “Enter” to confirm.

3.8.4 Set quantity

Press “F7” to prompt a dialog box in the redeem interface to set and manage the total tools. The quantity of the tool in this system can be set 99.

3.9 Screw compensation

Press “Parameter” twice in parameter interface to enter screw compensation interface to set the screw compensation.

Screw compensation is used for automatical compensating the error of screw pitch, compensate the influence from the error of screw pitch to the prevision of operating lathe. The system adopts storage mode of screw compensation: Making the lathe's home point as the starting point when debugging, measured the error curve of screw, studied out the correctional curve according to the error curve, import the value of correctional curve into the correctional parameter and system is going to compensate according to the parameter in automatical running.

Man Con		N00000		2013-02-26 14:51	
Basic Parameter		No	Coord(mm)	Value(um)	
1, Standby:	0	1)	<160.000>	0	
2, Neg-point:	3	2)	<80.000>	0	
3, Pos-point:	2	3)	<0.000>	0	
4, multiple:	1.000	4)	<-80.000>	0	
5, distance(um):	80000	5)	<-160.000>	0	
		6)	<-240.000>	0	
		Program %99.TXT			
		Instruction code			
		G53			
		T01H0D0			
		Machine Status			
		M05 M09 M10			
		M78 M33 M70			
		G00 X100%			
		F120 X100%			
		S0 X100%			
		Machine Coord			
		X 0.000 A 0.000			
		Y 0.000 B 0.000			
		Z 0.000 SP360			
		PartTime 0: 0			
		PartNo 0			
		SPrpm 0			
No Alarm					
X-axis	Y-axis	Z-axis	A-axis	B-axis	ChePro CLEAR CANCEL

Screw compensation interface

Screw compensation by the axis as the unit to set storage, set X Y Z axis separately,

by pressing “F1” “F2” “F3” “F4” to switch; Every axis of screw compensation interface has two areas (basic parameter and set the compensation), by pressing “→” “←” to move the cursor to realize.

Basic parameter:

Press “↑ ↓” to select current basic parameter to set in basic parameter, press “Enter” to pop up a dialog box to import the error compensation of every axis and import the basic information of screw compensation.

Set compensation value:

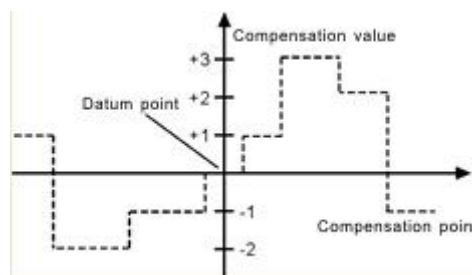
In the area of setting compensation, it will show the value of compensation and every axis' error compensation point of screw pitch. Press “↑ ↓ PgDn PgUp” to select current compensation point and press “Enter” to pop up a dialog box to import the value of compensation, import the value of current compensation point.

Test program generate automatically

Automatic generate a program of laser interferometer to check the screw compensation. Enter the screw pitch interface and set basic parameter, press checking program to pop up a dialog box and press “Enter” to generate corresponding checking program of screw compensation.

The number of compensation points can be set freely, the maximum number of each axis is 300. The basic parameter of every axis' error compensation of screw pitch includes as follows:

1. Reserve.
2. Backward checking points.
3. Forward checking points.
4. Compensation override.
5. The spacing of compensation points (um).



The system calculates every axis' error compensation points' positions of screw pitch according to basic parameter automatically, every axis's error compensation points' spacing is uniform, user can import compensation value of each point (This system requires importing absolute value, relating the of home point).

The compensation points are uniform, set the spacing into each axis.

For example:

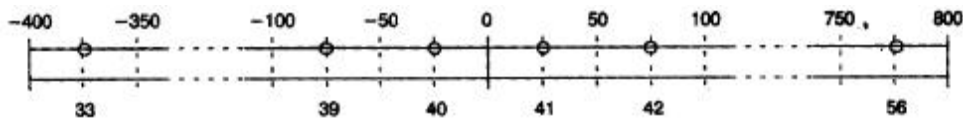
Example 1: Linear axis: when length of travel is -400mm ~ +800mm, spacing of points is 50mm, reference point compensation is No.40, it can figure out that Compensation point of farthest end in negative direction is:

Machine negative travel/point interval +1 = $40 - 400/50 + 1 = 33$.

Compensation point of farthest end in positive direction is:

Machine positive travel/point interval $+1=40+800/50=56$.

The corresponding relationship between machine coordinate and compensation point is:



output compensation value in 0 position

parameters set as follows:

Compensation point No. of reference point: 40

Compensation point No. of farthest end in negative direction: 30

Compensation point No. of farthest end in positive direction: 56

Compensation override: 1

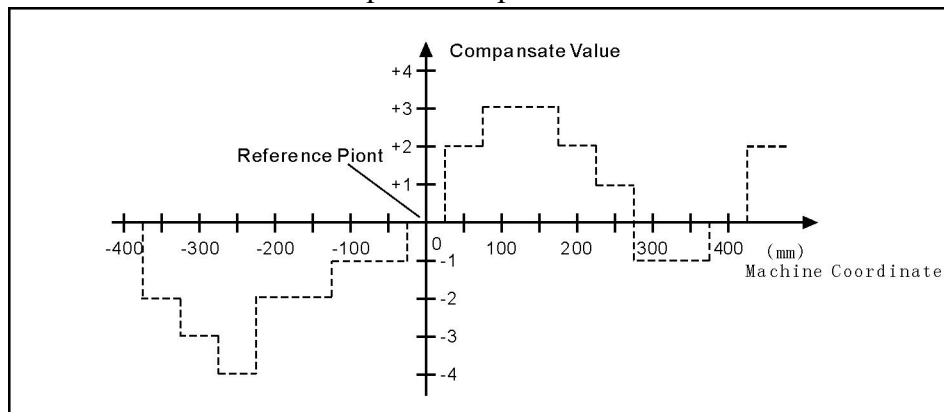
Compensation point interval: 50000

Corresponding compensation point and value:

The compensation value in corresponding compensation point:

No.	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	56
Value	+2	+1	+1	-2	0	-1	0	-1	+2	+1	0	-1	-1	-2	0	+1	+2	+1

The contrasted chart of compensation points and value as follows:

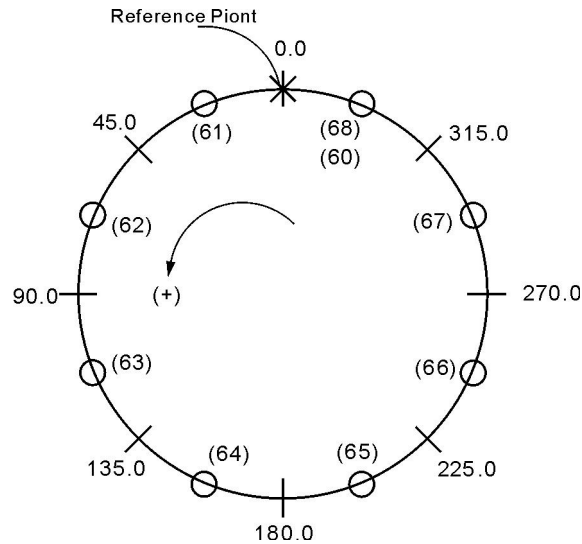


Example 2: rotor axis: when movement per revolution is 360° , interval of points 45° , reference point compensation NO. 60, Compensation point NO. of farthest end in negative direction is usually same as reference point compensation point NO.

Compensation point NO. of farthest end in positive direction is:

Reference compensation point NO.+ movement per revolution/compensation point interval $=60+360/45=68$.

Machine coordinate and compensation point NO. correspondence is:

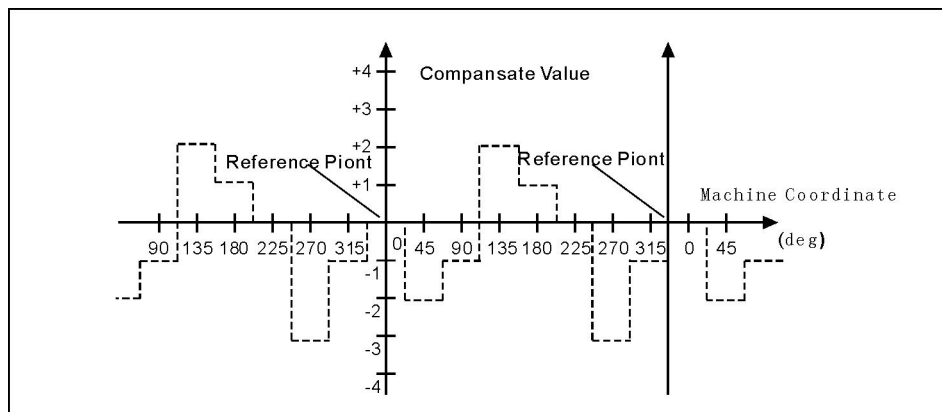


Note: input value in small circle. If the total amount from 61 to 68 doesn't equal 0, accumulated pitch error per revolution will deviate, so same value shall be put in 60 and 68.

Output compensation value at corresponding point:

NO.	60	61	62	63	64	65	66	67	68
VALUE	+1	-2	+1	+3	-1	-1	-3	+2	+1

Compensation point and value contrast:



3.10 Diagnosis

Press "Diagnosis" to enter the diagnosis interface in parameter interface.

System diagnosis interface(Input signal)

Press "F2" and "Pgup Pgdn" to check the status of input and output, press "F4" to check alarm information.

Man Con		N00000		2013-02-26 15:05																																																																																																																																																																																	
Input point <table border="1"> <tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>X00</td><td>X01</td><td>X02</td><td>X03</td><td>X04</td><td>X05</td><td>X06</td></tr> <tr><td>T01</td><td>T02</td><td>T03</td><td>T04</td><td>T05</td><td>T06</td><td>T07</td></tr> <tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>X08</td><td>X09</td><td>X10</td><td>X11</td><td>X12</td><td>X13</td><td>X14</td></tr> <tr><td>M34/A0</td><td>-L</td><td>+L</td><td>M36/Y0</td><td>X0</td><td>Z0</td><td>KRUN</td></tr> <tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>X16</td><td>X17</td><td>X18</td><td>X19</td><td>X20</td><td>X21</td><td>X22</td></tr> <tr><td>X20</td><td>Z20</td><td>KLEFT</td><td>KRIGHT</td><td>STOP</td><td>TOK</td><td>ALM</td></tr> <tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>X24</td><td>X25</td><td>X26</td><td>X27</td><td>X28</td><td>X29</td><td>X30</td></tr> <tr><td>ALM2</td><td>M28</td><td>M24</td><td>M22</td><td>M18</td><td>M12</td><td>M14</td></tr> <tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr> <tr><td>X32</td><td>X33</td><td>X34</td><td>X35</td><td>X36</td><td>X37</td><td>X38</td></tr> <tr><td>HX</td><td>HY</td><td>HZ</td><td>HA</td><td>HX1</td><td>HX10</td><td>HX100</td></tr> <tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>X40</td><td>X41</td><td>X42</td><td>X43</td><td>X44</td><td>X45</td><td>X46</td></tr> <tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td></tr> <tr><td>X60</td><td>X61</td><td>X62</td><td>X63</td><td>X64</td><td>X65</td><td>X66</td></tr> <tr><td>DS3</td><td>DS2</td><td>DS1</td><td>DS0</td><td>DK3</td><td>DK2</td><td>DK1</td></tr> <tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>X67</td><td>X68</td><td>X69</td><td>X70</td><td>X71</td><td>X72</td><td>X73</td></tr> <tr><td>DK0</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> </table>				0	0	0	0	0	0	0	X00	X01	X02	X03	X04	X05	X06	T01	T02	T03	T04	T05	T06	T07	0	0	0	0	0	0	0	X08	X09	X10	X11	X12	X13	X14	M34/A0	-L	+L	M36/Y0	X0	Z0	KRUN	0	0	1	0	0	0	0	X16	X17	X18	X19	X20	X21	X22	X20	Z20	KLEFT	KRIGHT	STOP	TOK	ALM	0	0	0	0	0	0	0	X24	X25	X26	X27	X28	X29	X30	ALM2	M28	M24	M22	M18	M12	M14	1	1	1	1	1	1	1	X32	X33	X34	X35	X36	X37	X38	HX	HY	HZ	HA	HX1	HX10	HX100	0	0	0	0	0	0	0	X40	X41	X42	X43	X44	X45	X46	1	1	1	1	0	1	0	X60	X61	X62	X63	X64	X65	X66	DS3	DS2	DS1	DS0	DK3	DK2	DK1	0	0	0	0	0	0	0	X67	X68	X69	X70	X71	X72	X73	DK0							Program %99.TXT Instruction code G53 T01H0D0 Machine Status <table border="1"> <tr><td>M05</td><td>M09</td><td>M10</td></tr> <tr><td>M78</td><td>M33</td><td>M70</td></tr> </table> G00 X100% F120 X100% S0 X100% Machine Coord <table border="1"> <tr><td>X</td><td>0.000 A</td><td>0.000</td></tr> <tr><td>Y</td><td>0.000 B</td><td>0.000</td></tr> <tr><td>Z</td><td>0.000</td><td>SP360</td></tr> </table> PartTime 0: 0 PartNo 0 SPrpm 0		M05	M09	M10	M78	M33	M70	X	0.000 A	0.000	Y	0.000 B	0.000	Z	0.000	SP360
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Checking interface of output signal

In the interface of output or input, No.0 or No.1 stands for status, 1 means effective, 0 means no effect.

AutoCon Stop		鑫科数控 N00000		2019-04-08 10:44	
Clear history press DEL key, Reset Alarm pre			Program 99.TXT		
Spindle encode check: 0 [No 1 encode]			Progress 		
1, [Curent Alarm]: NO			G53		
2, [2019-4-8 10:43]: Emergent Stop			T00H0D0		
			TrueFeed F 0.000		
			M05 M09 M10		
			M78 M33 M41		
			G00 X100%		
			F500 X100%		
			S0 X100% SP000		
			Machine Coord		
			X 206.579		
			Y -4.179		
			Z 200.000		
			PartTime 12:29		
			PartNo 62		
			SPrpm 0		
No Alarm					
F2 I/O		F3 LAD		F4 ALARM	
F5 EdLad		F6 Reset		F8 CANCEL	

Alarm information interface

The first line in this interface shows the number of spindle encoder, the number of current and historical alarm information is record total 10, the superfluous part is clear automatically, only shows 10 alarm information recently.

PLC interface and edition:

Press “F3” to check PLC ladder, press “F5” to edit PLC ladder, press “S” to search when editing or diagnose PLC ladder. After editing and saving PLC, new PLC will be executed when the controller restart. If you want PLC to be executed immediately, please press “R” in PLC diagnosis interface.

Man Con		N00000		2013-02-26 14:51			
File/File folde in system			Program %99.TXT				
CurDir: /NC			Instruction code				
%99.TXT Z2 LATHER~ MILL TSXT990MDA			G53 T01H0D0				
			Machine Status				
			M05 M09 M10 M78 M33 M70				
			G00 X100% F120 X100% S0 X100%				
			Machine Coord				
			X 0.000 A 0.000 Y 0.000 B 0.000 Z 0.000 SP360				
Compile-P; Receive-R, Tansmit-T, DNC-L			PartTime 0: 0				
			PartNo 0				
			SPrpm 0				
No Alarm							
New/Sek	COPY	RENAM	INFOR	LAST	USBdisk	EXEC	CANCEL

Center part of screen for program display, current program is showed by reverse display, press “PgUp”、 “PgDn” to choose program, and then press “Enter” to edit current program. Functional keys “F1、 F2、 F3、 F4、 F5、 F6、 F7、 F8” include: “new file/search”、 “copy”、 “rename”、 “information”、 “last grade” “USB disc”、 “execute program”、 “cancel”.

3.11.1 Editing

Select “New file/search” to pop up a dialog box to import the name of program, if the name is existent, the quondam program is called up; If the name is inexistent, the system will build a new file.

The name of program can be number, letter or mix, the length is 100 bits.

The system doesn’t allow the namesake, build a new program or select a program and press “Enter” to enter the editing interface.

Man Con		N00000		2013-02-26 14:51			
File/File folde in system			Program %99.TXT				
CurDir: /NC			Instruction code				
%99.TXT Z2 LATHER~ MILL TSXT990MDA			G53 T01H0D0				
			Machine Status				
			M05 M09 M10 M78 M33 M70				
			G00 X100% F120 X100% S0 X100%				
			Machine Coord				
			X 0.000 A 0.000 Y 0.000 B 0.000 Z 0.000 SP360				
Compile-P; Receive-R, Tansmit-T, DNC-L			PartTime 0: 0				
			PartNo 0				
			SPrpm 0				
No Alarm							
New/Sek	COPY	RENAM	INFOR	LAST	USBdisk	EXEC	CANCEL

The file name and the Chinese input of program content:

The directory interface of the "new / search", "copy", "rename", "to copy into the system", "to copy into the USB" can enter the Chinese characters; change the input method according to the prompt of system. Also can input Chinese characters in program editing interface according to the prompt of system.

Man Con		N00000		2013-02-26 14:51			
			Program %99.TXT				
			Instruction code				
			G53 T01H0D0				
			Machine Status				
			M05 M09 M10 M78 M33 M70				
			G00 X100% F120 X100% S0 X100%				
			Machine Coord				
			X 0.000 A 0.000 Y 0.000 B 0.000 Z 0.000 SP360				
			PartTime 0: 0				
			PartNo 0				
			SPrpm 0				
N1 M03 N2 G00 X200 Z200 N3 G00 U-44 N4 W-5 N5 G01 U4 W-2 F300 N6 W-11 N7 U-4 W-2 N8 W-5 N9 U10 W-10 N10 W-6 N11 G02 U-6 W-9 I12 K-9 N12 U10 W-15 I25 K0							
No.1 /Total127							
No Alarm							
COMPIL	FrELine	TeachIn	POS		DelLine	> >	CANCEL

The screen prompt the editing program name at the top left corner in the editing status;

The left is the content, the right is the information for lathe status, the operation in the editing status as follows:

(1) The current cursor locate:

Press “↑↓←→” to move the cursor to any position of program content

Press "Pgup" to the last page.

Press "Pgdn" to the next page.

(2) Character modification: Delete the character at the position of the cursor, then enter the new character.

(3) The character insertion: Enter a new direct character at the cursor position. When the input is the letter, the letter in front of automatically generating space. If you want to enter a space, first enter a letter, and then delete this letter.

(4) The character deletion: Press "Del" directly at the cursor position

(5) Inset the line: Press "Enter" directly, inset a line in front of the current line if the cursor is at the first line, otherwise insert a line after the current line.

(6) "Fast" superposition key operation:

The first function:

A, "F1": compile the program.

B, "F2": to the first line or last line of the program.

C, "F3": teaching function, enter the handwheel status; Press "F4" to read the tool coordinate in the current workpiece coordinate system according to the corresponding axis X/Y/Z/A/B/C/ (all the axis).

D, "F4": located to the specified line.

E, "F5": no operation.

F, "F6": delete the current line.

G, "F7": the first or second page selection.

H, "F8": Chinese characters and character input conversion.

The second function:

A, "F1": delete the program block.

B, "F2": copy the specified program block.

C, "F3": sort the program.

D, "F4": to find the specified character.

E, "F5": replacing the specified character.

F, "F6": all the content to be replaced by the specified character.

G, "F7": the first or second page selection.

H, "F8": Chinese characters and character input conversion.

3.11.2 Copy

Press “↑ ↓” in program main interface to select program which need to copy and press “F2” to prompt a dialog box to import a new name of program, to copy which is the same content but different name so that to modify, rename and back-up copy.

3.11.3 Delete

Press “↑ ↓” in program main interface to select program which need to delete and

press “Del” to delete the program.

The operation of delete need to be careful, it can’t be recovery.

3.11.4 Rename

Press “↑ ↓” in program main interface to select program which need to rename and press “R” to pop up a dialog box to import a new name.

3.11.5 Information

Press “↑ ↓” in program main interface to select program which need to check and press “Q” to pop up a dialog box to check the size of program and the remainder space of the system.

3.11.6 Checking program

Press “↑ ↓” in program main interface to select the checking program and press “P”, the system will check the form and grammar of program. Prompting when finding mistake.

3.11.7 Folder management

You can build a file in this system, Press “N” in program main interface to import a file name and press “.” to build a folder and it will prompt a “folder” after the name.

Move the cursor to the file name and press “Enter” to open to build a new file or folder in it.

Press “A” go to the last folder.

Move the cursor to the file name and press “Del” to delete the folder.

3.11.8 Select automatic program to run

Press “↑ ↓” in program main interface to select a program and press “C” to select the program and switch into the last interface.

3.11.9 Program communication

The system could adopt the RS232 serial port to deliver files.

Delivery (Transport)

Deliver the selected program in this system to another system or to PC to save.

Press “↑ ↓” in program main interface to select program and press “J” to deliver, press “Reset” to interrupt in the deliver process.

Reception

Receive the selected program in another system or PC (Must be text file form).

Press “K” to import a name of received program into the dialog box in program main interface, press “Reset” to interrupt in the receive process.

Pay attention: 1. Using the exclusive communication software to deliver

program in User's PC.

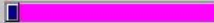
2. The speed of deliver of PC must be the same as the speed of receive, defeat otherwise.
3. The length of RS232 can't greater than 10 meters.
4. The number of serial port must be the same as the system setting.
5. Editing program of PC must be text file form.

3.11.10 U-disk management

To exchange files of parameter or program with other system or PC by U-disk. It also can upgrade or back-up the software or parameter in system.

Pay attention: The name of folder can't contain of space symbol.

Press "B" to enter the U-disk management interface in program main interface when U-disk connects the USB port. Press "B" again to back to the system interface.

Man Con		鑫科瑞数控 N00000		2019-03-19 14:57			
File/File folde in USB disk			Program 99.TXT				
CurDir: /USB			Progress 				
1000M参数界面.BMP 1000M文件管理.BMP 990M参数界面.BMP 990TPLC界面.BMP 990T参数.BMP 990T参数界面.BMP 990T文件界面.BMP CURBMP0.BMP CURBMP1.BMP CURBMP10.BMP CURBMP11.BMP CURBMP2.BMP			G53 T0000 TrueFeed F0.000 M05 M09 M10 M78 M33 M41 G00 X100% F2000 X150% S100 X100% SP000				
Compile-P: Input-R, Output-T, DNC-L			Machine Coord X 164.430 A 0.000 Z 157.845 C 0.000				
No Alarm			PartTime 0:0 PartNo 171 SPrpm 0				
F1 Backup	F2 Restore	F3 Export	F4 Import	F5 LAST	F6 SYSTM	F7 EXEC	F8 CANCEL

A. The processing program management

Copy the files or folder of U-disk into system

After connecting the U-disk, press "F6" to enter the U-disk directory in program main interface. Press "↑ ↓" to move cursor to select file or folder to copy and press "F4" to pop up a dialog box to import name, press "Enter" to confirm. If there is the same name of program in the system, it will pop up a dialog box to ask if cover the file or folder or not.

Press "R" to copy all the program in USB into system.

Copy the files or folder of system into U-disk

Press "↑ ↓" to move cursor to select file or folder and press "F6", press "F3" to pop up a dialog box to import name in U-disk interface and press "Enter" to confirm. If

there is the same name of program in the system, it will popup a dialog box to ask if cover the file or folder or not.

Press “T” to copy all the program in system into USB.

Pay attention: Before unplugging the U-disk must return to the display system of program files directory interface. (Exit U-disk interface)

Otherwise the data which is copied just now will be loss.

The name of folder can't have space symbol when using U-disk.

B. Using U-disk to manage parameter and system software

The system could use U-disk to deliver files or system software to upgrade and update, back-up files and parameter, the method of operation is as follows:

Using U-disk to copy parameter and system software into system(Upgrade, update).

First U-disk inserts the USB port and press “Program” to enter program main interface, press “B” to show the files in U-disk. Press “↑ ↓” to move the cursor to select a folder which is going to be copied into system and press “Enter” to open it, press “F2” to import code when appearing the files and press “Enter” to confirm, wait for seconds to copied the parameter successfully. Press “F6” to exit U-disk after copying successfully, restart the system, the system will reloads the new files to upgrade the parameter.

Pay attention: The parameter is better to be derived into a separate folder in U-disk to defend from the error operation to destroy the system files.

To derive or back-up parameter files by U-disk

First U-disk inserts the USB port and press “Program” to enter program main interface, press “B” to show the files in U-disk. Press “N” to import the code and press “Enter” to confirm, wait for seconds to derive successfully. The parameter in system is already derived into U-disk. Press “B” to exit U-disk.

Pay attention: The U-disk is empty better to arrange the files (Parameter files is lots of about several dozens) so that derive parameter or create a folder on your computer first, open the folder before deriving to derive the parameters into the folder.

3.11.11 Convert DXF files into G code.

In program directory, choose the DXF file then press“-”, it will be converted into G code file, whose name extension is .CNC. The format of the DXF file input into controller should be .DXF or .dxf. Parameter refers to process parameter No.400-402.

During generation of G code files, system will generate corresponding head and end code according to if there are HEADDXF.TXT and ENDDXF.TXT (or headdxf.txt and enddxf.txt) under current file directory.

Attention: head code file and end code file must be under current file directory.

Chapter 4 Programming

CNC machine is highly effective automatic equipment according to programmed program to process work piece. Programming is using the CNC system control language according to the requirement and drawing of the work piece to describe the processing trajectory and the assistant action. Ideal system not only could promise process qualified work piece, but also make the functions of cnc machine reasonable application and fully use, so it is very important to programming, this chapter will introduce many kinds of instructions and usage of CNC program, please read carefully.

4.1 Basic concepts

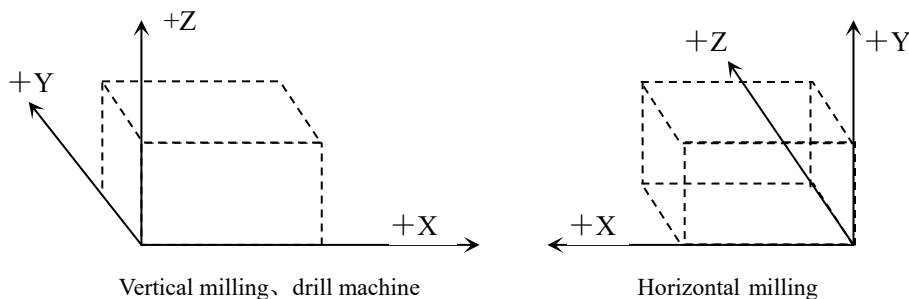
Program Segment: It is a complete command line consisted of instruction segment and data segment.

Program: is a congregation of program segment by machining logic structure in order to complete the machining of workpiece.

Machine Coordinate System : The establishment of coordinate is based on machine's zero point. The milling machine coordinate axis and its direction should follow to "ISO841" standard. The method as follow: Through right hand rule we can make the program coordinate, The Z axis is parallel as spindle, The X axis is horizontal, The Y axis is determined by right hand rule. The A, B, C are rotated axis or assistant axis which parallel as X,Y,Z axis. Furthermore, The coordinate axis direction is the increasing workpiece dimension direction.

As no work coordinate, make machine coordinate as work coordinate.

Machine Coordinate & direction
sketch map



Working Coordinate System: Work piece processing uses the coordinate system is called as the work piece coordinate system, it is set by CNC. The work piece coordinate system could change to move its zero point.

Set the work piece coordinates:

1. Use G54 ~ G59: Use operating parameter set coordinate system may set 6 work

piece coordinate system.

2.G52 instruction: Set a value in front of G52 instruction to set the workpiece coordinate system in program.

With absolute value instruction ,it must use the above method to establish the work piece coordinate system

Local coordinate system: Set local coordinate system of workpiece coordinate system in order to programme easily when programming in workpiece coordinate system.

Absolute Programming : It is confirmed coordinates data programming mode based on established absolute coordinate system. It is settlements by “G90”。

Relative Programming(increment programming): It is distance and direction of operation end point ,compared with starting point. It is settlements by “G90”。

Mode Instruction: The instruction which can remain the function in the program.It works both in this program and program in the future.

In the same operation, there may be several mode instruction, such as M03(spindle clockwise),M04(spindle counter clockwise),M05(spindle stop).They are all mode used to control spindle.The mode of same kind are categorized into one mode group.At any time it must be one of them,and there is only one of them.The original chosen mode is called mode origin.In the above mode group,M05 is such a mode origin 。

Suspending Mode (destroying mode): It is instruction which can turn mode instruction into mode origin or destroy the mode.Such as M20(program ending instruction),meaning the end of operation and returning to original status.

None Mode instruction : It is the instruction which has no function to store,and only works in the segment of program.

4.2 General description of program

G02, T02, H02, D02, M02, S04, F04, X-043, Y-043, Z-043, A-043, I-043, J-043, K-043, L04, P4, R043。

Note 1:“-”means this data can be use.

Note 2:In front of the numeral is 0, indicated this data only write the effective data。

Note 3:The digital presentation is a figure, when is two, top digit expression integer figure biggest figure, after low position expresses decimal point most imperial throne.

4.3 Program instruction

Introduce all the functions and using method of instruction code in this system.

4.3.1 Functional meaning of address symbol,data list

Functions	Address symbol	meaning	Data range
Document No.		File name of machining workpiece	0-9、A-Z
Program segment No.	N	No. of program segment	0000-9999
Preparation function	G	Content and mode of designated instruction operation	00-99
Auxiliary function	M	Auxiliary operation instruction	00-99
Tool chosen	TDH	No.of Tool.	01-99
Tool compensation	H D	The length compensates No.of the radius compensates of the parameter	1- 4
Spindle function	S SP	The spindle speed; spindle localization	00-99999
Cutting speed	F	Speed per minute	1-3000mm/min
Coordinates character	X Y Z A(B/C/ U/V/W)	The coordinates value of X Z and 4th axes.	δ 9999.999mm
Core coordinates	I J K	X Z axes and Z axes core coordinate increment value	δ 9999.999mm
Step length	R	Circular arc radius	0.001-999.999m m
Delay time	P.X	Delay time of designated delay	0.001-99.999s
Program entrance	P	Entrance of calling program name	0000-9999
Repeat times	L	Times of cycle or subprogram calling	1-9999
Line skip	/	Not to carry out when “/” is in front of program line	No.12 parameter in other parameter can shield the function

4.3.2 Program instruction

Table 1 G Instruction-code and functions

G code	function
G00	Fast positioning

G01	The straight-line interpolation	
G31	No alarm when checking skip	
G311	Alarm when checking skip	
G02	Interpolation along the circle interpolation/the spiral line interpolation CW: The spiral motion spiral line interpolation makes up the 2 circular arcs insert makes up the axis synchronization migration other axes. The instruction method only is simply adds on is not the circular arc inserts makes up the axis the shifting shaft	
G03	The counter circle interpolation/ the spiral line interpolation CCW	
G01(G00)X I	Beveling automatically	
G01(G00)Y I	Beveling automatically	
G01(G00)Z I	Beveling automatically	
G01(G00)X R	Smoothing automatically	
G01(G00)Y R	Smoothing automatically	
G01(G00)Z R	Smoothing automatically	
G04	Delay	
G15	Polar coordinate instruction cancellation	
G16	Polar coordinate instruction: The polar coordinate (radius and angle), the angle to is anti-clockwise changes of the first axis positive direction on the plane, but the negative direction is clockwise changes .	
G17	Choose the Xp Yp plane	X : X axis or its parallel axis Y : Y axis or its parallel axis Z : Z axis or its parallel axis
G18	Choose the Z X plane	
G19	Choose the Y Z plane	

G20	Inch input(British system)
G21	Millimeter input(Metric system)
G28/G281/G282/G283/G284	Go to machine zero point
M882, M883	Drop set of Z axis automatic tool setting
M884-M887	Aromatically finding edge of X, Y axis
M880	Set tool automatically
G30/G301/G302/G303/G304	Go to coordinate zero point
G26	ZXY axis go to program original point
G25	Memory current coordinate of X Y Z A
G261	X axis go to program original point
G262	Y axis go to program original point
G263	Z axis go to program original point
G264	A axis go to program original point
G61/G611/G612/G613/G614	Back to the memory point by G25
G40	Cancel tool radius compensation
G41	tool radius compensate(left)
G42	tool radius compensate(right)
G43	Positive tool length compensation
G44	Negative tool length compensation
G49	Cancel tool length compensate
G45	Tool offset adding
G46	Tool offset subtract
G47	Double tool offset adding
G48	Double tool offset subtract
G37	Cancel scale zoom
G36	Enable scale zoom : format : G36 X_Y_Z_R_
G12	Cancel programmable mirror
G11	Enable programmable mirror

G50/G52	Set coordinate system: the programmer in work coordinate system is easy program, subcoordinate system is permitted in workpiece coordinate system.	
G53	Choose machine coordinate system	
G54.1/G54.48	Choose work coordinate 1	Note: These six work coordinates are saved in CNC , user may choose one of them
G55	Choose work coordinate 2	
G56	Choose work coordinate 3	
G57	Choose work coordinate 4	
G58	Choose work coordinate 5	
G59	Choose work coordinate 6	
G60	exactitude stop positioning	
G64	Continue path work	
G68	rotate coordinate。format: G17 } G18 } G68 a-b- R-; R:Angle G19 }	
G69	Cancel rotate coordinate	
G73	Drill deep hole loop: intermittent feed, rapidly back.	
G80	Cancel cycle mode/cancel external operation	
G81	Drill cycle: cutting feed, boring cycle or external operation, rapidly back	
G82	Chip removal drill cycle or reverse boring cycle: cutting feed, stop tool at the bottom of hole, rapidly back.	
G83	Drill small deep hole cycle: intermittent feed, rapidly back.	
G85	Boring cycle:cutting feed, back cutting feed	
G86	Boring cycle:cutting feed, stop spindle at the bottom of hole, rapidly back.	
G89	Boring cycle: stop tool at the bottom of hole, back cutting feed	

G74	Left Tap cycle
G84	Right Tap cycle
G90	Absolute program
G91	Incremental program
G98	Go back to original point: used in final drill
G99	Go back to R point : used in the first drill
G22	Program cycle instruction
G800	Cancel Program cycle instruction

Table 2 M code and function

Spindle	M03	Spindle CW	M7053/M7054 P xxxx; It will stop after spindle CW or CCW for a time, the time is determined by "P". Unit: Millisecond. Example: M7053 P2000; Means it will delay 2 seconds to stop spindle.
	M04	Spindle CCW	
	M05	Stop Spindle	
	M203	The 2 nd spindle CW	
	M204	The 2 nd spindle CCW	
	M205	Stop 2 nd spindle	
Cooling	M08	Turn on cool	
	M09	Turn off cool	
Chuck	M10	Tighten tool	
	M11	Loosen tool	
Huff	M58	Turn off huff	Controls M59 signal
	M59	Turn on huff	
Lubrication	M32	Turn on lubricate	Controls M32 signal
	M33	Turn off lubricate	
User-defined	M79	User self-defined 7 output turn on	Double signal output
	M78	User self-defined 7 output turn off	
User-defined output	M61	User self-defined2 output turn on	Controls M61 signal
	M60	User self-defined2 output turn off	
	M63	User self-defined3 output turn on	Controls M63 signal
	M62	User self-defined3 output turn off	
	M65	User self-defined4 output turn on	Controls M65 signal

	M64	User self-defined4 output turn off	Controls M67 signal
	M67	User self-defined5 output turn on	
	M66	User self-defined5 output turn off	
	M69	User self-defined6 output turn on	Controls M69 signal
	M68	User self-defined6 output turn off	
	M71	User self-defined7 output turn on	Controls M71 signal
	M70	User self-defined7 output turn off	
	M75	User self-defined8 output turn on	Controls M75 signal
	M74	User self-defined8 output turn off	
Spindle shifting	M41	SP Speed first gear	Output S01,S02,S03,S04, to adjust analog voltage of spindle
	M42	SP Speed second gear	
	M43	SP Speed third gear	
	M44	SP Speed fourth gear	
User-defined input	M12 M13	Check M12 input valid Check M12 input invalidate	To skip when conditions are tenable Example: M12 P120 Program skips to 120th line to execute.
	M14 M15	Check M14 input valid Check M14 input invalidate	
	M16 M17	Check M16 input valid Check M16 input invalidate	
	M18 M19	Check M18 input valid Check M18 input invalidate	
	M28 M29	Check M28 input valid Check M28 input invalidate	

	M22 M23	Check M22 input valid Check M22 input invalidate	
	M24 M25	Check M24 input valid Check M24 input invalidate	
Subprogram	M97	Program skip	L=1-99 P is the line number of transferring program
	M98	Invoke subprogram	
	M99	Cancel invoke subprogram	
Program controlling	M87	Number of workpiece plus 1	other parameter P10=0 is set not to symptomatically plus 1, instruction M87 to make workpiece number plus 1
	M00	Suspend program	
	M01	Suspend program,input M22 effective suspend Program is over	
	M02	Program end	
	M30	M05,M09 program is over	
	M20	Program is over, automatic repeat run the program according to the parameter which set the running times, be used for the debugging	
Speed of spindle	S	Set speed of the first spindle	S=0-99999 SS=0-99999
	SS	Set speed of the second spindle	
	SC	Set speed of the third spindle	
	SD	Set speed of the fourth spindle	

Read the position of absolute motor	M500 M501 M502 M503 M504	M500: read absolute motor position of all the feeding axis and reset the current machine coordinate. M501: read absolute motor position of X axis and reset the current machine coordinate. M502: read absolute motor position of Y axis and reset the current machine coordinate. M503: read absolute motor position of Z axis and reset the current machine coordinate. M504: read absolute motor position of the fourth axis and reset the current machine coordinate.	
Clear workpiece coordinate	M312 M313 M314 M315 M316 M317 M318 M319 M320	M312:Clear C axis workpiece coordinate M313:Clear Xs axis workpiece coordinate M314:Clear Ys axis workpiece coordinate M315:Clear A axis workpiece coordinate M316:Clear B axis workpiece coordinate M317:Clear X axis workpiece coordinate M318:Clear Y axis workpiece coordinate M319:Clear Z axis workpiece coordinate M320:Clear all axis workpiece coordinate	

Clear machine coordinate	M412 M413 M414 M415 M416 M417 M418 M419 M420	M412:Clear C axis machine coordinate M413:Clear Xs axis machine coordinate M414:Clear Ys axis machine coordinate M415:Clear A axis machine coordinate M416:Clear B axis machine coordinate M417:Clear X axis machine coordinate M418:Clear Y axis machine coordinate M419:Clear Z axis machine coordinate M420:Clear all axis machine coordinate	
Bus driver instruction	M133 M135	M133:driver runs with specified speed (multiple of 30) Driver parameter P30=0 means constant speed, P30=1 means weave, referring to axis parameter. M135: enable specified driver	Example: M133 X300; M135 Z0: turn off Z axis driver; M135 X1:enable X axis driver;

M1xxx: waiting for auxiliary relay effective, for example, M1076 means wait for M76 on;

M2xxx: waiting for auxiliary relay ineffective, for example, M1078 means wait for M76 off;

M3xxx: set auxiliary relay effective, for example, M3330 means set M330 on;

M4xxx: set auxiliary relay ineffective, for example, M4331 means set M331 off;

M38xx: set output effective, for example, M3809 means set Y09 on;

M48xx: set output ineffective, for example, M4807 means set Y07 off;

M18xx: waiting for input point Xxx effective then go to next, for example, M1809 means waiting for X09 effective, then go to next step.

M28xx: waiting for input point Xxx ineffective then go to next, for example, M2807 means waiting for X07 ineffective, then go to next step.

M18xx Pxx: check input point Xxx effective then go to next, otherwise skip to Pxx line. for example, M1809 P234 means if X09 effective, then go to next step, if X09

ineffective, skip to P234 line.

M28xx Pxx: check input point Xxx ineffective then go to next, otherwise skip to Pxx line. for example, M2807 P456 means if X07 ineffective, then go to next step, if X07 effective, skip to P456 line.

4.4 Programme instruction and usage

4.4.1 Programme convention

(1). Multiply instruction exist in a segment simultaneously: one program line allows multiply instructions in order to reduce the lines, but the same group of instruction can not share one segment.

(2). Within a program segment, instruction and parameters can be arranged optionally.

Such as: M03 G01 X20 Y-30 can be written: G01 Y-30 X20 M03

(3). No repeat of instruction within a program segment.

(4). It can't be irrelative parameters and operation in a segment.

(5). "0" before a instruction is allowed to delete, such as: G01 G03 can be written as G1 G3.

(6). The command of optional point, line start or that after tool changing instruction must be programmed by absolute coordinates.

(7). Non mode command only in specified program line is effective, such as: G04.

(8). Mode instruction is always effective before appearing the same instruction.

For example: N0000 G01 X300 F100; G01 instruction

N0001 X260 ; G01 instruction

N0002 G00 Z200 ; G00 instruction, G01 is invalid

4.4.2 Instruction of G function

(1) Choose instruction of coordinate system.(G53/G54/G55/G56/G57/G58/G59)

Format: G53 (G54/G55/G56/G57/G58/G59) Mode after setting

G53 machine coordinate

G54 work coordinate 1

G55 work coordinate 2

G56 work coordinate 3

G57 work coordinate 4

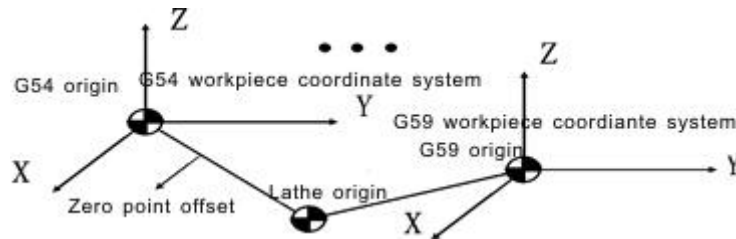
G58 work coordinate 5

G59 work coordinate 6

G53 machine coordinate is decided by machine reference point. The default coordinate is G53. Suggest not to adjust the value of G53, all the workpiece coordinate will have offset.

G54/G55/G56/G57/G58/G59 work coordinate have offset relative to machine

coordinate which can be set in parameter.



Example: G00 G54 X50 Y60 Z70

Move to X50 Y60 Z70 of G54 with speed of G00.

(2) Local coordinate system instruction

Set local coordinate system when programming in workpiece coordinate system is to programme easily. G50 is used to set workpiece coordinate system, and G52 is used local coordinate system setting.

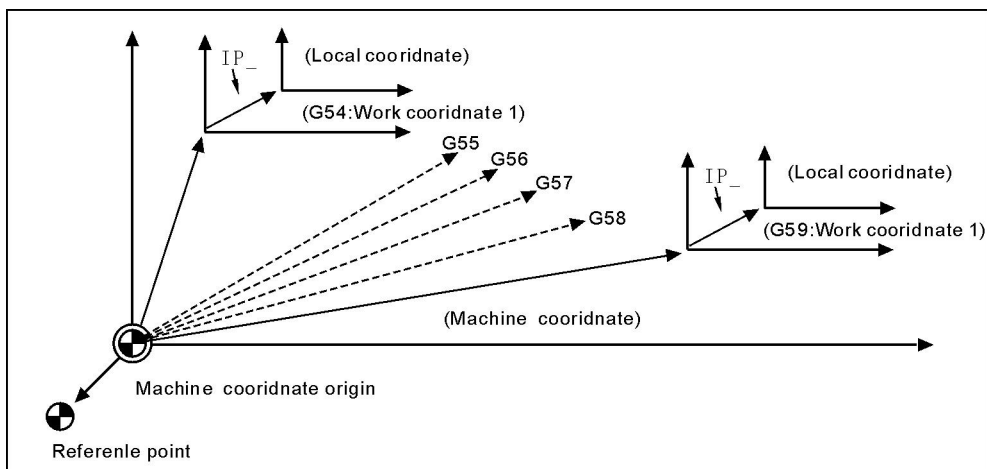
Format: G52 X_ Y_ Z_ ; Set local coordinate system (Mode)

G52 X0 Y0 Z0; Cancel local coordinate system

Use G52 to set local coordinate system in lathe coordinate system(G53) or workpiece coordinate system(G54~G59). The origin of local coordinate system is set by X_ Y_ Z_.

Once local coordinate system is done, the instructions by G90 will move in the local coordinate system. G50/G52 are used to specify the new zero point, changing position of local coordinate system. Local coordinate system should share the same zero point with workpiece coordinate system in order to cancel local coordinate and specify coordinate in workpiece coordinate system.

Local coordinate system setting does not change the workpiece coordinate system and machine coordinate system. When using the G52 instruction to set workpiece coordinate system, if it is not the instruction of all the axis coordinate value, the local coordinate system of non-specified axis will remain unchanged. Use the absolute mode to specify motion instruction next the G50/G52 program segment.



(3)Program method(G90/G91)

Two methods to move tools: Absolute value instruction and incremental value instruction. Programme coordinate value of end point with absolute value; while programme incremental value of moving distance with incremental value. G90 and G91 are respectively used to point out the coordinate with absolute value or incremental value.

Format: G90 (Mode,initial) ;Absolute
G91 (Mode) ;Increment

Pay attention:The rotating axis use nearest calculation and move accordingly when using absolute coordinate to programme, using programming calculation to work when using relative coordinate to programme.

(4)Select Plane(G17/G18/G19)

Format: G17 (Mode, Original) ;Set XY Plane
G18 (Mode) ;Set ZX Plane
G19 (Mode) ;Set YZ Plane

Used to point out arc interpolation plane.

Note: this instruction does not produce motion.

(5)Rapid motion(G00)

Tool moves to workpiece coordinate system by instruction according to G00 speed in parameter.

In absolute method, use endpoint coordinate to program;

In incremental method, use moving distance of tool to program.

Format: G00 X- Y- Z- A- (Mode, original)

Note: X, Y, Z, A, means motion axis.The data point out motion distance and direction by absolute or incremental method.

In G00 instruction, each axis moves to endpoint separately with the specified speed. Each axis stops moving till the aim point, and next axis keep moving to instruction position. All of axis can move simultaneously by other parameter P9, D6=0 means linked moving.

Moving speed is determined by Speed parameter of G00.

(6) Line interpolation(G01)

Used for single axis motion or 2,3,4 axis interpolation processing..

Format: G01 X_ Y_ Z_ A_ F_ (Mode)

Note: X, Y, Z, A means motion axis.The data point out motion distance and direction by absolute or incremental method.Motion speed is determined by F word.

The F instruction is mode.

The feeding speed of F in G01 can be adjusted by feeding override on the panel, the range is 0%~150%.

G01 instruction also can be written G1.

(7) Arc interpolation(G02/G03)

In the program plane, G02 clockwise and G03 counter-clockwise are used to execute arc interpolation processing

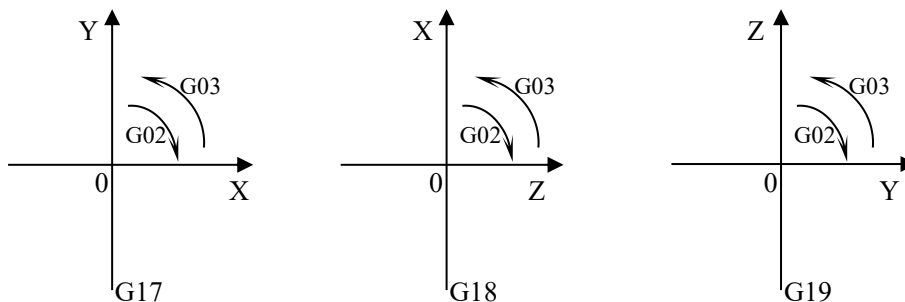
Format: .G17 $\left\{ \begin{matrix} G02 \\ G03 \end{matrix} \right\} X_Y_ \left\{ \begin{matrix} I_J_ \\ R_ \end{matrix} \right\} F_ ;XY \text{ plane}(\text{Mode})$

G18 $\left\{ \begin{matrix} G02 \\ G03 \end{matrix} \right\} X_Z_ \left\{ \begin{matrix} I_K_ \\ R_ \end{matrix} \right\} F_ ;ZX \text{ plane}(\text{Mode})$

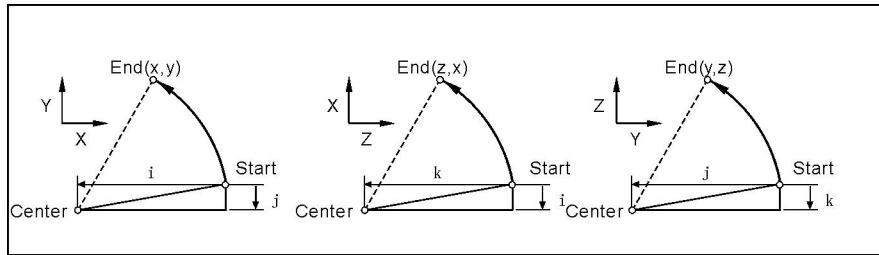
G19 $\left\{ \begin{matrix} G02 \\ G03 \end{matrix} \right\} Y_Z_ \left\{ \begin{matrix} J_K_ \\ R_ \end{matrix} \right\} F_ ;ZY \text{ plane}(\text{Mode})$

Note: Arc interpolation must point out interpolation plane, the X、Y、Z word point out the arc end coordinate value, I、J、K is incremental value from original point(speed parameter P47 =+4 means endpoint) of X, Y, Z axis to center point. In another word, let the original point be zero point, and if the center is in the forward direction of original point, the value will be positive, if the center is in the backward direction of original point, the value will be negative. I J K are used to define center point coordinate. Besides, R program is also used to define center point coordinate, the R is negative when arc angle is over 180 degree, but it is not applicable for whole cycle. The trend of arc goes from positive to negative direction that is perpendicular to arc interpolation plane in right-handed rectangular coordinates.

The arc track of different arc interpolation plane as follow:



I J K are used to define the arc center position, the value behind I J K are vector component from original point to arc center. I J K must be marked as positive or negative according to directions. I J K can be omitted, then endpoint and original point is the same one, and the arc will be whole cycle when center is specified by I J K, it will always be incremental as following no matter G90 or G91.



The arc interpolation speed is determined by F word.

Attention: I, J, K and R are the non-modality instruction.

Demonstration:

1) absolute programming

N0000 G92 X200 Y40 Z0;	set start point absolute coordinate;
N0010 G90 G03 X160 Y40 I-20 J0;	count-clockwise arc;
N0020 G02 X120 Y40 R20;	clockwise arc;
N0030 G02 X120 Y40 R20;	whole cycle;
N0040 G26 M02;	end and back to start point.

2) Relative programming

N0000 G91 G17 G03 X-40 Y0 R20 F300;	count-clockwise arc;
N0010 G02 X-40 Y0 R20;	clockwise arc;
N0020 G02 X0 Y0 R20;	whole cycle;
N0030 G26 M02;	end and back to start point.

Two methods have the same result.

(8) Spiral interpolation (G02/G03)

Spiral interpolation means arc interpolation adding another linear-axis interpolation, F instruction defines arc interpolation speed. therefore, the feed speed of line interpolation axis is as follow:

$$F \times \frac{\text{Lengthen of line axis}}{\text{Lengthen of arc}}$$

Format : G17 {G02
G03} X_ Y_ {I_ J_
R_} Z_ F_ ;XY plane(Mode)

G18 {G02
G03} X_ Z_ {I_ K_
R_} Y_ F_ ;ZX plane(Mode)

G19 {G02
G03} Y_ Z_ {J_ K_
R_} X_ F_ ;YZ plane(Mode)

Spiral line enhancement: screw pitch K(J or I) can be inserted into program to process multi-turn spiral line.

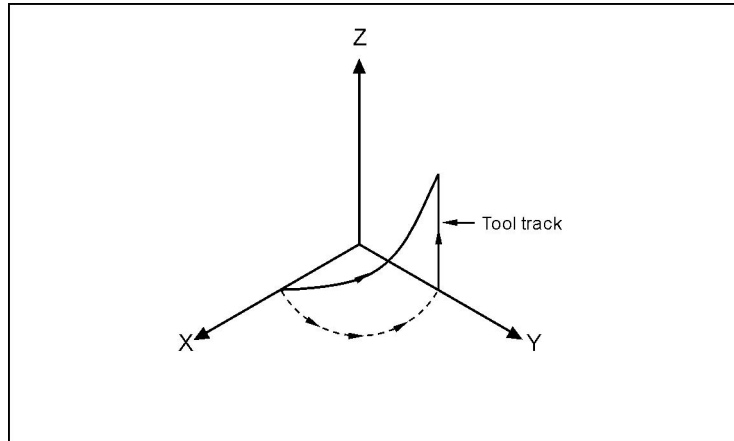
G02(G03) X_ Y_ I_ J_ F_ Z_ K_ (G17, K screw pitch, unit:mm);

G02(G03) X_ Z_ I_ K_ F_ Y_ J_ (G18, J screw pitch,unit:mm);

G02(G03) Y_ Z_ J_ K_ F_ X_ I_ (G19, I screw pitch, unit:mm);

G02(G03) X__ A__ I__ J__ F__ Z__ K__ (G177/XA plane, K screw pitch, unit:mm);
 G02(G03) Z__ A__ K__ I__ F__ Y__ J__ (G188/ZA plane, J screw pitch, unit:mm);
 G02(G03) Y__ A__ J__ K__ F__ X__ I__ (G199/YA plane, I screw pitch, unit:mm);
 The cutting tool radius compensates only carries on to the circular arc, cutting tool bias and the cutting tool length compensate cannot be inserted in the segment of spiral line program.

In the spiral interpolation section, cannot use tool length and radius compensation.



```
N0001 G90 G17 G54
N0002 G01 X20 Y0 Z0 F600
N0003 G03 X0 Y20 R20 Z15 F180
.....
```

(9) Three dimension space arc instruction(G06)

Format: G06 X__ Y__ Z__ I__ J__ K__ F__

Function: if center point and radius of three-dimension space arc are unknown, but the coordinates of three points in arc are known, then G06 can be used to find direction of arc by the middle point between start point and endpoint.

Definition of instruction: G06 is mode G code;

I: relative coordinate value(X direction) of middle point relative to start point on arc(radius value and with direction);

J: relative coordinate value(Y direction) of middle point relative to start point on arc(with direction);

K: relative coordinate value(Z direction) of middle point relative to start point on arc(with direction).

Attention:

- 1) Middle point: refers to any point in the arc except for start point and endpoint;
- 2) if there are three col-linear points, it will alarm;
- 3) when I is omitted, it defaults to I=0; when K is omitted, it defaults to K=0; when J is omitted, it defaults to J=0. but if I, J, K are omitted at the same time, it will alarm.
- 4) IJK of G06 is just similar to the displacement value IJK of center point relative to start point coordinate in G02/G03.

- 5) G06 cannot be used to process whole cycle.
 6) the instructions requires a lot of math operation, therefor it works perfect in bus type controller, while it may not work fluently as Bus type in other type controllers.

Example:

```
G0 X10 Y28 Z10
G06 X30 Y98 Z10 I5 J-6 K-5 F100
X130 Y198 Z120 I55 J-86 K-65
G0X0Z0
M02
```

(10) Delay instruction(G04)

Used to delay a certain time for the requirements of processing technique.

Format: G04 P_ ;or
 G04 X_ ;or
 G04 U_ ;

Function: Every axis is stop and mode instruction is still working when execute this instruction, after delaying the specified time to execute the next program segment.

Instruction introduction:

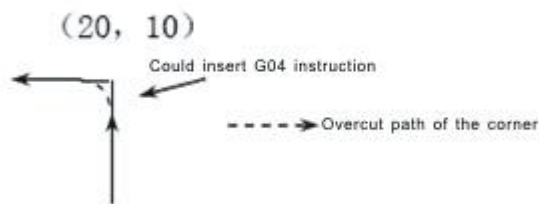
- The unit of P delay time is ms(Millisecond).
- The unit of X and U delay time are S (second) .

c.Example:

```
G04 X1;      delay 1s.
G04 P1000;   delay 1s.
G04 U1;      delay 1s.
```

d.Special application:G04 can be accurate stop instruction, such as processing corner kinds of workpiece, it appears over cutting sometimes, if use G04 instruction around the corner, it will eliminate the over cutting.

Example as follows:



Program:

```
N150 G01 X20 Z10 F100;
N160 G04 P150; (Clear the over cutting)
N170 G01 W-10;
```

.....

Pay attention: Set No.21 parameter in “N”(processing) parameter to clear the over cutting.

(11) Mirror instruction(G11/G12)

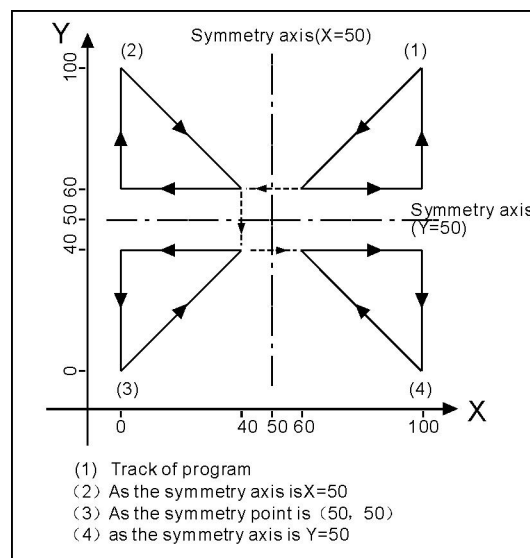
In order to decrease program codes, be used for machining symmetry workpiece.

Format: G11 X_ Y_ (Z_ X_) (Y_ Z_)(mode)
 according to XYZ symmetry axis
 G12 (mode, original) ;Cancel Mirror。

Note: mirror processing perform symmetric machining of the axis by changing feeding direction of axis. The axis changing direction is determined by address character of current interpolation plane G17(X_ Y_); G18 (Z_ X_); G19(Y_ Z_). the value behind X_ Y_ Z_ is coordinate of current workpiece coordinate system.

In programmable mirror mode, such instruction about back reference point as G26,G61,G28,G30 and instruction about changing coordinate system like G52,G53/G59 cannot be employed. These G code cannot be specified until cancel the programmable mirror mode.

For example:



The mirror procedure gives an example:

Sub program

KG11

N10 G00 G90 X60.0 Y60.0;

N20 G01 X100.0 F100;

N30 G01 Y100.0;

N40 G01 X60.0 Y60.0;

N50 M99;

Main program

N10 G00 G90;

N20 M98 PKG11;

N30 G11 X50.0;

N40 M98 PKG11;


```

N50 G11 X50.0 Y50.0;
N60 M98 PKG11;
N70 G11 Y50.0;
N80 M98 PKG11;
N90 G12;
N100 M02

```

(12) Proportions scale instruction(G36/G37)

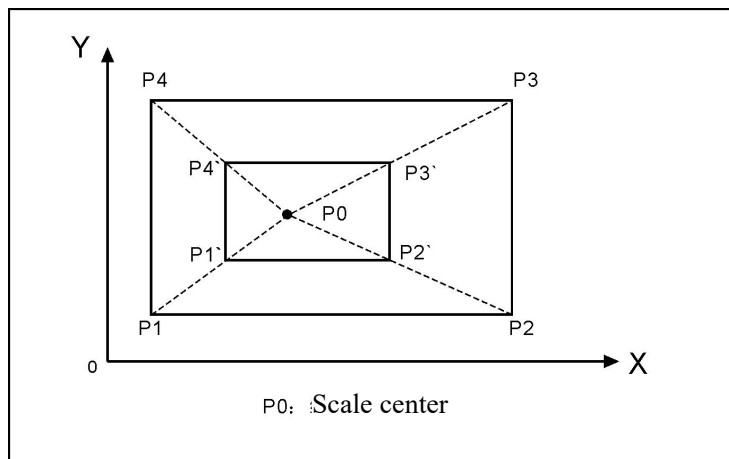
Used for unchanging the processing program, to process proportionally reduced or enlarged workpiece. Programming shape is magnified and reduced (scaling), use X_, Y_ and Z_ to specify scaling center. If an axis is not specified, the axis will not carry out scaling. The number behind the X_ Y_ Z_ is the coordinate of current workpiece coordinate system.

Format: G36 X_Y_Z_R_ (mode) ;Carry out
 G37 (mode, original) ;Cancel

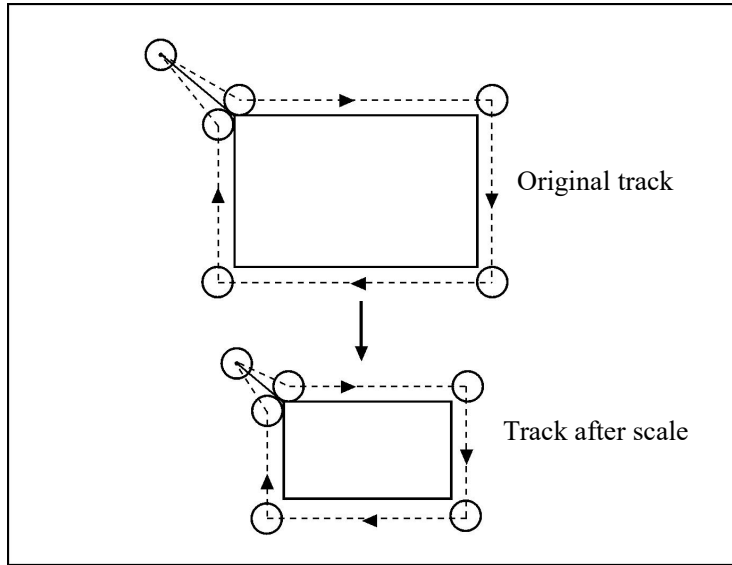
Note: behind R is the scale coefficient is

For example:

P1' P2' P3' P4' magnify to P1 P2 P3 P4, $R = P0P4' / P0P4$. When P1 P2 P3 P4 reduce to P1' P2' P3' P4', $R = P0P4 / P0P4'$. So: $R < 1$ when magnifying, $R > 1$ when reducing, $R = 1$ can be default.



In the proportions scale section, cannot use tool length and radius compensation:



(13) Coordinate rotate(G68/G69)

Programming shape can rotate, the rotating instruction can make workpiece rotate for specified angle. Anyway, if the shape of workpiece is comprised of many same graphics, the graphics unit can be compiled to subprogram, then the subprogram is on tap for rotation instruction of main program. This can simplify the programming, saving storage space.

Format: G68 X- Y- R- (mode) ; enable
 G68 Z- X- R- (mode) ; enable
 G68 Y- Z- R- (mode) ; enable
 G69 (mode, original) ; disable

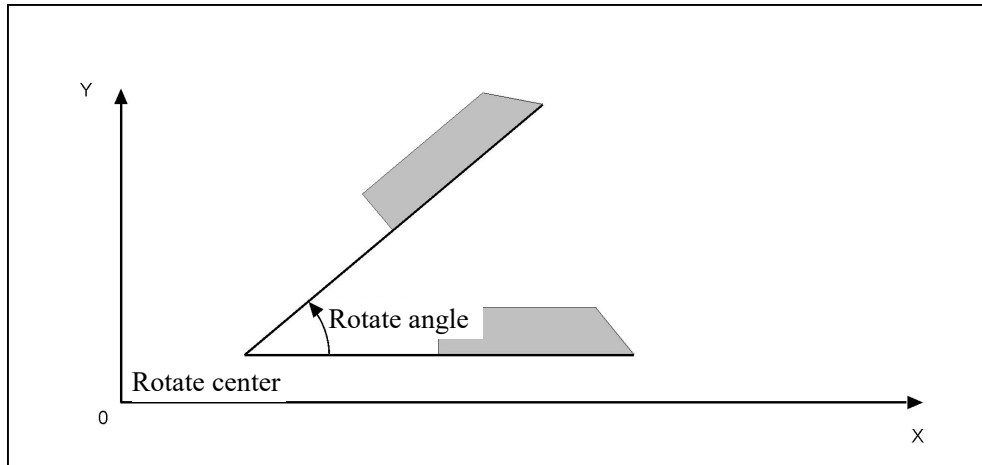
Note: The (G17)X-Y- or (G18)Z-X- or (G19) Y-Z- behind G68 is used for pointing out rotate center.

R word is used for pointing out rotate angle, unit is °(degree), the value behind X_Y_Z_ is coordinates of current coordinate system;

positive R means rotate CCW;

Negative R means rotate CW.

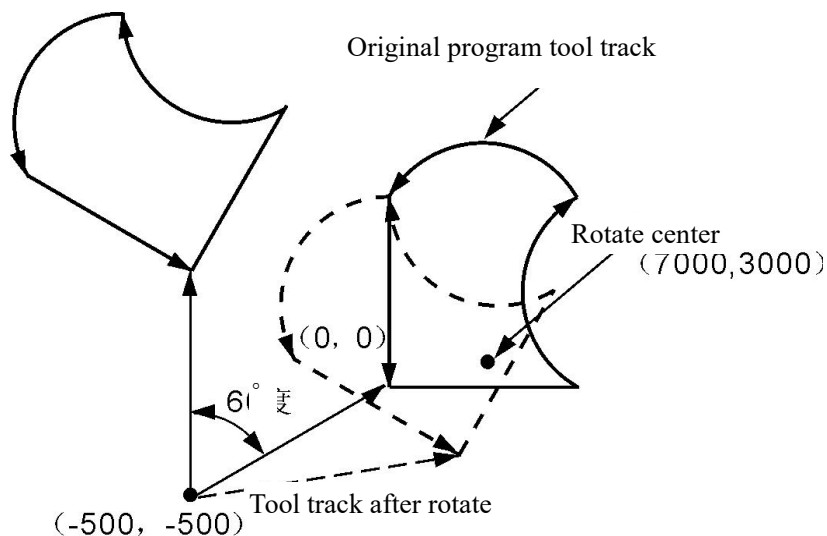
Example:



To specify plane and select code(G17 G18 or G19) before the G68 code program segment, select the code cannot be specified in the mode of rotating coordinate system. Cancel the mode G69 code must occupy a program segment separately. After rotating the coordinate system to carry out tool radius compensation, tool length compensation, tool offset and other compensation.

In the mode of rotating coordinate system, the G code (G28 G26 G31 G30 etc) which is related with backing to the home point and those related with the the G code (G52, G54/G59) of coordinate system cannot be specified, if need these G codes, must specify after canceling coordinate rotation mode. The first moving instruction must use absolute value instruction after the coordinate system rotation cancel G69, if use incremental instruction, it will not to carry out correct motion.

Example:



```

N1 G92 X-500Y-500F20000 G17;
N2 G68 X700Y300R60;
N3 G90 G01 X0 Y0 F20000;
N4 G91 X100
N5 G02 Y100 R100
N6 G03X-100I-50J-50;
N7 G01Y-100

```

```

N8 G69
N9 G90 X-500Y-500
N10 M02;

```

(14)Return reference point(Zero point)
(G28/G281/G282/G283/G284/G301/G302/G303/G304)

Return Reference instruction means tool go to reference point according to appointed axis.

Format:	G28 X(U)_ Y(V)_ Z(W)_ A_ B_ ;ZXYAB return to reference
G281	;only X return to reference
G282	;only Y return to reference
G283	;only Z return to reference
G284	;only A return to reference
G285	;only B return to reference
G301	;X axis return to zero
G302	;Y axis return to zero
G303	;Z axis return to zero
G304	;A axis return to zero

Pay attention: Should clear tool radius compensation and tool length

(15)Tool length compensation instruction(G43/G44/G49)

The difference between programming tool length and actual using tool length can be compensated with this function so that not to adjust the program. Use G43 or G44 to specify the direction of offset, import the corresponding H code address to select tool length offset value from the offset table.

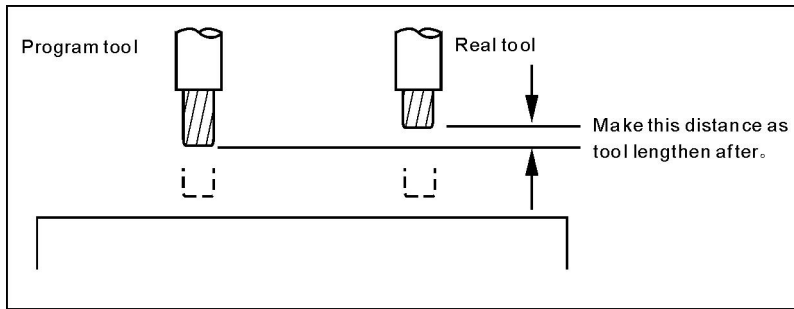
On the plane G17(XY), length compensation applies to Z axis, so tool should be installed in Z axis;

On the plane G18(ZX), length compensation applies to Y axis, so tool should be installed in Y axis;

On the plane G19(YZ), length compensation applies to X axis, so tool should be installed in X axis.

Format: G43 H_ ;Add tool length compensate。 G44 H_ ;subtract tool length compensate。 G49 or H0 ;cancel tool length compensate.

Note:tool length compensation refers to add or subtract the value of tool length compensation on Z axis or other axis. Invoking number is H1-H99. compensation value of parameter starts from the tool position before compensation.



Example: N0000 G43 H2 X10 (H2 value is 5)

N0010 G44 H3 X20 (H3 value is 10)

Executing first section, tool length add 5. Executing second section, tool length subtract 10 (real running is $10+5=15$), backing to original position.

(16) Offset of tool radius instruction (G45/G46/G47/G48)

Used for processing groove of the workpiece, programming according to the size of workpiece drawing, the instruction line with the one of this set of instruction, then it can work out correct production in different tool radius.

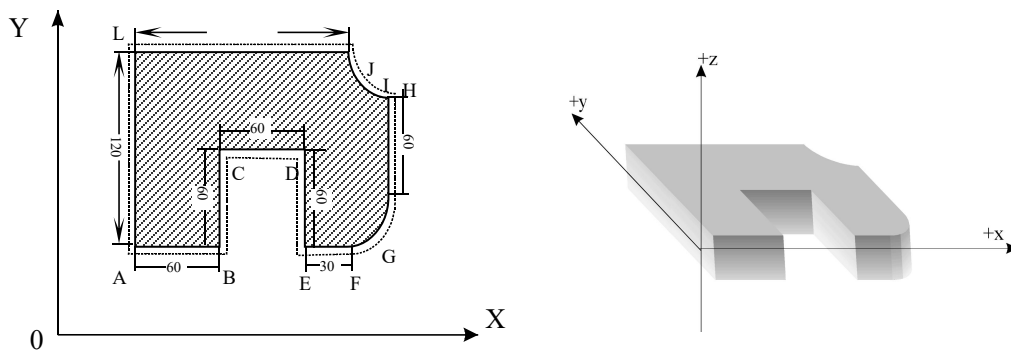
Format:	G45 T_	;Add one radius。
	G46 T_	;subtract one radius。
	G47 T_	;Add two radius。
	G48 T_	;subtract two radius。

Note: The instruction of increasing or decreasing the tool radius is increasing or decreasing one or two tool radius parameter values with T number in the execution. And be used with XY (G17) plane of the G00, G01, G02, G03 instruction. Can not be used with tool radius compensation instruction (G41, G42).

Increase or decrease one or two tool radius in the direction of axis for G00 and G01.

For G02 and G03 is increasing or decreasing one or two tool radius in the direction of arc radius.

G45/G46/G47/G48 are non-modal instructions.



Program as follows:

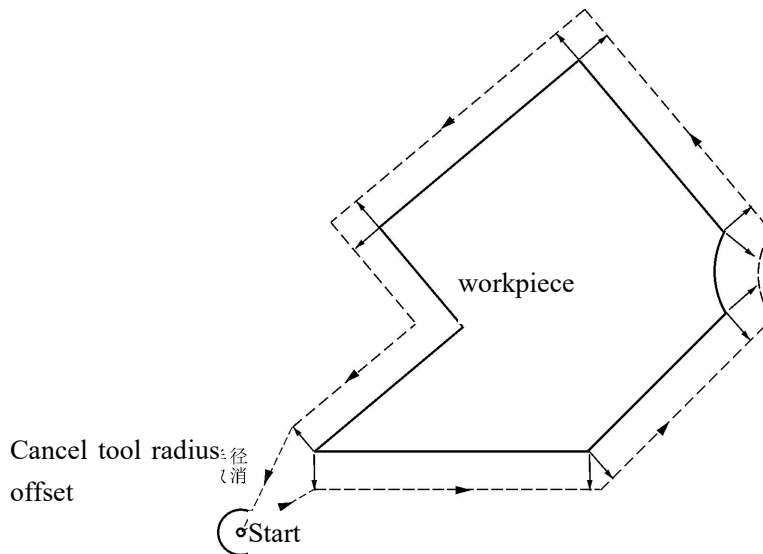
N0000 G01 Z-20 F400 G91 ; Z point starts

N0010 G46 T01 X55 Y55 decrease one tool radius(T01)	; Point location to A point, X and Y axis both
N0020 G47 G01 X60 F200	; A-B X axis increase two tools radius
N0030 Y60	; B-C length not change
N0040 G48 X60	; C-D X axis decrease two tools radius
N0050 Y-60	; D-E length not change
N0060 G45 X30	; E-F X axis increase one tools radius
N0070 G45 G03 X30 Y30 R30	; F-G CCW radius increase one tool radius
N0080 G45 G01 Y60	; G-H Y axis increase one tool radius
N0090 G46 X0 negative direction of X axis	; H-I Move for one tool radius along the
N0100 G46 G02 X-30 Y30 R30	; I-J CW radius decrease one tool radius
N0110 G45 G01 Y0 positive direction of Y axis	; J-K Move for one tool radius along the
N0120 G47 X-150	; K-L X axis increase two tools radius
N0130 G47 Y-120	; L-M Y axis increase two tools radius
N0140 G46 X-55 Y-55 radius	; M-N X and Y axis both decrease one tool
N0150 G26	; Program backs to the starting point to end
N0151 M02	

The relationship between Plus(+), Minus(-) and tool radius in above example is addition of term of the same sign, subtraction of term of different sign. Coordinate can be zero, used to specify coordinate axis, run length is 1 or 2 radius. The direction is specified by signs of instruction and radius.

(17) Tool radius compensate instruction(G40/G41/G42)

When the tool is moving, tool track can offset a radius. In order to offset a radius, CNC establish offset vector whose length equal tool radius. Offset vector is vertical to tool track. Completed machining, it needs to cancel tool radius compensation.



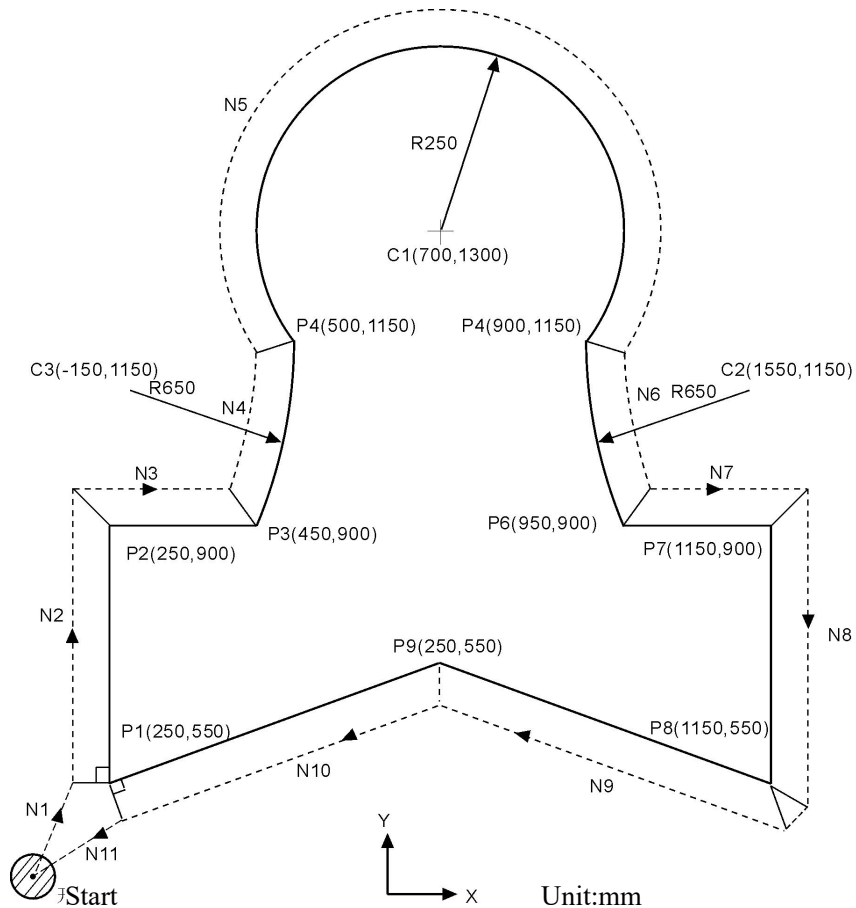
Format:	G40	(mode, original)	;Cancel compensation。
	G41 T—	(mode)	;tool locate to Left offset of workpiece。
	G42 T—	(mode)	; tool locate to Right offset of workpiece。

Note:

Left or right side of G41/G42 is following in the direction of cutting, tool is in the left or right of workpiece. tool radius is invoked by T1-T99. execution of offset starts from the line of G41/G42. It will generate transition line at sharp corner, in order to transit from the tool radius offset vector in the end of last segment to the offset vector in the beginning of next segment. Tool offset vector refers to describing value and direction of tool offset. radius vector is radius of tool. For arc, its direction is the radius direction. And for line, it direction is vertical direction of line.

Tool radius compensation establish and cancel have two type: A type and B type, which can set in other parameter. Furthermore, Tool radius compensation establish and cancel must be executed in line section.

For example:



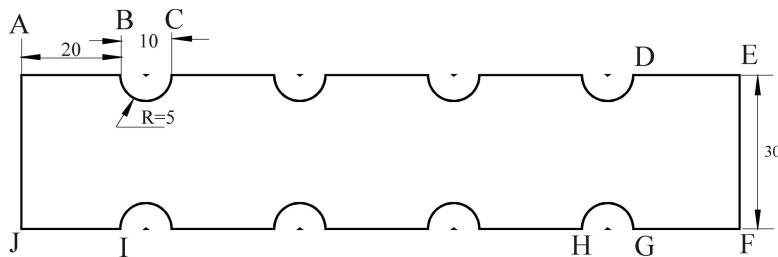
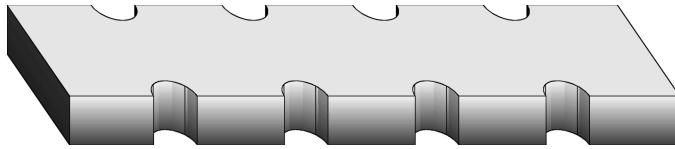
G54 X0 Y0 Z0;.....
 N1 G90 G17 G00 G41 T15 D2 X250.0 Y550.0; establish compensation
 N2 G01 Y900.0 F150; from P1 to P2
 N3 X450.0;from P2 to P3
 N4 G03 X500.0 Y1150.0 R650.0;from P3 to P4
 N5 G02 X900.0 R-250.0;from P4 to P5
 N6 G03 X950.0 Y900.0 R650.0;from P5 to P6
 N7 G01 X1150.0;from P6 to P7
 N8 Y550.0;from P7 to P8
 N9 X700.0 Y650.0;from P8 to P9
 N10 X250.0 Y550.0;from P9 to P1
 N11 G00 G40 X0 Y0; cancel compensation

(18) Program circulation instruction (G22--G800)

G22 is program circulation instruction, G800 is an instruction to end circulate. But G22 must be used with G800 for repeated processing. L means circulation times, the range is 1-99999. The circulation instruction can nest.

Format:	G22 L2		;begin
	:	}	
	:		
	:		
	G800		;end

For example



Program as follows:

```

N0000 G17 G90 X0 Y0 F250 M03    ;
N0001 G91 G01 Z-10
N0010 G22 L4                      ;
N0020 G01 X20                    ;
N0030 G03 X10 I5 J0 Y0          ;
N0040 G800                      ;
N0050 G01 X20                    ;D-E line
N0060 Y-30                      ;E-F line
N0070 G11 X140 Y-30             ;set mirror
N0080 G22 L4                    ;circulation begin
N0090 G01 X20                    ;F-G line
N0100 G03 X10 I5                ;G-H arc
N0110 G800                      ;circulation end
N0120 G01 X20                    ;I-J line
N0130 G01 Y-30                  ;cancel mirror
N0140 G12                      ;J-A line
N0150 G26                      ;go back to program begin point
N0160 M02                      ;over

```

(19) Accurate localization/Continual way processing (G60/G64)

According to requirement of processing, we can set program section connection way by the G60/G64 instruction.

Format: G60 ; accurate stop (mode)
 G64 ; continue section (mode, original)

(20) Circle instruction (G73、G74、G80~G89)

Using Circle instruction, we can shorten the program length, make the program more simple.

Circle instruction table

G code	Feed method	Motion in the bottom of hole	withdraw	application
G73	Intermission feed	No	Rapid move	High speed drill deep hole
G80	Continue feed	No	No	Cancel fixed cycle
G81	Continue feed	No	Rapid move	Drill cycle
G82	Continue feed	Stop	Rapid move	Drill cycle
G83	Intermission feed	No	Rapid move	Drill deep hole cycle
G85	Continue feed	No	Cutting feed	Bore hole cycle
G86	Continue feed	Spindle stop	Rapid move	Bore hole cycle
G89	Continue feed	Spindle stop	Cutting feed	Bore hole cycle
G74	Continue feed	Stop-Spindle rotate CW	Cutting feed	Left tap cycle
G84	Continue feed	Stop-Spindle rotate CCW	Cutting feed	Right tap cycle

Cycle instruction is consist of six motions

Motion 1 location of X and Y axis

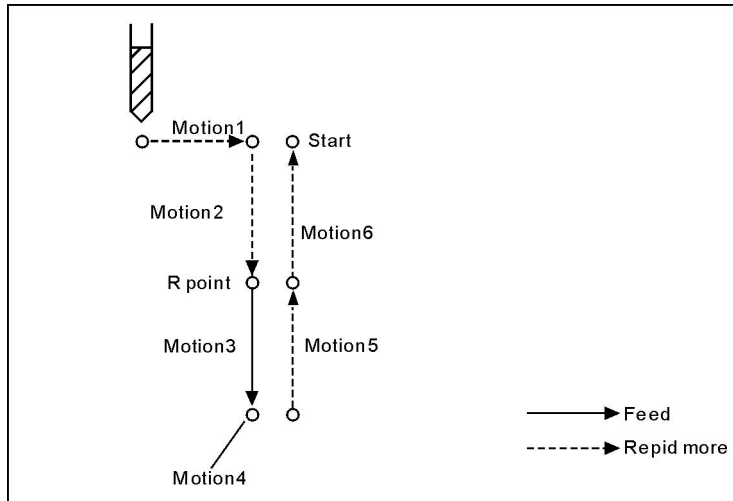
Motion 2 rapid move to R point

Motion 3 machining hole

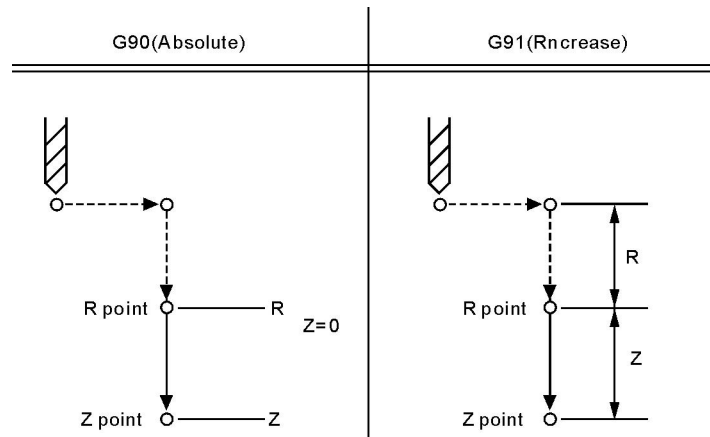
Motion 4 action in the bottom of hole

Motion 5 withdraw to R point

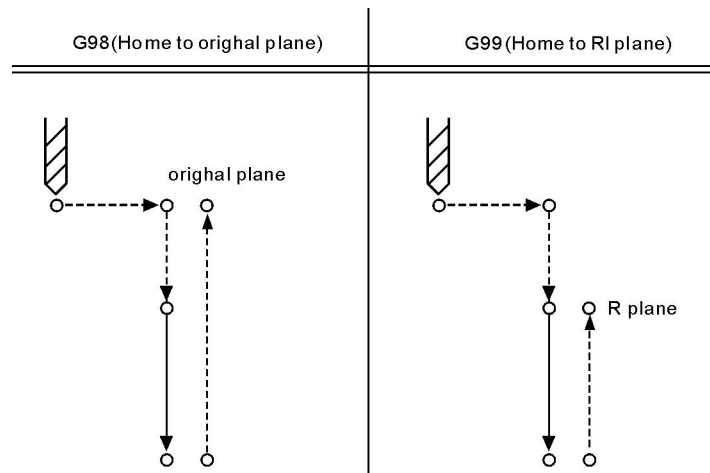
Motion 6 rapid move to original point



The difference of G90 and G91 as follow:



When tool touched the bottom of hole, tool will back to plane where R point stands, or original plane. Generally, G99 is used in first drilling, and G98 is used in final drilling.the original plane will not change even by G99, The difference of G98 and G99 as follow:



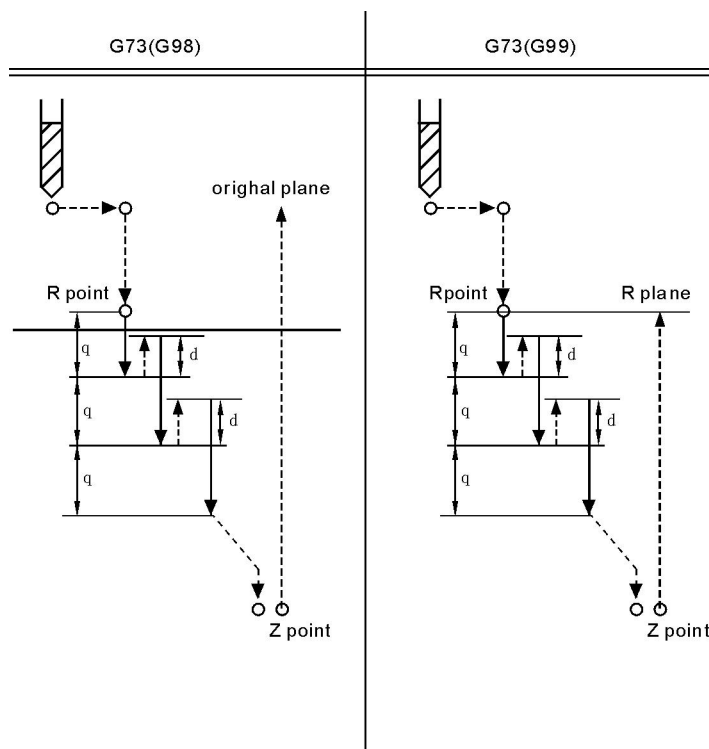
Use the L word to set cycle time, and L is just valid in specified segment. the maximum value is 9999, the default value is 1;

Orientation plane is determined by G17(XY)/G18(ZX)/G19(YZ). Besides, speed parameter P41 arc backlash compensation is set as 2 or 10.

(21) High speed drill deep hole(G73)

This cycle execute high speed drilling deep hole until reaching to bottom, at the same time, remove the cutting trifling from hole.

Format: G73 X_Y_Z_R_Q_F_L_ ;
 X_Y_:hole position data
 Z_:the distance(G91) or coordinate(G90) from R point to hole bottom
 R_: the distance(G91) or coordinate(G90) from original point to R point
 Q_:cutting depth every time
 F_:cutting speed
 L_:repeated times



Note: High speed drilling cycle along the Z axis to carry out intermittent feeding, when using this cycle, the swarf can be easily discharged from the hole and can set the smaller value when backing. This allows to carry out drilling effectively. No.1 parameter in "N"(Processing) parameter is to set the value of the tool withdrawal(d).

Using auxiliary function to rotate spindle before specifying G73 (M code).

Set the offset of tool length(G43 G44 or G49) in changeless cycle which is located

to R point to plus offset, the offset of tool radius is ignored.

For example:

M3 S2000

Spindle starts rotating

G90 G99 G73 X300. Y-250. Z-150. R-100. Q15. F120. Drill the first hole and return to R point

Y-550.;

Drill the second hole and return to R point

G98 Y-750.;

Drill the third hole and return to R point

G80

M30;

The end

(22) Drilling cycle, point drilling cycle(G81)

The cycle is used for normal drilling, execute the feeding cut to hole bottom. Then, the tool moves fast return form hole bottom.

Format: G81 X-Y-Z-R-F-L- ;

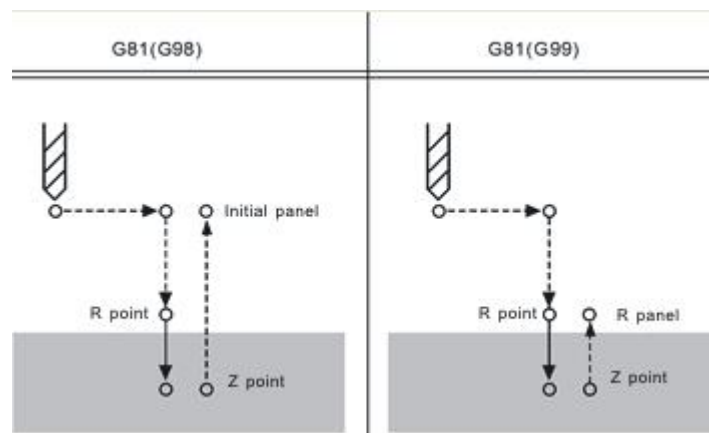
X-Y-: Data of hole position

Z-: Distance form R point to hole bottom (G91)
or coordinate value (G90)

R-: Distance form initial point to R point (G91) or
coordinate value (G90)

F-: Cutting speed

L-: Repeated times



Note: Move fast to R point along the X and Y axis location, execute drilling process from R point to Z point, then tool returns with rapid move.

Use auxiliary function M code to rotate spindle before specifying G81

Locate to R point plus the offset when specifying the offset of tool length(G43 G44 or G49) in fixed cycle, the offset of tool radius is ignored.

Example:

M3 S2000; spindle begins to rotate

G90 G99 G81 X300 Y-250 Z-150 R-100 F120; drill the first hole, and then return to R point

Y-550; drill the second hole, and then return to R point

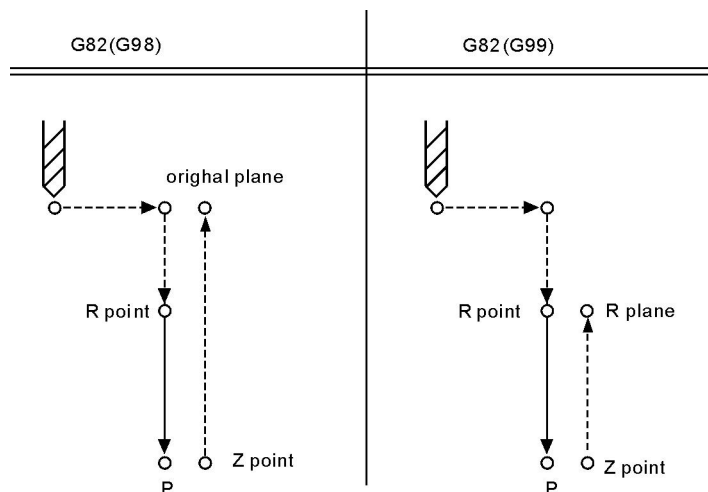
G98 Y-750; drill the third hole, and then return to panel of the initial position

G80
M30; end

(23) Drilling cycle, Reaming cycle(G82)

The cycle is used to drill hole, servo cut to the bottom and pause, then tool recede rapidly.

Format: G82 X-Y-Z-R- P -F-L- ;
 X-Y-: hole position data
 Z-: the distance(G91) or coordinate(G90) from R point to hole bottom
 R-: the distance(G91) or coordinate(G90) from original point to R point
 P-: pause time
 F-: cutting speed
 L-: repeat time



Note: orientating in the direction of X axis and Y axis, rapidly move to R, then drill from R to Z point, when tool arrive at the bottom, execute pause, then tool recede rapidly.

Rotate spindle with M code before G82 specified.

Set offset in the location of R point, while specify tool length offset(G43,G44,G49) in fixed cycle. Tool radius offset is ignored.

For example:

M3 S2000

G90 G99 G82 X300. Y-250. Z-150. R-100. P1000 F120.

Y-550.;

G98 Y-750.;

G80

M30;

(24) Intermission drill cycle (G83)

It applies to chip removal from hole during the execution of intermittent feeding cut to bottom of hole.

Format : G83 X_ Y_ Z_ R_ Q_ F_ L_ ;

X_ Y_ : hole position data

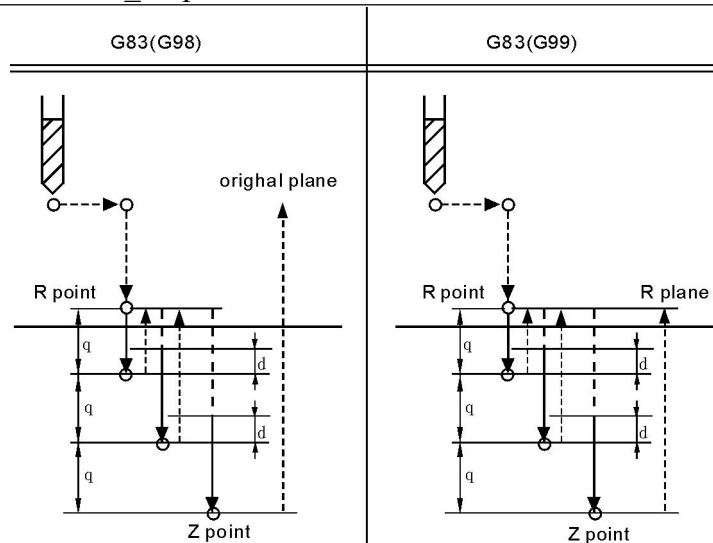
Z_ : the distance(G91) or coordinate(G90) from R point to hole bottom

R_ : the distance(G91) or coordinate(G90) from original point to R point

Q_ :cutting depth every time

F_ :cutting speed

L_ :repeated times



Note: Q means the cutting depth of each cutting feed, it must be specified in the incremental value. Cutting feed in the second and later must be executed quickly to move to d point to execute again before the end of last drilling, No.2 parameter in processing parameter set the cutting feed d, must specify the positive value in Q, negative is ignored.

Use the auxiliary functions M code to rotate spindle before specifying G83.

Set offset in the location of R point, while specify tool length offset(G43,G44,G49) in fixed cycle. Tool radius offset is ignored.

For example:

M3 S2000

spindle starts;

G90 G99 G83 X300. Y-250. Z-150. R-100. Q15. F120. drill the first hole, back to R;

Y-550.; drill the second hole, back to R;

Y-750.; drill the third hole, back to R;

G98 Y-600.; drill the fourth hole, back to original plane;

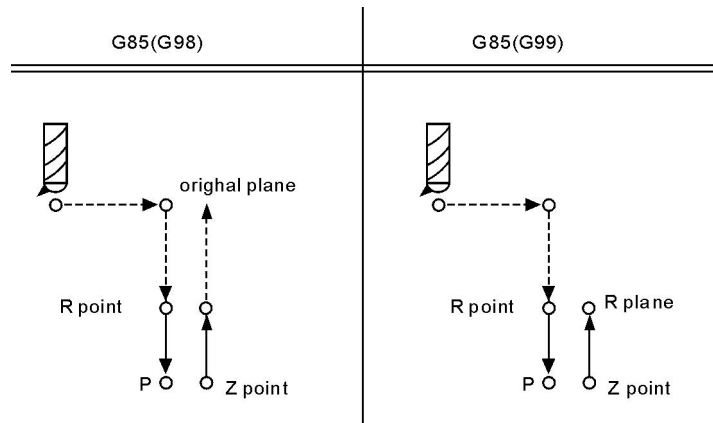
G80

M30;

End.

(25) Boring cycle(G85)

Format: G85 X_ Y_ Z_ R_ F_ L_ ;
 X_ Y_ : Hole position data
 Z_ : the distance(G91) or coordinate(G90) from R point to hole bottom
 R_ : the distance(G91) or coordinate(G90) from original point to R point
 F_ :cutting speed
 L_ :repeated times



Note: Along the X and Y axis to locate, move fast to R point and carry out boring from R point to Z point, carry out cutting feed and return to R point when arriving the bottom of hole.

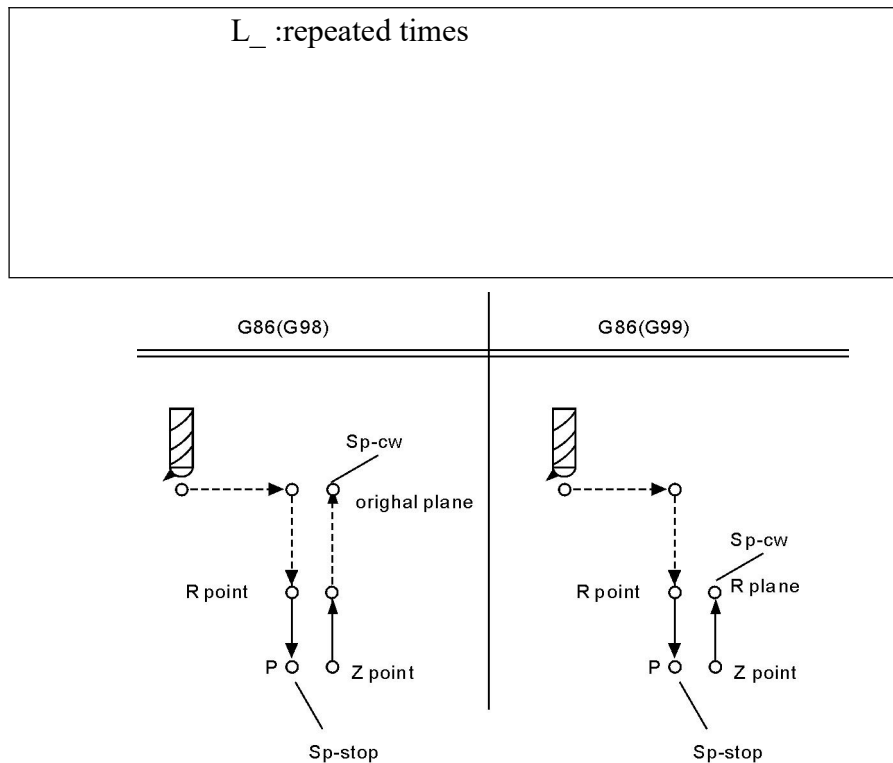
Set offset in the location of R point, while specify tool length offset(G43,G44,G49) in fixed cycle. Tool radius offset is ignored.

For example:

M3 S100	spindle starts
G90 G99 G85 X300. Y-250. Z-150. R-120. F120.	bore the first hole, back to R;
Y-550;	bore the second hole, back to R;
Y-750;	bore the third hole, back to R;
G98 Y-600;	bore the fourth hole, back to original plane
G80	
M30;	End

(26) Boring cycle(G86)

Format: G86 X_ Y_ Z_ R_ F_ L_ ;
 X_ Y_ : hole position data
 Z_ : the distance(G91) or coordinate(G90) from R point to hole bottom
 R_ : the distance(G91) or coordinate(G90) from original point to R point
 F_ :cutting speed



Note: Along the X and Y axis to locate, move fast to R point and carry out boring from R point to Z point. The tool will return quickly when spindle stopping at the bottom of hole.

Set offset in the location of R point, while specify tool length offset(G43,G44,G49) in fixed cycle. Tool radius offset is ignored.

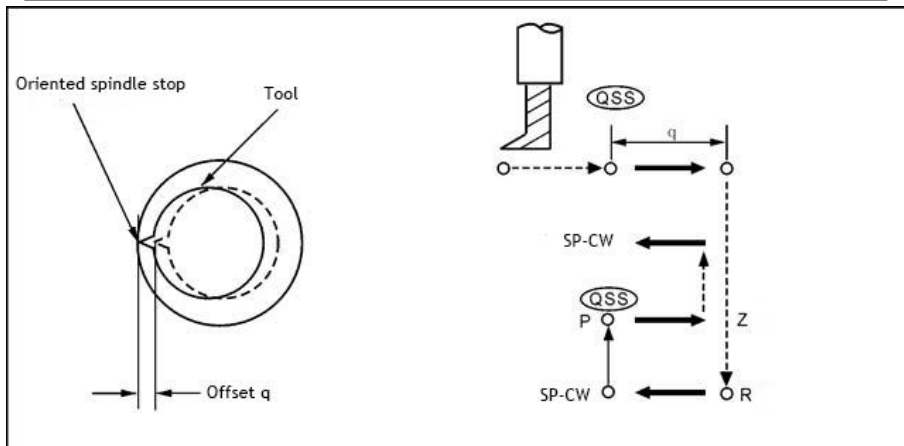
For example:

M3 S2000	Spindle starts
G90 G99 G86 X300. Y-250. Z-150. R-100. F120.	bore the first hole, back to R;
Y-550.;	bore the second hole, back to R;
Y-750.;	bore the third hole, back to R;
G98 Y-600.;	bore the fourth hole, back to original plane
G80	
M30;	End.

(27) Boring cycle, Back boring cycle(G87)

G87 only works in machine center controller(NEW1000MiC/NEW18iMi), because it requires that spindle has orientation function(output M61, check M22). During finishing boring, when tool reaches bottom of hole, spindle stops, and tool will leave the surface of work piece according to direction defied by Processing parameter P4.

Format: G87 X_ Y_ Z_ R_ Q_ P_ F_ L_ ;
 X_ Y_ : hole position data
 Z_ : the distance(G91) or coordinate(G90) from
 R point to hole bottom
 R_ : the distance(G91) or coordinate(G90) from
 original point to R point
 Q_ : offset value in hole bottom, direction is
 defined by processing parameter P4;
 P_ : pause time unit:s
 F_ : cutting speed
 L_ : repeat times



Note: after positioning in the X and Y axis direction, spindle stops at fixed degree. Tool moves rapidly in reversed direction(defined by process parameter P4,P5) of tool endpoint, then position at R point of hole bottom. Then tool moves in direction of tool endpoint and spindle rotate CW. Boring to Z point in the Z positive direction. Spindle stops at fixed degree when reaches Z point, tool moves rapidly in reversed direction(defined by process parameter P4,P5) of tool endpoint, afterwards tool retract to initial position, tool shifts in tool endpoint direction, spindle rotates CW, go to next segment.

please use M code to rotate spindle before G87. specifying tool length offset(G43,G44,G49) during fixed circle while positioning at R point, tool radius offset will be ignored.

Sample:

```

N10 M3 S100 ; spindle starts
N20 G90 G99 G87 X300 Y-250 Z-150 R-100 Q5 ; position, boring 1 hole,
then back to R point, orientation at hole bottom then move 5mm
N30 Y-350 ; boring 2 hole, then back to R point
N40 Y-550 ; boring 3 hole, then back to R point
N50 Y-750 ; boring 4 hole, then back to R point
N60 G80
N70 M30
  
```

(28) Boring cycle(G89)

Format: G89 X_ Y_ Z_ R_ P_ F_ L_ ;

X_ Y_ : hole position data

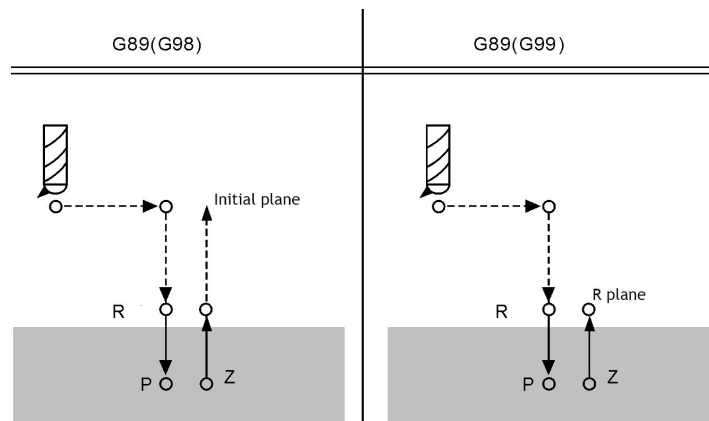
Z_ : the distance(G91) or coordinate(G90) from R point to hole bottom

R_ : the distance(G91) or coordinate(G90) from original point to R point

P_ :pause time

F_ :cutting speed

L_ :repeated times



Note:It is the same as G85 unless suspending at the bottom of hole.

Rotate spindle with M code before G82 specified.

Set offset in the location of R point, while specify tool length offset(G43,G44,G49) in fixed cycle. Tool radius offset is ignored.

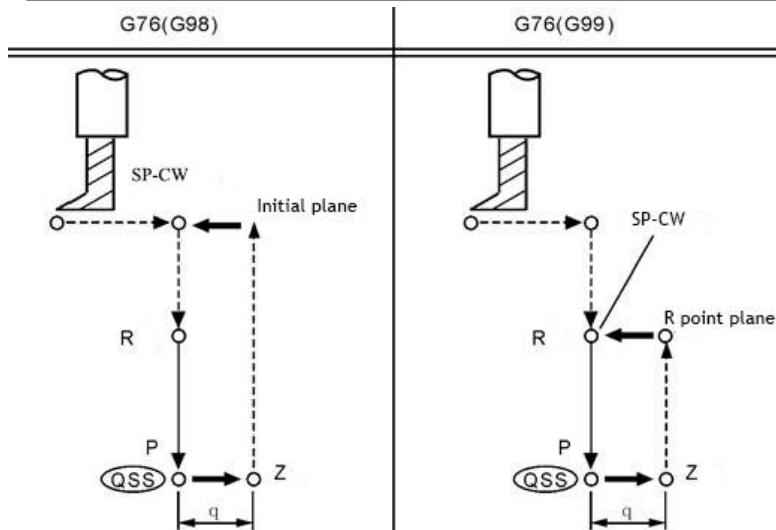
For example:

M3 S100	Spindle starts;
G90 G99 G89 X300. Y-250. Z-150. R-120. P1000 F120.	bore the first hole, back and pause for 1 second at the bottom;
Y-550.;	bore the second hole, back to R;
G98 Y-750.;	bore the third hole, back to original plane
G80	
M30;	End

(29) Finishing boring cycle(G76)

G74 only works in machine center controller(NEW1000MiC/NEW18iMi), because it requires that spindle has orientation function(output M61, check M22). During finishing boring, when tool reaches bottom of hole, spindle stops, and tool will leave the surface of work piece according to direction defied by Processing parameter P3.

Format: G76 X_ Y_ Z_ R_ Q_ P_ F_ L_ ;
 X_ Y_ : hole position data
 Z_ : the distance(G91) or coordinate(G90) from
 R point to hole bottom
 R_ : the distance(G91) or coordinate(G90) from
 original point to R point
 Q_ : offset value in hole bottom, direction is
 defined by processing parameter P3;
 P_ : pause time unit:s
 F_ : cutting speed
 L_ : repeat times



Note: when reaches hole bottom, spindle tops in fixed degree, and tool will return in revered direction of tool point, which will make sure process surface not be damaged. Spindle orientation degree is defined by process parameter P5. please use M code to rotate spindle before G76. specifying tool length offset(G43,G44,G49) during fixed circle while positioning at R point, tool radius offset will be ignored.

Sample:

```

N10 M3 S100 ;spindle start
N20 G90 G99 G76 X300 Y-250 Z-150 R-100 Q5 ; position, boring 1 hole, then
back to R point, orientation at hole bottom then move 5mm.
N30 Y-350 ; position, boring 2 hole, then back to R point,
N40 Y-550 ; position, boring 3 hole, then back to R point,
N50 Y-750 ; position, boring 4 hole, then back to R point,
N60 G80
N70 M30
  
```

(30) Left tap cycle(G74)

Tapping has two kinds of methods: Tracking the spindle encoder (P411=2, spindle must assemble encoder) and interpolation of Z axis and spindle servo (P405=0,

P410=95, P411=3). No.404~No.413 parameter in axis parameter to set.

Pay attention:

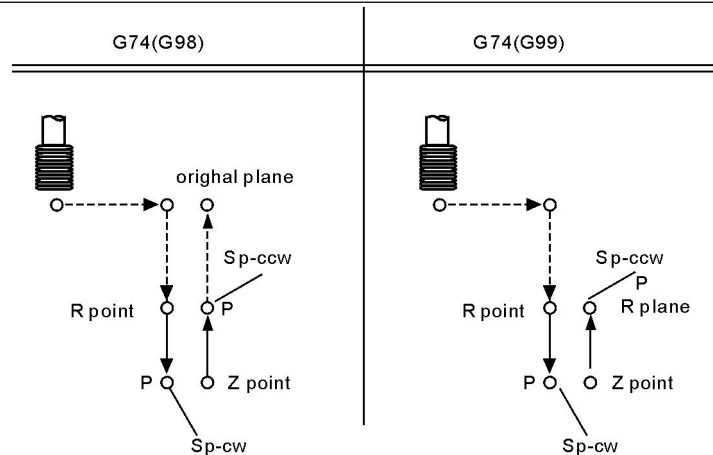
When spindle and encoder do not drive as 1:1, please modify the No.412 No.413 parameter in axis parameter;

412, the number of spindle teeth (requirement: less than or equal to the number of encoder teeth, must match our keysets when greater than the number of encoder teeth);

413, the number of encoder teeth

The loop is executed with tapping left, in the left tapping cycle, the depth Q of feeding, CCW spindle back distance d(by process parameter P7), type of chip removal set by process parameter P6(0 means high-speed removal and back by d, 8 means normal removal and back to R). the spindle rotates CW when arriving at the bottom of hole.

Format: G74 X_ Y_ Z_ R_ P_ K_ S_ L_ ;
 X_ Y_ : hole position data
 Z_ : the distance(G91) or coordinate(G90) from
 R point to hole bottom
 R_ : the distance(G91) or coordinate(G90) from
 original point to R point
 P_ : pause time unit:s
 K_ : screw pitch
 S_ : spindle rotate speed
 L_ : repeated times



Note: The speed of coordinate axis is determined by speed of spindle and screw pitch when processing thread, it is not a matter with speed F. The system will limit the speed within maximum feeding speed.

Spindle override switch and feeding axis override switch are invalid when processing thread.

Should specify the screw pitch K value in every processing program segment of thread, otherwise not through compile.

Use spindle CCW rotation to carry out tapping. Should process a negative thread in order to return spindle CW rotation when arriving at the bottom of hole. In the left tapping period, feeding suspension does not stop lathe until the return action

completed.

Use auxiliary function to rotate spindle CCW before specifying G74. When the spindle is pulse controlling mode without using auxiliary functions M code.

Set offset in the location of R point, while specify tool length offset(G43,G44,G49) in fixed cycle. Tool radius offset is ignored.

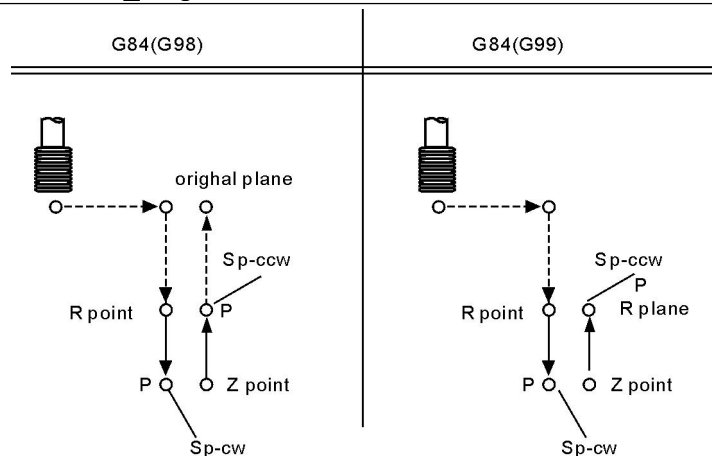
For example:

N1 M4 S100	Spindle starts;
N2 G90 G99 G74 X300. Y-250. Z-150. R-100. K5 S100	tap the first hole, then back to R point;
N3 G98 Y-550. K5;	tap the first hole, then back to R point;
N4 G80;	
N5 M30;	End;

(31) Right tap cycle(G84)

The same as G74 unless direction.

Format: G84 X_ Y_ Z_ R_ P_ K_ L_ ; X_ Y_ : hole position data Z_ : the distance(G91) or coordinate(G90) from R point to hole bottom R_ : the distance(G91) or coordinate(G90) from original point to R point P_ :pause time K_ :screw parameter S_ :spindle rotate speed L_ :repeated times



For example:

M3 S100	Spindle starts.
G90 G99 G84 X300 Y-250 Z-150 R-120 P300 K5 S100	
	Tap the first hole, then back to R;
Y-550. K5;	Tap the second hole, then back to R;
G98 Y-750. K5;	tap the third hole, then back to original plane;
G80	

M30;

End.

(32) Cancel cycle instruction (G80)

Cancel cycle instruction.

Format: G80 ;

Note: Cancel all cycle instruction and execute normal operation.

(33) Pole coordinate instruction(G15/G16)

Pole coordinate instruction inquire user provide radius and angle, the positive direction of angle is anticlockwise of positive direction of the first axis in plane, and negative direction is clockwise. Radius may use absolute and increase type(G90, G91), Angle only use absolute type(G91).

Format :

G15 Cancel Pole coordinate;
(G17/G18/G19) (G90/G91) G16 IP_ ;establish

Note:

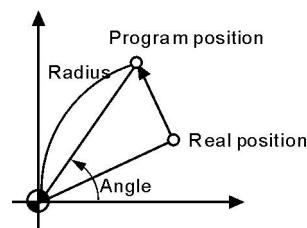
- 1.G17/G18/G19 specify the panel of pole coordinate instruction.
- 2.G90 specify the zero point of workpiece coordinate system as the origin of pole coordinate, measure radius from this point.
- 3.G91 specify current position as the origin of pole coordinate system, measure radius from this point.
- 4.IP_ specify the panel axis address and value of pole coordinate system panel selection.

First axis: Radius value of pole coordinate.

Second axis: Value of pole angle.

5.G90 set the zero point of workpiece coordinate system as origin of pole coordinate system:

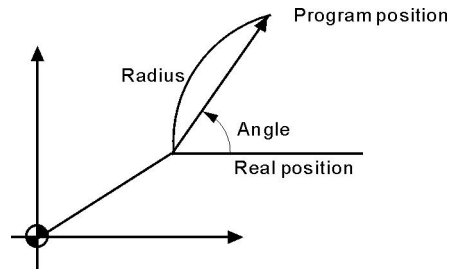
Use programming instruction of absolute value to specify the radius(The distance between zero point and programming point).When using local coordinate system(G52), the origin of local coordinate system changes into the center of pole coordinate system, the angle uses absolute value. As the follow shown:



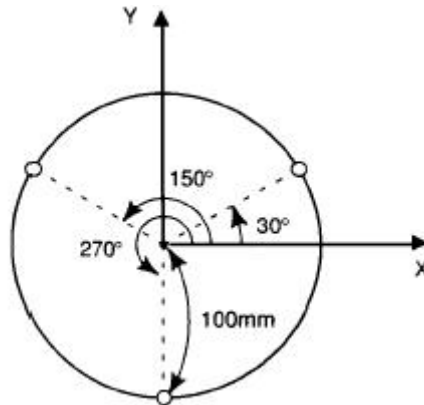
6.G91 set the current position as the origin of pole coordinate system:

Using programming instruction of incremental value to specify the radius(The distance between the current position and programming point).The angle use absolute

value.As the follow shown:



For example:



G17 G90 G16 X0 Y0; specify polar coordinate instruction, set the zero point of XY plane coordinate system as the original point of polar coordinate system;

G81 X100.0 Y30.0 Z-20.0 R-5.0 F200.0; specify distance of 100mm and angle of 30°;

Y150.0; specify distance of 100mm and angle of 150°

Y270.0; specify distance of 100mm and angle of 270°

G15 G80; cancel polar coordinate system.

(34) Metric system and British system(G20/G21), Feeding method(G94/G95)

Format:

G20 ; inch;
G21 ; millimeter;
G94: minute feeding;
G95: rotate feeding.

Note:The G code must be compiled in the beginning of the program, using separate program segment to specify before setting the coordinate system.Switch the unit of input data into minimal inch or millimeter after G code of switching inch or millimeter specifying,the angle of data input unit keeps unchanged, change the units of value as follows after switching the inch or millimeter:

- The feeding speed is specified by F code
- Position instruction

- Offset value of workpiece zero point
- Compensation value of tool
- The unit of manual pulse generator
- The distance in incremental feeding

Specification of unit under G20/G21(it is necessary to reset the unit in parameter):

- 1) The minimum unit of British system G20 is 0.0001inch, the minimum unit of Metric system is 0.001mm.
- 2) Under British system G20, unit of all instructions related to position is inch;
- 3) Under British system G20, unit of feeding speed F instruction is inch/min;
- 4) Under British system G20, unit of tool compensation and offset of workpiece coordinate system G54-G59 is inch;
- 5) Under British system G20, unit of parameters related to distance(such as soft limit and compensation) change from mm of Metric system into 0.1 inch or from um into 0.001inch;
- 6) Under British system G20, unit of parameters related to speed change from mm/min into 0.1inch/min ;
- 7) Under British system G20, unit of parameters related to accelerated speed change from mm/min/s into 0.1inch/min/s;
- 8) Under British system G20, handwheel incremental value are 0.0001inch, 0.001inch, 0.01 inch;
- 9) Under British system G20,if pulse equivalent changes, system will adjust automatically, electronic gear will accords to metric system;
- 10) Under British system G20, coordinate displays four numbers behind the decimal point;
- 11) G20/G21 should be written in the beginning of program, instead of middle of program;
- 12) Under British system G20, rotary axis rotating 1 degree accords to 0.1 inch;
- 13) two systems can be converted by executing G20/G21 in MDI, and restating controller.
- 14) It is necessary to set tool after conversion between G20 and G21;
- 15) As to system prompt information, you just need to modify the parameter configuration file cncsystemen. For example, the parameter about speed: change mm/min into 0.1inch/min, parameter about position:change mm or degree into 0.1 inch, change um into 0.0001 inch.

(35) Back to start point of program(G26/G261-G268)

Format :	G26	; ZXY all go back.
	G261	; X go back.
	G262	; Y go back.
	G263	; Z go back.
	G264	; A go back.
	G265	; B go back.
	G266	; C go back.
	G267	; Xs go back.
	G268	; Ys go back.

Note: G26 motion is according to linkage type.

(36) Memory the current point(G25)

Format: G25	;	To remember the coordinate of X Y Z A B
-------------	---	---

(37) Return to the memorial point(G61/G611-G618)

Format:	G61	;Return to X Y Z of memorial point
	G611	;Return to X of memorial point
	G612	;Return to Y of memorial point
	G613	;Return to Z of memorial point
	G614	;Return to A of memorial point
	G615	;Return to B of memorial point
	G616	;Return to C of memorial point
	G617	;Return to Xs of memorial point
	G618	;Return to Ys of memorial point

Note:the way that G61 back to the point memorized by G25 is by G00.

(38) Check skip(G31、 G311)

Format: G31 X_ Y_ Z_ A_ F_ P_ ;No alarm
 G311 X_ Y_ Z_ A_ F_ P_ ;alarm

P: N line+(X00/X39+1000 or 2000), 1000 means skip when availability, 2000 means skip when invalidation.

For example: G31 X50 Z100 F100 P331022 ;if X22 availability then go to N33.

G311 X50 Z100 F100 P2021 ;if X21 invalidation then go to next line. And it will alarm till X21 is invalid.

(39) Automatic beveling (I) and smoothing(R)

The acquiescent panel of milling is G17

Format for G17:

G01(G00) X I automatic beveling, the coordinate in the next program segment must be G01(G00) Y.

G01(G00) Y I automatic beveling, the coordinate in the next program segment must be G01(G00) X.

G01(G00) X R automatic smoothing, the coordinate in the next program segment must be G01(G00) Y.

G01(G00) Y R automatic smoothing, the coordinate in the next program segment must be G01(G00) X.

Format for G18:

G01(G00) X I automatic beveling, the coordinate in the next program segment must be G01(G00) Z.

G01(G00) Z I automatic beveling, the coordinate in the next program segment must be G01(G00) X.

G01(G00) X R automatic smoothing, the coordinate in the next program segment must be G01(G00) Z.

G01(G00) Z R automatic smoothing, the coordinate in the next program segment must be G01(G00) X.

Format for G19:

G01(G00) Y I automatic beveling, the coordinate in the next program segment must be G01(G00) Z.

G01(G00) Z I automatic beveling, the coordinate in the next program segment must be G01(G00) Y.

G01(G00) Y R automatic smoothing, the coordinate in the next program segment must be G01(G00) Z.

G01(G00) Z R automatic smoothing, the coordinate in the next program segment must be G01(G00) Y.

Pay attention:

1. The address of I and R are specified with radius model. The running distance of this line and the next line must be greater than the length of beveling or radius of smoothing, otherwise the system will decrease the length of beveling or radius of smoothing to minimal running distance of this line and the next line automatically.

2. The two adjacent lines must be 90 degrees.

For example:

0 G54 G0 X-50 Y-50 Z20

N1 M03 S500

N2 G01 G42 D01 X0 Y0 F200

N3 G01 Z-5

N4 X100 I4

; Beveling4x4

N5 Y40 R6

; SmoothingR6

```

N6 X47 R5                ; SmoothingR5
N7 Y70 I3                ; Beveling3x3
N8 X15

N9 X0 Y40
N10 Y0
N11 G0 X-50 Y-50 G40
N12 Z50
N13 M30

```

(40) Program call(M97 M98 M99)

Unconditional jump

The line running unconditional jump to the line which is specified by P; P4 stands for using four field of digital specify the program to the entrance line of the calling main program (mark line).

Subroutine call

In this system the subroutine should be an independent program.

M98 P L unconditional call subroutine instruction. P is to specify the name and path of subroutine call, L refers to the calling times address of subroutine.

The M98 instruction can be omitted without writing, format: PP file name, the file name can be hidden files, the first character of hidden files must be "HIDEFILE" at the beginning. Such as the file "HIDEFILE01", this program in the program area is not displayed, can use the instruction M98 PHIDEFILE01 or M98 P*01 or PP*01 or PPHIDEFILE01 when calling.

For example:

P sub/1390 means subroutine is tmp/NC/sub/1390

Note:

- 1.tmp/NC/ is the system's default path, sub is a folder for the following
- 2.The subroutine must be a independent program.
- 3.Method of the main program in USB calls the subroutine in USB: P[or P].

For example:

M98 P[A1234 means calling the subroutine A1234 in USB;

M98 P]SS12 means calling the subroutine SS12 in USB;

PP[FFDE means calling the subroutine FFDE in USB;

It needs to write the path of file if call the subroutine in folder of USB.

There must be space in front of L(Subroutine calling times). Return to the next program segment of main program when subroutine running to the end.If the program contains a fixed sequence or repeated pattern, then the sequence or pattern can be compiled to subroutine to save in memory storage in order to programme easily, the subroutine can be called by main program which is also can be called by another subroutine.

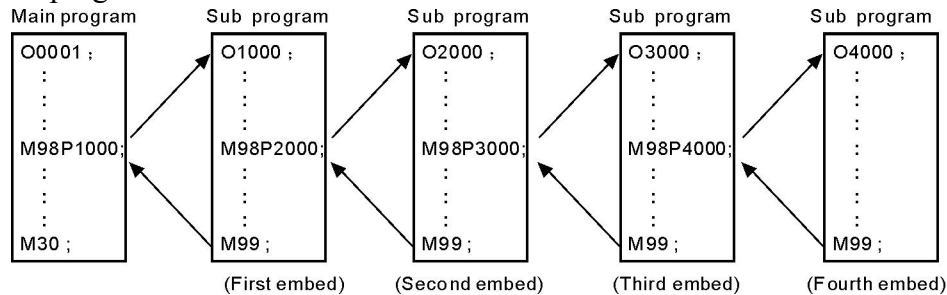
M99 is an instruction of ending subroutine return, must have this instruction to end the subroutine.

Pay attention:

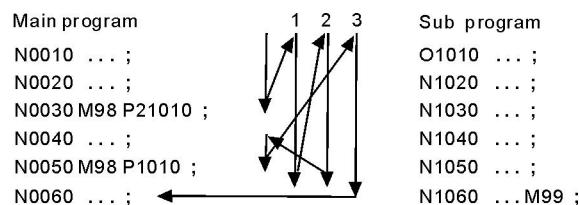
- 1) M99 in the main program is the same as M02;
- 2) M99 with P in the main program is the same as M97;
- 3) M99 returns to main program call in subroutine is in the next line;
- 4) M99 with P in the subroutine returns to P program line in main program;

Subprogram can call another subprogram(embedded). no matter the program called is main program or subprogram, it will back to the program line next to main calling program. When the subprogram is called by the main program, it is consider as first-level subroutine.

The Sub-program can embedded call as follow :



For example:



The calling instruction can be used for 9999 times in the most.

Conditional wait, jump instruction

The system of M code is used for detecting the external input signal as the condition, as follows:

Conditions wait

M12 M13 instruction are used to detect the input signal M12, M12 in program line is to detect M12 input signal is effective to execute the next program line , M13 means to detect M12 input signal is invalid to execute the next program line.The instruction is in an independent line.

M14 M15 instruction are used to detect the input signal M14, M14 in program line is to detect M14 input signal is effective to execute the next program line, M15 means to detect M14 input signal is invalid to execute the next program line.The instruction is in an independent line.

M16 M17 instruction are used to detect the input signal M16, M16 in program line is to detect M16 input signal is effective to execute the next program line, M17 means to detect M16 input signal is invalid to execute the next program line.The instruction is in an independent line.

M18 M19 instruction are used to detect the input signal M18, M18 in program line is to detect M18 input signal is effective to execute the next program line, M19 means to detect M18 input signal is invalid to execute the next program line.The instruction is in an independent line.

M22 M23 instruction are used to detect the input signal M22, M22 in program

line is to detect M22 input signal is effective to execute the next line program, M23 means to detect M22 input signal is invalid to execute the next program line. The instruction is in an independent line.

M24 M25 instruction are used to detect the input signal M24, M24 in program line is to detect M24 input signal is effective to execute the next line program, M25 means to detect M24 input signal is invalid to execute the next program line. The instruction is in an independent line.

M28 M29 instruction are used to detect the input signal M28, M28 in program line is to detect M28 input signal is effective to execute the next line program, M29 means to detect M28 input signal is invalid to execute the next program line. The instruction is in an independent line.

Conditional jump

Plus Pxxxx (number of program line) in front of the M12 /M13 /M14 /M15 /M16 /M17 /M22 /M23 /M28 /M29 instruction. Shifting if the condition success, otherwise execute the next.

For example: M14 P0120

When the program running to this line and the system detecting the M14 input signal effectively, program will jump to the 120th line of program (the marking line), execute the next instruction if the M14 input signal is invalid.

M186/M187 welding follow instruction(optional for Bus type controller):

- (1) A axis is rotary axis, parameter set as all reduction, for tool rotation.
- (2) C axis is rotary axis, parameter set as all reduction, for tool rotation.

C axis positive direction and XYZ coordinate should be agreed with right hand coordinate principle.

(3) in order to perform C axis rotation degree between sections in program, the C axis command speed is G00 speed, and moves nearby.

- (4) Command:

M186: open welding follow function

M187: close welding follow function

- (5) sample:

```
G54 G0X10Y60Z30
M3S200
M186                ; open follow function
G1 Z50 F500         ; Z axis lift
G1X10Y63F200        ; XY axis tool endpoint align
G1 Z0               ;Z axis move to welding start point
X0Y-23
X10Y50
G2X-23.20Y130R90
M187                ;cancel follow function
M5
M30
```

(41) Feeding speed F function

It is the mode, actual running speed is the setting speed times the trimming rate of speed,

F is used for specify the processing speed of feeding instruction G01 G02 G03.

The range is 0.01-15000mm/min, feeding speed is Fx trimming speed, F has mode function.

Executing the F instruction at the first, and then execute the motion instruction when the F instruction and motion instruction are in the same line.

(42) T/H/D function

The T/H/D function is means that tool length and radius compensate, which is mode, used by code in program.

The tool code is from T01 to T99, every tool have four tool compensate value, which is length compensation from H1 to H99 and radius compensation from D1 to D99.

(43) Spindle speed S, SS

The system offers two ways spindle controlling modes.

The first spindle speed is specified by S, the first spindle has two kinds of gear controlling mode:

(1) The first is four gear spindle speed electrical control, output four bits code of step speed change, M41-M44 instruction control corresponds to S01-S04 output code, step speed change. Use Axis parameter P50, P51, P52, P53 and P54 to set the mode of shifting.

(2) The second uses four gears + step-less speed, M41-M44 instruction control, correspond the output S01-S04 code. Use speed parameter P42, P43, P44, P45 to set the maximum speed of corresponding gear, use Axis parameter P50, P51, P52, P53 and P54 to set the mode of shifting.

Stepless speed, the range is 0-99999, output 0-10V variable-frequency voltage. The output voltage trims x10V of maximum speed of specified spindle.

Second spindle speed is specified by SS, the highest speed is controlled by the speed parameter P46, output 0-10V variable-frequency voltage.

(44) Macro program instruction(G65,G66,G67)**1.Input instruction: WAT**

Waiting for the input port X valid or invalid instruction

Format: WAT+ (-) X

Note: "+" to means the input is effective;

"-" means the input is invalid;

"X" means the input port X00-X55; see the I/O diagnosis;

2.The output instruction: OUT

Set the output port Y is valid or invalid instruction

Format: OUT +(-)Y

Note: "+" means the output is effective;

"-" means the output is invalid;

"Y" means the output port Y00-Y31; see the I/O diagnosis;

3. Variable and assignment: =

1) #0--#20 local variable: local variables only can be used to store data in macro program, such as a result of operation, when power is off, the local variables are initialized to the empty. The argument assignment to the local variable when calling the macro program.

2) #21--#600 global variables: The meanings are the same in different macro program.

When power is off, the variable #21--#100 is initialized to zero, the variable #101--#600 data is saved not to loss even if the power is off.

3) #1000-- system variable: the system variables are used to change various data when reading the running CNC. For example, the current position and the compensation of tool.

Special note: macro variables #100--#155 and #190--#202 have been used by the system, users can not use.

4) The macro variables #1001--#1099 corresponds the X axis offset value of lathe T1--T99(Unit: micron)

The macro variables #1401--#1499 corresponds the Z axis offset value of lathe T1--T99(Unit: micron)

Could read the value, for example: #200=#1003; To read the X axis offset value of the third tool into macro variables #200.

Could modify the value, for example: #1003=23000; To modify the X axis offset value of the third tool to 23000 micron.

#1003=#1003+50; To increase the X axis offset value of the third tool 50 micron.

5)The I/O variables:

#1800: X00-X07 (D0-D7)

#1801: X08-X15 (D0-D7)

#1802: X16-X23 (D0-D7)

#1802: X16-X23 (D0-D7)

#1803: X24-X31 (D0-D7)

#1804: X32-X39 (D0-D7)

#1805: X40-X47 (D0-D7)

#1806: X60-X67 (D0-D7)

#1808: Y00-Y15 (D0-D15)

#1809: Y16-Y31 (D0-D15)

Format:#i=Expression

6) Save all global macro variables with user program format.

FILEMS(AABBCC) or FILEMS[AABBCC]

To save #21--#999 into AABBCC files.

If you want to read these variables, just need to execute the file or call it as a

subprogram.

4. The arithmetic and logic operation

Table:

Function	Format	Note
Definition	#i = #j	
Addition	#i = #j + #k ;	
Subtraction	#i = #j - #k ;	
Multiplication	#i = #j * #k ;	
Division	#i = #j / #k ;	
Sin	#i = SIN(#j) ;	
Asin	#i = ASIN(#j);	
Cos	#i = COS(#j) ;	90.5 degrees mean 90 degrees 30 minutes
Acos	#i = ACOS(#j);	
Tan	#i = TAN(#j);	
Atan	#i = ATAN(#j);	
Square root	#i = SQRT(#j);	
Absolute value	#i = ABS(#j) ;	
Rounding off	#i= ROUND(#j);	
Round down	#i = FIX(#j);	
Round up	#i = FUP(#j);	
Natural logarithm	#i = LN(#j);	
Exponential function	#i = EXP(#j);	
Or	#i = #j OR #k ;	Executing with binary system
Exclusive or	#i = #j XOR #k ;	
And	#i = #j AND #k ;	

5. Unconditional transfer: GOTO N

Transfer to the program line with sequence number appears error when specifying beyond the 1-99999, could use expression to specify the sequence number.

For example: GOTO 5, GOTO#100

6.Conditional transfer: IF (Conditional expression) GOTO or THEN

1) IF (Condition) GOTO or THEN

2) IF (condition)

<operation>

ELSE

<operation>

ENDIF

3) IF (condition)

<operation>

ELIF

<operation>

ELIF

<operation>

ENDIF

If the conditional expression specified meet, execute this segment; if the conditional

expression specified does not meet, execute the next segment.

For example:

IF (#100 EQ 2) THEN #100=5

IF (#101 GT 2) GOTO 6

Operation meaning:

EQ equal

NE not equal

GT greater than >

GE greater than or equal >=

LT less than <

LE less than or equal

7. Cycle: WHILE (conditional expression) DO 1, 2, 3

Specifies a conditional expression in front of WHILE. When the specified conditions are met, execute the program between DO and END. Otherwise, turn to the program line after END. Cycle of the embed is 3 at the most.

For example:

WHILE (#100 LT 3) DO 1

.....

WHILE (#103 EQ 5) DO 2

.....

WHILE (#200 GE 20) DO 3

.....

END 3

.....

END 2

.....

END 1

8.Non-mode to call macro program:G65

Format: G65 P- L- <A-B-C-..... Argument passing data >

P is the name of macro program, L is the calling times, A B C are argument, the name of argument as follows:

#0->A、#1->B、#2->C、#3->D、#4->E、#5->F、#6->H、#7->I、#8->J、#9->K、
#10->M、#11->Q、#12->R、#13->S、#14->T、#15->U、#16->V、#17->W、#18->X、
#19->Y、#20->Z.

Special attention: The address G、L、N、Q、P can't be used in argument.

For example:

Main program:9000

G00 X0 Z0

G65 P8000 L1 A5 B6

G0 X0 Z0

M30

Macro program:8000

N1 #2=#0+#1

N2 IF (#2 EQ 10) GOTO 4

```

N3 GOO X#2
N4 G00 Z#1
N5 M99                ; Return

```

9.Mode to call macro program:G66 G67

G67 instruction is to cancel G66 instruction.The format is the same as G65.

For example:

```

Main program:9000
G00 X0 Z0
G66 P8000 L2 A5 B6
A8 B1
A9 B10
G67
M30
Macro program:8000
N1 #2=#0+#1
N2 IF  (#2 EQ 10) GOTO 4
N3 GOO X#2
N4 G00 Z#1
N5 M99                ; Return

```

10. information prompt dialog:

Format: MSG(parameter) or MSG[parameter]: parameter is information character string, pause;

Notice: the instruction also applies to NC program(non-macro program).

When prompting information dialog, system will convert into pause mode automatically.

Format: STAF(parameter) or STAF[parameter], parameter is information character string, not pause.

11. automatic generation of processing program

1> open the file:

FILEON(parameter) or FILEON[parameter]

Example: FILEON(AABBCC) or FILEON[AABBCC]

Means establish or open a file AABBCC.

2> Close the file

FILECE means close the current file, if not edit the instruction, the system will close the file automatically when the program is finished.

3> Write a string of character into opening file:

FILEWD (parameter) or FILEWD[parameter]

Example: FILEWD(G54G0X0Z0) or FILEWD[G54G0X0Z0]

Means write the string G54G0X0Z0 into the opening file.

4> Write the absolute coordinates of current feeding axis into the opening file.

Format: FILEWC

Example:

```

G0X0Z0
FILEON[AABBCC]

```

FILEWD [G54G0X0Z0]

G1X45Z89

FILEWC

G1X99Z76

FILEWC

FILECE

After the program finished, system will generated a file AABBBCC under program directory. The contents is as following:

G54G0X0Z0

X45Z89

X99Z76

(45)User-defined macro instruction(G101-G170,M880-M889)

Every user-defined G code is corresponding to a macro program ProgramGxxx, the M code is corresponding to a macro program of ProgramUser0 --ProgramUser9, the user cannot programme the macro program in NC system, must edit the macro code in the computer, and then copy into the system.

For example, defines the G152 function: the arc model porous drilling cycle. (must copy the macro program ProgramG152 into system).

Format:G152 Xx Yy Zz Rr Ii Aa Bb Hh Ff;

X: The X coordinate with absolute value or incremental value of center to specify.

Y: The Y coordinate with absolute value or incremental value of center to specify.

Z: Hole depth

R: Approaching fast to the point coordinate

F: Cutting feed speed

I: Radius

A: The angle of the first hole

B: Incremental angle specify(CW when negative)

Macro program ProgramG152 as follows:

#80=#0

#81=#1

#82=#2

#83=#3

#84=#4

#85=#5

#86=#6

#87=#7

#88=#8

#89=#9

#90=#10

#91=#11

#92=#12

#93=#13

```

#94=#14
#95=#15
#96=#16
#97=#17
#98=#18
#99=#19
#100=#20
#30=#4003
#31=#4014
G90
IF[#30 EQ 90] GOTO 1
G53
#98=#5001+#98
#99=#5002+#99
N1 WHILE[#86 GT 0] DO 1
#35=#98+#87*COS[#80]
#36=#99+#87*SIN[#80]
G81X#35Y#36Z#100R#92F#85
#80=#80+#81
#86=#86-1
END 1
G#30 G#31 G80
M99

```

(46) Rough milling of groove in circle G110/G111:

Format: G99 G110/G111 X_Y_R_Z_I_W_Q_K_V_E_D_F_

Application: only the absolute type closed-loop controller support the function of G110-G142;

Function: starting from the center of circle, execute arc interpolation repeatedly in the way of spiral, till process the circle groove.

Definition of instruction: please refer to fixed cycle instruction in chart3-2.

G110: clockwise rough milling of circle groove;

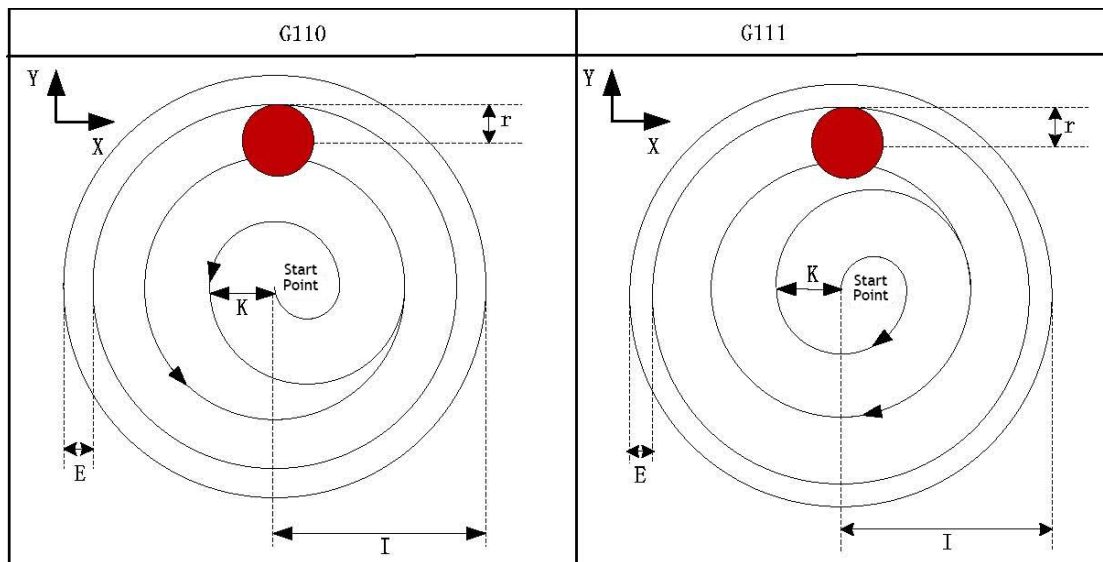
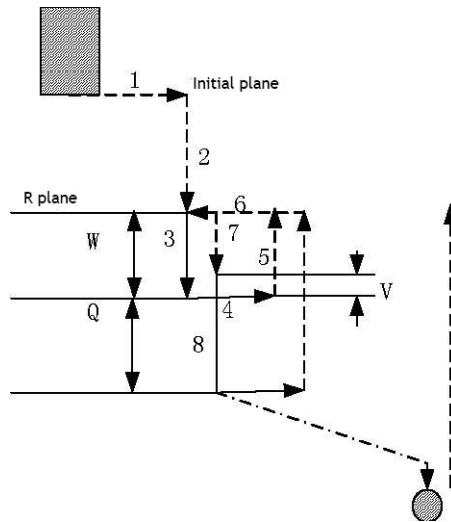
G111: anticlockwise rough milling of circle groove.

1. I: radius of circle groove, I should be more radius of current tool;
2. W: the first cutting depth in the direction Z axis, the distance down from R reference plane, it should be more 0.(if the first cutting depth is more than groove bottom, processing can start from the bottom)
3. Q: increment of each cutting depth in the direction of Z axis.
4. K: increment of width cutting on the XY plane, it should be less than tool diameter, and more than 0;
5. V: the distance from the end processing plane when declining tool, it should be more than 0;
6. E: processing margin of rough milling of circle groove.(more than or equal to 0, negative value is absolute)

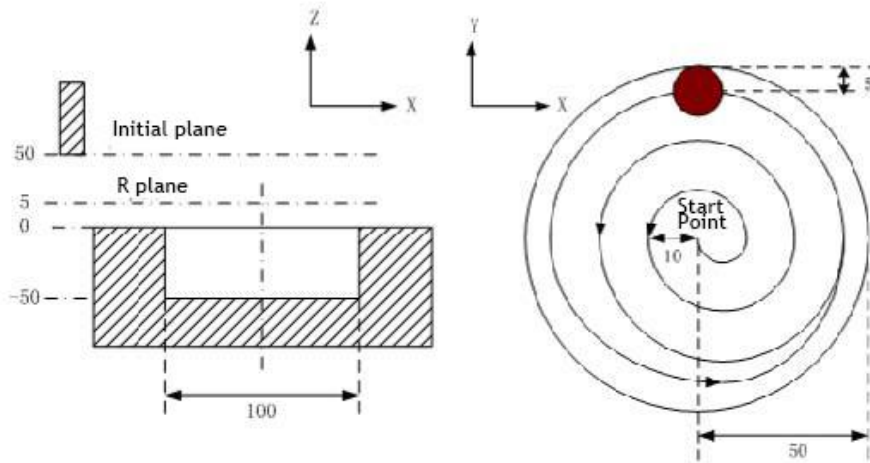
7.D: number of tool radius, range is 0-32, D0 default as 0, calling the current tool radius according to number.

Cycle process:

- (1) Rapidly locating at the XY plane;
- (2) Rapidly dropping to R point plane;
- (3) Cutting by depth W by the cutting speed;
- (4) Milling the disc of radius I and spread incrementally by K value from the center.
- (5) Z axis returns to R reference plane rapidly;
- (6) X and Y axis locate at the center;
- (7) Z axis drops rapidly by the distance V to end processing plane;
- (8) Z axis cuts by the depth Q+V;
- (9) Loop step 4-step 8 till processing of disc is finished.
- (10) Return to R point.
- (11) Return to X Y hole orientation position. Instruction tracking as following.



Example: rough milling of circle groove by fixed cycle G110, as following:



G90 G00 X50 Y50 Z50; (G00 rapid locating)

G99 G110 X25 Y25 R5 Z-50 I50 W20 Q10 K10 E0 V10 F800 D1 ; (rough milling of circle groove D1=5)

G80 X50 Y50 Z50; (cancel fixed cycle, return from R plane)

M30;

(47) Finishing milling cycle of inner circle G112/G113

Format: G99 G112/G113 X_ Y_ R_ Z_ I_ J_ D_ F_

Function: The tool mill inside a whole circle with the direction and radius I which is specified, return after the accurate milling.

Note: The relative instruction according to the table 3-2.

G112: finishing milling cycle CCW.

G113: finishing milling cycle CW.

I: Circle radius of accurate milling, the range -99999999~99999999 times the minimum unit, take absolute value if negative.

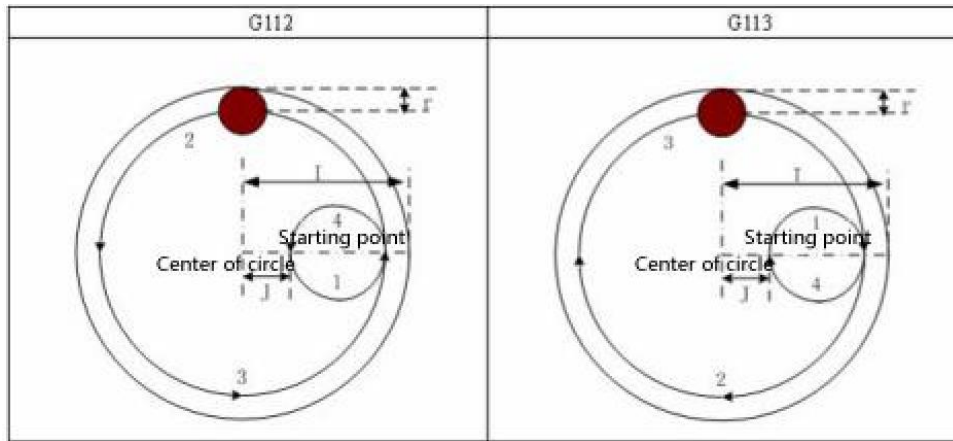
J: The distance between the starting point of accurate milling and the center of accurate milling circle, the range $J \leq I$ - the radius value of tool, take absolute value if negative.

D: The number of tool, the range 1~32, D0 default is 0. Take out the radius of the current tool according to the number.

Cycle process:

- (1) Orient to the position of XY plane
- (2) Move down to the R point plane
- (3) Cutting feed to the hole bottom
- (4) Process the arc interpolation according to the track of transition arc 1
- (5) Whole circle interpolation according to the arc 2, arc 3
- (6) Arc interpolation to the starting point according to the track of transition arc 4
- (7) Fixed to return R point

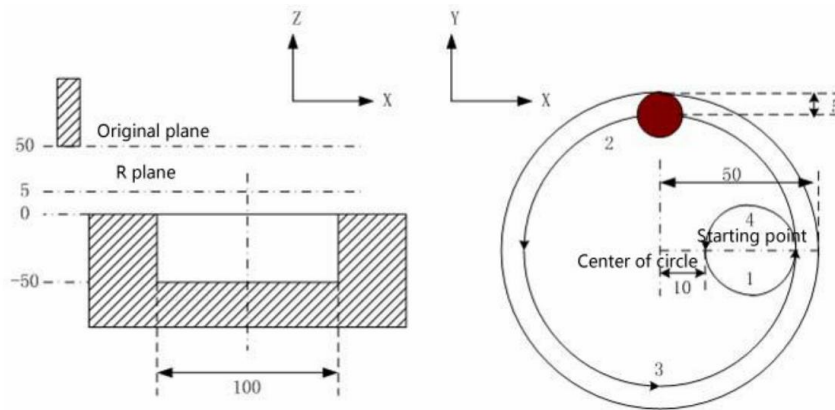
The track:



Note:

Q P L invalid under this cycle, but it will keep the value of Q P as the fixed mode cycle value to save.

Example: Use G112 to accurate milling the circle groove which has rough milled.



G90 G00 X50 Y50 Z50; (G00 orient with rapid move)

G99 G112 X25 Y25 R5 Z-50 I50 J10 F800 D1; (Start fixed cycle, move to the hole bottom and process accurate milling cycle in circle D1=5)

G80 X50 Y50 Z50; (Cancel the fixed cycle, return from R point plane)

M30;

(48) Finishing milling cycle of outer circle G114/G115

Format: G99 G114/G115 X_ Y_ R_ Z_ I_ J_ D_ F_

Function: The tool accurate milling a whole circle at excircle with the specified radius and direction, return after finishing.

Note: The relative instruction refers to the table 3-2.

G114: finishing milling cycle excircle CCW.

G115: finishing milling cycle excircle CW.

I: Circle radius of accurate milling, the range -99999999~99999999 times the minimum unit, take absolute value if negative.

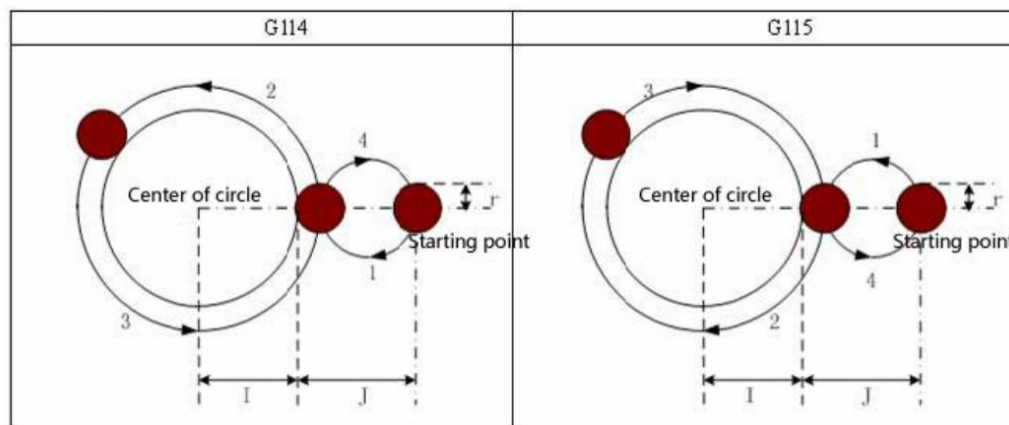
J: The distance between the starting point of accurate milling and the center of accurate milling circle, the range $J \geq D$ (radius of tool), take absolute value if negative.

D: The number of tool, the range 1~32, D0 default is 0. Take out the radius of the current tool according to the number.

Cycle process:

- (1) Orient to the position of XY plane
- (2) Move down to the R point plane
- (3) Cutting feed to the hole bottom
- (4) Process the arc interpolation according to the track of transition arc 1
- (5) Whole circle interpolation according to the arc 2, arc 3
- (6) Arc interpolation to the starting point according to the track of transition arc 4
- (7) Fixed to return R point

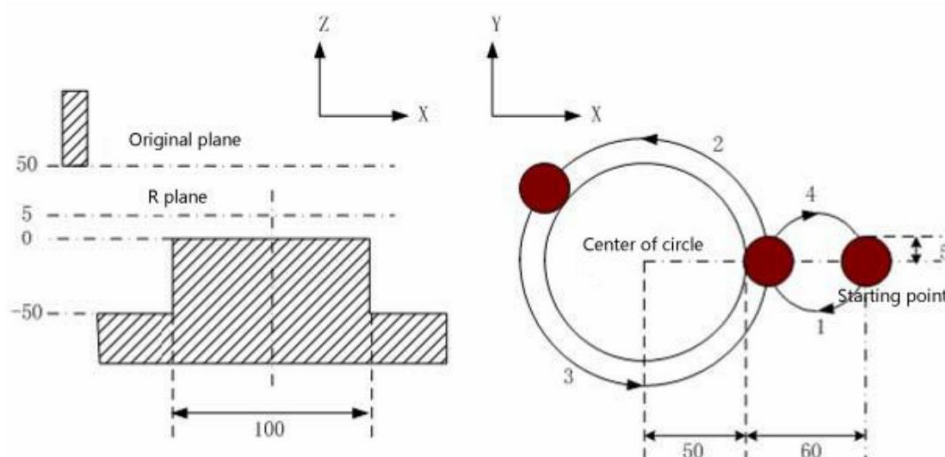
The track:



Note:

- (1) The direction of transition arc and finishing milling arc is different when finishing milling excircle, the interpolation direction means the interpolation direction of accurate milling arc.
- (2) Q P L invalid under this cycle, but it will keep the value of Q P as the fixed mode cycle value to save.

Example:



G90 G00 X50 Y50 Z50; (G00 orient)

G99 G114 X25 Y25 R5 Z-50 I50 J60 F800 D1; (Starts fixed cycle, move down to

the hole bottom and process accurate milling cycle of excircle D1=5)
 G80 X50 Y50 Z50; (Cancel the fixed cycle and return from R point plane)
 M30;

(49) Rough milling cycle of excircle G116/G117

Format: G99 G116/G117 X_ Y_ Z_ R_ I_ J_ W_ Q_ K_ C_ E_ D_ F_

Function: Start from starting point, the tool process whole circle interpolation until process the size which is programmed.

Note: The relative instruction refers to the table 3-2.

G116: Rough milling cycle of excircle CCW

G117: Rough milling cycle of excircle CW

I: The radius of rough milling circle (should be greater than 0, take absolute value if negative)

J: The radius of workpiece (should be greater than 0, take absolute value if negative);

E: The allowance of rough milling excircle in XY plane (should be greater than 0 or equal to 0, take absolute value if negative);

W: The first time cutting depth with the direction of Z axis, the distance from the R plane move down, should be greater than 0, take absolute value if negative(if the first time cutting is deeper than the groove bottom, so process from the groove bottom)

Q: Every incremental depth with direction of Z axis, take absolute value if negative

K: The cutting incremental width in XY plane(should be smaller than the tool radius, greater than 0, take absolute value, if not to specify K, the default is the tool radius D)

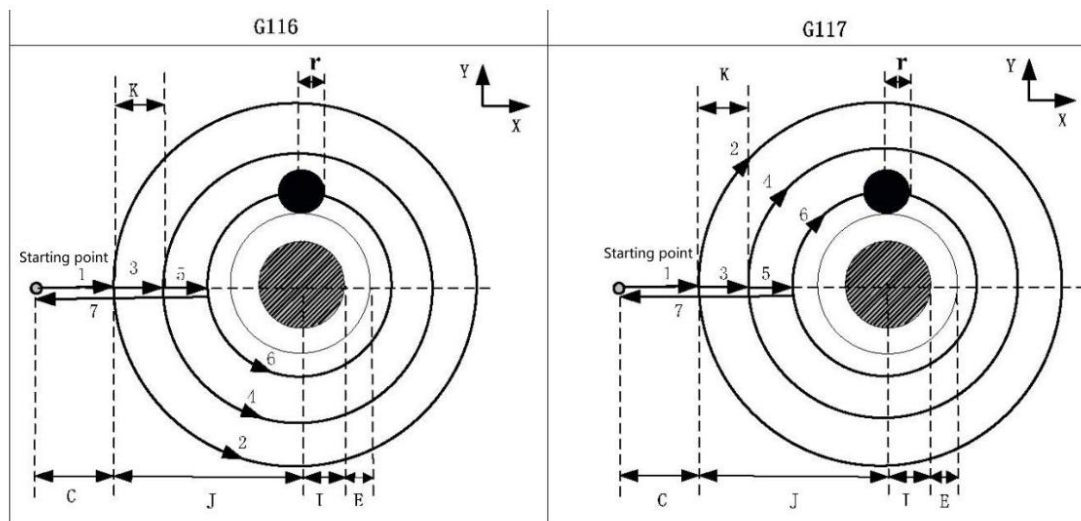
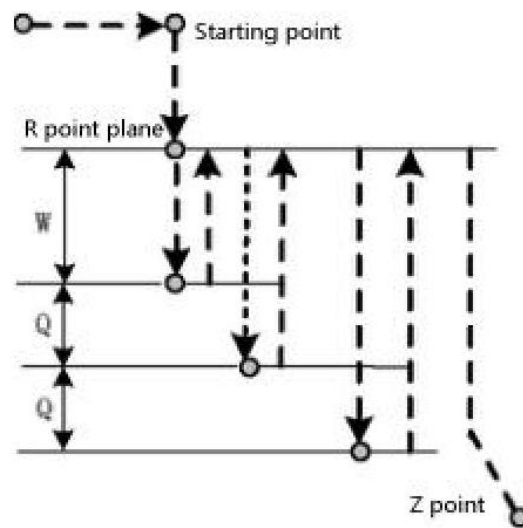
C: The first time feed cutting (C should be greater or equal to the tool radius +2.0, X axis move positive direction when greater than 0, the workpiece is located at the positive direction. When smaller than 0, X axis move negative direction, the workpiece is located at the negative direction)

D: The number of tool, the range 0~32, D0 the default is 0. Take out the current tool radius according to the number which is specified

Cycle process:

- (1) Orient to the position of XY plane
- (2) Move down to the R point plane
- (3) Move down to W depth
- (4) Process straight interpolation to feed tool with the first time feeding value C of X axis and track of straight 1.
- (5) Process whole circle interpolation with track of arc 2
- (6) Every time according to the K value to increase to mill the (I+E) radius circle from outside to the center
- (7) Orient Z axis back to R plane
- (8) Orient to starting position(XY plane)
- (9) Z axis rapid move to the position (unmachined plane + Q)
- (10) Cycle 5~9 motion until finish all the cutting depth Z
- (11) Fixed return to R point

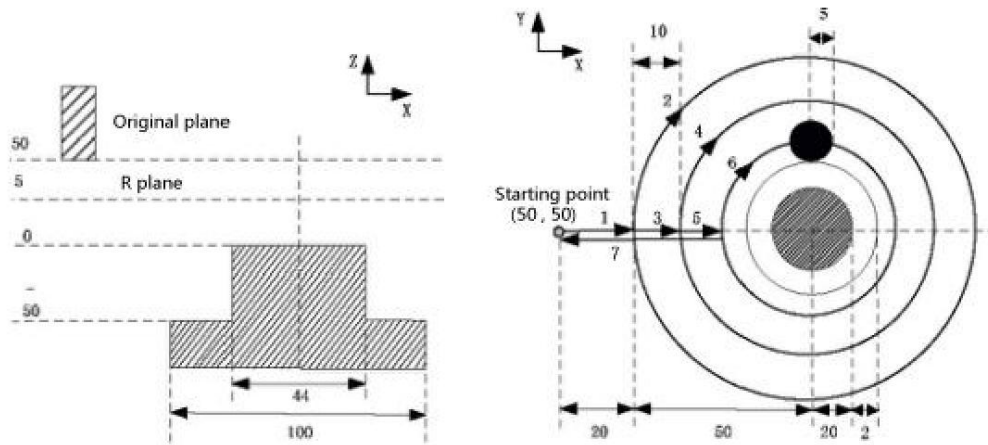
(12) Return to starting position of XY



Note: P and L are invalid under the cycle, but it will keep the value of P as the fixed mode

cycle value to save.

Example: Use G117 to rough mill



G90 G00 X0 Y0 Z50; (G00 orient)

G99 G117 X50 Y50 R5 Z-50 I20 J50 W20 Q10 K10 C20 E2 F800 D1; (process rough milling excircle cycle D1=5)

G80 X50 Y50 Z50; (cancel the fixed cycle, return from R point plane)

M30;

(50) Rough milling cycle of outer rectangle G132/G133

Format: G99 G132/G133 X_ Y_ Z_ R_ I_ J_ A_ B_ W_ Q_ K_ C_ E_ D_ F_

Function: Start from starting point to cycle the straight cutting according to the parameter until process the size which is programmed.

Note: The relative instruction refers to the table 3-2.

G132: Rough milling outer rectangle CCW

G133: Rough milling outer rectangle CW

XY: The coordinate position of processing point

I: The width of rough milling outside rectangle in direction of X axis (should be greater than 0. Take absolute value if negative)

J: The width of rough milling outside rectangle in direction of Y axis (should be greater than 0. Take absolute value if negative)

A: The width of blank in direction of X axis (should be greater than 0. Take absolute value if negative)

B: The width of blank in direction of Y axis (should be greater than 0. Take absolute value if negative)

E: The allowance of rough milling outside rectangle in XY plane (should be greater than or equal to 0. Take absolute value if negative)

W: The first time depth in direction of Z axis which is from R plane to downward position, should be greater than 0, take absolute value if negative (if the first time cutting is deeper than the groove bottom, so process from the groove bottom)

Q: Every time the incremental cutting depth in direction of Z axis, take absolute value if negative

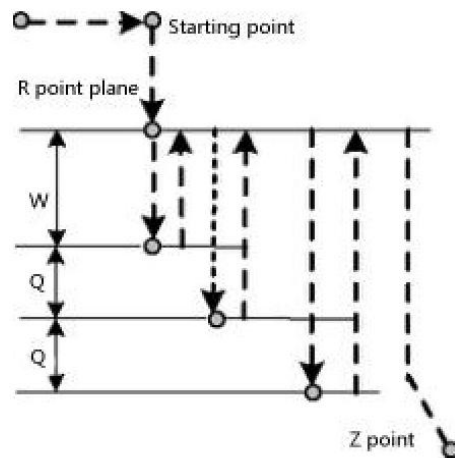
K: The incremental cutting width in XY plane (should smaller than the tool radius, greater than 0, take absolute value if negative, it not specify K, the default is K=D tool radius)

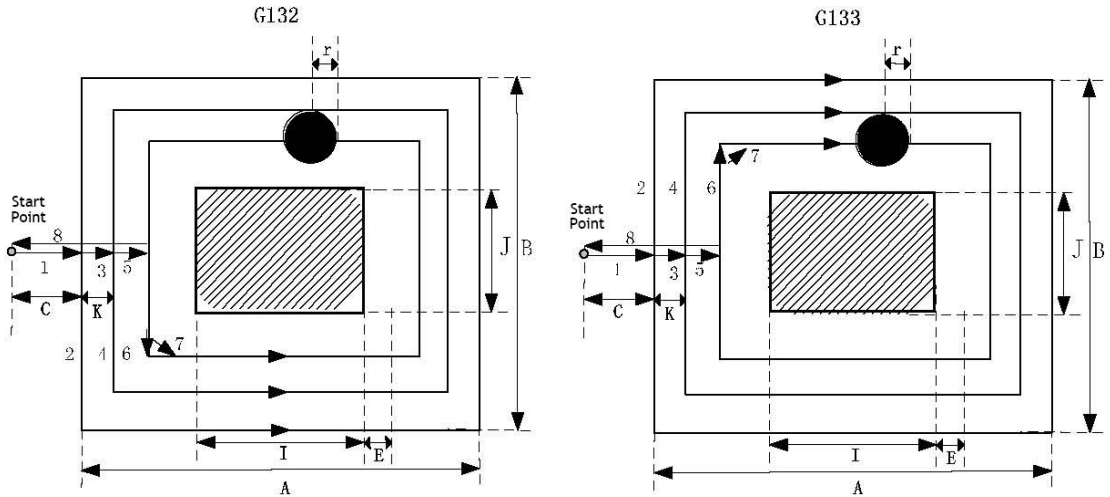
C: The first time feed cutting in direction of X (C should be greater or equal to the tool radius +2.0, X axis move positive direction when greater than 0, the workpiece is located at the positive direction. When smaller than 0, X axis move negative direction, the workpiece is located at the negative direction)

D: The number of tool, the range 0~32, D0 the default is 0. Take out the current tool radius according to the number which is specified

Cycle process:

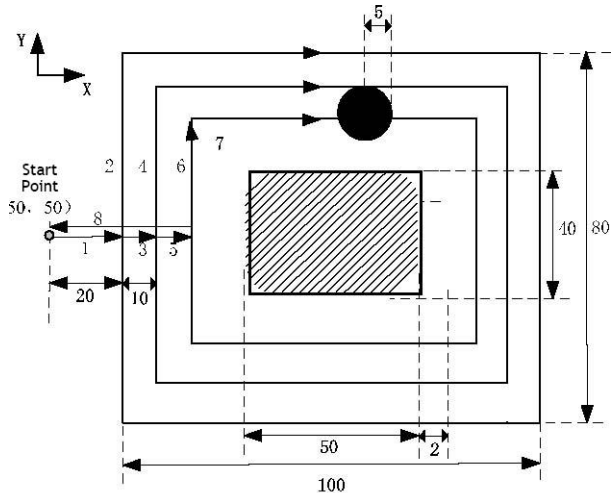
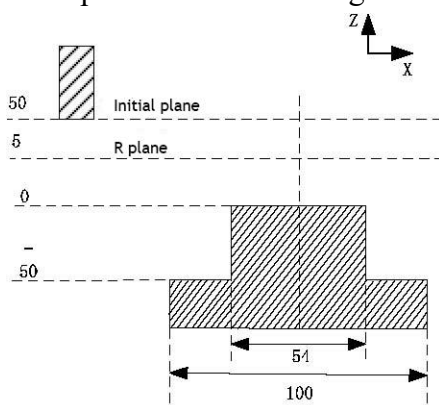
- (1) Orient to the position of XY plane
- (2) Move down to the R point plane
- (3) Move down to W depth
- (4) Process straight interpolation to feed tool with the first time feeding value C of X axis and track of straight 1.
- (5) Process straight interpolation with track 2.
- (6) Every time according to the K value to increase to mill the rectangle with (I+2E) long , (J+2E) width.
- (7) Arc angle of milling rectangle
- (8) Z axis returns to R point plane
- (9) X and Y axis orient to the starting point plane
- (10) Z axis rapid move to the position (end processing plane + Q)
- (11) Cycle 4~9 motion until finish all the cutting depth Z
- (12) Fixed return to R point
- (13) Return to starting position of XY





Note: P L invalid under the cycle, but it will keep the value of P as the fixed mode cycle value to save.

Example: Use G133 to rough mill



G90 G00 X0 Y0 Z50; (G00 to orient)

G99 G133 X50 Y50 R5 Z-50 I50 J40 A100 B80 W20 Q10 K10 C20 E2 U5 F800

D1; (Process the rough milling groove cycle D1=5 in rectangle)

G80 X50 Y50 Z50;

M30;

(51) Rough milling cycle of rectangle groove G134/G135

Format: G99 G134/G135 X_ Y_ Z_ R_ I_ J_ K_ W_ Q_ E_ V_ D_ F_

Function: Start from the center of rectangle to straight cutting cycle with the specified parameter data until process the rectangle groove which is programmed.

Note: The relative instruction refers to the table 3-2.

G134: Rough milling the rectangle groove CCW.

G135: rough milling the rectangle groove CW.

I: The width of rectangle groove in direction of X axis.

J: The width of rectangle groove in direction of Y axis.

K: The cutting incremental width in X Y plane, should be less than the diameter of tool, greater than 0.

W: The first time depth in direction of Z axis which is from R plane to downward position, should be greater than 0, take absolute value if negative (if the first time cutting is deeper than the groove bottom, so process from the groove bottom)

Q: Every time the incremental cutting depth in direction of Z axis, take absolute value if negative

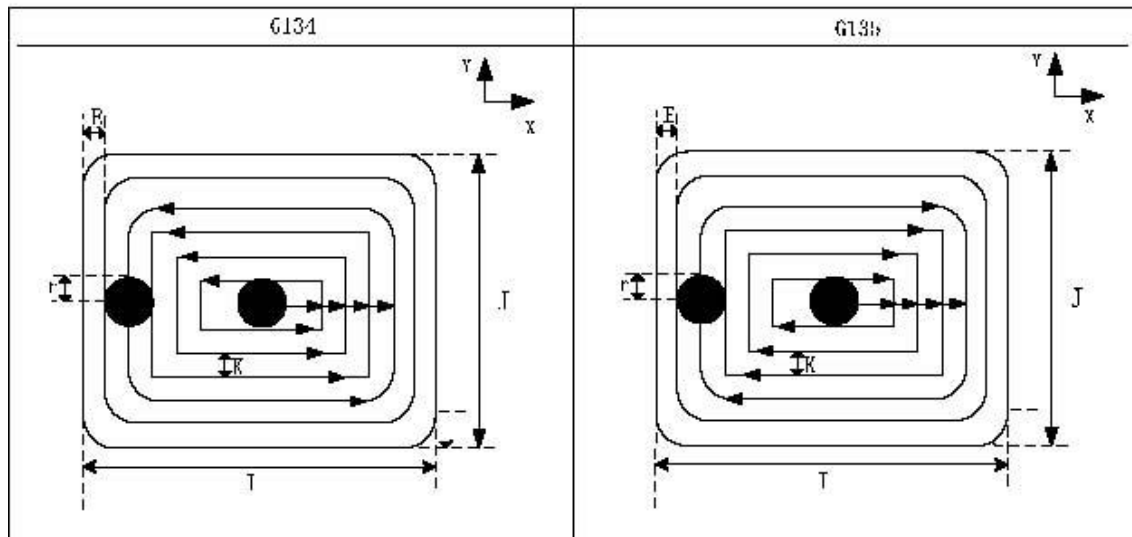
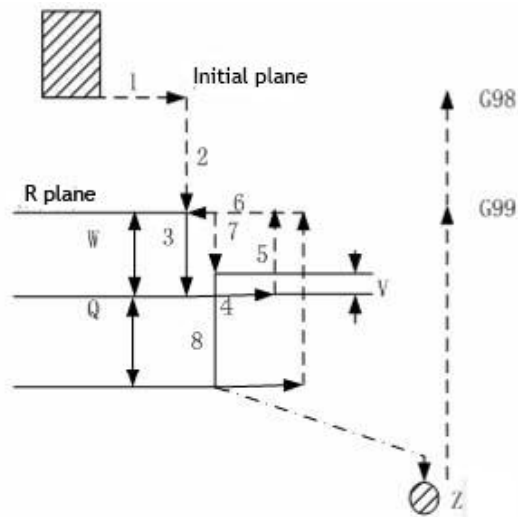
V: The distance from the tool to the processing plane when tool starts moving, should be greater than 0

E: The allowance of rough milling rectangle groove in XY plane (should be greater than or equal to 0. Take absolute value if negative, take the parameter data when not to be specified)

D: The number of tool, the range 0~32, D0 the default is 0. Take out the current tool radius according to the number which is specified

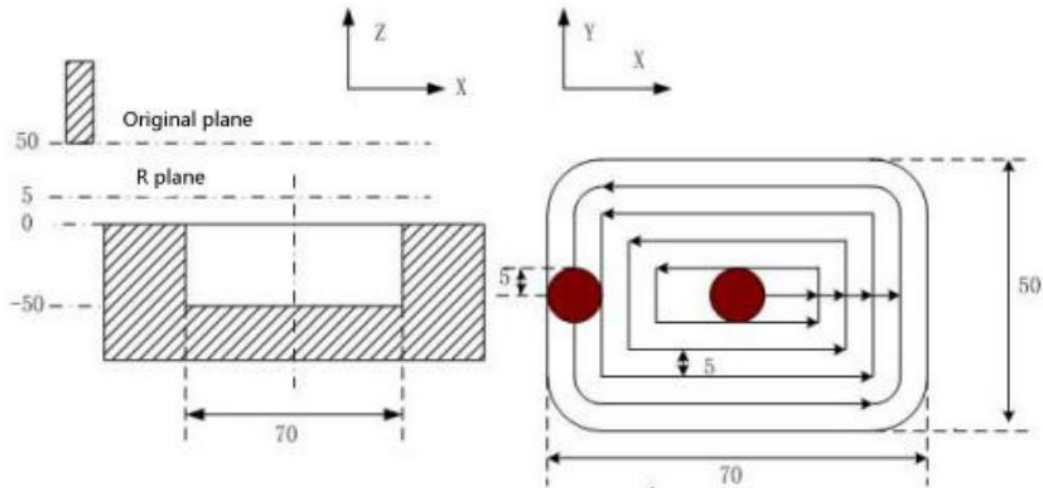
Cycle process:

- (1) Orient to the position of XY plane
- (2) Move down to the R point plane
- (3) Move down to W depth
- (4) milling rectangle plane incrementally by K value from the center.
- (5) Z axis returns to R reference plane.
- (6) X and Y axis orient to the center of rectangle;
- (7) Z axis rapid move to the position (end processing plane + V)
- (8) Z axis cuts down by the distance(Q+V)
- (9)Cycle 4~8 motion until finish all the cutting depth;
- (10) Fixed return to R point
- (11) Return to starting position of XY
- (12) instruction tracking:



Note: P and L is invalid under the cycle, but it will keep the value of P as the fixed mode cycle value to save.

Example: Use the fixed cycle G134 to rough milling groove in rectangle



G90 G00 X50 Y50 Z50; (G00 to orient)
 G99 G134 X25 Y25 R5 Z-50 I70 J50 W20 Q10 K5 E0 V10 U10 F800 D1;
 (Process the rough milling groove rectangle cycle D1=5);
 G80 X50 Y50 Z5; (Cancel the fixed cycle, return from R point plane)
 M30;

(52) Finishing milling cycle of inner rectangle groove G136/G137

Format: G99 G136/G137 X_ Y_ R_ Z_ I_ J_ D_ K_ U_ F_ ;

Function: The finishing milling in rectangle with specified width and direction, return after finishing.

Note: The relative instruction refers to the table 3-2.

G136: Finishing milling cycle in rectangle groove CCW.

G137: Finishing milling cycle in rectangle groove CW.

I: The width of rectangle in direction of X axis, the range: -99999999~99999999 times the minimum unit. Take absolute value when negative.

J: The width of rectangle in direction of Y axis, the range: -99999999~99999999 times the minimum unit. Take absolute value when negative. The value must greater than or equal to the tool radius D, otherwise it will alarm.

D: The number of tool, the range 0~32, D0 the default is 0. Take out the current tool radius according to the number which is specified

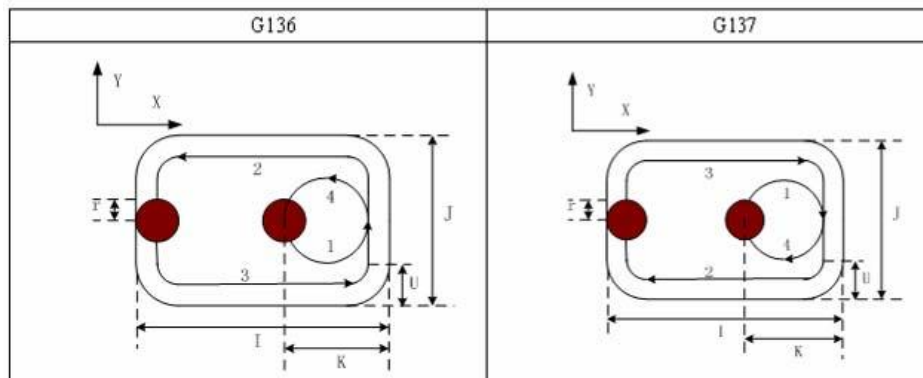
K: The distance from the starting point to the rectangle in direction of X axis, the range -99999999~99999999 times the minimum unit. Take absolute value when negative.

U: Corner radius of arc, omit means no corner arc to transition. When omitting U or U=0 and the tool radius greater than 0, it will alarm.

Cycle process:

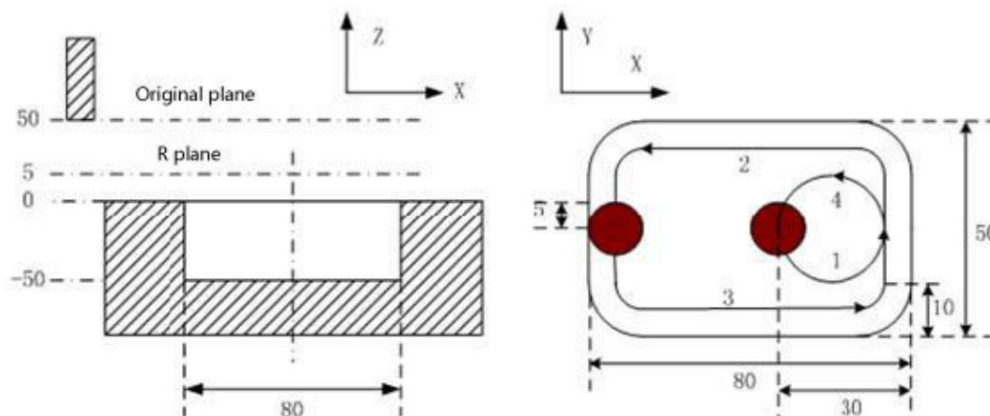
- (1) Orient to the position of X Y plane;
- (2) Move to the R point plane;
- (3) Cutting feed to the hole bottom;
- (4) Start from starting point to arc interpolation with transition arc 1;
- (5) Process straight line and arc interpolation with 2-3;

- (6) Process arc interpolation to back to starting point with transition arc 4;
 (7) Fixed back to R point



Note: P and L is invalid under the cycle, but it will keep the value of P as the fixed mode cycle value to save.

Example: Use the fixed cycle G136 to do finishing milling.



G90 G00 X50 Y50 Z50; (G00 rapid orientation)

G136 X25 Y25 R5 Z-50 I80 J50 K30 U10 F800 D1; (Process accurate milling D1=5 at the bottom of hole under the fixed cycle)

G80 X50 Y50 Z50; (Cancel the fixed cycle and return from R point plane)

M30;

(53) Finishing milling cycle of outer rectangle groove G138/G139/G141/G142

Format: G99 G138/G139/G141/G142 X_ Y_ R_ Z_ I_ J_ D_ K_ U_ F_

Function: The tool finishing milling on the outer the rectangle with specified width and direction, return after finishing.

Note: The relative instruction refers to the table 3-2.

G138: Finishing milling outside the rectangle CCW.

G139: Finishing milling outside the rectangle CW.

G141: Finishing milling of outer lateral rectangle or square (I>J).

G142: Finishing milling of outer vertical rectangle or square (I<J).

I: The rectangle width in direction of X axis, the range -99999999~99999999 times the minimum unit. Take absolute value when negative.

J: The rectangle width in direction of Y axis, the range -99999999~99999999 times the minimum unit. Take absolute value when negative.

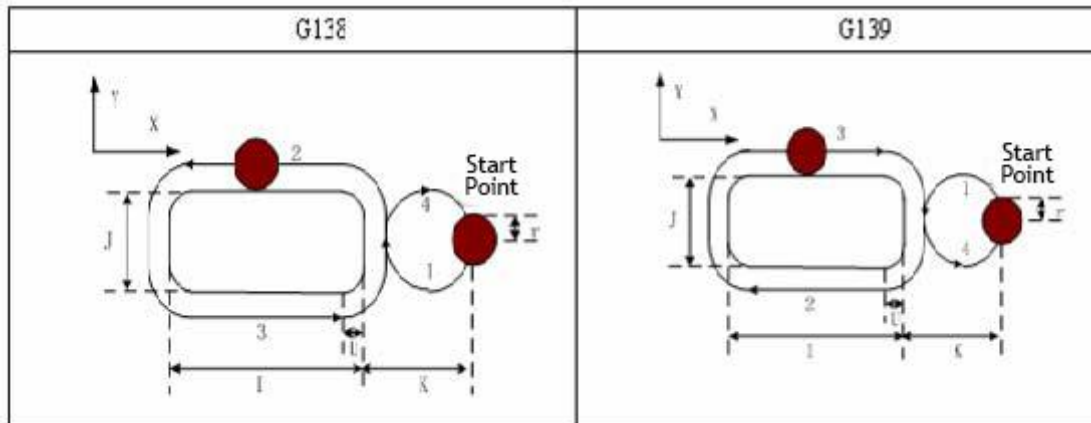
D: The number of tool, the range 0~32, D0 the default is 0. Take out the current tool radius according to the number which is specified

K: The distance from the starting point to the rectangle in direction of X axis, the range -99999999~99999999 times the minimum unit. Take absolute value when negative.

U: Corner radius of arc, omit means no corner arc to transition.

Cycle process:

- (1) Orient to the position of X Y plane;
- (2) Move to the R point plane;
- (3) Cutting feed to the hole bottom;
- (4) Start from starting point to arc interpolation with transition arc 1;
- (5) Process straight line and arc interpolation with 2-3;
- (6) Process arc interpolation to back to starting point with transition arc 4;
- (7) Fixed back to R point.

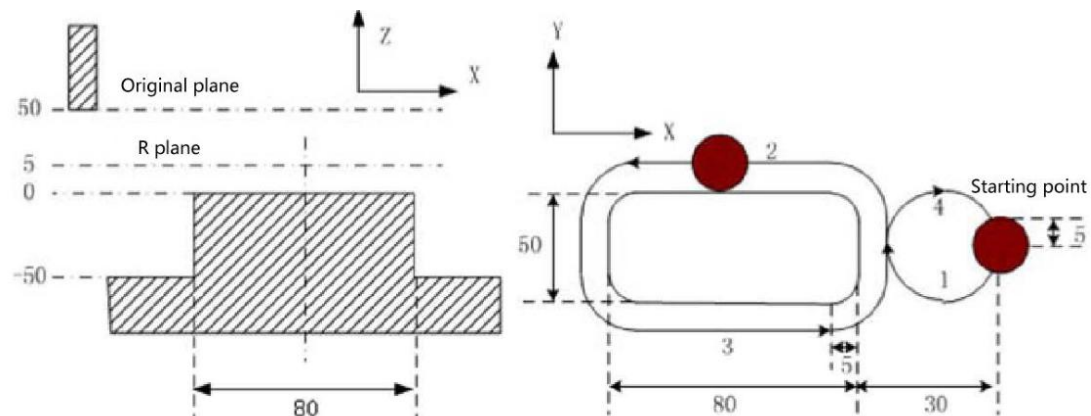


Note:

(1) The interpolation direction of transition arc and finishing milling arc is different, in finishing milling outside the rectangle, the interpolation direction means the direction of finishing milling arc.

(2) Q, P, L are invalid under the cycle, but it will keep the value of Q, P as the fixed mode cycle value.

Example: Use the fixed cycle G138 to do finishing milling.



G90 G00 X50 Y50 Z50; (G00 to orient)
G99 G138 X25 Y25 R5 Z-50 I80 J50 K30 U5 F800 D1; (Process finishing milling
outside the rectangle at the bottom of hole under the fixed cycle)
G80 X50 Y50 Z50; (Cancel the fixed cycle and return from R point plane)
M30;

(54) Programmable data input instruction G10/G710

Programmable input instruction is used to modify work piece coordinate in
program, or offset work piece coordinate.

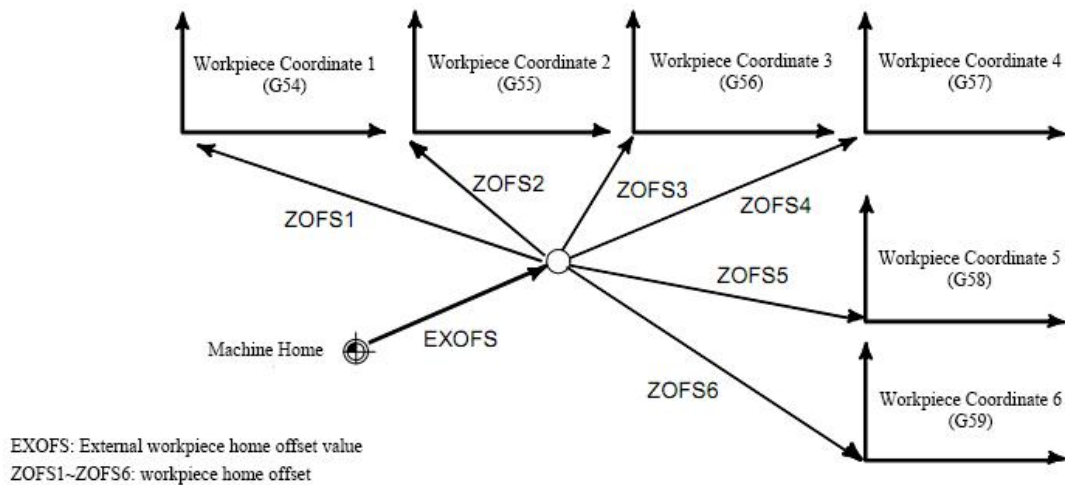
G10 Lxx ;open programmable data input
 Px Rxx ;execute data modification
 Px Ixx ;execute data modification
 X_Y_Z_A_ ;set current coordinate of specified work piece coordinate
G710 ;close programmable data input

Note:

G10 means open programmable data input, modal instruction;
G710 means close programmable data input, modal instruction;
L address specifies data type, 54--59 matches G54-G59 offset value,
5401--5448 matches G54.1-G54.48 offset value, modal instruction;
P address specifies axis which will be modified, 1-8 corresponds to
XYZABCXsYs, modal instruction;
R address specifies modify value, absolute value;
I address specifies modify value, incremental value;

Sample:

G10 L55 ;open programmable data input, set G55 coordinate offset value
P2 R6 ;Y axis of G55 coordinate offset set as 6mm
L56 X1Y2Z3;set G56 coordinate as X1Y2Z3
P1 I5 ;X axis of G56 coordinate add 1.5mm incrementally
G710 ;open programmable data input

(55) Programmable data input instruction G10

<1> when call L2, it means set external work piece coordinate or G54-G59 coordinate home offset value:

G10 L2 P_X_Y_Z_A_B_C_.....

P=0 means set external work piece coordinate home offset.

Method 1: P0 set external work piece coordinate home offset;

Method 2: when main interface shows G53 status, press F7 to set tool, and set external work piece coordinate offset.

Method 3: In coordinate interface, set P1-0=53, then P1-1/P1-8 can be used to set each axis offset value in external work piece coordinate.

When P=1--6, it corresponds to G54-G59 home offset value.

If current coordinate status is G90, then coordinate offset is absolute value. If current coordinate status is G91, then coordinate offset is incremental value.

When call L2, G10 is non-modal instruction, G10 L2 must be added in each line.

Sample:

G10 L2 P1 X0Y0Z0 G90 :G54 coordinate coincide external coordinate.

G10 L2 P3 X11Y22Z33 ;G56 and external coordinate offset X11mm, Y22mm, Z33mm.

<2> when call L20, it means set external work piece coordinate or G54.1-G54.48 coordinate home offset value:

G10 L20 P_X_Y_Z_A_B_C_.....

When P=1--48, it corresponds to G54.1-G54.48 home offset value.

If current coordinate status is G90, then coordinate offset is absolute value. If current coordinate status is G91, then coordinate offset is incremental value.

When call L20, G10 is non-modal instruction, G10 L20 must be added in each line.

Sample:

G10 L20 P1 X0Y0Z0 G90 :G54.1 coordinate coincide external coordinate.

G10 L20 P3 X11Y22Z33 ;G54.3 and external coordinate offset X11mm, Y22mm, Z33mm.

Note: when call L2 or L20, instruction is compatible with Fanuc system.

<3> External coordinate system setting and modification

1. manual mode, choose coordinate G53, then set coordinate.
2. in coordinate parameter, choose G53, then modify offset.
3. in tool compensation mode, modify external coordinate.

(56) Ellipse instruction(G602/G603)

Format: G17 G602/G603 X(U)_Y(V)_I_J_Q_
 G18 G602/G603 Z(W)_Y(V)_K_I_Q_
 G19 G602/G603 X(U)_Z(W)_J_K_Q_

I: semi-axis distance in X axis direction;

J: semi-axis distance in Y axis direction;

Q: oblique angle(unit:degree).

No matter diameter or radius program, address IJK are radius value, unite:mm.

G602 is clockwise ellipse, G603 is count-clockwise ellipse.

When semi-axis value defined by IJK are positive, it means central angle of ellipse is less than or equal to 180 degrees; when one or two value among IJK is negative, it means central angle of ellipse is bigger than or equal to 180 degrees.

Q defines ellipse rotary oblique angle around ellipse center, CCW direction of right-hand coordinate system is positive, unit: degree, without Q means no rotary oblique angle.

G602/G603 are mode instruction, and IJKQ are modeless.

(57) Parabola instruction(G702)

Format:

G17 G702 X(U)_Y(V)_R_Q_ corresponds to equation: $Y^2 = 2 \cdot R \cdot X$ (X:radius);

G18 G702 Z(W)_X(U)_R_Q_ corresponds to equation: $X^2 = 2 \cdot R \cdot Z$ (X:radius);

G19 G702 Y(V)_Z(W)_R_Q_ corresponds to equation: $Z^2 = 2 \cdot R \cdot Y$.

R is parabola equation coefficient, means double of distance between focus and vertex, allowing positive or negative, unit:mm.

Q is rotary angle of parabola around vertex, CCW direction of right-hand coordinate system is positive, unit: degree, without Q means no rotary oblique angle.

G702 are mode instruction, and RQ are modeless.

Attention: R value allows positive or negative, corresponding to different direction of parabola, positive R means rightward parabola, negative R means leftward parabola.

4.5 Radius compensation C of tool

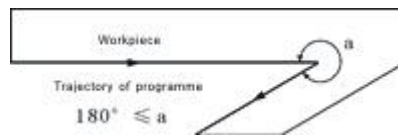
C means the system calculates the tool trajectory of radius compensation according

to the last program line and the next program line.

4.5.1 Inside and outside

It calls inside when the included angle of tool trajectory is over 180 degrees which is built by two program segments, it calls outside when the included angle is between 0 and 180 degrees. As the follows:

Inside:



Outside:



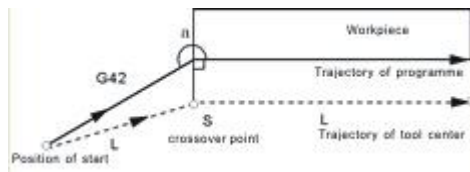
4.5.2 Tool motion when starting

The radius compensation without tool builds tool radius compensation

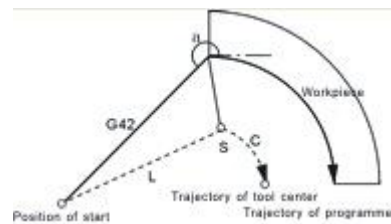
(1) Tool motion around the inside corner ($\alpha \geq 180$)

The tool center will move to the tool vector radius vertex of the starting point in next program line.

Straight line->Straight line



Straight line->Arc

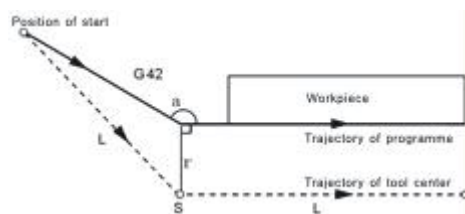


(2) The tool motion around the outside corner of obtuse angle ($90 \leq \alpha < 180$)

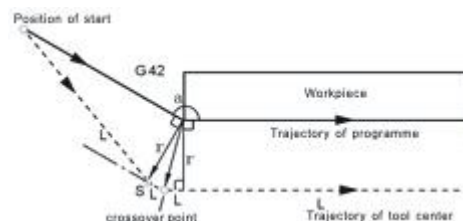
The tool center will move to the tool vector radius vertex of the end point in this program line.

Straight line->Straight line

A type



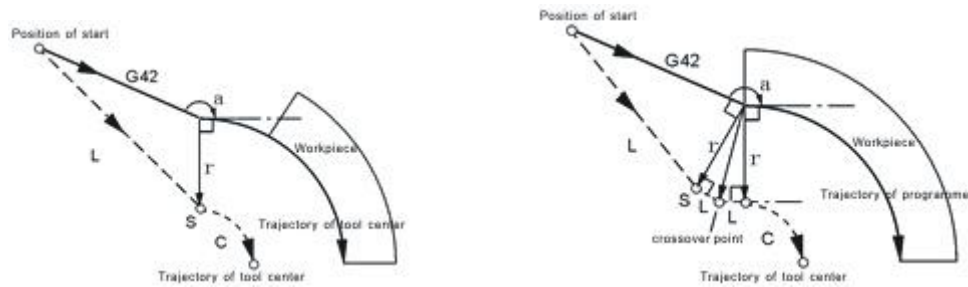
B type



Straight line->Arc

A type

B type

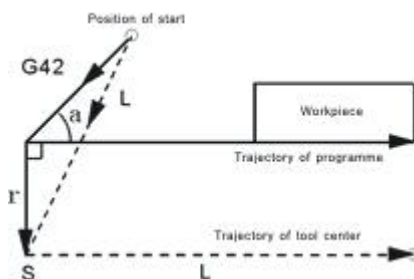


(3) The tool motion around the outside corner of acute angle ($\alpha < 90^\circ$)

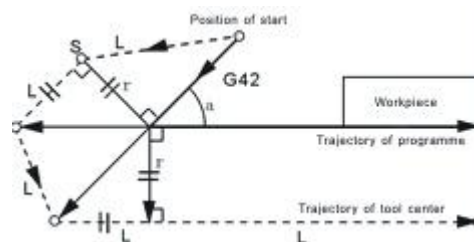
The tool center will move to the tool vector radius vertex of the end point in this program line.

Straight line→Straight line

A type

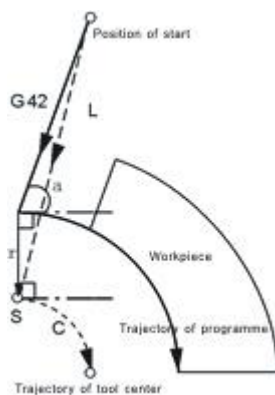


B type

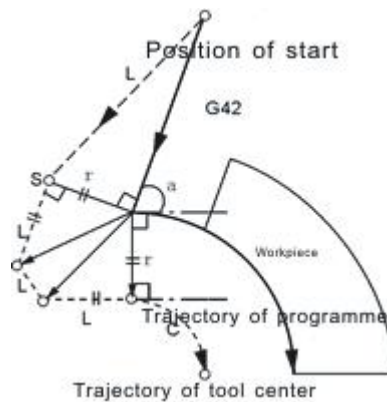


Straight line→Arc

A type



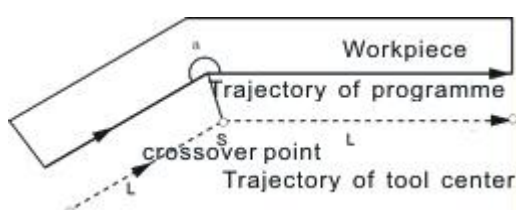
B type



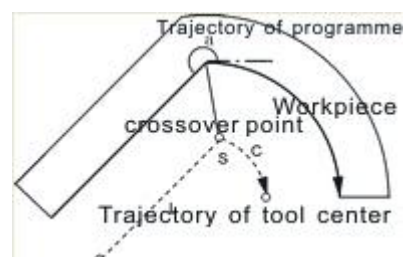
4.5.3 Tool motion in offset mode

(1) Tool motion around the inside corner ($180 \leq \alpha$)

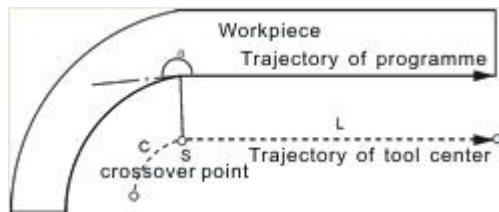
Straight line→Straight line



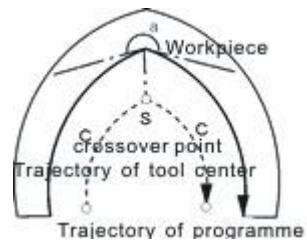
Straight line→Arc



Arc-> Straight line

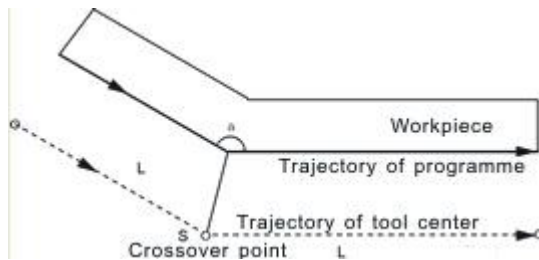


Arc->Arc

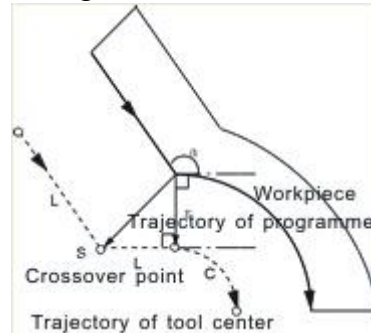


(2)The tool motion around the outside corner of obtuse angle ($90 \leq \alpha < 180$)

Straight line ->Straight line



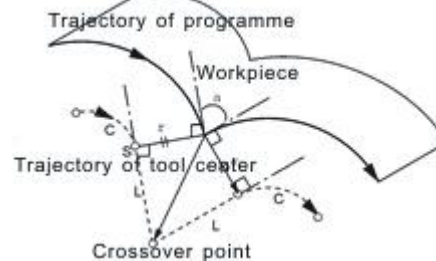
Straight line ->Arc



Arc-> Straight line

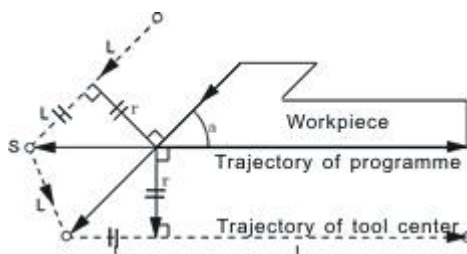


Arc->Arc

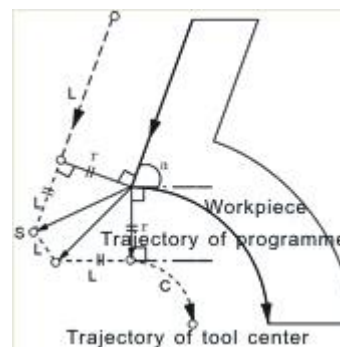


(3)The tool motion around the outside corner of acute angle ($\alpha < 90$)

Straight line->Straight line

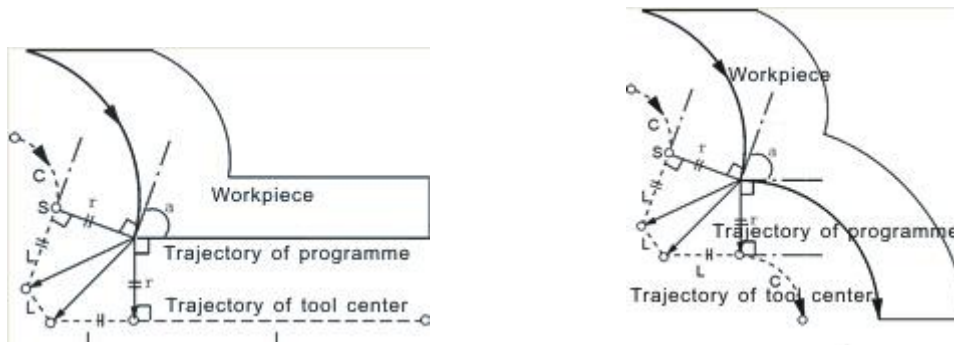


Straight line->Arc



Arc-> Straight line

Arc->Arc

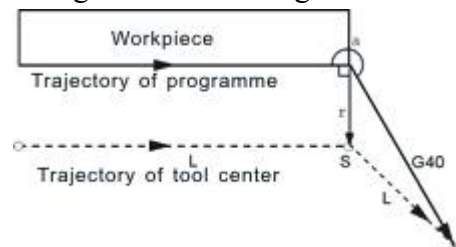


4.5.4 Tool motion in offset-cancel mode

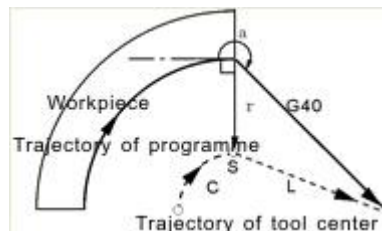
(1) Tool motion around the inside corner ($180 \leq \alpha$)

The tool center will move to the tool vector radius vertex of the end point in this program line.

Straight line $\rightarrow$ Straight line



Arc $\rightarrow$ Straight line

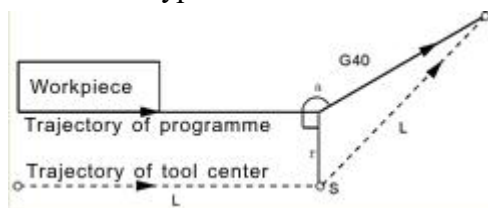


(2) The tool motion around the outside corner of obtuse angle ($90 \leq \alpha < 180$)

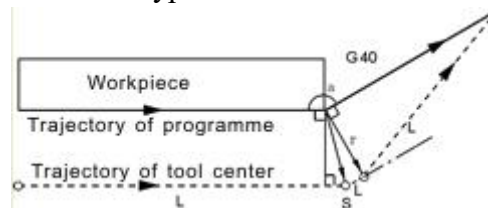
The tool center will move to the tool vector radius vertex of the starting point in next program line.

Straight line $\rightarrow$ Straight line

A type

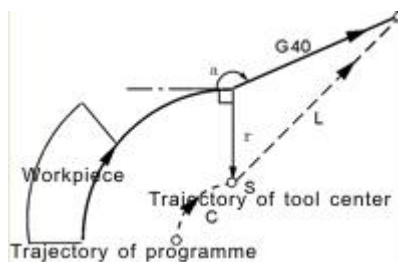


B type

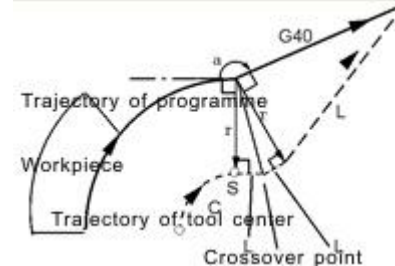


Arc $\rightarrow$ Straight line

A type



B type

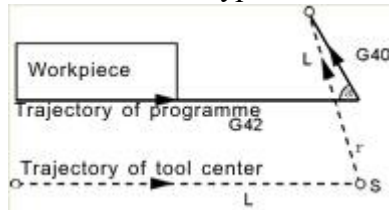


(3) The tool motion around the outside corner of acute angle ($\alpha < 90$)

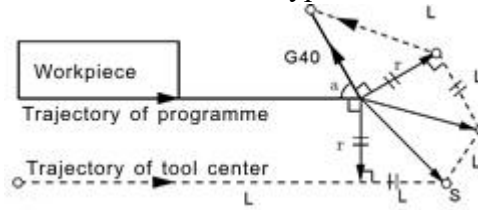
The tool center will move to the tool vector radius vertex of the starting point in next program line.

Straight line—>Straight line

A type



B type

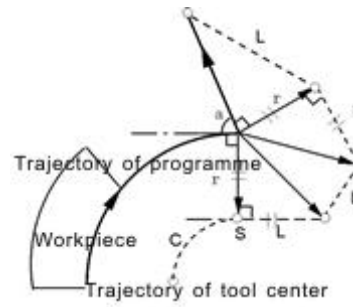


Arc—>Straight line

A type



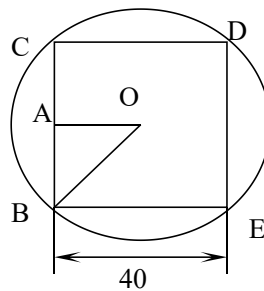
B type



4.6 Comprehensive examples

In the actual programming, must according to the drawings and processing requirements to select the install method and suitable tool correctly, combined with the actual working performance of lathe to select the correct cutting allowance, for example:

Example 1: Cut square and cut circle



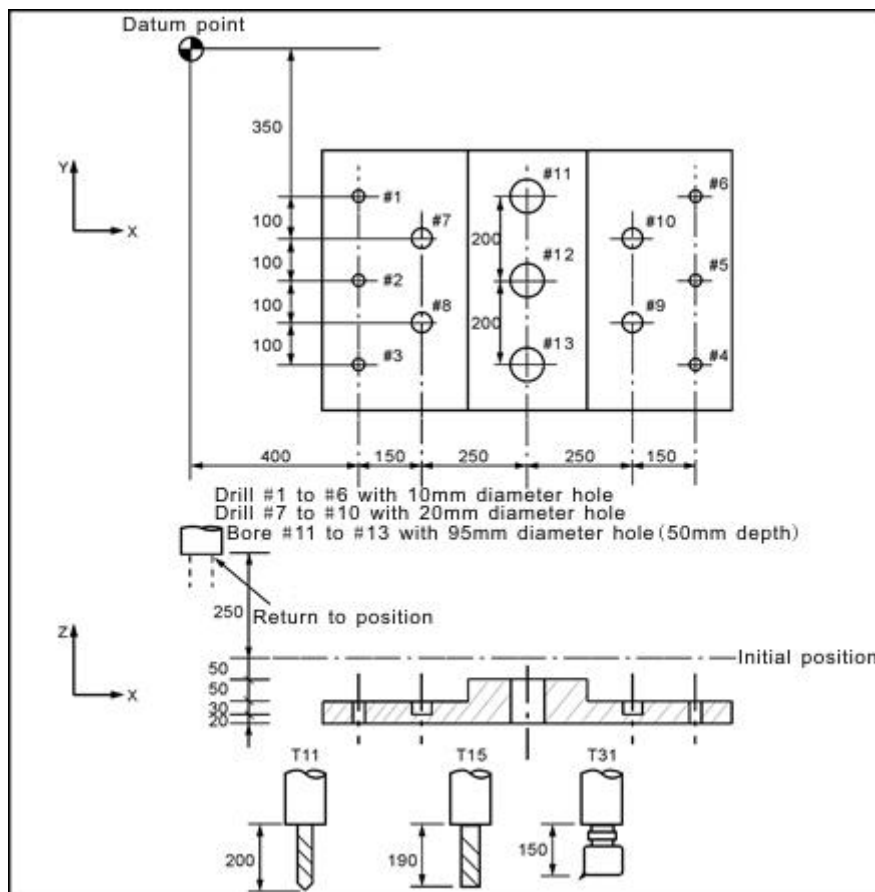
Beginning from the center, the center coordinate is G54 X0 Y0 Z50

The tool radius in the D01 of the T01 parameter

N0 G54 G00 X0 Y0 Z50	Starting point of processing
N1 M03 S1000	Turn on spindle
N2 G00 Y-40 X-40 G90	Move outside of square
N3 Z10	
N4 G01 Z-10 F200	Start cutting from Z axis
N5 G01 G41 T01 D01 X-20 Y-20 F400	Move to B point

N6 Y20	Tool cuts BC line at the left of the workpiece
N7 X20	Closed angle transition in C point, cut CD line
N8 Y-20	Closed angle transition in D point, cut DE line
N9 X-25	Closed angle transition in E point, cut CB line is for smooth, move 5mm more
N10 G00 X-40 Y-40 G40	Move outside of circle
N11 G01 G41 X-20 Y-20 F500	Lengthen T01 value along A-B
N12 Z-20 F100	Start cutting from Z axis
N13 G91 G02 I20 J20 Y0 X0	Circle of contact
N14 G00 Z50	End cutting
N15 G40 G00 X0 Y0	Cancel tool compensation
N16 M05	Turn off spindle
N17 M02	Program ends.

Example 2:



Assume to used 3 tools T11 T15 T31. The tool length compensation values were 200 (H1), 190 (H1), 150 (H1), entered into the tool parameter. According to the processing requirements of parts drawing, the program is as follows:

N1 G54 X0 Y0 Z0	Set workpiece coordinate at the home point
-----------------	--

N2 G90 G00 Z250.0 T11	Tool exchange
N3 G43 Z0 H1	The offset length in initial position
N4 S30 M3	Turn on spindle
N5 G99 G81 X400.0 Y-350.0 Z-153.0 R-97.0 F120	To drill #1 hole
N6 Y-550.0	To drill #2 hole and return to initial position
N7 G98 Y-750.0	To drill #3 hole and return to initial position
N8 G99 X1200.0	To drill #4 hole and return to initial position
N9 Y-550.0	To drill #5 hole and return to initial position
N10 G98 Y-350.0	To drill #6 hole and return to initial position
N11 G00 X0 Y0 M5	Return to the home point and stop spindle
N12 G49 Z250.0 T15	Cancel the tool length offset to change tool
N13 G43 Z0 H1	Tool length offset in initial position
N14 S20 M3	Turn on spindle
N15 G99 G82 X550.0 Y-450.0 Z-130.0 R-97.0 F500	To drill #7 hole and return to initial position
N16 G98 Y-650.0	To drill #8 hole and return to initial position
N17 G99 X1050.0	To drill #9 hole and return to initial position
N18 G98 Y-450.0	To drill #10 hole and return to initial position
N19 G00 X0 Y0 M5	Return to the home point and stop spindle
N20 G49 Z250.0T31	Cancel the tool length offset to change tool
N21 G43 Z0 H1	Tool length offset in initial position
N22 S10 M3	Turn on spindle
N23 G85 G99 X800.0 Y-350.0 Z-153.0 R47.0 F150	To bore #11 hole and return to initial position
N24 G91 Y-200.0	To bore #12 hole and return to initial position

N25 Y-200.0	To bore #13 hole and return to initial position
N26 G90 G28 X0 Y0 M5	Return to the home point and stop spindle
N27 G49 Z0 G80	Cancel the tool length offset
N28 M02	Program ends.

4.7 Automatic tool setting gauge

1. Note for parameter:

Define macro variables of the automatic tool setting gauge function are as follows (corresponding to Other parameters P380 - P389):

#380: The X axis machine coordinate of initial position when automatic tool setting;(Unit:mm)

#381: The Y axis machine coordinate of initial position when automatic tool setting;(unit:mm)

#382: The Z axis machine coordinate of initial position and returning point with automatic tool setting;(unit:mm)

#383: The negative speed of automatic tool setting;(mm/min)

#384: The positive speed of automatic tool setting;(mm/min)

#385: The Z axis coordinate of work piece surface in current work piece coordinate system after automatic tool setting;(mm)

#386: The speed which is rapid move to locating position with automatic tool setting;(mm/min)

#387: Automatic tool setting mode (1 means fixed point, 0 means floating point).

#388: The minimal machine coordinate value of Z axis (mm);

#389: The gap value of Z axis [The height which is the gauge surface relative to the work piece surface(mm)];

Fixed point gauge means putting the gauge in a fixed position, every time the X Y Z axis are automatic running to the fixed point first in tool setting; But the floating point gauge search the tool setting gauge signal along negative of the Z axis.

The input point X25 is default to be the checking point of automatic tool setting gauge to input.

2.The instruction:

M880 (corresponding to ProgramUser0) automatic tool setting instruction;

M882 (corresponding to ProgramUser2),

M883 (corresponding to ProgramUser3) set the gap of Z axis.

3. Automatic tool setting steps:

1)Set the No.380--No.388 parameter in other parameter;

2)Set the No.389 parameter in other parameter to set the drop value of Z axis: this operation needs to be set only once.

A.Run M882 instruction in MDI to set the gap of Z axis;

B.Manual run Z axis to move the tool nose to the work piece surface;

C.Run M883 instruction in MDI to automatic set the gap of Z axis No.389

parameter in other parameter;

- 3) MDI choose the work piece coordinate system G54/G59;
- 4) Automatic tool setting: MDI running the M880 instruction, automatic set the Z axis offset of the current work piece coordinate system.

4.8 Automatic edge finder

1. The X axis edge: M884(Corresponding to ProgramUser4)

- 1) Choose the current coordinate system such as G54;
- 2) Manually moving the X to the negative terminal of workpiece; run M884 instruction in MDI;
- 3) Manually moving the Y to the positive terminal of workpiece; run M884 instruction in MDI, automatically divide the center of Y axis and set the middle point of workpiece as current coordinate system of Y axis origin, that's automatically setting the current coordinate system, such as the coordinate offset value of Y axis in G54.

2. The Y axis edge:M885 (Corresponding to ProgramUser5)

- 1) Choose the coordinate system such as G54;
- 2) Manually moving the Y to the negative terminal of workpiece; Run M885 instruction in MDI;
- 3) Manually moving the Y to the positive terminal of workpiece; Run M885 instruction in MDI, automatically divide the center of Y axis and set the center point of workpiece as the current coordinate system of Y axis origin,that's automatically setting the current coordinate system, such as the offset value of Y axis in G54.

3. Interior X axis edge: M886(Corresponding to ProgramUser6)

- 1) Choose the current coordinate system such as G54;
- 2) Manually moving X axis into interior of workpiece;
- 3) Run M885 instruction in MDI, controller will detect input point X31, then divide X axis coordinate automatically, and set middle point of work piece as X zero point of current coordinate system, finally move to X zero point.

4. Interior Y axis edge: M887(Corresponding to ProgramUser7)

- 1) Choose the current coordinate system such as G54;
- 2) Manually moving X axis into interior of workpiece;
- 3) Run M885 instruction in MDI, controller will detect input point X31, then divide Y axis coordinate automatically, and set middle point of work piece as Y zero point of current coordinate system, finally move to Y zero point.

Attention: when run M884-M887, external coordinate offset value should be 0.

4.9 Instruction of follow-up axis

1.Method A: the coordinate of follow-up axis will not follow, while the fourth axis can be hidden in screen. When axis parameter P414 is 7, the fourth axis is following X axis in both automatic and manual mode. When P414 is 8, the fourth axis is following Y axis in both automatic and manual mode. When P414 is 9, the fourth axis is following Z axis in both automatic and manual mode.

Besides, when P414 is 17/18/19, the C axis is following X/Y/Z axis in both automatic and manual mode. When P414 is 27/28/29, the B axis is following X/Y/Z axis in both automatic and manual mode.

2. Method B: it is applied to specify follow-up axis with instructions in automatic mode, but not follow in manual mode. In this way, is coordinate following when axis is following.

M123: turn on the follow-up Y axis following X axis;

M124: turn off the follow-up Y axis following X axis;

M125: turn on the follow-up A axis following X axis;

M126: turn off the follow-up A axis following X axis.

4.10 4 axis/5 axis RTCP function

(5 axis and above bus type controller only)

4.10.1、 User Commands:

M601 Enables RTCP compensation for one rotary axis in 5-axis mode.

When machine structure parameter equals 10: Only compensates rotary B axis.

When machine structure parameter equals 11: Only compensates rotary A axis.

When machine structure parameter equals 12: Only compensates rotary A axis.

When machine structure parameter equals 13: Only compensates rotary A axis.

When machine structure parameter equals 14: Only compensates rotary A axis.

When machine structure parameter equals 15: Only compensates rotary B axis.

M602/G43.4 Enables RTCP compensation for both rotary axes in 5-axis mode, compensating both rotary axes.

M600/G49 Disables 5-axis RTCP function.

In M601 or M602 mode, manual movement of Z axis follows the tool direction for forward/backward movement, rather than moving along the machine coordinate Z axis alone.

Auxiliary relay M1060: Toggles between M600 and M602 states cyclically. Can be programmed in ladder logic to enable manual quick switching between M600 and M602 states via button.

4.10.2、 Machine Structure Type, Axis Naming and Direction Definition Rules:

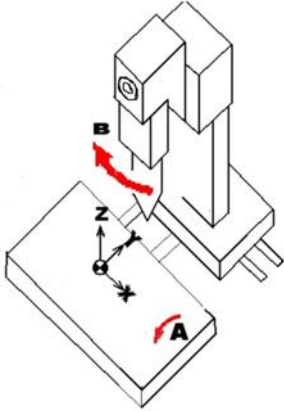
A axis is the rotary axis around linear X axis, with positive rotation direction following right-hand coordinate system rule.

B axis is the rotary axis around linear Y axis, with positive rotation direction following right-hand coordinate system rule.

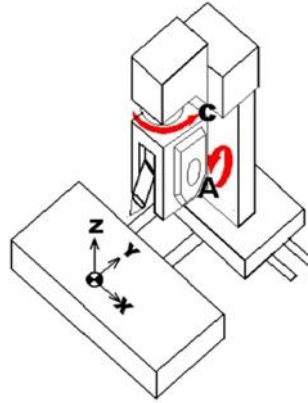
C axis is the rotary axis around linear Z axis, with positive rotation direction following right-hand coordinate system rule.

Notes:

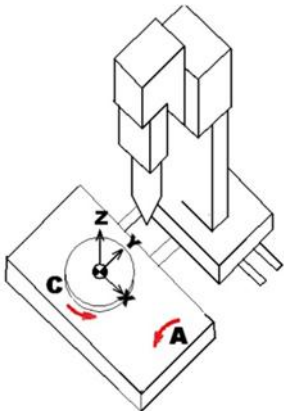
- 1) The movement direction of linear axes is also defined by tool movement relative to workpiece;
- 2) The rotation direction of rotary axes attached to tool is defined by tool rotation relative to workpiece;
- 3) The rotation direction of rotary axes attached to worktable is defined by rotation around the worktable itself.



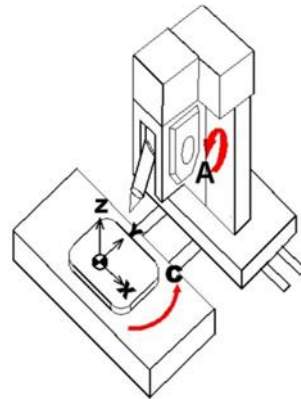
B-axis tool with A-axis table 5-axis machine (Type 10)



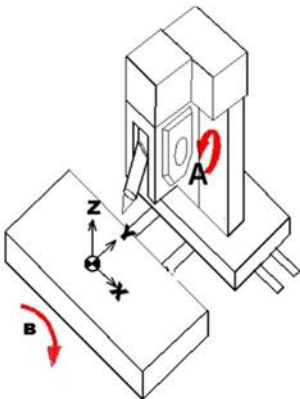
AC-axis tool-rotary 5-axis machine (Type 11)



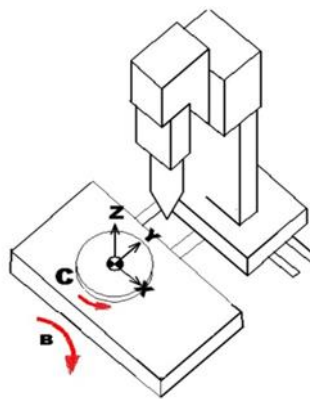
AC-axis tool-table 5-axis machine (Type 12)



A-axis tool with C-axis table 5-axis machine (Type 13)



A-axis tool with B-axis table 5-axis machine (Type 14)



BC-axis tool-table 5-axis machine (Type 15)

4.10.3、 Definition Rules for Rotary Axis Machine Coordinate Zero Positions in

Various 5-Axis Machine Tools:

B-axis tool with A-axis table hybrid 5-axis CNC machine (10):

The position aligned with the positive Y-axis direction is defined as the A-axis machine coordinate zero point.

The position where the tool tip aligns with the negative Z-axis direction is defined as the B-axis machine coordinate zero point.

AC-axis tool rotary 5-axis CNC machine (11):

The position where the tool tip aligns with the negative Z-axis direction is defined as the A-axis machine coordinate zero point.

The position aligned with the negative Y-axis direction is defined as the C-axis machine coordinate zero point.

AC-axis table rotary 5-axis CNC machine (12):

The position aligned with the positive Y-axis direction is defined as the A-axis machine coordinate zero point.

The position aligned with the positive X-axis direction is defined as the C-axis machine coordinate zero point.

A-axis tool with C-axis table hybrid 5-axis CNC machine (13):

The position where the tool tip aligns with the negative Z-axis direction is defined as the A-axis machine coordinate zero point.

The position aligned with the positive X-axis direction is defined as the C-axis machine coordinate zero point.

A-axis tool with B-axis table hybrid 5-axis CNC machine (14):

The position where the tool tip aligns with the negative Z-axis direction is defined as the A-axis machine coordinate zero point.

The position aligned with the positive Z-axis direction is defined as the B-axis machine coordinate zero point.

BC-axis table rotary 5-axis CNC machine (15):

The position aligned with the positive Z-axis direction is defined as the B-axis machine coordinate zero point.

The position aligned with the positive X-axis direction is defined as the C-axis machine coordinate zero point.

4.10.4、Machining Parameters:

551,5-axis CNC machine structure type (10-99)

10 represents B-axis tool with A-axis table hybrid 5-axis CNC machine;

11 represents AC-axis tool rotary 5-axis CNC machine (bus system);

12 represents AC-axis table rotary 5-axis CNC machine (bus system);

13 represents A-axis tool with C-axis table hybrid 5-axis CNC machine (bus system);

14 represents A-axis tool with B-axis table hybrid 5-axis CNC machine (bus system);

15 represents BC-axis table rotary 5-axis CNC machine (bus system);

552,RTCP compensation reverse and coordinate system coincidence (reverse

10/14:B+4,A+8;11/12/13:A+4,C+8;coordinate system coincidence+16:[12A axis,13C axis,14B axis] corresponds to programming and workpiece coordinate system

coincidence;+32:12C axis corresponds to programming and workpiece coordinate system coincidence)

Type 10/14(B+4;A+8):B axis compensation reverse+4;A axis compensation reverse+8;

Type 11/12/13(A+4;C+8):A axis compensation reverse+4;C axis compensation reverse+8;

Type 15(B+4;C+8):B axis compensation reverse+4;C axis compensation reverse+8;
+16 means:Type 12A axis/13C axis/14B/15B axis machine coordinates correspond to programming coordinate system and workpiece coordinate system coincidence,otherwise indicates that when the rotary table axis machine coordinate is 0 it corresponds to programming coordinate system and workpiece coordinate system coincidence;

+32 means:Type 12/15 rotary table C axis machine coordinates correspond to programming coordinate system and workpiece coordinate system coincidence,otherwise indicates that when C axis machine coordinate is 0 it corresponds to programming coordinate system and workpiece coordinate system coincidence;

553,5-axis CNC rotary table center machine coordinate 1(um)

When machine structure parameter equals 10:A axis rotation center Y axis machine coordinate.

When machine structure parameter equals 11:No meaning.

When machine structure parameter equals 12:A axis rotation center Y axis machine coordinate.

When machine structure parameter equals 13:C axis rotation center X axis machine coordinate.

When machine structure parameter equals 14:B axis rotation center X axis machine coordinate.

When machine structure parameter equals 15:B axis rotation center X axis machine coordinate.

554,5-axis CNC rotary table center machine coordinate 2(um)

When machine structure parameter equals 10:A axis rotation center Z axis machine coordinate.

When machine structure parameter equals 11:No meaning.

When machine structure parameter equals 12:A axis rotation center Z axis machine coordinate.

When machine structure parameter equals 13:C axis rotation center Y axis machine coordinate.

When machine structure parameter equals 14:B axis rotation center Z axis machine coordinate.

When machine structure parameter equals 15:B axis rotation center Z axis machine coordinate.

555,Measuring 5-axis CNC rotary table center tool length(um)

When machine structure parameter equals 10:Measuring A axis rotation center tool length.

When machine structure parameter equals 11:No meaning.

When machine structure parameter equals 12:Measuring rotary table center tool compensation value H1.

When machine structure parameter equals 13:Measuring C axis rotation center tool length.

When machine structure parameter equals 14:Measuring B axis rotation center tool length.

When machine structure parameter equals 15:Measuring rotary table center tool compensation value H1.

556,AC/BC axis table rotary 5-axis machine C axis rotation center X/Y axis machine coordinate(um)

This parameter is valid when machine structure parameter equals 12.

557,AC-axis tool rotary type AC axis line offset(um)

This parameter is valid when machine structure parameter equals 11.At AC axis machine coordinate zero position,when A axis center line is in front of C axis center line (Y axis direction),the offset value (Y axis direction) is set negative,otherwise set positive.

558,AC-axis tool rotary type C axis and spindle X direction offset(um)

This parameter is valid when machine structure parameter equals 11.At AC axis machine coordinate zero position,C axis center line and spindle center line offset value (X axis direction) can be set positive or negative.

4.10.5. RTCP Tool Setting Operation

Press 'H' in the system tool compensation interface for automatic tool length measurement:

Note: During tool setting, the C-axis must remain at 0 degrees to ensure the A-axis rotates precisely around the X-axis.

(1) When machine structure parameter equals 10:

Used for automatically measuring the distance from tool tip to B-axis rotation center.

First, select the tool number and position the tool tip at a marked point on the workpiece. Press 'H' and input 1 to set the first point. Then lift the tool, rotate the B-axis (rotation angle should not be too small), and slowly move X and Z axes to reposition the tool tip at the same marked point. Press 'H' and input 2 to set the second point. The system will automatically calculate the distance (um) from tool tip to B-axis rotation center and save it to H2 at the current cursor position. During RTCP B-axis compensation calculation, the system will automatically call the H2 value corresponding to the current tool number.

(2) When machine structure parameter equals 11, 13 or 14:

Used for automatically measuring the distance from tool tip to A-axis rotation center.

First, select the tool number and position the tool tip at a marked point on the workpiece. Press 'H' and input 1 to set the first point. Then lift the tool, rotate the A-axis (rotation angle should not be too small), and slowly move Y and Z axes to reposition the tool tip at the same marked point. Press 'H' and input 2 to set the second point. The system will automatically calculate the distance (um) from tool tip to

A-axis rotation center and save it to H2 at the current cursor position. During RTCP A-axis compensation calculation, the system will automatically call the H2 value corresponding to the current tool number. This measurement must be repeated when changing tools.

Note: Since H2 stores the distance from tool tip to rotation center for RTCP compensation calculation G43/G44 tool length compensation cannot use H2 and must use H1 for each tool. When adjusting Z offset in G54 or H1 values, do not modify H2 values as they represent actual tool lengths and must remain accurate for proper compensation. The program must specify which tool is currently selected for the system to call the corresponding H2 value. G43/G44 tool length compensation can be used simultaneously with M601/M602 RTCP compensation commands.

If tool setting is inaccurate or tools are worn, both H1 and H2 values need incremental adjustment:

In the tool compensation interface, prefix the input with 'H' to modify RTCP H2 tool length values.

Example: Inputting HE123.56 sets H2 to 123.56mm absolutely; inputting H8.6 adjusts H2 incrementally by adding 8.6mm to original value.

Press 'R' on the main interface to automatically measure machine coordinates of rotary table centers, automatically determining axis parameters 553, 554, 555, and 556:

(1) When machine structure parameter equals 10:

Automatically measures Y(P553) and Z(P554) machine coordinates of A-axis rotation center.

First select tool number and position B-axis at zero machine coordinate (perpendicular to X-axis). Move Y and Z axes to position tool tip at a marked point on workpiece (mounted on A-axis). Press 'R' and input 1 for first point. Lift tool, rotate A-axis (preferably by a large angle), then slowly adjust Y and Z to reposition tool tip at same point. Press 'R' and input 2 for second point, then 3 for third point. The system will automatically calculate Y and Z machine coordinates of A-axis rotation center and record current tool length in parameter 555. This measurement is only required during initial installation or A-axis reinstallation, not during tool changes. Measurement of distance from tool tip to B-axis rotation center must be completed first.

Parameter 555 records current tool's H2 length value.

(2) When machine structure parameter equals 11:

This operation is not required.

(3) When machine structure parameter equals 12:

Step 1: Automatically measures Y(P553) and Z(P554) machine coordinates of A-axis rotation center.

Select tool number and move Y/Z axes to position tool tip at marked point on workpiece (A-axis mounted). Press 'R'+1 for first point. Lift tool, rotate A-axis substantially, then readjust Y/Z to same point. Press 'R'+2 for second point, then 3 for third point. System calculates Y/Z coordinates of A/B-axis rotation center. For multiple tools, set H1 compensation values first.

Parameter 555 records current tool's H1 compensation value. If not using G43 tool length compensation, set H1 to zero. Only required during initial installation or A-axis reinstallation.

Step 2: Automatically measures X(P556) machine coordinate of C-axis rotation center.

Select tool number and position A-axis at zero (C-axis horizontal). Move X/Y to position tool tip at marked point (C-axis mounted). Press 'R'+A for first point. Lift tool, rotate C-axis substantially, then readjust X/Y to same point. Press 'R'+B for second point, then C for third point. System calculates X coordinate of C-axis rotation center. Only required during initial installation or C-axis reinstallation. For multiple tools, set H1 values first.

Parameter 555 records current tool's H1 compensation value.

(4) When machine structure parameter equals 15:

Step 1: Automatically measures X(P553) and Z(P554) machine coordinates of B-axis rotation center.

Select tool number and move X/Z to position tool tip at marked point (B-axis mounted). Press 'R'+1 for first point. Lift tool, rotate B-axis substantially, then readjust X/Z to same point. Press 'R'+2 for second point, then 3 for third point. System calculates X/Z coordinates of B-axis rotation center. For multiple tools, set H1 values first.

Parameter 555 records current tool's H1 compensation value. If not using G43, set H1 to zero. Only required during initial installation or B-axis reinstallation.

Step 2: Automatically measures Y(P556) machine coordinate of C-axis rotation center.

Select tool number and position B-axis at zero (C-axis horizontal). Move X/Y to position tool tip at marked point (C-axis mounted). Press 'R'+A for first point. Lift tool, rotate C-axis substantially, then readjust X/Y to same point. Press 'R'+B for second point, then C for third point. System calculates Y coordinate of C-axis rotation center. Only required during initial installation or C-axis reinstallation. For multiple tools, set H1 values first.

Parameter 555 records current tool's H1 compensation value.

Note 1: For BC-type RTCP compensation (not using B-axis), only Step 2 for measuring C-axis X/Y coordinates is needed.

Note 2: B-axis zero must be set when worktable is horizontal, while C-axis zero can be set arbitrarily.

Note 3: In Fanuc mode, G43 command calls H1 for T1, H2 for T2, etc.

(5) When machine structure parameter equals 13:

Automatically measures X(P553) and Y(P554) machine coordinates of C-axis rotation center.

Select tool number and position A-axis at zero (perpendicular to Y-axis). Move X/Y to position tool tip at marked point (C-axis mounted). Press 'R'+1 for first point. Lift tool, rotate C-axis substantially, then readjust X/Y to same point. Press 'R'+2 for second point, then 3 for third point. System calculates X/Y coordinates of C-axis rotation

center and records current tool length in parameter 555. Only required during initial installation or C-axis reinstallation. Measurement of distance from tool tip to A-axis rotation center must be completed first.

Parameter 555 records current tool's H2 length value.

(6) When machine structure parameter equals 14:

Automatically measures X(P553) and Z(P554) machine coordinates of B-axis rotation center.

Select tool number and position A-axis at zero (perpendicular to Y-axis). Move X/Z to position tool tip at marked point (B-axis mounted). Press 'R'+1 for first point. Lift tool, rotate B-axis substantially, then readjust X/Z to same point. Press 'R'+2 for second point, then 3 for third point. System calculates X/Z coordinates of B-axis rotation center and records current tool length in parameter 555. Only required during initial installation or B-axis reinstallation. Measurement of distance from tool tip to A-axis rotation center must be completed first.

Parameter 555 records current tool's H2 length value.

Key considerations when machine structure parameter equals 10 (similar principles apply to other types):

1> Define B-axis zero machine coordinate where it aligns with negative Z-direction, with positive direction following right-hand rule.

2> Define A-axis zero machine coordinate where it aligns with positive Z-direction, with positive direction following right-hand rule.

3> When setting G54 workpiece coordinate Z-axis, first position B-axis at zero machine coordinate (perpendicular to X-axis).

4> Since H2 stores tool tip to B-axis center distance for RTCP B-axis compensation, G43/G44 must use H1 values instead. H2 values must remain accurate to actual tool lengths. Programs must specify current tool number for proper H2 reference. G43/G44 can be used with M601/M602.

5> In MDI mode, entering M601/M602 activates RTCP (indicated on interface). Manual B/A-axis movement will trigger automatic XYZ-axis compensation.

6> RTCP direction verification:

- Set B-axis zero correctly (Z perpendicular to X)
- Measure tool tip to B-axis center distance
- Measure A-axis Y/Z center coordinates
- In MDI, enter M602
- Align tool tip with marked point, then manually rotate B-axis - correct compensation will maintain alignment (adjust parameter 552 if needed)
- Repeat test with A-axis rotation checking Y/Z movement

7> In RTCP mode, machine coordinates show actual positions while workpiece coordinates show virtual positions (matching programmed coordinates).

8> For workpiece coordinate setup, position rotary axes at zero machine coordinates for easier programming.

9> When programming M601/M602, first position XYZAB axes at a safe point in non-RTCP mode, e.g.:

G54 T1

```
M600
G0 X_ Y_ Z_ A_ B_
M601 or M602
G0 X_ Y_ ....
G1 X_ A_ B_ ....
....
M600
M30
```

4.10.6、 Four-axis system RTCP function usage method:

1>When having XYZ linear axes and one A-axis rotating around X-axis (when having XYZ linear axes and one C-axis rotating around Z-axis, you can define Z-axis as X-axis, X-axis as Y-axis, Y-axis as Z-axis, and rotating axis as A-axis):

Enable four axes XYZA and set machine type parameter to 14.

Use M601 to enable RTCP function, use M600 to disable RTCP function.

Press 'H' in system tool compensation interface for automatic tool length measurement.

Automatically measure distance from tool tip to A-axis rotation center: First select tool number, position tool tip at marked point on workpiece, press 'H' and input 1 to set first point. Then lift tool, rotate A-axis (angle should not be too small), slowly move Y and Z axes to reposition tool tip at same marked point, press 'H' and input 2 to set second point. System will automatically calculate distance (um) from tool tip to A-axis rotation center and save to H2 at current cursor position. During RTCP A-axis compensation calculation, system will automatically call H2 value corresponding to current tool number.

2>When having XYZ linear axes and one B-axis rotating around Y-axis:

Enable five axes XYZAB and set machine type parameter to 10.

Since there is no actual A-axis, clear A-axis machine coordinate to zero, then disable A-axis operation by writing PLC ladder diagram.

Use M601 to enable RTCP function, use M600 to disable RTCP function.

Press 'H' in system tool compensation interface for automatic tool length measurement.

Automatically measure distance from tool tip to B-axis rotation center: First select tool number, position tool tip at marked point on workpiece, press 'H' and input 1 to set first point. Then lift tool, rotate B-axis (angle should not be too small), slowly move X and Z axes to reposition tool tip at same marked point, press 'H' and input 2 to set second point. System will automatically calculate distance (um) from tool tip to B-axis rotation center and save to H2 at current cursor position. During RTCP B-axis compensation calculation, system will automatically call H2 value corresponding to current tool number.

4.10.7、 RTCP function testing method:

In manual M602 mode, align tool tip with a point on worktable, rotate A-axis or C-axis, XYZ axes will automatically follow. If tool tip maintains alignment with the point on worktable, it indicates RTCP function is working properly.

Chapter 5 The CNC Machining Center

Note: the function of chapter 5 is only available in NEW1000/1500MiC series controller.

5.1 I/O ports

I/O	Signal	Umbrella type tool magazine	Mechanical hand tool magazine
X00	T01	Magazine motor overload alarm detection	Magazine motor overload alarm detection
X01	T02	Spindle loosening tool detection	Spindle loosening tool detection
X02	T03	Lubricant oil level detection	Lubricant oil level detection
X03	T04	Coolant level detection	Coolant level detection
X04	T05	Spindle tightening tool detection	Spindle tightening tool detection
X05	T06	Magazine forward ready detection	
X06	T07	Magazine backward ready detection Requirement: magazine must backward ready before spindle start.	
X07	T08	Tool count	Tool count
X08	M36/A0	A axis home signal	A axis home signal
X23	ALM1	Spindle alarm	Spindle alarm
X24	ALM2	Cooling motor overload alarm	Cooling motor overload alarm
X25	M28/C0	C axis home signal	C axis home signal
X26	M24/B0	B axis home signal	B axis home signal
X27	M22	Spindle orientation ready/home ready	Spindle orientation ready/home ready
X29	M12	Door switch detection Requirement: Other parameter P7=1	Door switch detection Requirement: Other parameter P7=1
X30	M14	Pressure alarm of compressed air	Pressure alarm of compressed air
X31	M16	Remote button of spindle loosen tool Requirement: Other parameter P22=1	Remote button of spindle loosen tool Requirement: Other parameter P22=1
X40		Chip removal motor overload alarm	Chip removal motor overload alarm

X41			Magazine positioning detect signal, if magazine without positioning signal, please connect with T08
X42			Mechanical hand stop detection;
X43			1. mechanical hand home signal, invalid before changing tool; 2. servo magazine loosening/tighten ready input
X44			Tool-case raising ready detection input. Requirement: must raise up ready before changing tool
X45			Tool-case falling down ready detection input
X46		Pause instruction M01 detect input	Pause instruction M01 detect input
X47		Spindle fan motor alarm	Spindle fan motor alarm
Y00	M61	Spindle home output, detect home ready M22	Spindle home output, detect home ready M22
Y01	M63	Magazine rotate CW output, detect T08 to count tool;	Magazine rotate CW output, detect T08 to count tool;
Y02	M65	Yellow lamp (Stop)	Yellow lamp (Stop)
Y03	M67	Red lamp (Fault)	Red lamp (Fault)
Y04	M69	Green lamp (Running)	Green lamp (Running)
Y05	M71	Magazine rotate CCW output, detect T08 to count tool;	Magazine rotate CCW output, detect T08 to count tool;
Y06	M73	Magazine forward output, T06 is forward ready input;	Magazine forward output, T06 is forward ready input;
Y07	M59	M59 huff output	M59 huff output
Y08	M32	Lubrication output	Lubrication output
Y10	M10	Loosening tool output: Loose tool: output M10, T02 is loose ready input; Tighten tool: cancel M10, T05 is tighten ready input;	Loosening tool output: Loose tool: output M10, T02 is loose ready input; Tighten tool: cancel M10, T05 is tighten ready input;
Y11	M08	Cooling output	Cooling output
Y12	M05	M05	M05
Y13	M04	Spindle rotate CCW output	Spindle rotate CCW output
Y14	M03	Spindle rotate CW output	Spindle rotate CW output
Y15	M75	Spindle control mode switch	Spindle control mode switch

Y24		Tool magazine backward controlling output, checking backward enough T07	Mechanical hand motor control output
Y25			Servo magazine loosen/tighten output
Y26			Tool case raise-up output; X44 is raise-up ready input
Y27			Tool case fall-down output; X44 is fall-down ready input; Magazine positioning ready input X41 must be valid.
Y28			
Y29		Retract tool signal during rigid tapping	Retract tool signal during rigid tapping
Y30			
Y31	SPEN2	Spindle servo enable signal output	Spindle servo enable signal output

5.2 Parameter of CNC machining center

(A) Axis parameter

P200, the number of encoder pulse per turn(4 times value of encoder resolution)
P200-1,SP encode pulse alarm (diagnostics)[>10:Valid]
P200-2,SP encode pulse[4 times encode thread](When Encode:SP is not 1:1)
P400, Check SP-encode feed back(0 Yes,1 No)
P401, check angle when spindle orientation
P402, check error tolerance during spindle orientation
P404, controlling signal of spindle pulse (0 means negative, 1 means normal)
P405, spindle whether or not to use electronic gear (0 means use, 1 means not to use)
P406, molecular of spindle low-gear electronic gear (1-32767)
P407, the denominator spindle low-gear electronic gear (1-32767)
P408, molecular of spindle high-gear electronic gear (1-32767)
P409, the denominator of spindle high-gear electronic gear (1-32767)

(B) Tool parameter

The total number of tool: press "F7" set in "Redeem"(tool compensation) interface.
10, Change tool function[1 Yes,0 No]
11, Go home axis when change tool[3 mean ZX; 4 mean ZY; 5 mean ZXY;others means Z]
12, Go home reference point number when change tool [1,2,3,4]
13, Check SP fix position [1 Yes,0 No]
14, SP go to fix position when change tool[1 Yes,0 No]
15, Tools magazine type [0 mechanical hand,1 Line tools,2 Umbrella]
16, Special Tools magazine[0:Standard,16:Special umbrella,32:Servo magazine,64:Special mechanical hand,128:linear tool]

Note:

- +2: spindle stops after tool changing;
- +4: mask K5 button;
- +8: simultaneous tool unloading structure;
- +16: special simplified umbrella type magazine, controlled through ProgramTool;
- +128: Linear tool, or special tool magazine, execute T command through "ProgramTool" macro program;
- +256: mechanical hand magazine need to move forward or backward;
- +512: special tool magazine, whose tool plate rotation is controlled by ProgramTool;
- +1024: exchange function of M06 and M106;
- +2048: execute M06 and T command in one line according to sequence.

For example: M06Txx, TxxM06, TxxM06Txx.

- 17, Z-axis lift distance when change change tool in umbrella type magazine
- 17-1, Z axis G00 lift distance (mm) when change change tool in umbrella type magazine
- 18, Z-axis speed When change change tool in umbrella type magazine
- 19, SP back tool delay (0.1s) during tool change in umbrella type magazine
- 20, SP loose tool delay (0.1s) during tool change in umbrella type magazine
- 20-1, Delay (0.1s) before tighten tool when Z axis fall ready during tool change in umbrella type magazine
- 21, Tool seat control delay(ms)
- 22, Interlock between Z-axis move & umbrella magazine(0 mean no,1 mean Yes)
- 23, Check Z axis position before instruction M71 is valid(1 mean no,0 mean Yes)
- 24, Maximum tool number (setup this parameter ,must initialize tool table)
- 25, Tool counting input signal [rise"1000+input number",drop"2000+input number"]
- 26, Magazine positioning output signal [1000+output number]
- 27, Magazine rotate CW output signal [1000+output number]
- 28, Magazine rotate CCW output signal [1000+output number]
- 29, Mechanical hand brake input signal [1000+input number]
- 30, Mechanical hand motor output signal [1000+output number]
- 32, Tool selection/number tool signal[D0=0:one way selection tool,D0=1:Two-way selection tool,D1-D14:check manual]
- D0=1:Tool section method is random in 2 direction;
- D1=1(+2): tool counting signal reverse direction;
- D2=1(+4): filter tool counting signal;
- D3=1(+8): filter mechanical hand stop signal;
- D4=1(+16): tool counting signal and mechanical hand stop signal use 8ms filter method[otherwise 4ms filter];
- D5=1(+32): in mechanical hand magazine, after T command, tool seat fall down automatically. when this function is on, after M6, tool seat will not lift again automatically, until execute T command again.
- D6=1(+64): in mechanical hand magazine, after execute last step of tool changing, no need to detect raise-up ready signal before run next section.

D7=1(+128): when execute T code, no need to wait tool seat up before running next section.

D8=1(+256): do 2ms filter for tool position signal in lathe machine;

D9=1(+512): do 4ms filter for tool position signal in lathe machine;

D10=1(+1024): do 8ms filter for tool position signal in lathe machine; D8-D10 can be 1 at the same time;

D11=1(+2048): do 2ms filter for WAT input signal;

D12=1(+4096): do 4ms filter for WAT input signal;

D13=1(+8192): do 8ms filter for WAT input signal; D11-D13 can be 1 at the same time;

D14=1(+16384): mechanical hand magazine is special designed type, no need to set.

34, Set up tool counting signal filtering time (ms)[>8]

35, The interval between magazine rotation and tool counting signal(ms)[>8]

70, Servo magazine go home speed(degree/min)

71, Servo magazine go home reverse speed(degree/min)

72, Servo magazine go home offset(0.001degree)

73, Servo magazine choose tool speed(degree/min)

74, Servo magazine axis(1=A; 2=B; 3=C)

75, Servo magazine acceleration(degree/min/s)

76, Servo magazine go home max distance for zero signal(0.001degree)

77, Servo magazine go home check motor zero signal(0:is No, 1:is Yes, 367: Servo motor)

78, Servo magazine one tool-position pulse output

79, Servo magazine go home tool No.

100, Reference point 1 X(mm)

101, Reference point 1 Y(mm)

102, Reference point 1 Z(mm)

103, Reference point 1 A(mm)

104, Reference point 2 X(mm)

105, Reference point 2 Y(mm)

106, Reference point 2 Z(mm)

107, Reference point 2 A(mm)

108, Reference point 3 X(mm)

109, Reference point 3 Y(mm)

110, Reference point 3 Z(mm)

111, Reference point 3 A(mm)

112, Reference point 4 X(mm)

113, Reference point 4 Y(mm)

114, Reference point 4 Z(mm)

115, Reference point 4 A(mm)

(B) Speed parameter

230, SP G00 speed(0.1rpm)

231, SP G01 mode[+4: F, +8: G90/G91, +16: show value]

- 0: spindle home is controlled by pulse;
- +4: when SP command run G01, speed is current F speed[degree/min], otherwise follow P230 speed.
- +8: SP command is controlled by current G90/G91 status, if G90, it means absolute coordinate and reduction, G91 means incremental and without reduction.
- +16: start to calculate endpoint coordinate based on current value.
- 232,SP orientation direction under interpolation mode[0: Positive,1: Negative]
Only works in G90 or parameter setting out control of G90/G91.
- 233,SP home control mode(1:pulse,2: Output M61,check M22)
1: pulse control;
2: driver control, controller output M61, and detect M22;
16/32: firstly stop spindle, then controller outputs M75, and check M22(if 32, not check M22). after that,controller will calculate current spindle position according to encoder feedback, then position to degree defined by P234 with P230 speed. If position degree offset is big, try to set Axis parameter P56=16 to reverse encoder feedback direction.
- 234,SP stop angle(0.1)
- 235,SP back to Zero speed(0.1 rpm)

5.3 Debugging of CNC machining center

Definition of "Tab HxDx cdTef" on the screen, "ab," means current tool number in spindle, "cd" means the current tool case position number in tool magazine, "ef" means the current tool number in tool magazine.

5.3.1 The standard umbrella type magazine

1, the manual button operation

K1 is spindle orientation (lamp K1 will brighten after finishing spindle orientation): output M61, check M22 ;

K2 is Z axis go to tool change point;

K3 is setting the tool case number ;

"Tool magazine rotate CW": output M63, check T08 ;

"Tool magazine rotate CCW": output M71, check T08 ;

2, M instruction

M71 is tool magazine forward, cancel Y24, output M73, tool parameter P23=1;

M73 is tool magazine backward, cancel M73, output Y24, tool parameter P23=1;

M881 is the same as "K1 function";

3, Tool changing operation

Txx: Move the tool of spindle back to the tool case and move the Txx to spindle.

M36 Txx: Move the Txx to spindle according to the step in single step mode. After the executing a step to be in a suspended state, continue to execute the next step after pressing button "Run", mainly use for debugging.

4, Set the offset of spindle orientation(spindle encoder signal must access into

system)

- 1) P400=1, check spindle position feedback when changing tool, set to "0" not to check the position feedback;
- 2) set P401, check angle when spindle orientation, manual press K1 to import SP value in P401 after finishing spindle orientation;
- 3) set P402, check the error of angle when spindle orientation;

5, Tool changing process

- 1) Z axis to changing tool point (Z axis must return to zero first);
- 2) Spindle orientation: output M61, check the backing zero in place M22;
- 3) Tool goes forward: output M73, check going forward in place T06;
- 4) Spindle tool loosen: output M10, check loosening tool in place of T02;
- 5) Z axis lifting;
- 6) Tool is rotating CW (or CCW): output M63 (or M71), check tool position counting T08;
- 7) spindle loose tool: output M10, detection of loosen tool in place of T02;
- 8) Z axis falls down to the tool changing point;
- 9) Spindle tighten tool: cancel M10, check tightening tool in place T05;
- 10) Tool backward: output Y24, check backward in place T07;

6, Tool parameter

- P10=1, start the tool changing program;
- P11=0, just Z back to the tool changing point;
- P12=2, feeding axis back to the second datum point when changing tool;
- P13=1, check spindle orientation and the tool changing point of feeding axis before changing tool, check M22 signal ;
- P14=1, spindle orientation when changing tool, output signal M61;
- P15=2, the tool magazine type is "bamboo hat type";
- P16=0, the standard tool magazine;
- P17=120, the lifting height of Z axis when changing tool(mm);
- P18=2000, the raising speed of Z axis when tool magazine changing tool (mm/min);
- P19=5, bamboo hat type to back tool delay (0.1 sec), tool magazine goes forward check the delay time after T06 is in the place and spindle tighten tool check the delay time after T05 is in the place;
- P20=5, delay time(0.1 sec) after loosening tool when changing tool, loosen tool check the delay time after T02 is in the place;
- P21=5, delay time after tool case lifting (ms);
- P22=1, whether the Z axis motion and main hand tool / bamboo hat type tool magazine is interlocking [1 means yes, 0 means no], set to "1" means check the signal T07 of tool magazine backward in place;
- P23=1, the tool goes forward instruction M71 is to check whether the position of Z axis[0 means yes, 1 means no], when setting to "0", to check the Z axis is at the tool changing position;
- P24=0, the maximum tool number of fixed tool position area [after setting the parameter, please initialize the tool case table];

P25=1007 (X07), signal of tool magazine count [rising along "1000+ number", "down along "2000+ number"];

P26=1041 (X41), signal of tool magazine locating[1000+ number];

P27=1001 (Y01), output point of tool magazine rotating CW [1000+ number];

P28=1005 (Y05), output point of tool magazine rotating CCW [1000+ number];

P100=0, reference point1 X (mm) ;

P101=0, reference point1 Y (mm) ;

P102=0, reference point1 Z (mm) ;

P103=0, reference point1 A (mm) ;

P104=0, reference point2 X (mm) ;

P105=0, reference point2 Y (mm) ;

P106=50, reference point2 Z (mm) ;

P107=0, reference point2 A (mm) ;

Special attention:

1. When power off suddenly or emergency happening cause the tool case is in a mess in tool changing process, please rotate the tool case for some tool position in manual mode and use K3 to set the current tool case number;
2. Pay attention to check position of Z axis in case for accident when using instruction M71 to make tool case going forward;
3. No.T0 tool should not have tool, otherwise may be an accident;
4. The first time to install tool, spindle must have tool (if the display is T00, usually can initialize the tool case table, spindle should be No.T01 tool).

5.3.2 The standard mechanical hand tool magazine

1. The manual button operation

K1 is spindle orientation (pilot lamp K1 will brighten after finishing spindle orientation): output M61, check M22 ;

K2 is the backing changing point of Z axis;

K3 is setting the tool case number of the current tool case;

"Tool magazine rotate CW": output M63, check T08, the tool case must lift to position X44;

"Tool magazine rotate CCW": output M71, check T08, the tool case must lift to position X44;

2. M instruction

M71 is tool magazine going forward, cancel Y26, output M27, tool case locating check the X41 must in place;

M73 is tool magazine backward, cancel M27, output Y26;

M65 is unconditional rotation step of mechanical hand for debugging,input Y24;

M881 is the same as "K1 function";

3. Tool changing operation

M06: Move the tool of current tool case to spindle.

Txx: Move the Txx of instruction to the current tool changing position.

M06 Txx: Change the tool of current tool case into spindle and move the Txx of instruction into tool changing position for the next time tool changing(First choose

tool and then change tool).

M106 Txx: Move the Txx of instruction into tool changing position and then change the current tool of tool case into spindle.(First change tool and then choose tool)

M36: Move the Txx to spindle according to the step in single step mode. After the executing a step to be in a suspended state, continue to execute the next step after pressing button "Run", mainly use for debugging.

4. Set the offset of spindle orientation(spindle encoder signal must access into system)

1) P400=1, check spindle position feedback when changing tool, set to "0" not to check the position feedback;

2) set P401, check angle when spindle orientation, manual press K1 to import SP value in P401 after finishing spindle orientation;

3) set P402, check the error of angle when spindle orientation;

5. Tool changing process

1) Z axis to changing tool point (Z axis must return to zero first);

2) Spindle orientation: output M61, check the backing zero in place M22;

3) Fallen tool case: output Y27, check fallen in place X45, tool magazine locating check X41 must be in place;

4) Mechanical hand clasp tool: output Y24, check clasp tool in place X42;

5) Spindle loose tool: output M10, detection of loosen tool in place of T02;

6) Mechanical hand take tool and change tool: output Y24, check take tool in place X42;

7) Spindle tighten tool: cancel M10, check tightening tool in place T05;

8) Mechanical hand return: output Y24, check return in place X42;

9) Raise up tool case: output Y26, check raising in place X44, if not detect X44, it will alarm "Tool magazine fault";

Special attention:

1. When power off suddenly or emergency happening cause the tool case is in a mess in tool changing process, please rotate the tool case for some tool position in manual mode and use K3 to set the current tool case number;

2. For this kind of tool magazine, could set the No.24 parameter in tool parameter to set the maximum tool number of fixed tool position area, it's better to initialize the tool case table after setting the parameter. For example: Set to 8 stand for just could from 1 to 8 tool case table to put in the tool with corresponding number by one-one, that's not random, it's fixed to install. This function could be used for installing tool of large diameter cutter;

3. No.T0 tool should not have tool, otherwise may be an accident;

4. M65 doesn't check the condition, only stop in emergency, pay more attention when using, otherwise to cause the safe accident.

6. Tool parameter setting

P10=1, start the tool changing program;

P11=0, just Z back to the tool changing point;

P12=2, feeding axis back to the second datum point when changing tool;

P13=1, check spindle orientation and the tool changing point of feeding axis before changing tool, check M22 signal ;

P14=1, spindle orientation when changing tool, output signal M61;

P15=2, the tool magazine type is "bamboo hat type";

P16=0, the standard tool magazine;

P17=120, the lifting height of Z axis when changing tool(mm);

P18=2000, the raising speed of Z axis when tool magazine changing tool (mm/min);

P19=5, bamboo hat type to back tool delay (0.1 sec), tool magazine goes forward check the delay time after T06 is in the place and spindle tighten tool check the delay time after T05 is in the place;

P20=5, delay time(0.1 sec) after loosening tool when changing tool, loosen tool check the delay time after T02 is in the place;

P22=1, whether the Z axis motion and main hand tool / bamboo hat type tool magazine is interlocking [1 means yes, 0 means no], set to "1" means check the signal T07 of tool magazine backward in place;

P23=1, the tool goes forward instruction M71 is to check whether the position of Z axis[0 means yes, 1 means no], when setting to "0", to check the Z axis is at the tool changing position;

P24=0, the maximum tool number of fixed tool position area [after setting the parameter, please initialize the tool case table];

P25=1007 (X07), signal of tool magazine count [rising along "1000+ number", "down along "2000+ number"];

P26=1041 (X41), signal of tool magazine locating[1000+ number];

P27=1001 (Y01), output point of tool magazine rotating CW [1000+ number];

P28=1005 (Y05), output point of tool magazine rotating CCW [1000+ number];

P29=1042 (X42), braking output point of mechanical hand [1000+ number]

P30=1024 (Y24), rotating output point of mechanical hand [1000+ number]

P100=0, reference point1 X (mm) ;

P101=0, reference point1 Y (mm) ;

P102=0, reference point1 Z (mm) ;

P103=0, reference point1 A (mm) ;

P104=0, reference point2 X (mm) ;

P105=0, reference point2 Y (mm) ;

P106=50, reference point2 Z (mm) ;

P107=0, reference point2 A (mm) ;

5.3.3 Special umbrella type tool magazine

1. Parameter setting: tool parameter P16=16, tool case is set to "special bamboo hat".

2. Editing tool changing program ProgramTool deliver into system. ProgramTool programming from the tool changing process of the third) step begins, other operation is the same as standard bamboo hat type tool magazine.

5.3.4 Special mechanical hand tool magazine

1. Parameter setting: tool parameter P16=64, tool magazine is set to "special mechanical hand"; tool parameter P29=0, P30=0.
2. Editing tool changing program ProgramM6 deliver into system. ProgramM6 programming from the tool change process of the third) step begins, other operation is the same as standard mechanical hand tool magazine.

5.3.5 Linear type tool magazine

When start up controller, just need to press Enter, and controller will pop up a dialog for linear tool atc configuration.

5.3.6 Match mechanical disk type of mechanical hand

1. Parameter setting: tool parameter P16=64, tool case is set to "special mechanical hand"; tool parameter P29=0 P30=0.
2. Editing tool changing program ProgramM6 deliver into system. ProgramM6 programming from the tool change process of the third) step begins, other operation is the same as standard mechanical hand tool magazine.

3. M instruction

M71 is tool case fallen down, cancel Y26, output M27, tool case locating check the X41 must in place;

M73 is tool case raising up, cancel M27, output Y26;

OUT+Y30 is an unconditional step for rotating of mechanical hand, be used for debugging, output Y30;

M881 is the same as "K1 function";

4. The signal connection table of tool magazine(input signal of tool magazine through 2803 negative):

I/O	Signal	Mechanical hand type of tool magazine	mechanical disk type of tool magazine
X00	T01	Detection for the motor of tool magazine overload	
X01	T02	Detection for loosening tool of spindle	
X02	T03	Detection for lubricating oil level	
X03	T04	Detection for lacking coolant	
X04	T05	Detection for tightening tool of spindle	
X05	T06		Confirm signal of clasping tool S6 (Right)
X06	T07		Confirm signal of the original point S7 (Left)

X07	T08	Tool magazine count	Tool magazine count S1
X23	ALM1	Driver of spindle alarm	
X24	ALM2	Detection for cooling motor overload	
X27	M22	Spindle finish locating/Detection for backing to zero position	
X29	M12	Detection for lathe door switch Requirement: P7=1 in other parameter	
X30	M14	Detection for pressure alarm of compressed air	
X31	M16	Button for spindle loosen tool remotely Requirement: P22=1 in other parameter	
X40		Detection for motor of chip removal overload alarm	
X41		Detection for tool magazine locating	Locating signal of tool magazine S2
X42		Stop checking the Mechanical hand tool magazine:	Confirm signal of braking S5 (Middle)
X43			
X44		Detection for tool-case raising up Requirement: Raise up enough before changing tool	Locating signal of tool returning S4
X45		Detection for tool-case falling down	Locating signal of tool fallen S3
X46		Stop instruction M01 input	
X47		Detection for fan of spindle overload	
Y00	M61	Backing to zero of spindle output, detection for backing to zero enough M22	
Y01	M63	Tool magazine rotate CW controlling output, Checking tool count of T08;	Motor of tool changing motor rotate CW
Y02	M65	Yellow indicator lamp (Stop)	
Y03	M67	Red indicator lamp (Fault)	
Y04	M69	Green indicator lamp(Running)	
Y05	M71	Tool magazine rotate CCW controlling output, Checking tool count of T08;	Motor of tool changing motor rotate CCW

Y06	M73		
Y07	M59	M59 Huff	
Y08	M32	Lubrication controlling output	
Y10	M10	The control output elastic tool: Spindle loose tool: output M10, detection for loosening enough T02; Spindle tighten tool: cancel M10, detection for tightening enough T05;	
Y11	M08	Cooling controlling output	
Y12	M05	M05	
Y13	M04	Spindle rotate CCW controlling output	
Y14	M03	Spindle rotate CW controlling output	
Y15	M75	Switch controlling mode of spindle output	
Y24			
Y25			
Y26		Tool case raise up controlling output, checking raise up in place X44	Solenoid valve of tool returning S10
Y27		Tool case fall down controlling output, checking fall down in place X45, detection for tool magazine locating must be in place;	Solenoid valve of tool fallen S8
Y28		Feeding servo driver with energy	Feeding servo driver with energy
Y29		Servo driver of spindle with energy	Servo driver of spindle with energy
Y30		Motor controlling output of mechanical hand	Motor of the tool magazine structure
Y31			

5.3.7 Example: Umbrella type ATC

If equip with spindle servo of GSK, pay attention as follows:

System M75 connects servo position of speed switching signal;

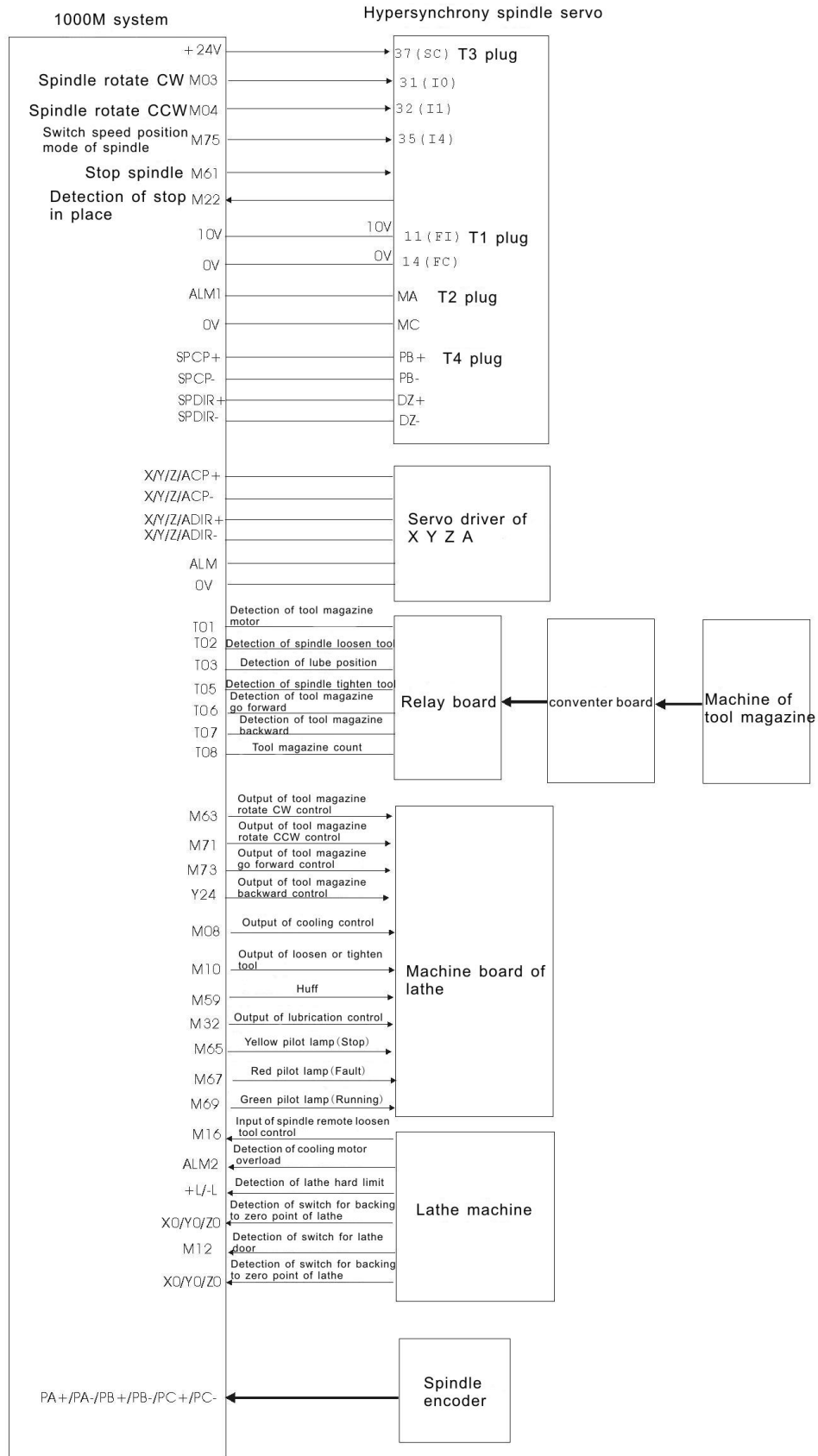
System M22 connects servo position of finish speed switching signal;

System M61 connects servo the starting orientation signal;

System M03/M04/M05/+10V/ALM1/+24V/0V connects corresponding signal of driver;

Servo drive parameter should be set to the position / speed controlling mode.

Connection of 1000M bamboo hat type of CNC machining center





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